# Amath753 Advanced PDEs

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This note is based on the book "Partial Differential Equations" by Lawrence C. Evans [2] and the course AMATH753 in Winter 2025.

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## Contents

**1 Preliminaries** **3**
- 1.1 Introduction . . . 3
- 1.2 Metric Spaces and Complete Spaces . . . 4
  - 1.2.1 Compactness . . . 5
- 1.3 Banach Spaces . . . 5
- 1.4 Hilbert Spaces . . . 5
- 1.5 Bounded linear operators . . . 8
  - 1.5.1 Dual Space . . . 9
  - 1.5.2 Adjoint Operator . . . 11
  - 1.5.3 Compact Operators . . . 12
- 1.6 Function Spaces . . . 14
  - 1.6.1 Continuous functions . . . 14
  - 1.6.2 Lebesgue Spaces . . . 15

**2 Sobolev Spaces** **18**
- 2.1 Holder Spaces . . . 18
- 2.2 Convolution and Mollification . . . 18
- 2.3 Weak Derivative and Sobolev Spaces . . . 21
  - 2.3.1 Weak Derivatives . . . 21
  - 2.3.2 Sobolev Spaces . . . 26
- 2.4 Smooth Approximation . . . 29
- 2.5 Extensions . . . 34
- 2.6 Traces . . . 37
- 2.7 Weak and classical derivatives . . . 38
- 2.8 Sobolev Inequalities . . . 40
- 2.9 Compactness . . . 42
- 2.10 Poincare Inequalities . . . 46
- 2.11 Sobolev Spaces with Negative Orders . . . 48
- 2.12 Difference Quotients . . . 50
- 2.13 Weak Derivative Calculus . . . 53
- 2.14 Fourier Transform . . . 57
- 2.15 Sobolev Spaces with Real Orders . . . 59

**3 Elliptic PDEs** **63**
3.1 Weak Solutions . . . 63
3.2 Existence of weak solution . . . 64
3.2.1 First Existence Theorem . . . 64
3.2.2 More Existence Theorems . . . 68
3.3 Regularity . . . 74

**4 Parabolic PDEs** **84**
4.1 Spaces Involving Time . . . 84
4.1.1 Bochner Spaces . . . 84
4.1.2 Sobolev Spaces In Time . . . 88
4.2 Second Order Parabolic Equations . . . 95
4.3 Galerkin Method . . . 96

**5 Semi-group Theory** **107**

# 1 Preliminaries

See more in Amath731-Functional Analysis Notes from Prof. Giang Tran, and my Pmath651 Measure Theory and Pmath753 Functional Analysis Notes.

## 1.1 Introduction

**Definition 1.1.** We will use the following notations:

- $C$ means a positive constant.
- $U \subset \mathbb{R}^n$ is open.
- If $u : U \to \mathbb{R}$ is a function, we write $u(x) := u(x^1, \dots, x^n)$ for $x = (x^1, \dots, x^n) \in U$.
- A function $u$ is **smooth** if $u \in C^\infty(U)$.
- For $1 \le i \le n$, we write $\partial_i u := u_{x^i} := u_i := D_i u := \frac{\partial}{\partial x^i} u := \frac{\partial u}{\partial x^i}$.
- Let $\alpha = (\alpha_1 \dots, \alpha_n) \in \mathbb{N}^n$. We write $|\alpha| := \sum_{i=1}^n \alpha_i$, and
$$D^\alpha u := \frac{\partial^{|\alpha|} u}{\partial_{x^1}^{\alpha_1} \cdots \partial_{x^n}^{\alpha_n}} = \partial_{x^1}^{\alpha_1} \cdots \partial_{x^n}^{\alpha_n} u.$$
- If $k \in \mathbb{N}$, we let $D^k u(x) := \{D^\alpha u(x) : |\alpha| = k\}$
- When $k = 1$, we write $Du := D_x u := (u_{x^1}, \dots, u_{x^n})^T = \nabla u$ to be the **gradient**.
- When $k = 2$, we write $D^2 u := \begin{pmatrix} u_{x^1,x^1} & \cdots & u_{x^1,x^n} \\ \vdots & & \vdots \\ u_{x^n,x^1} & \cdots & u_{x^n,x^n} \end{pmatrix}$ to be the **Hessian** matrix.
- $\Delta u := \sum_{i=1}^n u_{x^i,x^i} = div\ Du = tr(D^2 u)$ is the **Laplacian** of u.
- Let $x \in \mathbb{C}^n$, we write $x^\alpha := \prod_{j=1}^n (x^j)^{\alpha_j}$.

**Example 1.1.1.** Consider a body $U \subset \mathbb{R}^3$ and let $U_0 \subseteq U$ with boundary $\partial U_0$, which does not change over time.
The Conservation of Energy states that the rate of change of total energy in $U_0$ is the inflow of heat through the boundaries plus heat produced by the source in $U_0$.
Let $e(x,t) \in \mathbb{R}$ be the density of internal energy, then the total energy is $\int_{U_0} e dx$.
Let $j(x,t) \in \mathbb{R}^3$ be the heat flux (vector pointing at the direction that heat is flowing).
Let $n$ denote the exterior unit normal on $\partial U_0$.
The net outflow of the heat through $\partial U_0$ is $\int_{\partial U_0} j \cdot n ds$.
Let $p(x,t) \in \mathbb{R}$ be the power density of the source. Heat production in $U_0$ is $\int_{U_0} p dx$.
Thus we have
$$\frac{d}{dx} \int_{U_0} e dx = -\int_{\partial U_0} j \cdot n ds + \int_{U_0} p dx.$$
By divergence theorem, we have $\int_{\partial U_0} j \cdot n ds = \int_{U_0} div\ j dx$.
Thus we have
$$\int_{U_0} (\partial_t e + div\ j - p) dx = 0.$$
Since $U_0$ is arbitrary, we must have
$$\partial_t e + div\ j - p = 0.$$
Assume that $e$ depends linearly on temperature $T$ as $e = e_0 + \sigma u$, where $e_0$ is a constant reference internal energy, and $u = T - T_0$, where $T_0$ is a constant reference temperature, and $\sigma$ is the specific heat capacity.
A generalized form of Fourier's law states that:

- Heat flow is proportional to the temperature gradient.
- Heat is transformed by convection with heat flux $be$, where $b(x,t) \in \mathbb{R}^3$ is a given convection velocity.

Namely, $j = -aDu + be$, where $a(x)$ is a known heat conductivity.
Thus we have

$$\sigma \partial_t u + div(b\sigma u) - div(aDu) = p - div(be_0).$$

**Definition 1.2.** We consider the operator

$$Lu := -\sum_{i,j=1}^{n} \left(a^{ij} u_{x^i}\right)_{x^j} + \sum_{i=1}^{n} b^i u_{x^i} + cu,$$

for given coefficients $a^{ij}, b^i, c$.

- The second-order elliptic boundary-value problems are $\begin{cases} Lu = f & \text{in } U \\ u = 0 & \text{on } \partial U \end{cases}$
- The second-order parabolic boundary-value problems are $\begin{cases} u_t + Lu = f & x \in U, t \in (0,T] \\ u = 0 & \text{on } \partial U, t \in (0,T] \\ u = u_0 & \text{on } \partial U, t = 0 \end{cases}$

**Example 1.1.2.** Some special cases are

- Laplace equation: $-\Delta u = 0$
- Poisson's equation: $-\Delta u = f$
- Heat equation: $u_t - \Delta u = 0$

## 1.2 Metric Spaces and Complete Spaces

**Definition 1.3.** A **metric space** is a set $X$ that has a (distance) metric:

$$\begin{aligned} d(\cdot,\cdot) : X \times X \to \mathbb{R}, \text{ such that } \forall x, y, z \in X \\ d(x,x) = 0 \\ \forall x \neq y, d(x,y) > 0 \\ d(x,y) = d(y,x) \\ d(x,z) \leq d(x,y) + d(y,z) \end{aligned}$$

**Definition 1.4.** Given a metric space $(X,d)$, a sequence $(x_n)_{n=1}^{\infty}$ in $X$ has a **limit point** $x \in X$ if $\lim\limits_{n\to\infty} d(x,x_n) = 0$. In this case, we say $(x_n)_{n=1}^{\infty}$ is a **convergent sequence**, and write $x = \lim_{n\to\infty} x_n$.

**Definition 1.5.** A sequence $(x_n)_{n=1}^{\infty}$ is a **Cauchy sequence** in a metric space $(X,d)$ if

$$\forall \epsilon > 0, \exists N \in \mathbb{N}_+, \text{ such that } \forall m, n \in \mathbb{N},\ m, n > N \implies d(x_m, x_n) < \epsilon.$$

**Definition 1.6.** A metric space $X$ is **complete** if every Cauchy sequence $(x_i)_{i=1}^{\infty}$ converges to a limit point in $X$. i.e. $\exists x \in X, \lim\limits_{i\to\infty} x_i = x$.

**Proposition 1.1.** *Let $(X,d)$ be a metric space, then every convergent sequence is Cauchy.*

**Proposition 1.2.** *Let $(X,d)$ be a metric space. If $(x_n)_{n=1}^{\infty}$ is a Cauchy sequence and has a convergent subsequence such that $\lim_{k\to\infty} x_{n_k} = x \in X$, then $\lim_{n\to\infty} x_n = x$.*

### 1.2.1 Compactness

*Remark.* See the definition of compactness and more in Section 2.9 of AMATH731 Notes from Prof. Tran.

**Definition 1.7.** Let $(X, d)$ be a metric space. A set $S \subseteq X$ is **sequentially compact** if every sequence $(x_i)_{i=1}^{\infty}$ in $S$ has a convergent subsequence whose limit is in $S$. Namely, $\exists x \in S$, such that $x = \lim_{j\to\infty} x_{i_j}$ for some choice of $i_j$'s.

*Remark.* In metric spaces, sequentially compact and compact are equivalent, so we will just use this as the definition for compactness.

**Definition 1.8.** Let $(X, d)$ be a metric space. A set $S \subseteq X$ is **relatively compact**, or **pre-compact** if its closure $\overline{S}$ is compact in $X$.

**Proposition 1.3.** *Let $(X, d)$ be a metric space, then $S \subseteq X$ is relatively compact iff for any sequence $(x_n)_{n=1}^{\infty} \subseteq S$, it has a convergent subsequence $(x_{n_k})_{k=1}^{\infty}$, such that $x_{n_k} \to x$ for some $x \in X$.*

## 1.3 Banach Spaces

**Definition 1.9.** A **normed vector space** is a vector space $(X, \|\cdot\|)$ over a field $\mathbb{F}$ endowed with a norm (length) function: $\|\cdot\| : X \to [0, \infty)$, such that $\forall x, y \in X, a \in \mathbb{F}$, it satisfies

1. subadditivity (triangular inequality); i.e. $\|x + y\| \leq \|x\| + \|y\|$,
2. absolute homogeneity; i.e. $\|a \cdot x\| = |a|\|x\|$, and
3. positive definiteness; i.e. if $x \neq 0$, we must have $\|x\| > 0$.

**Proposition 1.4.** *For every **normed space** with $\|\cdot\|$, there is a metric $d(x, y) = \|x - y\|$.*

*Proof.*

$$\begin{aligned} d(x,x) &= \|x - x\| = \|0\| = 0 \\ \forall x \neq y, d(x,y) &= \|x - y\| > 0 \\ d(x,y) &= \|x - y\| = \|-(y - x)\| = |-1|\|y - x\| = \|y - x\| = d(y,x) \\ d(x,z) &= \|x - z\| = \|x - y + y - z\| \geq \|x - y\| + \|y - z\| = d(x,y) + d(y,z) \end{aligned}$$

Thus $d(x, y) = \|x - y\|$ is a metric. □

**Definition 1.10.** A normed space is called a **Banach space** if it is complete.

**Definition 1.11.** Let $(X, \|\cdot\|)$ be a Banach space, a subset $A \subseteq X$ is **dense** in $X$ if the closure $\bar{A} = X$.

**Definition 1.12.** A Banach space is **separable** if there is a dense countable subset of it.

## 1.4 Hilbert Spaces

**Definition 1.13.** An **inner product space** is a vector space $H$ that has an inner product: $\langle \cdot, - \rangle : H \times H \to \mathbb{C}$, such that $\forall u, v, w \in H, a, b \in \mathbb{C}$, it satisfies

1. conjugate symmetry; i.e. $\langle v, w \rangle = \overline{\langle w, v \rangle}$,
2. linearity in the first argument; i.e. $\langle au + bw, v \rangle = a\langle u, v \rangle + b\langle w, v \rangle$, and
3. positive definiteness; i.e. $\langle v, v \rangle \geq 0$, and if $v \neq 0$, we must have $\langle v, v \rangle > 0$.

**Lemma 1.5.** *For every inner product space with $\langle \cdot, - \rangle$, and $x, y \in H$, we have*

$$\langle x, y \rangle + \langle y, x \rangle = 2\Re(\langle x, y \rangle) = 2\Re(\langle y, x \rangle),$$

*which is twice the real part of $\langle x, y \rangle$. Similarly,*

$$\langle x, y \rangle - \langle y, x \rangle = 2\Im(\langle x, y \rangle) = -2\Im(\langle y, x \rangle),$$

*which is twice the imaginary part of $\langle x, y \rangle$.*
*Also, we have*

$$\langle x, y \rangle \langle y, x \rangle = |\langle x, y \rangle|^2.$$

*Proof.*

$$\begin{aligned}
\langle x, y \rangle + \langle y, x \rangle &= \langle x, y \rangle + \overline{\langle x, y \rangle} \\
&= 2\Re(\langle x, y \rangle) \\
\langle x, y \rangle - \langle y, x \rangle &= \langle x, y \rangle - \overline{\langle x, y \rangle} \\
&= 2\Im(\langle x, y \rangle) \\
\langle x, y \rangle \langle y, x \rangle &= \langle x, y \rangle \langle y, x \rangle \\
&= |\langle x, y \rangle|^2
\end{aligned}$$

□

**Theorem 1.6** (Cauchy-Schwarz)**.** *For every inner product space $H$,*

$$\forall u, v \in H, |\langle u, v \rangle| \leq \|u\| \|v\|,$$

*where we define $\|x\| = \sqrt{\langle x, x \rangle}$ or any $x \in H$.*
*In particular, when $\|u\| \neq 0, \|u\|^2 \|v\|^2 - |\langle u, v \rangle|^2 = \|z\|^2$, where $z := \|u\| v - \frac{\langle u, v \rangle}{\|u\|} u$.*

*Proof.* Notice that this is trivially true and equality holds to be zero when $u = 0$.
Now we assume $u \neq 0$, then $\|u\| = \sqrt{\langle u, u \rangle} > 0$.

$$\begin{aligned}
\|z\|^2 &= \langle z, z \rangle \\
&= \left\langle \|u\| v - \frac{\langle u, v \rangle}{\|u\|} u, \|u\| v - \frac{\langle u, v \rangle}{\|u\|} u \right\rangle \\
&= \|u\|^2 \langle v, v \rangle - \langle u, v \rangle \langle u, v \rangle - \overline{\langle u, v \rangle} \langle v, u \rangle + \frac{\langle v, u \rangle \langle u, v \rangle}{\cancel{\|u\|^2}} \cancel{\langle u, u \rangle}^{\|u\|^2} \\
&= \|u\|^2 \|v\|^2 - |\langle u, v \rangle|^2 - \cancel{|\langle v, u \rangle|^2} + \cancel{|\langle v, u \rangle|^2} \\
&= \|u\|^2 \|v\|^2 - |\langle u, v \rangle|^2.
\end{aligned}$$

Now $\|z\|^2 = \langle z, z \rangle \geq 0$, we have the result.

□

**Proposition 1.7.** *For every inner product space with $\langle \cdot, - \rangle$, there is a norm $\|x\| = \sqrt{\langle x, x \rangle}$.*

*Proof.* Consider any $x \in H, a \in \mathbb{C}$,

$$\begin{aligned}
\|a \cdot x\| &= \sqrt{\langle ax, ax\rangle} = \sqrt{a^* a \langle x, x\rangle} = \sqrt{|a|^2}\sqrt{\langle x, x\rangle} = |a|\|x\| \\
\forall x \neq 0, \|x\| &= \sqrt{\langle x, x\rangle} > 0 \\
\|0\| &= \sqrt{\langle 0, 0\rangle} = 0 \\
\|x + y\|^2 &= \langle x + y, x + y\rangle = \langle x, x\rangle + \langle y, y\rangle + \langle x, y\rangle + \langle y, x\rangle \\
&= \|x\|^2 + \|y\|^2 + 2\Re(\langle x, y\rangle) \\
&\leq \|x\|^2 + \|y\|^2 + 2|\langle x, y\rangle| \\
&\leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| \\
&\leq (\|x\| + \|y\|)^2.
\end{aligned}$$

Thus $\|x\| = \sqrt{\langle x, x\rangle}$ is a norm. □

**Corollary 1.8.** *For every inner product space, there is a metric* $d(x, y) = \sqrt{\langle x - y, x - y\rangle}$

**Proposition 1.9.** *If* $\forall v, \langle v, u\rangle = 0$*, then* $u = 0$*.*

**Proposition 1.10.** *For an Inner product space* $H, \forall y, x = \lim_{i\to\infty} x_i \in H$*, we have*

$$\langle x, y\rangle = \lim_{i\to\infty} \langle x_i, y\rangle.$$

*Proof.* Given any $\epsilon > 0$, let $\epsilon_0 = \frac{\epsilon}{\|y\|}$.
Since $x = \lim_{i\to\infty} x_i$, we can find $N > 0,$ such that $\forall n > N, \|x - x_n\| < \epsilon_0$,
thus $|\langle x, y\rangle - \langle x_n, y\rangle| = |\langle x - x_n, y\rangle| \leq \|x - x_n\|\|y\| < \epsilon_0\|y\| = \epsilon$ □

**Corollary 1.11.** *For an Inner product space* $H, \forall y, x = \lim_{i\to\infty} x_i \in H$*, we have* $\langle y, x\rangle = \lim_{i\to\infty} \langle y, x_i\rangle$*.*

**Definition 1.14.** An inner product space $\mathcal{H}$ is called a **Hilbert space** if it is complete.

**Definition 1.15.** Let $H$ be an inner product space. Two vectors $u, v \in H$ are called **orthogonal** if $\langle u, v\rangle = 0$.

**Definition 1.16.** Let $H$ be an inner product space. A set $\{e_i\}_{i\in I} \subseteq H$ is called an **orthonormal set** if

$$\forall i, j \in I, \langle e_i, e_j\rangle = \delta_{ij}.$$

**Definition 1.17.** Let $H$ be an inner product space. A set $\{e_i\}_{i\in I} \subseteq H$ is called an **orthonormal set** if

$$\forall i, j \in I, \langle e_i, e_j\rangle = \delta_{ij} := \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}.$$

**Theorem 1.12** (generalized Fourier series)**.** *Let* $\mathcal{H}$ *be a Hilbert space, and* $\{e_i\}_{i\in I} \subseteq \mathcal{H}$ *be an orthonormal set, then the following are equivalent:*

1. $\{e_i\}_{i\in I}$ *is an orthonormal basis.*
2. *If* $\forall i \in I, \langle x, e_i\rangle = 0$*, then* $x = 0$*.*
3. $\forall x \in \mathcal{H}, x = \sum_{i\in I} \langle e_i, x\rangle e_i$*. (Fourier series)*
4. $\forall x \in \mathcal{H}, \|x\|^2 = \sum_{i\in I} |\langle e_i, x\rangle|^2$*. (Parseval Identity)*

**Theorem 1.13.** $\mathcal{H}$ *is a separable Hilbert Space if and only if there is a maximal orthonormal set in* $\mathcal{H}$*. Moreover, in this case, every maximal orthonormal set is at most countable.*

**Definition 1.18.** Let $\mathcal{H}$ be a Hilbert space. $S \subseteq \mathcal{H}$, the subspace **orthogonal** to $S$ is

$$S^{\perp} := \{u \in \mathcal{H} : \langle u, v\rangle = 0, \forall u \in S\}.$$

**Lemma 1.14.** *Let $\mathcal{H}$ be a Hilbert space, $S \subseteq \mathcal{H}$, we always have $S^{\perp}$ is a subspace of $\mathcal{H}$.*

**Definition 1.19.** Let $V$ be a vector space, and $U, W \subseteq V$ be two subspaces, we say $V = U \oplus W$, if $V = U + W = \{v + w : v \in V, w \in W\}$, and $V \cap W = \{0\}$.

**Proposition 1.15.** *Let $V$ be a vector space, and $U, W \subseteq V$ be two subspaces. $V = U \oplus W$, if and only if $\forall v \in V$, it can be uniquely written as $v = u + w$, where $u \in U, w \in W$.*

**Theorem 1.16.** *Let $\mathcal{H}$ be a Hilbert space, if $S \subseteq \mathcal{H}$ is a subspace, then*

$$\mathcal{H} = \bar{S} \oplus S^{\perp}.$$

## 1.5 Bounded linear operators

**Definition 1.20.** Let $X, Y$ be vector spaces, $A : X \to Y$ is a linear operator if $\forall c \in \mathbb{R}, u, v \in X$,

$$A(u + cv) = Au + cAv.$$

**Definition 1.21.** Let $X, Y$ be linear normed spaces, the **operator norm** of a linear operator $T : X \to Y$ is

$$\|T\| := \sup_{\|u\|_X \leq 1} \|Tu\|_Y = \sup_{\|u\|_X < 1} \|Tu\|_Y = \sup_{\|u\|_X = 1} \|Tu\|_Y = \sup_{u \neq 0 \in X} \frac{\|Tu\|_Y}{\|x\|_X}.$$

**Definition 1.22.** Let $X, Y$ be normed spaces, a linear operator $A : X \to Y$ is **bounded** if $\|A\| < \infty$.

**Definition 1.23.** Let $X, Y$ be normed spaces, we denote

$$B(X, Y) := \{A : X \to Y | A \text{ is bounded linear opeartor}\}.$$

**Theorem 1.17.** *The set $B(X, Y)$ is a normed linear space with the operator norm.*

**Proposition 1.18.** *Let $X, Y, Z$ be normed spaces, if $A : X \to Y, B : Y \to Z$ are both linear bounded operators, then so is $B \circ A$, with*

$$\|B \circ A\| \leq \|B\|\|A\|.$$

**Theorem 1.19.** *Let $X, Y$ be normed spaces, a linear operator $A : X \to Y$ is bounded if and only if it is continuous.*

**Definition 1.24.** Let $X, Y$ be normed spaces, a linear operator $A : X \to Y$ is **closed** if $\forall u_k \to u$ in $X$ and $Au_k \to v$ in $Y$, we have $Au = v$.

**Theorem 1.20** (Closed graph)**.** *Let $X, Y$ be Banach spaces, if a linear operator $A : X \to Y$ is closed, it is also bounded.*

**Theorem 1.21.** *(Bounded inverse Theorem) Let $X, Y$ be normed spaces, if a bounded linear operator $A : X \to Y$ is bijective, then $A^{-1}$ is continuous and bounded as well.*

**Proposition 1.22.** *Let $Y$ be a Banach space, $S$ be a dense subset of a normed space $X$. For any bounded linear operator $E : S \to Y$, we can extend it to $\tilde{E} : X \to Y$, such that $\tilde{E}$ is also bounded and linear, with $\left\|\tilde{E}\right\| = \|E\|$, and $\tilde{E}|_S = E$.*

*Proof.* Consider any $x \in X$. Since $S$ is dense in $X$, we know $\forall m \in \mathbb{N}_+, \exists x_m \in S$, such that $\|x - x_m\|_X \leq \frac{1}{m}$.

Since $E$ is linear on $S$, we have that

$$\begin{aligned}\|Ex_m - Ex_l\|_Y &= \|E(x_m - x_l)\|_Y \\ &\leq \|E\|\|x_m - x_l\|_X \\ &= \|E\|\|(x_m - x) + (x - x_l)\|_X \\ &\leq \|E\|\|x - x_m\|_X + \|E\|\|x - x_l\|_X \\ &\leq \|E\|\left(\frac{1}{m} + \frac{1}{l}\right).\end{aligned}$$

Thus given any $\epsilon > 0$, for any $m, l \geq \lceil \frac{2\epsilon}{\|E\|} \rceil$, we can make $\|Ex_m - Ex_l\|_Y < \epsilon$. Thus $(Ex_m)_{m=1}^{\infty}$ is a Cauchy sequence in $Y$. Since $Y$ is a Banach space, $\exists y^* \in Y$, such that $Ex_m \to y^*$ in $Y$.

We claim that $y^*$ is independent of choice of the sequence $(x_m)_{m=1}^{\infty}$. Indeed, consider any other sequence $(v_m)_{m=1}^{\infty} \subseteq C^{\infty}(\bar{x})$, such that $\forall m \in \mathbb{N}_+, \|x - x_m\|_X \leq \frac{1}{m}$,

$$\begin{aligned}\|y^* - Ev_m\|_Y &\leq \|y^* - Ex_m\|_Y + \|Ex_m - Ev_m\|_Y \\ &\leq \|y^* - Ex_m\|_Y + \|E\|\|x_m - v_m\|_X \\ &\leq \|y^* - Ex_m\|_Y + \|E\|\|x_m - x\|_X + \|E\|\|x - v_m\|_X.\end{aligned}$$

Since all three terms on the right go to 0 when $m \to \infty$, we have that $Ev_m \to y^*$ in $Y$. Thus we can uniquely define $\tilde{E}x := y^*$. In addition,

$$\begin{aligned}\left\|\tilde{E}x\right\|_Y &= \left\|\lim_{m\to\infty} Ex_m\right\|_Y \\ &= \lim_{m\to\infty} \|Ex_m\|_Y \\ &\leq \lim_{m\to\infty} \|E\|\|x_m\|_X \\ &= \|E\|\left\|\lim_{m\to\infty} x_m\right\|_X \\ &= \|E\|\|x\|_X.\end{aligned}$$

Thus $\left\|\tilde{E}\right\| = \|E\|$. □

### 1.5.1 Dual Space

**Definition 1.25.** Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{F}$. A **functional** is an operator that maps into $\mathbb{F}$.

**Definition 1.26.** Let $(X, \|\cdot\|)$ be a normed vector space. The **dual norm** of a (anti-)linear functional $\phi$ is defined to be

$$\|\phi\|_{X^\dagger} := \|\phi\|_{X^*} := \sup_{\|x\|\leq 1} |\phi(x)| = \sup_{x\in X \text{ such that } x\neq 0} \frac{|\phi(x)|}{\|x\|}.$$

$\phi$ is **bounded** if $\|\phi\|_{X^*} < \infty$.

**Definition 1.27.** Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{F}$. The **dual space** of $X$ is the collection of bounded linear functionals on $X$, denoted

$$X^* := B(X, \mathbb{F}) = \{\phi : X \to \mathbb{F} : \phi \text{ is linear and bounded}\}.$$

The **antidual space** of $X$ is the the collection of bounded anti-linear functionals on $X$, denoted $X^\dagger$.

**Proposition 1.23.** *There is a canonical anti-linear isometric isomorphism between $X^*$ and $X^\dagger$, defined by $\bar{\phi}(x) := \overline{\phi(x)}$.*

**Definition 1.28.** Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{F}$. For a subspace $Y \subseteq X^*$, we define the **duality pairing** to be $\langle\cdot|\cdot\rangle_{Y,X} : \bar{Y} \times X \to \mathbb{F}$ by $\left\langle\bar{\phi}\middle|x\right\rangle_{Y,X} := \phi(x)$, the **action** of $\phi$ on $x$.

We sometimes also write $\langle \phi | x \rangle_{Y,X}$ when the context is clear, but notice that the duality pairing is anti-linear in the first entry. In the literature, the duality pairing $\langle \phi | x \rangle$ is often directly defined to be $Y \times X \to \mathbb{F}$, which is bilinear. They are the same when $\mathbb{F} = \mathbb{R}$, but when $\mathbb{F} = \mathbb{C}$, one has to be clear which definition they are using. Our sesquilinear definition aligns with the ket-bra notation, commonly used in Physics, and also aligns with the inner product when $X$ is a Hilbert space, as we will see later.

**Definition 1.29.** A Banach space $X$ is **reflexive** if $i(X) = X^{**}$, where $i : X \hookrightarrow X^{**}$ is the canonical linear isometric embedding by $i(x) := x^{**},\ x^{**}(\phi) := \phi(x)$.

**Definition 1.30** (Ket-bra Notation)**.** In Ket-bra notation, we denote an element in $X$ as a **ket** $|x\rangle$, and an element in $X^*$ as a **bra** $\langle \phi |$. The action of $\langle \phi |$ on $|x\rangle$ is just $\langle \phi | x \rangle$. We sometimes abuse the notation, and use $|x\rangle$ to represent both $x \in X$ and $i(x) \in X^{**}$, where $i$ is the canonical linear isometric embedding $i : X \hookrightarrow X^{**}$ by $i(x) := \hat{x},\ \hat{x}(\phi) := \phi(x)$.

**Theorem 1.24** (Riesz-Frechet Representation theorem)**.** *Let $\mathcal{H}$ be a Hilbert space, then for each $u^* \in \mathcal{H}, \exists! u \in \mathcal{H}$, such that $\forall v \in \mathcal{H}, \langle u^* | v \rangle_{\mathcal{H}^*, \mathcal{H}} = \langle u, v \rangle_{\mathcal{H}}$, and $\|u^*\|_{\mathcal{H}^*} = \|u\|_{\mathcal{H}}$.*

**Corollary 1.25.** *Let $\mathcal{H}$ be a Hilbert space, then the map $\Phi : \mathcal{H} \to \mathcal{H}^*;\ u \mapsto \langle u, \cdot \rangle_{\mathcal{H}}$ is an **canonical bijective isometric antilinear isomorphism**.*

**Corollary 1.26.** *Every Hilbert space is reflexive.*

**Definition 1.31** (Ket-bra Notation)**.** We thus abuse the notation and denote the canonical bijective isometric anti-linear isomorphism by $\langle u | := u^* := \Phi(u)\ \forall u \in \mathcal{H}$, and $|u\rangle := (u^*)^* := \Phi^{-1}(u^*)\ \forall u^* \in \mathcal{H}^*$. Notice that by definition

$$\forall u = |u\rangle \in \mathcal{H},\ \langle u |^* = (u^*)^* = u = |u\rangle,$$

and

$$\forall u^* = \langle u | \in \mathcal{H}^*,\ |u\rangle^* = ((u^*)^*)^* = u^* = \langle u |.$$

**Definition 1.32.** Let $X$ be a Banach Space, we say $(u_k)_{k=1}^{\infty} \subset X$ converges to $u \in X$ weakly, denoted $u_k \rightharpoonup u$, if

$$\forall v^* \in X^*, \langle v^* | u_k \rangle \to \langle v^* | u \rangle$$

as real numbers.

**Proposition 1.27.** *Let $X$ be a Banach Space, $(u_k)_{k=1}^{\infty} \subset X$ be a sequence, then*

1. *If $u_k \to u$, we always have $u_k \rightharpoonup u$.*
2. *If $u_k \rightharpoonup u$, we have that $u$ is unique.*
3. *If $u_k \rightharpoonup u$, we have $(u_k)_{k=1}^{\infty}$ is bounded.*
4. *If $u_k \rightharpoonup u$, every subsequence $(u_{k_j})_{j=1}^{\infty}$ also converges weakly to $u$.*

*Proof.* See A5Q1 for 1. □

**Theorem 1.28** (Weakly compact for reflexive Banach Space)**.** *Let $X$ be a reflexive Banach Space, and $(u_k)_{k=1}^{\infty} \subset X$ be a bounded sequence, then $\exists (u_{k_j})_{j=1}^{\infty}$ a subsequence, and $u \in X$, such that $u_{k_j} \rightharpoonup u$.*

**Proposition 1.29.** *Let $\mathcal{H}$ be a Hilbert space, then $u_k \rightharpoonup u$ if and only if $\forall v \in \mathcal{H}, \langle v, u_k \rangle \to \langle v, u \rangle$ as real numbers.*

*Proof.* Suppose $u_k \rightharpoonup u$.
Notice that for all $v \in \mathcal{H}$, we have that $v^* \in \mathcal{H}^*$, and thus $\langle v, u_k \rangle = \langle v^* | u_k \rangle \to \langle v^* | u \rangle = \langle v, u \rangle$.
Now suppose $\forall v \in \mathcal{H}, \langle v, u_k \rangle \to \langle v, u \rangle$.
Notice that for any $f \in \mathcal{H}^*$, by Riesz-Frechet Representation theorem 1.24, there is some $f^* \in \mathcal{H}$, such that

$$\langle f | u_{k_j} \rangle = \langle f^*, u_{k_j} \rangle \to \langle f^*, u \rangle = \langle f | u \rangle.$$

Thus, $u_{k_j} \rightharpoonup u$. □

**Proposition 1.30.** *Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces, $T : \mathcal{H}_1 \to \mathcal{H}_2$ be a bounded operator, and $(u_k)_{k=1}^{\infty} \subset \mathcal{H}_1$ be a sequence. If $u_k \rightharpoonup u \in \mathcal{H}_1$, then $Tu_k \rightharpoonup Tu \in \mathcal{H}_2$.*

*Proof.* Let $y_k := Tu_k, y := Tu \in \mathcal{H}_2$.
Consider any $g \in \mathcal{H}_2^*$, we define $f := g \circ K \in \mathcal{H}_1^*$.
Since $u_k \rightharpoonup u$, we must have

$$\begin{aligned} \lim_{k\to\infty} f(u_k) &= f(u) \\ \lim_{k\to\infty} g(Ku_k) &= g(Ku) \\ \lim_{k\to\infty} g(y_k) &= g(y). \end{aligned}$$

We thus have $y_k \rightharpoonup y$. □

**Proposition 1.31.** *Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces, $K : \mathcal{H}_1 \to \mathcal{H}_2$ be a compact operator, and $(u_k)_{k=1}^{\infty} \subset \mathcal{H}_1$ be a sequence. If $u_k \rightharpoonup u \in \mathcal{H}_1$, then $Ku_k \to Ku \in \mathcal{H}_2$.*

*Proof.* Let $y_k := Ku_k, y := Ku \in \mathcal{H}_2$.
Since $K$ is compact, it is bounded, so $y_k \rightharpoonup y$.
Now suppose for contradiction $\lim_{k\to\infty} \|y_k - y\| \neq 0$.
Then there is some $\epsilon > 0$ and a subsequence $(u_{k_j})_{j=1}^{\infty}$ such that $\forall j \geq 1, \left\|y_{k_j} - y\right\| \geq \epsilon$.
Since $u_k \rightharpoonup u \in \mathcal{H}$, we have $(u_k)_{k=1}^{\infty}$ is bounded, and thus $(u_{k_j})_{j=1}^{\infty}$ is bounded.
Since $K$ is compact, there is some further subsequence $(u_{k_{j_m}})_{m=1}^{\infty}$ such that $\lim_{m\to\infty} Ku_{k_{j_m}} = \tilde{y} \in \mathcal{H}_2$.
Thus $Ku_{k_{j_m}} \rightharpoonup \tilde{y}$. Since weak convergence, we must have $\tilde{y} = y$.
Thus $\lim_{m\to\infty} Ku_{k_{j_m}} = y$, which is a contradiction. □

### 1.5.2 Adjoint Operator

**Definition 1.33.** Let $X, Y$ be normed spaces, the **dual operator** of a linear operator $A : X \to Y$ is

$$A^* : Y^* \to X^*;\ f \mapsto f \circ A.$$

**Proposition 1.32.** *Let $X, Y, Z$ be normed spaces, $S \in B(X, Y), T \in B(Y, Z)$, then $(S \circ T)^* = T^* \circ S^*$.*

*Proof.* Consider any $f \in Z^*$, and any $x \in X$, we have

$$\begin{aligned} (T^* \circ S^*)(f)(x) &= (S^*)(f)(Tx) \\ &= (f)(S(T(x))) \\ &= (f \circ (S \circ T))(x) \\ &= (S \circ T)^*(f)(x). \end{aligned}$$

Thus $(T^* \circ S^*)(f) = (S \circ T)^*(f)$. □

**Definition 1.34.** Let $\mathcal{H}$ be a Hilbert space, and $T : \mathcal{H} \to \mathcal{H}$ be a bounded linear operator, the **Hilbert adjoint operator** of $T$ is $T^* : \mathcal{H} \to \mathcal{H}$ such that $\langle x, Ty\rangle = \langle T^*x, y\rangle \forall x, y \in \mathcal{H}$.

**Theorem 1.33.** *Let $\mathcal{H}$ be a Hilbert space, and $T : \mathcal{H} \to \mathcal{H}$ be a bounded linear operator, $T^*$ always exists, and is given by $T^* = \Phi^{-1} \circ T^* \circ \Phi$, where $\Phi : \mathcal{H} \to \mathcal{H}^*;\ u \mapsto \langle u, \cdot\rangle_{\mathcal{H}}$ is the canonical bijective isometric antilinear isomorphism, and $T^*$ is the dual operator of $T$. In addition, $T^*$ is also a bounded linear operator, with $\|T^*\| = \|T\|$, and $(T^*)^* = T$.*

*Proof.* $\forall y \in \mathcal{H}$, we have that

$$\begin{aligned} \langle T^*x, y\rangle &= \left\langle (\Phi^{-1} \circ T^* \circ \Phi)(x), y \right\rangle \\ &= ((T^* \circ \Phi)(x))(y) \\ &= (\Phi(x))(Ty) \\ &= \langle x, Ty\rangle. \end{aligned}$$

Now consider any $x, y, z \in \mathcal{H}, c \in \mathbb{C}$, we have that

$$\begin{aligned}\langle T^*(x+cz), y\rangle &= \langle x+cz, Ty\rangle \\ &= \langle x, Ty\rangle + \bar{c}\langle z, Ty\rangle \\ &= \langle T^*x, y\rangle + \bar{c}\langle T^*z, y\rangle \\ &= \langle T^*x + cT^*z, y\rangle.\end{aligned}$$

Since this holds for any $y \in \mathcal{H}$, we have that $T^*(x+cz) = T^*x + cT^*z$, and thus $T^*$ is linear.
Now given any $x \in \mathcal{H}$, we have that

$$\begin{aligned}\|T^*x\|^2 &= \langle T^*x, T^*x\rangle \\ &= \langle x, TT^*x\rangle \\ &\leq \|x\|\|TT^*x\| \\ &\leq \|x\|\|T\|\|T^*x\| \\ &\implies \\ \|T^*x\| &\leq \|x\|\|T\| \\ &\implies \\ \|T^*\| &= \sup_{x\neq 0\in\mathcal{H}} \frac{\|T^*x\|}{\|x\|} \\ &\leq \sup_{x\neq 0\in\mathcal{H}} \frac{\|x\|\|T\|}{\|x\|} \\ &= \|T\|.\end{aligned}$$

Thus $T^*$ is also a bounded linear operator.
Now $\forall x, y \in \mathcal{H},\ \langle x, T^*y\rangle = \overline{\langle T^*y, x\rangle} = \overline{\langle y, Tx\rangle} = \langle Tx, y\rangle$.
Thus $(T^*)^* = T$. □

*Remark.* $\forall x, y \in \mathcal{H},\ \langle (Tx)^*|y\rangle = \langle Tx, y\rangle = \langle x, T^*y\rangle = \langle x^*|T^*y\rangle$. We thus abuse the notation, and write $(Tx)^* = \langle x|T^*$

**Definition 1.35.** A bounded linear operator $T : \mathcal{H} \to \mathcal{H}$ is **delf-adjoint** if $T^* = T$.

### 1.5.3 Compact Operators

**Definition 1.36.** Let $X, Y$ be metric spaces, a linear operator $A : X \to Y$ is **compact** if for each bounded subset $S \subseteq X$, we have its image $A(S)$ is pre-compact in $Y$.

**Proposition 1.34.** *Let $X, Y$ be metric spaces, a linear operator $A : X \to Y$ is compact if and only if $A$ is bounded, and each bounded sequence $(x_n)_{n=1}^{\infty} \subseteq X$ has some subsequence $(x_{n_k})_{k=1}^{\infty}$ such that $(Ax_{n_k})_{k=1}^{\infty}$ converges to some $y \in Y$.*

**Definition 1.37.** Let $X, Y$ be Banach spaces and $X \subseteq Y$, then we say $X$ is **compactly embedded** in $Y$, denoted

$$X \subset\subset Y$$

if the inclusion map $i : X \hookrightarrow Y;\ x \mapsto x$ is compact.
Namely, $\exists C > 0,$ such that $\forall x \in X, \|x\|_Y \leq C\|x\|_X$, and each bounded sequence $(x_n)_{n=1}^{\infty} \subseteq X$ having some subsequence $(x_{n_k})_{k=1}^{\infty}$ that converges to some $y \in Y$.

**Proposition 1.35.** *Let $X, Y, Z$ be Banach spaces and $X \subset\subset Y$, if an operator $T : Z \to X$ is bounded, then $\tilde{T} := i \circ T : Z \to Y$ is compact.*

*Proof.* Consider any bounded set $S \subseteq Z$, such that $\forall z \in S, \|z\|_Z \leq M$.
We have $\|Tz\|_X \leq \|T\|\|z\|_M \leq M\|T\| < \infty$, and thus $T(S)$ is bounded in $X$.
Yet $i$ is compact, and thus $i(T(S))$ is pre-compact.
This shows $\tilde{T}(S) = (i \circ T)(S)$ is pre-compact for any bounded set $S \subseteq Z$.
Thus $\tilde{T}$ is compact. □

**Theorem 1.36.** *Let $\mathcal{H}$ be a Hilbert space, and $K : \mathcal{H} \to \mathcal{H}$ be a compact linear operator, then $K^*$ is also compact.*

*Proof.* $K^*$ is bounded by 1.34.
Let $(u_k)_{k=1}^{\infty}$ be any bounded sequence in $\mathcal{H}$.
By 1.28, we have that $\exists (u_{k_j})_{j=1}^{\infty}$ a subsequence, and $u \in X$, such that $u_{k_j} \rightharpoonup u$.
Notice that for any $f \in \mathcal{H}^*$, by Riesz-Frechet Representation theorem 1.24, there is some $f^* \in \mathcal{H}$, such that

$$\begin{aligned}\left\langle f \middle| K^*(u_{k_j} - u)\right\rangle &= \left\langle f^*, K^*(u_{k_j} - u)\right\rangle \\ &= \left\langle Kf^*, u_{k_j} - u\right\rangle \\ &= \langle Kf^*, u\rangle - \langle Kf^*, u\rangle \qquad \to 0,\end{aligned}$$

since $u_{k_j} \rightharpoonup u$ and by 1.29.
Since $\left\langle f \middle| K^*(u_{k_j} - u)\right\rangle \to 0 = \langle f|0\rangle$ for any $f \in \mathcal{H}^*$, we have that $K^*(u_{k_j} - u) \rightharpoonup 0$.
By 1.31, we have that $KK^*(u - u_{k_j}) \to 0$.

$$\begin{aligned}\left\|K^*u - K^*u_{k_j}\right\|^2 &= \left\langle K^*u - K^*u_{k_j}, K^*u - K^*u_{k_j}\right\rangle \\ &= \left\langle K^*(u - u_{k_j}), K^*(u - u_{k_j})\right\rangle \\ &= \left\langle KK^*(u - u_{k_j}), u - u_{k_j}\right\rangle \\ &\le \left\|KK^*(u - u_{k_j})\right\|\left\|u - u_{k_j}\right\| \\ &\to 0.\end{aligned}$$

Thus $K^*u_{k_j} \to K^*u \in \mathcal{H}$,
Since $(u_k)_{k=1}^{\infty}$ is any bounded sequence, we have that $K^*$ is compact by 1.34. □

**Theorem 1.37** (Fredholm alternative)**.** *Let $\mathcal{H}$ be a Hilbert space, and $K : \mathcal{H} \to \mathcal{H}$ be a compact linear operator, then*

1. $\mathrm{Ker}(I - K)$ *is finite dimensional.*
2. $\mathrm{Im}(I - K)$ *is closed.*
3. $\mathrm{Im}(I - K) = \mathrm{Ker}(I - K^*)^{\perp}$.
4. $\dim(\mathrm{Ker}(I - K)) = \dim(\mathrm{Ker}(I - K^*))$.
5. $\mathrm{Ker}(I - K) = \{0\} \iff \mathrm{Im}(I - K) = \mathcal{H}$.

**Corollary 1.38.** *Let $\mathcal{H}$ be a Hilbert space, and $K : \mathcal{H} \to \mathcal{H}$ be a compact linear operator, then exactly one of the following holds:*

1. $\forall v \in \mathcal{H},\ \exists! u \in \mathcal{H},$ *such that* $(I - K)u = v$.
2. $\exists u \neq 0 \in \mathcal{H},$ *such that* $(I - K)u = 0$.

*Proof.* When $\mathrm{Ker}(I - K) = \{0\}$, we have that $I - K$ is injective, and $\mathrm{Im}(I - K) = \mathcal{H}$.
Thus $\forall v \in \mathcal{H},\ \exists! u \in \mathcal{H},$ such that $(I - K)u = v$.
On the other hand, if 1. is true, we have that $(I - K)$ is surjective, so $\mathrm{Im}(I - K) = \mathcal{H}$, so $\mathrm{Ker}(I - K) = \{0\}$.
Thus $\mathrm{Ker}(I - K) = \{0\} \iff 1.$.
We also have that $\mathrm{Ker}(I - K) \neq \{0\} \iff \exists u \neq 0 \in \mathrm{Ker}(I - K) \iff 2.$. □

**Theorem 1.39** (Structure theorem for compact operators)**.** *Let $K : X \to X$ be a compact operator on an infinite-dimensional Banach space $X$, then*

1. $0 \in \mathrm{Spec}(K)$.
2. $\mathrm{Spec}(K) \setminus \{0\} = \mathrm{Spec}_p(K) \setminus \{0\}$.

3. $\mathrm{Spec}(K) \setminus \{0\}$ *is finite, or* $\mathrm{Spec}(K) \setminus \{0\} = (\lambda_k)_{k=1}^{\infty}$ *such that* $\lim_{k\to\infty} \lambda_k = 0$.

**Theorem 1.40** (Spectral theorem / Hilbert-Schmidt Theorem)**.** *Let $T : \mathcal{H} \to \mathcal{H}$ be a compact linear self-adjoint operator on a countably infinite-dimensional complex Hilbert space $\mathcal{H}$, then*

1. *There exists orthonormal eigenvectors $(\phi_k)_{k=1}^{\infty} \subset \mathcal{H}$ and eigenvalues $(\lambda_k)_{k=1}^{\infty} \subset \mathbb{R}$ such that $|\lambda_1| \geq |\lambda_2| \geq \ldots$, and*
$$T\phi_k = \lambda_k \phi_k, \lambda_k \neq 0, \ \forall k \in \mathbb{N}_+,$$
$$\forall v \in \mathcal{H}, Tv = \sum_{k=1}^{\infty} \lambda_k \langle \phi_k, v \rangle \phi_k = \sum_{k=1}^{\infty} \langle \phi_k, Tv \rangle \phi_k.$$
2. $\lim_{k\to\infty} \lambda_k = 0$,
3. $(\phi_k)_{k=1}^{\infty}$ *is an orthonormal basis for $\mathcal{H}$ iff 0 is not an eigenvalue for $T$.*

## 1.6 Function Spaces

### 1.6.1 Continuous functions

**Definition 1.38.** $u : U \to \mathbb{R}$ is **continuous** at $x \in U$ if
$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \forall y \in U, \|x - y\| < \delta \implies |u(x) - u(y) < \epsilon|.$$
A function $u$ is continuous if it is continuous at all $x \in U$.

- $C(U) := \{u : U \to \mathbb{R} : u \text{ is continuous}\}$
- $C^k(U) := \{u : U \to \mathbb{R} : u \text{ is k-times continuously differentiable}\}$
- $C^{\infty}(U) := \{u : U \to \mathbb{R} : u \text{ has continuous derivatives of all orders}\}$

**Definition 1.39.** $u : U \to \mathbb{R}$ is **uniformly continuous** if
$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \forall x, y \in U, \|x - y\| < \delta \implies |u(x) - u(y) < \epsilon|.$$

- $C(\bar{U}) := \{u : U \to \mathbb{R} : u \text{ is uniformly continuous on bounded subsets of } U\}$
- $C^k(\bar{U}) := \{u : U \to \mathbb{R} : \forall |\alpha| \leq k, D^{\alpha}u \text{ is uniformly continuous on bounded subsets of } U, \}$
- If $u \in C^k(\bar{U})$, then we can extend $D^{\alpha}u$ continuously to $\bar{U}$.

**Definition 1.40.** The support of $u : U \to \mathbb{R}$ is
$$\mathrm{Supp}(u) := \overline{\{x \in U : u(x) \neq 0\}}.$$

**Definition 1.41.** $u : U \to \mathbb{R}$ has compact support if $\mathrm{Supp}(u)$ is a compact subset of $U$.

**Definition 1.42.** We denote the functions in $C(U)$ and $C^k(U)$ with compact support by $C_c(U), C_c^k(U)$.

**Definition 1.43.** Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable, we define the set of **test functions**
$$C_c^{\infty}(U) := \{\phi \in C^{\infty}(U) : \mathit{Supp}(f) \subseteq U \text{ and is compact}\},$$
where
$$\mathit{Supp}(\phi) := \overline{\{x \in U : \phi(x) \neq 0\}}.$$

**Proposition 1.41** (Cut-off Function)**.** *For any $0 < r < R \in \mathbb{R}$, there is a $\eta \in C_c^{\infty}(\mathbb{R}^n)$, such that*
$$\eta(x) = \begin{cases} 1, & \|x\| \leq r \\ 0, & \|x\| \geq R \end{cases}.$$

*Proof.* Consider $f(t) := \begin{cases} e^{-\frac{1}{t}}, & t > 0 \\ 0, & t \leq 0 \end{cases} \in C^\infty(\mathbb{R})$. Define $\eta(x) := \frac{f(R-\|x\|)}{f(R-\|x\|)+f(\|x\|-r)}$. It is still $C^\infty(\mathbb{R}^n)$ since it is a composition of smooth functions, and the denominator is zero only when $R - \|x\| \leq 0, \|x\| - r \leq 0 \implies R \leq \|x\| \leq r$, which will never be the case. When $\|x\| \leq r$, we have that $\eta(x) = \frac{f(R-\|x\|)}{f(R-\|x\|)+0} = 1$, and when $\|x\| \geq R$, we have that $\eta(x) = \frac{0}{0+f(\|x\|-r)} = 0$. □

**Definition 1.44.** Consider a sequence of functions $\{u_m\}_1^\infty$ with $u_m : U \to \mathbb{R}$ and a function $u : U \to \mathbb{R}$, we have

- $u_m \to u$ point-wise on $U$ if

$$\forall x \in U,\ \delta > 0,\ \exists M \in \mathbb{N},\ \text{ such that } m > M \implies |u_m(x) - u(x)| < \delta.$$

- $u_m \to u$ uniformly on $U$ if

$$\forall \delta > 0,\ \exists M \in \mathbb{N},\ \text{ such that } \forall x \in U,\ m > M \implies |u_m(x) - u(x)| < \delta.$$

**Definition 1.45.** Let $(X, d)$ be a metric space, and $I \subseteq \mathbb{R}$ be an interval. A function $f : I \to X$ is **absolutely continuous** if $\forall \epsilon > 0, \exists \delta > 0$ such that for every finite disjoint collection $\{(a_k, b_k)\}_{k=1}^n$ of open intervals in $I$,

$$\sum_{k=1}^n (b_k - a_k) < \delta \implies \sum_{k=1}^n d(f(b_k) - f(a_k)) < \epsilon.$$

We denote the set of all absolutely continuous functions from $I$ to $X$ as $AC(I; X)$.

**Theorem 1.42.** *Suppose $I$ is compact, then $AC(I; X) \subseteq C\left(\bar{I}\right) = C(I)$. Namely, every absolutely continuous function on a compact interval is uniformly continuous, thus continuous.*

### 1.6.2 Lebesgue Spaces

See more in my Measure Theory Notes.

**Definition 1.46.** We denote the Lebesgue measure by $\lambda$ on $\mathbb{R}^n$. We denote $\int_A f d\lambda$ by $\int_A f(x)dx$ for any measurable set $A \subseteq \mathbb{R}^n$.

**Definition 1.47.** Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable. We define

$$\mathcal{L}^1(U) := \left\{ f : U \to \mathbb{R} \middle| \int_U |f(x)|dx < \infty \right\}.$$

**Definition 1.48.** Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable, and $1 \leq p < \infty$. We define

$$\mathcal{L}^p(U) := \left\{ f : X \to \mathbb{R} \middle| f^p \in \mathcal{L}^1(U) \right\} = \left\{ f : X \to \mathbb{C} \middle| f \text{ is measurable}, \int_U |f(x)|^p dx < \infty \right\}.$$

In addition, we define

$$\|f\|_{\mathcal{L}^p(U)} := \left( \int_U |f(x)|^p dx \right)^{\frac{1}{p}}.$$

**Definition 1.49.** The **essential supremum** of a function $u : U \to \mathbb{R}$ is

$$\operatorname{ess\,sup} f := \inf \{ M \in \mathbb{R} : |\{x : f(x) > M\}| = 0 \}.$$

**Definition 1.50.** Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable. We define

$$\mathcal{L}^\infty(U) := \{ f : U \to \mathbb{R} | \operatorname{ess\,sup} f < \infty \}.$$

In addition, we define

$$\|f\|_{\mathcal{L}^\infty} := \operatorname{ess\,sup} |f|.$$

**Definition 1.51.** Two measurable functions $f, g : U \to \mathbb{R}$ are said to be equal almost everywhere if $\{x \in U : f(x) \neq g(x)\}$ has measure zero.

*Remark.* We note that $\|\cdot\|_{\mathcal{L}^p(\mu)}$ is not a norm (though almost), but a shorthand that abuses notation.

**Proposition 1.43.** *For any $1 \leq p \leq \infty$, we have $\|f - g\|_{\mathcal{L}^p(\mu)} = 0 \iff f = g$ almost everywhere.*

**Definition 1.52.** For any $1 \leq p \leq \infty$, if we identify $f, g \in \mathcal{L}^p(U)$ by $f \sim g \iff f = g$ almost everywhere, we get the quotient space

$$L^p := \mathcal{L}^p/_\sim = \{[f] : f \in L^p(U)\}$$

to be the collection of all equivalence classes of functions in $\mathcal{L}^p$.

**Definition 1.53.** Let $(X, \mathcal{M}, \mu)$ be a measure space, and $1 \leq p \leq \infty$ we define the norm

$$\|[f]\|_{L^p(\mu)} := \|f\|_{\mathcal{L}^p(\mu)}$$

for any representative $f \in [f]$.

**Theorem 1.44** (Fischer-Riesz)**.** *For any $1 \leq p \leq \infty$, we have that $\|\cdot\|_{L^p(\mu)}$ is well-defined. Also, the space $(L^p, \|\cdot\|_{L^p(\mu)})$ is a Banach space.*

**Theorem 1.45.** *For any $1 \leq p < \infty$,*

- *$C_c(U)$ is dense in $L^p(U)$.*
- *$C(\bar{U})$ is dense in $L^p(U)$.*

**Definition 1.54.** Let $U, V \subseteq \mathbb{R}^n$ be open, we say that $V$ is **compactly contained** in $U$ if $V \subseteq \bar{V} \subseteq U$, and $\bar{V}$ is compact. We write this as $V \subset\subset U$.

**Definition 1.55.** The **locally summable spaces** are

$$L^p_{loc}(U) := \{u : U \to \mathbb{R} : \forall V \subset\subset U, u \in L^p(V)\}.$$

**Definition 1.56.** We say some property holds for $L^p_{loc}(U)$, if it holds $\forall L^p(V)$ such that $V \subset\subset U$.
For instance, let $(f_n)_{n=1}^\infty \subseteq L^p_{loc}(U)$, then $f_n \to f$ in $L^p_{loc}(U)$ if $f_n \to f$ in $L^p(V)$, $\forall V \subset\subset U$.

**Proposition 1.46.** *For any $1 \leq p \leq \infty$, we have*

$$L^p(U) \subseteq L^1_{loc}(U).$$

**Example 1.6.1.** Let $u(x) = \frac{1}{x}$ on $U = (0, 1)$.
We have $\int_0^1 |u| dx = \infty$, and thus $u \notin L^1(U)$. However, $u \in L^1_{loc}(U)$.

**Theorem 1.47** (Holder's Inequality)**.** *Assume $1 \leq p, q \leq \infty$, $\frac{1}{p} + \frac{1}{q} = 1$.*
*If $u \in L^p(U), v \in L^q(U)$, we have*

$$\int_U |uv| dx \leq \|u\|_p \|v\|_q.$$

*For $a, b \in \mathbb{R}^n$, we have*

$$\left|\sum_{k=1}^n a_k b_k\right| \leq \left(\sum_{k=1}^n |a_k|^p\right)^{1/p} \left(\sum_{k=1}^n |b_k|^q\right)^{1/q}$$

**Theorem 1.48** (Minkowski's Inequality)**.** *Assume $1 \leq p \leq \infty$.*
*Let $u, v \in L^p(U)$, we have*

$$\|u + v\|_p \leq \|u\|_p + \|v\|_p.$$

*For $a, b \in \mathbb{R}^n$, we have*

$$\left(\sum_{k=1}^n |a_k + b_k|^p\right)^{1/p} \leq \left(\sum_{k=1}^n |a_k|^p\right)^{1/p} + \left(\sum_{k=1}^n |b_k|^p\right)^{1/p}$$

**Theorem 1.49** (Lebesgue Monotone Convergence)**.** *Let $f_n : X \to [0,\infty]$ be measurable functions with $0 \leq f_1 \leq f_2 \leq \cdots \leq \infty$. Let $f(x) := \lim_{n\to\infty} f_n(x)$, then $f : X \to [0,\infty]$ is measurable, and*

$$\lim_{n\to\infty} \int_X f_n dx = \int_X f dx.$$

**Theorem 1.50** (Lebesgue Dominated Convergence)**.** *Let $f_n : X \to \mathbb{C}$ be measurable functions, defined almost everywhere on $X$, such that $f(x) := \lim_{n\to\infty} f_n(x)$ is defined almost everywhere for $x \in X$. If there is $0 \leq g(x) \in \mathcal{L}^1(X,\mu)$, such that for almost everywhere $x \in X, \forall n \in \mathbb{N},\ |f_n(x)| \leq g(x)$, then $f \in \mathcal{L}^1(X,\mu)$, and*

$$\lim_{n\to\infty} \int_X f_n d\mu = \int_X f d\mu, \lim_{n\to\infty} \int_X |f - f_n| d\mu = 0.$$

**Theorem 1.51.** *We have that*

$$L^q(U) \cong L^p(U)^*,$$

*where $\frac{1}{p} + \frac{1}{q} = 1$, and the isometric isomorphism $L^q(U) \xrightarrow{\sim} L^p(U)^*;\ u \mapsto u^*$ is defined to be*

$$\forall v \in L^p(U), \langle u^* | v \rangle := \int_U uv dx.$$

**Definition 1.57** (Ket-bra Notation)**.** We will write $\langle u|v\rangle := \int_U \bar{u}v dx$ with $u \in L^q(U)$ instead of $u^* \in L^p(U)^*$. Note that here $\langle u|$ is the anti-linear isometric isomorphism from $L^q(U) \xrightarrow{\sim} L^p(U)^*$ given by the above isometric isomorphism on $\bar{u}$. We will sometimes abuse this notation to indicate this integral without checking the spaces.

**Corollary 1.52.** *In particular, $L^2(U) \cong L^2(U)^*$ , with the isometric isomorphism $L^2(U) \to L^2(U)^*;\ u \mapsto u^*$ is defined to be*

$$\forall v \in L^2(U), \langle u^* | v \rangle = \int_U uv dx = \langle u, v \rangle_{L^2(U)}.$$

**Definition 1.58.** For $f : U \to \mathbb{R}^m$, we define

$$\|f\|_{L^p(U)} := \| \|f\|_m \|_{L^p(U)}.$$

**Theorem 1.53** (Fundamental Theorem of Lebesgue Integral Calculus)**.** *[3, Theorem 6.10, Cor 6.12] $f : [a,b] \to \mathbb{R}$ is absolutely continuous if and only if there is a Lebesgue integrable function $g$, such that $\forall x \in [a,b],\ f(x) = f(a) + \int_a^x g(t)dt$. In this case, $f$ is differentiable a.e., and $f'(x) = g(x)$ for a.e. $x \in [a,b]$.*

# 2 Sobolev Spaces

This section follows Chapter 5 in Evan's book.

## 2.1 Holder Spaces

**Definition 2.1.** For $u : U \to \mathbb{R}$ be bounded and continuous, we write

$$\|u\|_{C(\bar{U})} := \sup_{x \in U} |u(x)|.$$

**Definition 2.2.** A function $u : U \to \mathbb{R}$ is **Holder continuous** with $0 < \gamma \leq 1$ if

$$\exists C, \text{ such that } \forall x, y \in U, \ |u(x) - u(y)| \leq C\|x - y\|^{\gamma}.$$

**Definition 2.3.** The $\gamma^{th}$**-Holder semi-norm** of $u : U \to \mathbb{R}$ is

$$[u]_{C^{0,\gamma}(\bar{U})} := \sup_{x,y \in U, x \neq y} \left( \frac{|u(x) - u(y)|}{\|x - y\|^{\gamma}} \right).$$

The $\gamma^{th}$**-Holder norm** of $u : U \to \mathbb{R}$ is

$$\|u\|_{C^{0,\gamma}(\bar{U})} := [u]_{C^{0,\gamma}(\bar{U})} + \|u\|_{C(\bar{U})}.$$

**Definition 2.4.** For $k \in \mathbb{N}, u \in C^k(\bar{U})$ we define

$$\|u\|_{C^{k,\gamma}(\bar{U})} := \sum_{|\alpha| \leq k} \|D^{\alpha} u\|_{C(\bar{U})} + \sum_{|\alpha| = k} [D^{\alpha} u]_{C^{0,\gamma}(\bar{U})}.$$

The **Holder Space** is

$$C^{k,\gamma}(\bar{U}) := \left\{ u \in C^k(\bar{U}) : \|u\|_{C^{k,\gamma}(\bar{U})} < \infty \right\}.$$

**Theorem 2.1.**

$$\left( C^{k,\gamma}(\bar{U}), \|\cdot\|_{C^{k,\gamma}(\bar{U})} \right)$$

*is a Banach Space.*

## 2.2 Convolution and Mollification

**Definition 2.5.** For $f, g : \mathbb{R}^n \to \mathbb{R}$, we define the **convolution** $f * g : \mathbb{R}^n \to \mathbb{R} \cup \{\pm\infty\}$ to be

$$(f * g)(x) := \int_{\mathbb{R}^n} f(x - y) g(y) dy.$$

**Proposition 2.2.** *For $f, g : \mathbb{R}^n \to \mathbb{R}$, we have $f * g = g * f$.*

*Proof.* Take $z := x - y$, we have $y = x - z$, and $d(z^i) = -d(y^i)$. We have that for any $x \in \mathbb{R}$,

$$\begin{aligned}
(f * g)(x) &= \int_{\mathbb{R}^n} f(x - y) g(y) dy \\
&= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x - y) g(y) d(y^1) \cdots d(y^n) \\
&= (-1)^n \int_{\infty}^{-\infty} \cdots \int_{\infty}^{-\infty} f(z) g(x - z) d(z^1) \cdots d(z^n) \\
&= (-1)^n (-1)^n \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(z) g(x - z) dz \\
&= \int_{\mathbb{R}^n} f(z) g(x - z) dz \\
&= (g * f)(x).
\end{aligned}$$

□

**Proposition 2.3.**

$$\text{Supp}(f * g) \subseteq \text{Supp}(f) + \text{Supp}(g).$$

*Proof.* Let $f^x(y) := f(x - y)$, we have $f * g(x) = \int_{\mathbb{R}^n} f^x(y)g(y)dy$.
Suppose $\text{Supp}(f^x) \cap \text{Supp}(g) = \emptyset$, then we have $(f * g)(x) = 0$.
In addition,

$$\begin{aligned} &\text{Supp}(f^x) \cap \text{Supp}(g) \neq \emptyset \\ \iff &\exists y, x - y \in \text{Supp}(f), y \in \text{Supp}(g) \\ \iff &x \in \text{Supp}(f) + \text{Supp}(g). \end{aligned}$$

Thus $\text{Supp}(f * g) \subseteq \{x \in \mathbb{R}^n : \text{Supp}(f^x) \cap \text{Supp}(g) \neq \emptyset\} = \text{Supp}(f) + \text{Supp}(g)$. □

**Proposition 2.4** (Young's Convolution Inequality)**.** *Let $f \in L^1(\mathbb{R}^n), g \in L^p(\mathbb{R}^n)$ with $1 \leq p \leq \infty$, then for a.e.$x \in \mathbb{R}^n$, the function $f(x - y)g(y)$ is integrable. Thus $f * g : \mathbb{R}^n \to \mathbb{R}$ is well-defineda.e.. In addition, $f * g \in L^p(\mathbb{R}^n)$, and*

$$\|f * g\|_{L^p(\mathbb{R}^n)} \leq \|f\|_{L^1(\mathbb{R}^n)} \|g\|_{L^p(\mathbb{R}^n)}.$$

**Definition 2.6.**

$$\bar{B}(x, r) = \{y \in \mathbb{R}^n : \|x - y\| \leq r\}$$

is the closed ball around $x$ of radius $r$, and

$$B(x, r) = \{y \in \mathbb{R}^n : \|x - y\| < r\}$$

is the closed ball around $x$ of radius $r$.

**Definition 2.7.** For $\epsilon > 0$,

$$U_\epsilon := \{x \in U : \text{dist}(x, \partial u) > \epsilon\}.$$

*Remark.* This definition does not require $U$ to be bounded.

**Definition 2.8.** The **standard mollifier** $\eta(x) \in C^\infty(\mathbb{R}^n)$ is defined as

$$\eta(x) := \begin{cases} C \exp\left(\frac{1}{|x|-1}\right), & |x| < 1 \\ 0, & o.w. \end{cases},$$

with $C$ such that $\int_{\mathbb{R}^n} \eta(x)dx = 1$.
For each $\epsilon > 0$,

$$\eta_\epsilon := \frac{1}{\epsilon^n} \eta\Big(\frac{x}{\epsilon}\Big).$$

**Proposition 2.5.** $\forall \epsilon > 0$, *we have*

1. $\eta_\epsilon \in C^\infty(\mathbb{R}^n)$,
2. $\int_{\mathbb{R}^n} \eta_\epsilon(x)dx = 1$,
3. $\text{Supp}(\eta_\epsilon) \subseteq \bar{B}(0, \epsilon)$.

**Definition 2.9.** Let $f \in L^1_{loc}(U), \epsilon > 0$, its **mollification** $f^\epsilon : U_\epsilon \to \mathbb{R}$ is defined as

$$f^\epsilon(x) := \eta_\epsilon * f := \int_U \eta_\epsilon(x - y)f(y)dy = \int_{\bar{B}(0,\epsilon)} f(x - z)\eta_\epsilon(z)dz.$$

*Remark.* When $U \subsetneq \mathbb{R}^n$, the mollification $\eta_\epsilon * f$ is not using the formal definition of convolution, but we will soon see the abuse of notation makes sense.

**Proposition 2.6.** *Let $f^\epsilon$ be defined as above, if we zero-extend $f$ outside of $U$ to be*

$$\bar{f}(x) := \begin{cases} f(x) & x \in U \\ 0 & x \in \mathbb{R}^n \setminus U \end{cases},$$

*we have $\forall\ x \in U_\epsilon$,*

$$\begin{aligned}(\eta_\epsilon * \bar{f})(x) &= \int_{\mathbb{R}^n} \eta_\epsilon(x-y)\bar{f}(y)dy \\ &= \int_U \eta_\epsilon(x-y)f(y)dy \\ &= f^\epsilon(x)\end{aligned}$$

**Theorem 2.7** (C.5.7 in Evans)**.** *Let $f^\epsilon$ be defined as above, we have:*

1. *$f^\epsilon \in C^\infty(U_\epsilon)$,*
2. *$D^\alpha(f^\epsilon) = (D^\alpha\eta_\epsilon) * f$ on $U_\epsilon$,*
3. *$f^\epsilon \to f$ a.e., as $\epsilon \to 0$,*
4. *If $f \in C(U)$, we have $f^\epsilon \to f$ uniformly on compact subsets of $U$,*
5. *If $1 \le p < \infty, f \in L^p_{loc}(U)$, we have $f^\epsilon \to f$ in $L^p_{loc}(U)$. Namely, $f^\epsilon \to f$ in $L^p(V)$, $\forall V \subset\subset U$.*
6. *$\mathrm{Supp}(f^\epsilon) \subseteq \mathrm{Supp}(f) + \mathrm{Supp}(\eta_\epsilon) = \mathrm{Supp}(f) + \bar{B}(0,\epsilon)$.*

**Proposition 2.8.** *Let $1 \le p \le \infty$. Let $u \in L^p(U)$, then for any $\epsilon > 0$, $U_\epsilon \supseteq V \supseteq \mathrm{Supp}(u^\epsilon)$, we have*

$$\|u^\epsilon\|_{L^p(V)} \le \|u\|_{L^p(V)}.$$

*Proof.* Notice that $u^\epsilon \in C^\infty(U_\epsilon) \subseteq L^1(U_\epsilon)$.
If we zero-extend $u$ outside of $U$ to be $\bar{u}(x) := \begin{cases} u(x) & x \in U \\ 0 & x \in \mathbb{R}^n \setminus U \end{cases}$, we have

$$\begin{aligned}\|\bar{u}\|_{L^p(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} |\bar{u}(x)|^p dx \\ &= \int_U |u(x)|^p dx + 0 \\ &= \|u\|^p_{L^p(U)} \\ &< \infty.\end{aligned}$$

Thus, $\bar{u} \in L^p(\mathbb{R}^n)$.
By 2.4, we have

$$\begin{aligned}\|u^\epsilon\|_{L^p(V)} &= \|\eta_\epsilon * \bar{u}\|_{L^p(V)} \\ &\le \|\eta_\epsilon * \bar{u}\|_{L^p(\mathbb{R}^n)} \\ &\le \|\eta_\epsilon\|_{L^1(\mathbb{R}^n)}\|\bar{u}\|_{L^p(\mathbb{R}^n)} \\ &= \left(\int_{\mathbb{R}^n} |\eta_\epsilon(x)|dx\right)\|u\|_{L^p(V)} \\ &= \|u\|_{L^p(V)}.\end{aligned}$$

□

## 2.3 Weak Derivative and Sobolev Spaces

### 2.3.1 Weak Derivatives

We would like to have a notion of derivative for $L^p(U)$ functions. However, since they are defined as function classes that are equal a.e., we can modify their value at any point, which is measurable, and still get the same element in $L^p(U)$. Thus, it does not make sense to talk about pointwise behaviour of those functions. In particular, continuity and derivatives. We thus need a better notion to generalize them, which preserves properties of classical derivatives, while ignorant of measure zero sets. is ignorant of measure zero sets.

**Theorem 2.9.** *Suppose $U$ has Lipschitz boundary, and $u, \phi \in C^{k-1}(U), |\alpha| = k$, with $D^\alpha u, D^\alpha \phi \in C(U)$, integration by parts gives:*

$$\int_U u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U D^\alpha u \phi dx.$$

**Definition 2.10.** Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable, and $\alpha \in \mathbb{N}^n$ is an $n$ tuple. For $u, v \in L^1_{loc}(U)$, we say $v$ is the $\alpha^{th}$-weak derivative of $u$ if

$$\forall \phi \in C_c^\infty(U), \int_U u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U v \phi dx,$$

where $D^\alpha \phi := \partial_{x_1}^{\alpha_1} \dots \partial_{x_n}^{\alpha_n} \phi$, and $|\alpha| := \sum_{i=1}^n \alpha_i$.
If such a $v$ exists, we say that $D^\alpha u = v$ in the weak sense. Otherwise, $u$ does not possess a $\alpha^{th}$ weak derivative.

**Theorem 2.10.** *Suppose $v \in L^1_{loc}(U)$ be such that*

$$\forall \phi \in C_c^\infty(U), \int_U \phi v dx = 0,$$

*we must have $v = 0$ a.e..*

*Proof.* By 2.7, we have $v^\epsilon \to v$ a.e., as $\epsilon \to 0$.
Now pick any such $y \in U$ where $v^\epsilon(y) \to v(y)$. Since $U$ is open, we can find $r > 0$, such that $\bar{B}(y, r) \subset U$.
Now we define the function $\phi_{y,\epsilon}(x) := \eta_\epsilon(y - x)$ for each $\epsilon \in (0, r)$.
Since $\text{Supp}(\eta_\epsilon) \subseteq \bar{B}(0, \epsilon)$, we have $\text{Supp}(\phi_{y,\epsilon}) \subseteq \bar{B}(y, \epsilon) \subset U$ is compactly contained in $U$.
Also, $\phi_{y,\epsilon} \in C^\infty(\mathbb{R}^n) \subset C^\infty(U)$.
This shows that $\phi_{y,\epsilon} \in C_c^\infty(U)$.
Now we have

$$\begin{aligned} 0 &= \int_U \phi_{y,\epsilon} v dx \\ &= \int_U \eta_\epsilon(y - x) v(x) dx \\ &= v^\epsilon(y). \end{aligned}$$

Since this holds for all $\epsilon \in (0, r)$, we must have $v(y) = \lim_{\epsilon \to 0} v^\epsilon(y) = 0$.
Since this holds for a.e. $y \in U$, we have that $v = 0$ a.e.. □

**Proposition 2.11.** *If a weak derivative $D^\alpha u$ exists, it is uniquely defined up to a set of measure zero.*

*Proof.* Suppose $v, \tilde{v}$ are both $D^\alpha u$, then $\forall \phi \in C_c^\infty(U)$,

$$\int_U u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U v \phi dx = (-1)^{|\alpha|} \int_U \tilde{v} \phi dx.$$

Thus $\forall \phi \in C_c^\infty(U), \int_U (v - \tilde{v}) \phi dx = 0$.
By the previous theorem, we have that $v = \tilde{v}$ a.e.. □

**Proposition 2.12.** *The weak derivative is linear. Namely, suppose both $u, v \in L^1_{loc}(U)$ possess the $\alpha^{th}$ weak derivative, we have that*

$$D^\alpha(\lambda u + v) = \lambda D^\alpha u + D^\alpha v, \ \forall \lambda \in \mathbb{R}.$$

**Proposition 2.13.** *If $u, v \in C(U)$ are both continuous, and $u = v$ a.e., then $\forall x \in U, u(x) = v(x)$.*

*Proof.* Consider any $x \in U$.
Since $U$ is open, we can find some $r > 0$, such that $B(x, r) \subseteq U$.
For any $i \geq \lceil \frac{1}{r} \rceil$, we must have some $x_i \in B(x, \frac{1}{i}) \subseteq U$, such that $u(x_i) = v(x_i)$.
Otherwise $\{x \in U : u(x) \neq v(x)\} \supseteq B(x, \frac{1}{i}) \cap U = B(x, \frac{1}{i})$ does not have measure 0.
Thus $\lim_{i\to\infty} x_i = x$.
Since $u, v$ are both continuous, we have that

$$\begin{aligned} u(x) &= u\Big(\lim_{i\to\infty} x_i\Big) \\ &= \lim_{i\to\infty} u(x_i) \\ &= \lim_{i\to\infty} v(x_i) \\ &= v(\lim_{i\to\infty} x_i) \\ &= v(x). \end{aligned}$$

This is true for any $x \in U$, which completes the proof. □

**Proposition 2.14** (Continuous classical derivative is weak derivative)**.** *Given any $u \in C^{k-1}(U)$, and any $|\alpha| \leq k$. Suppose its $\alpha^{th}$ classical derivative $D^\alpha u$ exists and is continuous, then for any $v = u$ a.e., we have that $D^\alpha u$ is an $\alpha^{th}$ weak derivative of $v$. In particular, this holds for all $u \in C^k(U)$.*

*Proof.* Consider any $\phi \in C_c^\infty(U)$, we have that

$$\begin{aligned} \int_U v D^\alpha \phi dx &= \int_U u D^\alpha \phi dx \\ &= (-1)^{|\alpha|} \int_U D^\alpha u \phi dx, \end{aligned}$$

where the second equality follows from integration by parts over some $\text{Supp}(\phi) \subseteq V \subseteq U$ with Lipschitz boundary. □

**Definition 2.11.** Let $k \in \mathbb{N}$. We define

$$W^k(U) := \left\{ u \in L^1_{loc}(U) : \forall |\alpha| \leq k, \ \exists v \in L^1_{loc}(U) \text{ such that } D^\alpha u = v \text{ in the weak sense} \right\}.$$

Namely, it is the set of functions in $L^1_{loc}(U)$, whose $\alpha^{th}$-weak derivative exists for each $|\alpha| \leq k$.

**Theorem 2.15.** *Assume $u, v \in W^k(U)$, then for any $|\alpha| \leq k, \lambda \in \mathbb{R}$,*

1. *$D^\alpha u \in W^{k-|\alpha|}(U)$;*
2. *$D^\beta(D^\alpha u) = D^\alpha(D^\beta u) = D^{\alpha+\beta} u, \forall \alpha, \beta$ such that $|\alpha| + |\beta| \leq k$;*
3. *$\lambda u + v \in W^k(U)$;*
4. *$\forall V \subseteq U$ be open, $u \in W^k(V)$.*

*Proof.* 1. This is by definition.

2. Consider any $\phi \in C_c^\infty(U)$, we have

$$\begin{aligned}\int_U D^\alpha(D^\beta u)\phi dx &= (-1)^{|\alpha|}\int_U D^\beta u D^\alpha \phi dx \\ &= (-1)^{|\alpha|}(-1)^{|\beta|}\int_U u D^\beta(D^\alpha\phi)dx \\ &= (-1)^{|\alpha+\beta|}\int_U u D^{\alpha+\beta}\phi dx \\ &= \int_U D^{\alpha+\beta}u\phi dx.\end{aligned}$$

Thus $D^{\alpha+\beta}u = D^\alpha(D^\beta u)$. Similarly, $D^{\alpha+\beta}u = D^\beta(D^\alpha u)$.

□

**Proposition 2.16** (Leibniz rule for weak derivatives)**.** *Assume $u \in W^k(U), |\alpha| \le k$. If $\xi \in C_c^\infty(U), \xi u \in W^k(U)$, and the Leibniz formula holds:*

$$D^\alpha(\xi u) = \sum_{\beta\le\alpha}\binom{\alpha}{\beta} D^\beta \xi D^{\alpha-\beta}u,$$

*where* $\binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha-\beta)!} = \binom{\alpha_1}{\beta_1}\cdots\binom{\alpha_n}{\beta_n}$, *and* $\alpha! := \alpha_1!\cdots\alpha_n!$.

*Proof.* We have $\forall\phi \in C_c^\infty(U), \int_U \xi u D^\alpha\phi dx = (-1)^{|\alpha|}\int_U D^\alpha(\xi u)\phi dx$.
We prove by induction:
The base case is $|\alpha| = 1$, we have by Leibniz rule on regular derivatives:

$$\begin{aligned}D^\alpha(\xi\phi) &= \xi D^\alpha\phi + \phi D^\alpha\xi \\ \int_U \xi u D^\alpha\phi dx &= \int_U u(D^\alpha(\xi\phi) - \phi D^\alpha\xi)dx \\ &= \int_U u D^\alpha(\xi\phi)dx - \int_U u\phi D^\alpha\xi dx \\ &= -\int_U \xi\phi D^\alpha u dx - \int_U u\phi D^\alpha\xi dx \\ &= -\int_U \phi(\xi D^\alpha u + u D^\alpha\xi)dx.\end{aligned}$$

Since this hold for any $\phi \in C_c^\infty(U)$, we have

$$\xi D^\alpha u + u D^\alpha\xi = D^\alpha(u\xi).$$

Now suppose $l < k$ and the result holds $\forall|\beta| \le l$.

Consider any $|\alpha| = l+1$, we have $\alpha = \beta + \gamma$ where $|\beta| = l, |\gamma| = 1$.

$$\begin{aligned}
\int_U \xi u D^\alpha \phi dx &= \int_U \xi u D^\beta (D^\gamma \phi) dx \\
&= (-1)^{|\beta|} \int_U D^\beta (\xi u) D^\gamma \phi dx \\
&= (-1)^{|\beta|} \int_U \sum_{\sigma \le \beta} \binom{\beta}{\sigma} D^\sigma \xi D^{\beta-\sigma} u D^\gamma \phi dx \\
&= (-1)^{|\beta|}(-1)^{|\gamma|} \int_U \sum_{\sigma \le \beta} \binom{\beta}{\sigma} D^\gamma (D^\sigma \xi D^{\beta-\sigma} u) \phi dx \\
&= (-1)^{|\beta+\gamma|} \int_U \sum_{\sigma \le \beta} \binom{\beta}{\sigma} \left(D^\sigma \xi D^\gamma D^{\beta-\sigma} u + D^{\beta-\sigma} u D^\gamma D^\sigma \xi\right) \phi dx \\
&= (-1)^{|\alpha|} \int_U \sum_{\sigma \le \beta} \binom{\beta}{\sigma} \left(D^\sigma \xi D^{\gamma+\beta-\sigma} u + D^{\beta-\sigma} u D^{\gamma+\sigma} \xi\right) \phi dx \\
&= (-1)^{|\alpha|} \int_U \sum_{\sigma \le \beta} \binom{\beta}{\sigma} \left(D^\sigma \xi D^{\alpha-\sigma} u + D^{\alpha-(\gamma+\sigma)} u D^{\gamma+\sigma} \xi\right) \phi dx \\
&= (-1)^{|\alpha|} \int_U \left( \sum_{\sigma \le \beta} \binom{\beta}{\sigma} D^\sigma \xi D^{\alpha-\sigma} u + \sum_{\rho \le \alpha, \rho_j \ge 1} \binom{\beta}{\rho - \gamma} D^{\alpha-\rho} u D^\rho \xi \right) \phi dx \\
&= (-1)^{|\alpha|} \int_U \left( \sum_{\sigma \le \alpha} \left( \mathbb{1}_{\sigma_j \le \alpha_j - 1} \binom{\beta}{\sigma} + \mathbb{1}_{\sigma_j \ge 1} \binom{\beta}{\sigma - \gamma} \right) D^\sigma \xi D^{\alpha-\sigma} u \right) \phi dx,
\end{aligned}$$

where $\gamma_i = \delta_{ij}$.
Now consider any $\sigma \le \alpha$.
If $\sigma_j = 0$, we have

$$\begin{aligned}
\binom{\beta}{\sigma} &= \frac{\beta!}{\sigma!(\beta-\sigma)!} \\
&= \frac{\beta!(\beta_j+1)}{\sigma!(\beta-\sigma)!(\beta_j+\sigma_j+1)} \\
&= \frac{(\beta+\gamma)!}{\sigma!(\beta-\sigma+\gamma)!} \\
&= \frac{\alpha!}{\sigma!(\alpha-\sigma)!} \\
&= \binom{\alpha}{\sigma}.
\end{aligned}$$

If $\sigma_j = \alpha_j$, we have

$$\begin{aligned}\binom{\beta}{\sigma-\gamma} &= \frac{\beta!}{(\sigma-\gamma)!(\beta-\sigma+\gamma)!} \\ &= \frac{\beta!\alpha_j}{\alpha_j(\sigma-\gamma)!(\alpha-\sigma)!} \\ &= \frac{\beta!(\beta_j+1)}{\sigma_j(\sigma-\gamma)!(\alpha-\sigma)!} \\ &= \frac{(\beta+\gamma)!}{(\sigma-\gamma+\gamma)!(\alpha-\sigma)!} \\ &= \frac{\alpha!}{\sigma!(\alpha-\sigma)!} \\ &= \binom{\alpha}{\sigma}.\end{aligned}$$

If $1 \leq \sigma_j \leq \alpha_j - 1$, we have

$$\begin{aligned}\binom{\beta}{\sigma} + \binom{\beta}{\sigma-\gamma} &= \frac{\beta!}{\sigma!(\beta-\sigma)!} + \frac{\beta!}{(\sigma-\gamma)!(\beta-\sigma+\gamma)!} \\ &= \frac{\beta!(\beta_j-\sigma_j+1)}{\sigma!(\beta-\sigma)!(\beta_j-\sigma_j+1)} + \frac{\beta!\sigma_j}{\sigma_j(\sigma-\gamma)!(\beta-\sigma+\gamma)!} \\ &= \frac{\beta!(\beta_j-\sigma_j+1)+\beta!\sigma_j}{\sigma!(\beta-\sigma+\gamma)!} \\ &= \frac{\beta!(\beta_j+1)}{\sigma!(\beta-\sigma+\gamma)!} \\ &= \frac{(\beta+\gamma)!}{\sigma!(\beta-\sigma+\gamma)!} \\ &= \frac{\alpha!}{\sigma!(\alpha-\sigma)!} \\ &= \binom{\alpha}{\sigma}.\end{aligned}$$

Thus we can see that

$$\int_U \xi u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U \left( \sum_{\sigma \leq \alpha} \binom{\alpha}{\sigma} D^\sigma \xi D^{\alpha-\sigma} u \right) \phi dx.$$

Since $\phi$ is arbitrary, we have that

$$D^\alpha(\xi u) = \sum_{\sigma \leq \alpha} \binom{\alpha}{\sigma} D^\sigma \xi D^{\alpha-\sigma} u.$$

Inductively, we can prove this for any $|\alpha| = n \geq 1$. □

**Proposition 2.17.** *For $u, v \in L^1_{loc}(U)$, suppose $v|_V = D^\alpha(u|_V)$ on every $V \subset\subset U$, then $v$ is the $\alpha^{th}$ weak derivative of $u$ on globally. i.e. $v = D^\alpha u$.*

*Proof.* Consider any $\phi \in C_c(U)$, we have that $\text{Supp}(\phi) \subset\subset U$, so we can find a $\text{Supp}(\phi) \subset\subset V \subset\subset U$. Now

$$\int_U \phi v dx = \int_V \phi v dx = (-1)^{|\alpha|} \int_V D^\alpha \phi v dx = (-1)^{|\alpha|} \int_U D^\alpha \phi v dx,$$

since $v|_V = D^\alpha(u|_V)$ and $\phi$ is constantly 0 outside of $V$. □

### 2.3.2 Sobolev Spaces

**Definition 2.12.** Let $k \in \mathbb{N}, 1 \leq p \leq \infty, u \in W^k(U)$. The **Sobolev norm** of $u$ is

$$\|u\|_{W^{k,p}(U)} := \begin{cases} \left(\sum_{|\alpha|\leq k} \|D^\alpha u\|^p_{L^p(U)}\right)^{1/p}, & 1 \leq p < \infty \\ \sum_{|\alpha|\leq k} \operatorname{ess\,sup}_{x\in U} |D^\alpha u(x)| \cong \max_{|\alpha|\leq k} \|D^\alpha u\|_{L^\infty(U)}, & p = \infty \end{cases}.$$

**Definition 2.13.** For $k = 1$, we write

$$\|Du\|^p_{L^p(U)} := \int_U \|Du\|^p_p dx = \int_U \sum_{i=1}^n |\partial_i u|^p dx = \sum_{i=1}^n \|\partial_i u\|^p_{L^p(U)}$$

for $1 \leq p < \infty$, and

$$\|Du\|_{L^\infty(U)} := \operatorname*{ess\,sup}_{x\in U} \|Du(x)\|_1 = \operatorname*{ess\,sup}_{x\in U} \sum_{i=1}^n |\partial_i u(x)| = \sum_{i=1}^n \|\partial_i u\|_{L^\infty(U)}$$

for $p = \infty$.
In this case,

$$\|u\|_{W^{1,p}(U)} = \begin{cases} \left(\|u\|^p_{L^p(U)} + \|Du\|^p_{L^p(U)}\right)^{1/p} & 1 \leq p < \infty \\ \|u\|_{L^\infty(U)} + \|Du\|_{L^\infty(U)} & p = \infty. \end{cases}.$$

**Proposition 2.18.** *Let $k \in \mathbb{N}, 1 \leq p \leq \infty, u \in L^1_{loc}(U)$. We have*

$$\forall |\alpha| \leq k, \|u\|_{W^{k,p}(U)} \geq \|D^\alpha u\|_{L^p(U)}.$$

**Definition 2.14.** Let $k \in \mathbb{N}, 1 \leq p \leq \infty$. The **Sobolev space** is defined as

$$W^{k,p}(U) := \left\{v \in L^1_{loc}(U) : \|v\|_{W^{k,p}(U)} < \infty\right\}.$$

**Proposition 2.19.** *Let $k \in \mathbb{N}, 1 \leq p \leq \infty, u \in L^1_{loc}(U)$, then $u \in W^{k,p}(U)$ if and only if*

$$\forall |\alpha| \leq k,\ D^\alpha u \in L^p(U),$$

*where $D^\alpha u \in L^p(U)$ is the $\alpha^{th}$ weak derivative of $u$.*

**Definition 2.15.** In the special case where $p = 2$, we write

$$H^k(U) := W^{k,2}(U) = \left\{v \in W^k(U) : \|v\|_{H^k(U)} < \infty\right\},$$

where

$$\|u\|_{H^k(U)} := \left(\sum_{|\alpha|\leq k} \|D^\alpha u\|^2_{L^p(U)}\right)^{1/2}.$$

*Remark.*

$$W^{0,1}(U) = H^0(U) = L^2(U).$$

**Definition 2.16.** Let $(u_m)_{m=1}^\infty, u \in W^{k,p}(U)$, then

- $u_m \to u$ in $W^{k,p}(U)$ if $\lim_{m\to\infty} \|u_m - u\|_{W^{k,p}(U)} = 0$.
- $u_m \to u$ in $W^{k,p}_{loc}(U)$ if $u_m \to u$ in $W^{k,p}(V)$ for all $V \subset\subset U$.

**Theorem 2.20.** *Assume $u, v \in W^{k,p}(U),\ |\alpha| \leq k$, then*

1. *$D^\alpha u \in W^{k-|\alpha|,p}(U)$.*

2. *$D^\beta(D^\alpha u) = D^\alpha(D^\beta u) = D^{\alpha+\beta}u, \forall \alpha, \beta$ such that $|\alpha| + |\beta| \leq k$.*

3. *$\lambda u + v \in W^{k,p}(U), D^\alpha(\lambda u + v) = \lambda D^\alpha u + D^\alpha v, \forall \lambda \in \mathbb{R}$.*

4. *$\forall V \subseteq U$ be open, $u \in W^{k,p}(V)$.*

5. *Assume $u \in W^{k,p}(U), |\alpha| \leq k$. If $\xi \in C_c^\infty(U), \xi u \in W^{k,p}(U)$, and the Leibniz formula holds:*

$$D^\alpha(\xi u) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta \xi D^{\alpha-\beta} u,$$

*where $\binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha-\beta)!} = \binom{\alpha_1}{\beta_1} \cdots \binom{\alpha_n}{\beta_n}$, and $\alpha! := \alpha_1! \cdots \alpha_n!$.*

*Proof.* This is direct from the previous result on weak derivatives, by noticing $\lambda, \binom{\alpha}{\beta}, \left\|D^\beta \xi\right\|_\infty < \infty$, and using triangle inequality and linearity of the $L^p(U)$ norm. □

**Theorem 2.21.** *$\left(W^{k,p}(U), \|\cdot\|_{W^{k,p}(U)}\right)$ is a Banach space for $k \in \mathbb{N}, 1 \leq p \leq \infty$.*

*Proof.* See A2 for the proof of $\|\cdot\|_{W^{1,\infty}(U)}$ is a norm.
Now for $1 \leq p < \infty$, we want to check:

1. If $\|u\|_{W^{k,p}(U)} = 0$, then $\|u\|_{L^p(U)} = 0$, and thus $u = 0$ a.e. on $U$.

2. If $u = 0$ a.e. on $U$, then $\forall \phi \in C_c^\infty(U)$, we have

$$\int_U D^\alpha u \phi dx = (-1)^{|\alpha|} \int_U u D^\alpha \phi dx = 0.$$

   Thus $D^\alpha u = 0$ a.e. for any $|\alpha| \leq k$.
   Thus $\|u\|_{W^{k,p}(U)} = 0$.

3. Let $\lambda \in \mathbb{R}$, we have

$$\begin{aligned}
\|\lambda u\|_{W^{k,p}(U)} &= \left( \sum_{|\alpha| \leq k} \|D^\alpha(\lambda u)\|_{L^p(U)}^p \right)^{1/p} \\
&= \left( \sum_{|\alpha| \leq k} \|\lambda D^\alpha(u)\|_{L^p(U)}^p \right)^{1/p} \\
&= \left( \sum_{|\alpha| \leq k} |\lambda|^p \|D^\alpha(u)\|_{L^p(U)}^p \right)^{1/p} \\
&= |\lambda| \left( \sum_{|\alpha| \leq k} \|D^\alpha(u)\|_{L^p(U)}^p \right)^{1/p} \\
&= |\lambda| \|u\|_{W^{k,p}(U)}.
\end{aligned}$$

4. Consider any $u, v \in W^{k,p}(U)$,

$$\begin{aligned}
\|u+v\|_{W^{k,p}(U)} &= \left(\sum_{|\alpha|\le k} \|D^\alpha(u+v)\|^p_{L^p(U)}\right)^{1/p} \\
&\le \left(\sum_{|\alpha|\le k} \left(\|D^\alpha u\|_{L^p(U)} + \|D^\alpha v\|_{L^p(U)}\right)^p\right)^{1/p} \\
&\le \left(\sum_{|\alpha|\le k} \|D^\alpha u\|^p_{L^p(U)}\right)^{1/p} + \left(\sum_{|\alpha|\le k} \|D^\alpha u\|^p_{L^p(U)}\right)^{1/p} \\
&= \|u\|_{W^{k,p}(U)} + \|v\|_{W^{k,p}(U)}.
\end{aligned}$$

Thus $\|\cdot\|_{W^{k,p}(U)}$ is a norm.
Consider any Cauchy sequence $(u_m)_{m=1}^\infty$.
Given any $\epsilon > 0, \exists N \ge 1,$ such that $\forall n, m \ge N, \|u_m - u_n\|_{W^{k,p}(U)} < \epsilon$.
Consider any $|\alpha| \le k$, we have

$$\|D^\alpha u_m - D^\alpha u_n\|_{L^p(U)} = \|u\|_{W^{k,p}(U)} \ge \|D^\alpha(u_m - u_n)\|_{L^p(U)} \le \|u_m - u_n\|_{W^{k,p}(U)} < \epsilon.$$

Thus $(D^\alpha u_n)_{n=1}^\infty$ must be a Cauchy sequence in $(L^p(U), \|\cdot\|_{L^p(U)})$ for any $|\alpha| \le k$.
Since $(L^p(U), \|\cdot\|_{L^p(U)})$ is complete, there must be some

$$u_\alpha \in L^p(U) \text{ such that } \lim_{n\to\infty} \|u_\alpha - D^\alpha u_n\|_{L^p(U)} = 0.$$

In particular, we have $u \in L^p(U),$ such that $\lim_{n\to\infty} \|u - u_n\|_{L^p(U)} = 0$.
Now consider any $|\alpha| \le k$.
Given any $\phi \in C_c^\infty(U)$, we have

$$\begin{aligned}
\left|\int_U u D^\alpha \phi dx - \int_U u_n D^\alpha \phi dx\right| &= \left|\int_U (u - u_n) D^\alpha \phi dx\right| \\
&\le \int_U |(u - u_n) D^\alpha \phi| dx \\
&\le \|u - u_n\|_{L^p(U)} \|D^\alpha \phi\|_{L^{\frac{p}{p-1}}(U)}, \\
\left|\int_U u_\alpha \phi dx - \int_U D^\alpha u_n \phi dx\right| &= \left|\int_U (u_\alpha - D^\alpha u_n) \phi dx\right| \\
&\le \int_U |(u_\alpha - D^\alpha u_n) \phi| dx \\
&\le \|u_\alpha - D^\alpha u_n\|_{L^p(U)} \|\phi\|_{L^{\frac{p}{p-1}}(U)}.
\end{aligned}$$

Since $u_n \to u, D^\alpha u_n \to u_\alpha$ in $L^p(U)$, and $\|\phi\|_{L^{\frac{p}{p-1}}(U)}, \|D^\alpha \phi\|_{L^{\frac{p}{p-1}}(U)} < \infty$, those two limits converges to 0.
Thus we have

$$\begin{aligned}
\int_U u D^\alpha \phi dx &= \lim_{n\to\infty} \int_U u_n D^\alpha \phi dx \\
&= \lim_{n\to\infty} (-1)^{|\alpha|} \int_U D^\alpha u_n \phi dx \\
&= (-1)^{|\alpha|} \int_U u_\alpha \phi dx.
\end{aligned}$$

Since this is true for any $\phi \in C_c^\infty(U)$, we have that $D^\alpha u = u_\alpha = \lim_{n\to\infty} D^\alpha u_n$ in $L^p(U)$.
Since this is true for any $|\alpha| \le k$, we have that $u_n \to u$ in $W^{k,p}(U)$. □

**Proposition 2.22.** *For any $1 \le s \le r < \infty, k \ge 1$, and bounded $U$, we have some constant $C := |U|^{\frac{1}{s}-\frac{1}{r}} m^{\frac{1}{s}-\frac{1}{r}}$, where $m = |\{\beta \in \mathbb{N}^n : |\beta| \le k\}|$, such that*

$$\forall u \in W^{k,r}(U),\ \|u\|_{W^{k,s}(U)} \le C\|u\|_{W^{k,r}(U)}.$$

*Thus, $W^{k,r}(U) \subseteq W^{k,s}(U)$.*

*Proof.* We have

$$\begin{aligned}
\|u\|^s_{W^{1,s}(U)} &= \sum_{|\beta|\le 1} \left\|D^\beta u\right\|^s_{L^s(U)} \\
&\le \sum_{|\beta|\le 1} \left(|U|^{\frac{1}{s}-\frac{1}{r}} \left\|D^\beta u\right\|_{L^r(U)}\right)^s \\
&= \left(|U|^{\frac{1}{s}-\frac{1}{r}}\right)^s \sum_{|\beta|\le 1} \left\|D^\beta u\right\|^{r*\frac{s}{r}}_{L^r(U)} \\
&\le \left(|U|^{\frac{1}{s}-\frac{1}{r}}\right)^s m^{1-\frac{s}{r}} \left(\sum_{|\beta|\le 1} \left\|D^\beta u\right\|^r_{L^r(U)}\right)^{\frac{s}{r}} \\
&= \left(|U|^{\frac{1}{s}-\frac{1}{r}}\right)^s m^{1-\frac{s}{r}} \left(\|D^\alpha u\|^r_{W^{1,r}(U)}\right)^{\frac{s}{r}} \\
&= \left(|U|^{\frac{1}{s}-\frac{1}{r}}\right)^s m^{1-\frac{s}{r}} \|D^\alpha u\|^s_{W^{1,r}(U)} \\
&\implies \\
\|u\|_{W^{1,s}(U)} &\le |U|^{\frac{1}{s}-\frac{1}{r}} m^{\frac{1}{s}-\frac{1}{r}} \|D^\alpha u\|_{W^{1,r}(U)}.
\end{aligned}$$

□

**Definition 2.17.**

$$W^{k,p}_0(U) = \overline{C^\infty_c(U)} = \left\{u \in W^{k,p}(U) : \exists (u_m)_{m=1}^\infty \subset C^\infty_c(U) \text{ such that } u_m \to u \text{ in } W^{k,p}(U)\right\}.$$

$$H^k_0(U) = W^{k,2}_0.$$

*Remark.* $W^{k,p}_0(U)$ are those $u \in W^{k,p}(U)$ such that $D^\alpha u = 0$ on $\partial U$ for all $|\alpha| \le k$.

## 2.4 Smooth Approximation

**Proposition 2.23** (weak derivative commute with mollification)**.** *Let $1 \le p \le \infty, k \ge 1$. For any $u$ with its $\alpha^{th}$ weak derivative $v := D^\alpha u$, and $\epsilon > 0$, we have that*

$$D^\alpha(u^\epsilon) = v^\epsilon = \eta_\epsilon * v$$

*on $U_\epsilon$.*

*Proof.* Fix any $x \in U_\epsilon$, we have

$$\begin{aligned}
D^\alpha u^\epsilon(x) &= D^\alpha(\eta_\epsilon * u)(x) \\
&= (D^\alpha \eta_\epsilon * u)(x) \\
&= \int_U D^\alpha \eta_\epsilon(x-y)u(y)dy,
\end{aligned} \tag{2.7}$$

Consider $\eta_{\epsilon,x}(y) := \eta_\epsilon(x-y)$, we can see $\forall i \in [n],\ \partial_i \eta_{\epsilon,x}(y) = -\partial_i \eta_\epsilon(x-y)$, thus we have

$$\begin{aligned} D^\alpha u^\epsilon(x) &= \int_U D^\alpha \eta_\epsilon(x-y)u(y)dy \\ &= (-1)^{|\alpha|}\int_U D^\alpha \eta_{\epsilon,x}(y)u(y)dy \\ &= \int_U \eta_{\epsilon,x}(y) D^\alpha u(y)dy \\ &= \int_U \eta_\epsilon(x-y) D^\alpha u(y)dy \\ &= (\eta_\epsilon * D^\alpha u)(x). \end{aligned}$$

Since this holds for any $x \in U_\epsilon$, we have proved our result. □

**Corollary 2.24.** *Let $1 \le p < \infty, k \ge 1$. Let $u \in W^{k,p}(U)$, then for any $\epsilon > 0,\ U_\epsilon \supseteq V \supseteq \mathrm{Supp}(u) + \bar{B}(0,\epsilon)$, we have that*

$$\|u^\epsilon\|_{W^{k,p}(V)} \le \|u\|_{W^{k,p}(V)}.$$

*Proof.* By 2.23, $\forall |\alpha| \le k$, we have $D^\alpha(u^\epsilon) = \eta_\epsilon * D^\alpha u$ on the entire $U_\epsilon$ and thus on $V$.
Since $\forall |\alpha| \le k,\ \mathrm{Supp}(D^\alpha u) \subseteq \mathrm{Supp}(u)$, we have $\mathrm{Supp}(\eta_\epsilon * D^\alpha u) \subseteq \mathrm{Supp}(u) + \bar{B}(0,\epsilon) \subseteq V$. By 2.8,

$$\begin{aligned} \|u^\epsilon\|^p_{W^{k,p}(V)} &= \sum_{|\alpha|\le k} \|D^\alpha u^\epsilon\|_{L^p(V)} \\ &= \sum_{|\alpha|\le k} \|\eta_\epsilon * D^\alpha u\|_{L^p(V)} \\ &\le \sum_{|\alpha|\le k} \|D^\alpha u\|_{L^p(V)} \\ &= \|u\|^p_{W^{k,p}(V)}. \end{aligned}$$

□

**Theorem 2.25** (Local Smooth Approximation)**.** *Let $1 \le p < \infty, k \ge 1$. Suppose $U$ is open, for any $u \in L^1_{loc}(U)$, we have that*

1. *$\forall \epsilon > 0, u^\epsilon \in C^\infty(U_\epsilon)$;*
2. *Suppose $u$ admits $\alpha^{th}$ weak derivative, then $D^\alpha u^\epsilon \to D^\alpha u$ in $L^p_{loc}(U)$ as $\epsilon \to 0$;*
3. *Suppose $u \in W^{k,p}(U)$, we have that $u^\epsilon \to u$ in $W^{k,p}_{loc}(U)$ as $\epsilon \to 0$.*

*Thus, $C^\infty_c(U)$ is dense in $W^{k,p}(V)$ for any $V \subset\subset U$.*

*Proof.* 1. $\forall \epsilon > 0, u^\epsilon \in C^\infty(U_\epsilon)$ by 2.7.1.

2. Given any $V \subset\subset U$, we can find some $\epsilon_V > 0$ such that $V \subset\subset U^{\epsilon_V}$. We have $D^\alpha u \in L^p(U) \subseteq L^p_{loc}(U)$. By 2.7.5, we have that $\eta_\epsilon * D^\alpha u \to D^\alpha u$ in $L^p_{loc}(U)$ as $\epsilon \to 0$, and thus $\eta_\epsilon * D^\alpha u \to D^\alpha u$ in $L^p(V)$.

   In addition, by 2.23, $\forall \epsilon > 0,\ D^\alpha(u^\epsilon) = \eta_\epsilon * D^\alpha u$ in $U^\epsilon$. Now $\forall 0 < \epsilon < \epsilon_V,\ V \subset\subset U^{\epsilon_V} \subseteq U^\epsilon$, and thus $D^\alpha u^\epsilon = \eta_\epsilon * D^\alpha u$ in $V$. Thus $D^\alpha u^\epsilon \to D^\alpha u$ in $L^p(V)$ as $\epsilon \to 0$. Since this holds for any $V \subset\subset U$, we have that $D^\alpha u^\epsilon \to D^\alpha u$ in $L^p_{loc}(U)$ as $\epsilon \to 0$.

3. Now suppose $u \in W^{k,p}(U)$. Pick any $V \subset\subset U$. We have that $D^\alpha u^\epsilon \to D^\alpha u$ in $L^p(V)$ as $\epsilon \to 0$ for any $|\alpha| \le k$. Thus,

$$\begin{aligned}\|u - u^\epsilon\|^p_{W^{k,p}(V)} &= \sum_{|\alpha|\le k} \|D^\alpha(u - u^\epsilon)\|^p_{L^p(U)} \\ &= \sum_{|\alpha|\le k} \|D^\alpha u - D^\alpha u^\epsilon\|^p_{L^p(U)} \\ &\to 0.\end{aligned}$$

We thus have $u^\epsilon \to u$ in $W^{k,p}(V)$. Since this holds for any $V \subset\subset U$, we have that $u^\epsilon \to u$ in $W^{k,p}_{loc}(U)$. □

**Corollary 2.26.** *Suppose $U$ is open, and $u \in L^1_{loc}(U)$ is compactly supported in $U$.*

1. *For small enough $\epsilon$, we have that $u^\epsilon \in C^\infty_c(U)$;*
2. *Suppose $u$ has an $\alpha^{th}$ derivative, then $D^\alpha u^\epsilon \to D^\alpha u$ in $L^p(U)$ as $\epsilon \to 0$;*
3. *Suppose $u \in W^{k,p}(U)$, then $u^\epsilon \to u$ in $W^{k,p}(U)$ as $\epsilon \to 0$. Thus, $u \in W^{k,p}_0(U)$.*

*Proof.* 1. Since $\mathrm{Supp}(u) \subset U$ is compact, we must have $r := \min\left\{\frac{1}{2}\operatorname{dist}(\mathrm{Supp}(u), \partial U), 1\right\} > 0$. Let $W := \overline{\mathrm{Supp}(u) + \bar{B}(0, r/2)} \subset U$. Notice that it is compact, and $\forall \epsilon < \frac{r}{2}$, $\mathrm{Supp}(u^\epsilon) \subseteq \mathrm{Supp}(u) + \bar{B}(0,\epsilon) \subseteq W$, which means $u^\epsilon \in C^\infty_c(U)$.

2. We have that $D^\alpha u^\epsilon \to D^\alpha u$ in $L^p_{loc}(U)$ as $\epsilon \to 0$. Now there is some $W \subset V \subset\subset U$, so $u^\epsilon \to u$ in $W^{k,p}(V)$. In addition,

$$\begin{aligned}\|D^\alpha u - D^\alpha u^\epsilon\|^p_{L^p(U)} &= \int_U |D^\alpha u - D^\alpha u^\epsilon)|^p dx \\ &= \int_V |D^\alpha u - D^\alpha u^\epsilon|^p dx \\ &= \|D^\alpha u - D^\alpha u^\epsilon\|^p_{L^p(V)}.\end{aligned}$$

Thus $\lim_{\epsilon\to 0} \|D^\alpha u - D^\alpha u^\epsilon\|_{L^p(U)} = \lim_{\epsilon\to 0} \|D^\alpha u - D^\alpha u^\epsilon\|_{L^p(V)} = 0$.

3. Since 2 holds for all $|\alpha| \le k$, we have that $u^\epsilon \to u$ in $W^{k,p}(U)$. Thus, $u \in \overline{C^\infty_c(U)} = W^{k,p}_0(U)$.

□

**Proposition 2.27** (Global Bounded Approximation)**.** *Suppose $U = \mathbb{R}^n$. For any $u \in L^1_{loc}(\mathbb{R}^n)$, we have a family of functions $(u_r)_{r>0}$, such that the following holds:*

1. *Each $u_r$ is compactly supported in $\bar{B}(0, 2r)$, and $u_r \equiv u$ on $\bar{B}(0, r)$.*
2. *Suppose $u \in W^k(\mathbb{R}^n)$, so is each $u_r$.*
3. *Suppose $u \in W^k(\mathbb{R}^n)$, and $D^\alpha u \in L^p(\mathbb{R}^n)$ for any $|\alpha| \le k$, then we also have $D^\alpha(u_r) \in L^p(\mathbb{R}^n)$. In addition, $D^\alpha(u_r) \to D^\alpha u$ in $L^p(\mathbb{R}^n)$.*
4. *Suppose $u \in W^{k,p}(\mathbb{R}^n)$, then $u_r \to u$ in $W^{k,p}(\mathbb{R}^n)$.*

*Thus, $W^{k,p}_0(\mathbb{R}^n) = W^{k,p}(\mathbb{R}^n)$.*

*Proof.* 1. By proposition 1.41, there is a cut-off function $\eta \in C^\infty_c(\mathbb{R}^n)$, such that $\eta(x) = 1$ when $\|x\| \le 1$, and $\eta(x) = 0$ when $\|x\| \ge 2$.Notice that $\eta \in C^\infty_c(\mathbb{R}^n)$. For any $r > 0$, let $u_r(x) := u(x)\eta(x/r)$. We notice that $u_r \equiv 0$ outside of $\bar{B}(0, 2r)$, and $u_r = u$ in $\bar{B}(0, r)$.

2. Pick any $|\alpha| \le k$. By Leibniz formula, $u_r$ has its $\alpha^{th}$ weak derivative, with
$$D^\alpha(u_r) = \sum_{\beta\le\alpha} \binom{\alpha}{\beta} \frac{1}{r^{|\beta|}} D^\beta\eta D^{\alpha-\beta}u = D^\alpha u + \sum_{0<\beta\le\alpha} \frac{1}{r^{|\beta|}} \binom{\alpha}{\beta} D^\beta\eta D^{\alpha-\beta}u.$$

3. Now suppose $D^\alpha u \in L^p(\mathbb{R}^n)$. Since $\eta \in C_c^\infty(\mathbb{R}^n)$, for each $0 < \beta \le \alpha$, we have $\left\|D^\beta\eta\right\|_\infty < \infty$, so
$$\left|\frac{1}{r^{|\beta|}} \binom{\alpha}{\beta} D^\beta\eta(x) D^{\alpha-\beta}u(x)\right| \le \frac{1}{r^{|\beta|}} \left\|D^\beta\eta\right\|_\infty \left|\binom{\alpha}{\beta}\right| \left|D^{\alpha-\beta}u(x)\right|,$$
which goes to 0 a.e. when $r \to \infty$. Since $D^{\alpha-\beta}u \in L^p(\mathbb{R}^n)$, so is $\left\|D^\beta\eta\right\|_\infty \left|\binom{\alpha}{\beta}\right| D^{\alpha-\beta}u$. Also, for any $r \ge 1$, we have that
$$\left|\frac{1}{r^{|\beta|}} \binom{\alpha}{\beta} D^\beta\eta(x) D^{\alpha-\beta}u(x)\right| \le \left\|D^\beta\eta\right\|_\infty \left|\binom{\alpha}{\beta}\right| \left|D^{\alpha-\beta}u(x)\right|.$$
When $p < \infty$, by Lebesgue's dominated convergence theorem,
$$\lim_{r\to\infty} \left\|\frac{1}{r^{|\beta|}} \binom{\alpha}{\beta} D^\beta\eta D^{\alpha-\beta}u\right\|_{L^p(\mathbb{R}^n)}^p = 0.$$
When $p = \infty$, we have that
$$\lim_{r\to\infty} \left\|\frac{1}{r^{|\beta|}} \binom{\alpha}{\beta} D^\beta\eta D^{\alpha-\beta}u\right\|_{L^\infty(\mathbb{R}^n)} \le \lim_{r\to\infty} \frac{1}{r^{|\beta|}} \left\|D^\beta\eta\right\|_\infty \left|\binom{\alpha}{\beta}\right| \left\|D^{\alpha-\beta}u\right\|_{L^\infty(\mathbb{R}^n)} = 0.$$
Since this holds for all $0 < \beta \le \alpha$, we have that $\lim_{r\to\infty} \sum_{0<\beta\le\alpha} \frac{1}{r^{|\beta|}} \binom{\alpha}{\beta} D^\beta\eta D^{\alpha-\beta}u = 0$ in $L^p(\mathbb{R}^n)$. Thus, $\lim_{r\to\infty} D^\alpha(u_r) = D^\alpha u$ in $L^p(\mathbb{R}^n)$.

4. Since above holds for all $|\alpha| \le k$, we have that $\lim_{r\to\infty} u_r = u$ in $W^{k,p}(\mathbb{R}^n)$.

This shows that $W_0^{k,p}(\mathbb{R}^n)$ is dense in $W^{k,p}(\mathbb{R}^n)$. Since $W_0^{k,p}(\mathbb{R}^n)$ is closed in $W^{k,p}(\mathbb{R}^n)$, they must be the same space. □

**Corollary 2.28** (Global Smooth Approximation)**.** *$C_c^\infty(\mathbb{R}^n)$ are dense in $W^{k,p}(\mathbb{R}^n)$.*

**Theorem 2.29** (Meyer-Serrin)**.** *Let $1 \le p < \infty, k \ge 1$. Suppose $U$ is open and bounded, and $u \in W^{k,p}(U)$. There exists $(u_m)_{m=1}^\infty \subseteq C^\infty(U) \cap W^{k,p}(U)$ such that $u_m \to u$ in $W^{k,p}(U)$.*

*Proof.* Let $\delta > 0$ be given.
Let $U_i := U_{\frac{1}{i}} = \left\{x \in U : \operatorname{dist}(x, \partial u) > \frac{1}{i}\right\}$ for $i \in \mathbb{N}_+$.
We have $U_1 \subseteq \bar{U_1} \subseteq U_2 \subseteq \bar{U_2} \subseteq U_3 \subseteq \cdots U$.
Indeed, for some $x \in \bar{U_i}$, we know that $\forall y \in \partial U, \|x - y\| \ge \frac{1}{i} > \frac{1}{i+1} \implies x \in U_{i+1}$.
Since $U$ is open, for any $x \in U$, we can find some $i \ge 1$, such that $B(x, \frac{1}{i}) \subseteq U$, which means $\operatorname{dist}(x, \partial U) \ge \frac{1}{i}$, and thus $x \in \bar{U_i} \subseteq U_{i+1}$. Thus we have $U = \bigcup_{i=1}^\infty U_i$.
Let $V_i := U_{i+3} \setminus \bar{U_{i+1}}$ for $i \in \mathbb{N}_+$. Since $U$ is bounded, we can choose $V_0 \subset\subset U$ with $V_0 \supset \bar{U_2}$, we claim that $\forall n \ge 1, \bigcup_{i=0}^n V_i = U_{n+3}$.
It is easy to see $\bigcup_{i=0}^n V_i \subseteq U_{i+3}$. For the other direction, we will prove by induction.
The base case $n = 1$, we can see that $V_0 \cup V_1 \supset \bar{U_2} \cup (U_4 \setminus \bar{U_2}) = U_4$.
Now suppose $n > 1$, and it holds for $n - 1$, we have that
$$\begin{aligned}
\bigcup_{i=0}^n V_i &= \left(\bigcup_{i=0}^{n-1} V_i\right) \cup (V_n) \\
&= U_{n-1+3} \cup \left(U_{n+3} \setminus \bar{U_{n+1}}\right) \\
&\supset U_{n+2} \cup (U_{n+3} \setminus U_{n+2}) \\
&= U_{n+3}.
\end{aligned}$$

By induction, we have that $\forall n \geq 1, \bigcup_{i=0}^{n} V_i = U_{n+3}$.
Notice that $\forall x \in U = \bigcup_{n=1}^{\infty} U_n, \exists n \geq 1,$ such that $x \in U_n \subseteq U_{n+3} \subseteq \bigcup_{i=0}^{n} V_i \implies \exists i \geq 0,$ such that $x \in V_i$.
Thus

$$U = \bigcup_{i=0}^{\infty} V_i.$$

Now let $W_i := U_{i+4} \setminus \bar{U}_i$ for $i \in \mathbb{N}_+$.
Since each $U_{i+4} \subseteq \bar{U}_{i+4} \subseteq U_{i+5} \subseteq U$, we also have $U_{i+4} \subset\subset U$ and thus

$$W_i \subset\subset U.$$

Notice that $\forall x, y \in U$,

$$\begin{aligned}
\operatorname{dist}(x, \partial U) &= \inf\{\|z - x\| : z \in \partial U\} \\
&= \inf\{\|z - y + y - x\| : z \in \partial U\} \\
&\leq \inf\{\|z - y\| + \|y - x\| : z \in \partial U\} \\
&= \inf\{\|z - y\| : z \in \partial U\} + \|y - x\| \\
&= \operatorname{dist}(y, \partial U) + \|y - x\|.
\end{aligned}$$

Similarly, $\operatorname{dist}(y, \partial U) \leq \operatorname{dist}(x, \partial U) + \|y - x\|$. Thus we have

$$\operatorname{dist}(y, \partial U) - \|y - x\| \leq \operatorname{dist}(x, \partial U) \leq \operatorname{dist}(y, \partial U) + \|y - x\|.$$

Consider any $0 < \epsilon < \frac{1}{i+3} - \frac{1}{i+4} < \frac{1}{i} - \frac{1}{i+1}$, we have that

$$\begin{aligned}
x \in \bar{B}(0, \epsilon) + V_i &\implies \exists y \in U_{i+3} \setminus \bar{U}_{i+1} \text{ such that } \|x - y\| \leq \epsilon \\
&\implies \exists y \in U \text{ such that } \frac{1}{i+3} < \operatorname{dist}(y, \partial U) < \frac{1}{i+1}, \|x - y\| \leq \epsilon \\
&\implies \exists y \in U \text{ such that } \frac{1}{i+3} - \|x - y\| < \operatorname{dist}(x, \partial U) < \frac{1}{i+1} + \|x - y\|, \|x - y\| \leq \epsilon \\
&\implies \frac{1}{i+3} - \epsilon < \operatorname{dist}(x, \partial U) < \frac{1}{i+1} + \epsilon \\
&\implies \frac{1}{i+4} < \operatorname{dist}(x, \partial U) < \frac{1}{i} \\
&\implies x \in W_i.
\end{aligned}$$

Thus we have

$$\forall 0 < \epsilon < \frac{1}{i+3}, \ \bar{B}(0, \epsilon) + V_i \subseteq W_i.$$

Finally, since $V_0 \subset\subset U$, we can choose $V_0 \subset\subset W_0 \subset\subset U$, such that $V_0 + B(0, \epsilon_0'') \subseteq W_i$ for some $\epsilon_0' > 0$.
Let $(\zeta_i)_{i=0}^{\infty}$ be a **smooth partition of unity subordinate to** $\{V_i\}_{i \in \mathbb{N}}$ such that

$$\forall x \in U \ \sum_{i=0}^{\infty} \zeta_i(x) = 1, \ \forall i \geq 0, \begin{cases} 0 \leq \zeta_i \leq 1, \\ \zeta_i \in C^{\infty}(U), \\ \operatorname{Supp}(\zeta_i) \subseteq V_i. \end{cases}$$

Since $\operatorname{Supp}(\zeta_i) \subseteq V_i$ is closed and bounded, we have that $\zeta_i \in C_c^{\infty}(U)$. Thus, $\forall u \in W^{k,p}(U)$, we have $\zeta_i u \in W^{k,p}(U)$ as well. Moreover, $\operatorname{Supp}(\zeta_i u) \subseteq V_i$.
Let $u_i^{\epsilon} := \eta_{\epsilon} * (\zeta_i u) \ \forall \epsilon > 0$.
By previous theorem, we have that $u_i^{\epsilon} \to \zeta_i u$ in $W_{loc}^{k,p}(U)$.
Thus for $W_i \subset\subset U$, we can find $\epsilon_i' > 0$ such that $\forall \epsilon < \epsilon_i', \ \|u_i^{\epsilon} - \zeta_i u\|_{W^{k,p}(W_i)} < \frac{\delta}{2^{i+1}}$.
Now pick $\epsilon_0 = \min(\epsilon_0'', \epsilon_0'), \forall i \in \mathbb{N}_+, \epsilon_i = \frac{1}{2} \min\left(\frac{1}{i+3} - \frac{1}{i+4}, \epsilon_i'\right) > 0$.
We have that by 2.7,

$$\operatorname{Supp}(u_i^{\epsilon_i}) \subseteq \operatorname{Supp}(\eta_{\epsilon_i}) + \operatorname{Supp}(\zeta_i u) \subseteq \bar{B}(0, \epsilon_i) + V_i \subseteq W_i,$$

and

$$\|u_i^{\epsilon_i} - \zeta_i u\|_{W^{k,p}(U)} = \|u_i^{\epsilon_i} - \zeta_i u\|_{W^{k,p}(W_i)} < \frac{\delta}{2^{i+1}}.$$

Now let $v := \sum_{i=0}^{\infty} u_i^{\epsilon_i}$.
Notice that $\forall x \in U, \exists V \subset\subset U_{\epsilon_i}$ be open, such that $x \in V$. Since $V \cap W_i \neq \emptyset$ for only finitely many $i$, and $\text{Supp}(u_i^{\epsilon_i}) \subseteq W_i$, we must have $v = \sum_{i=0}^{k} u_i^{\epsilon_i}$ on $V$ for some finite $k$.
In addition, by 2.7, each $u_i^{\epsilon_i} \in C^\infty(U_{\epsilon_i})$, thus infinitely differentiable at $x$.
Thus $v = \sum_{i=0}^{k} u_i^{\epsilon_i}$ is infinitely differentiable at $x$.
Since $x \in U$ is arbitrary, we have that $v \in C^\infty(U)$.
In addition,

$$\forall x \in U,\ u(x) = \sum_{i=0}^{\infty} \zeta_i(x)u(x) = \sum_{i=0}^{\infty} (\zeta_i u)(x)$$

by definition of partition of unity. Thus

$$u(x) - v(x) = \sum_{i=0}^{\infty} (\zeta_i u)(x) - \sum_{i=0}^{\infty} u_i^{\epsilon_i}(x) = \sum_{i=0}^{\infty} (\zeta_i u)(x) - u_i^{\epsilon_i}(x) = \sum_{i=0}^{\infty} (\zeta_i u - u_i^{\epsilon_i})(x).$$

Since this holds for all $x \in U$, we have that

$$u - v = \sum_{i=0}^{\infty} \zeta_i u - u_i^{\epsilon_i}.$$

Now we have

$$\begin{aligned}
\|v - u\|_{W^{k,p}(U)} &= \left\| \sum_{i=0}^{\infty} \zeta_i u - u_i^{\epsilon_i} \right\|_{W^{k,p}(U)} \\
&\leq \sum_{i=0}^{\infty} \|\zeta_i u - u_i^{\epsilon_i}\|_{W^{k,p}(U)} \\
&< \sum_{i=0}^{\infty} \frac{\delta}{2^{i+1}} \\
&= \delta.
\end{aligned}$$

□

**Definition 2.18.** Let $U \subseteq \mathbb{R}^n$ be open and bounded, then $\partial U$ is $C^k$ if $\forall z \in \partial U, \exists r > 0, \gamma \in C^k(\mathbb{R}^{n-1})$, such that

$$U \cap \bar{B}(z,r) = \left\{ x \in B(z,r) : x^n > \gamma(x^1, \ldots, x^{n-1}) \right\}.$$

**Theorem 2.30.** *Let $U$ be bounded, and $\partial U$ is $C^1$, then $\forall u \in W^{k,p}(U)$ for $1 \leq p < \infty$, there exists functions $u_m \in C^\infty(\bar{U})$ such that $u_m \to u$ in $W^{k,p}(U)$.*

*Proof.* See 5.3.3 in Eavan's book. □

## 2.5 Extensions

**Proposition 2.31.** *Let $U \subseteq \mathbb{R}^n$ be open and bounded, with $\partial U$ be $C^k$. Then $\forall z \in \partial U, \exists r > 0, \Phi \in C^k(B(z,r), \mathbb{R}^n)$ a diffeomorphism, such that $\Phi(\partial U \cap B(z,r))$ is in a flat hyperplane, and $\det(D\Phi) = \det(D\Psi) = 1$, for $\Psi := \Phi^{-1}$.*

*Proof.* Let

$$\Phi^i(x) := x^i\ \forall i \in [n-1],\ \Phi^n(x) := x^n - \gamma(x^1, \ldots, x^{n-1}),$$

and let

$$\Psi^i(y) := y^i\ \forall i \in [n-1],\ \Psi^n(y) := y^n + \gamma(y^1, \ldots, y^{n-1}).$$

□

**Theorem 2.32** (Sobolev Norm Equivalence Under Diffeomorphism)**.** *Let $W \subseteq \mathbb{R}^n$ and $\Phi$ is a $C^1(W)$ diffeomorphism, i.e., it has inverse $\Psi \in C^1(W)$. Let $v := u \circ \Psi$, then*

$$\exists C_0, C_1 \text{ such that } C_0\|u\|_{W^{1,p}(W)} \leq \|v\|_{W^{1,p}(\Phi(W))} \leq C_1\|u\|_{W^{1,p}(W)}.$$

**Lemma 2.33.** *Let $1 \leq p < \infty$. Assume $U$ is bounded, with $\partial U$ be $C^1$. Let $V$ be open and bounded, with $U \subset\subset V$, then there exists a bounded linear operator $E : C^1(\bar{U}) \to W^{1,p}(\mathbb{R}^n)$, such that $\forall u \in C^1(\bar{U})$ :*

1. *$Eu = u$ in $U$,*
2. *$\mathrm{Supp}(Eu) \subseteq V$,*
3. *$\exists C > 0$, such that $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)}$.*

*Proof.* Fix $z \in \partial U$.
In addition, we assume $\partial U$ is flat around $z$ on the plane $\{x^n = 0\}$.
Then there exists an open ball $B := B(z, r)$, such that

$$B^+ := B \cap \{x^n > 0\} \subseteq \bar{U}, B^- := B \cap \{x^n \leq 0\} \subseteq \mathbb{R}^n \setminus U.$$

Let $\bar{u}_z(x) := \begin{cases} u(x) & x \in B^+ \\ -3u(x^1, \ldots, x^{n-1}, -x^n) + 4u(x^1, \ldots, x^{n-1}, -\frac{1}{2}x^n) & x \in B^-. \end{cases}$
Then we claim $\bar{u}_z \in C^1(B)$.
Indeed, let $u^- := \bar{u}_z|_{B^-}, u^+ := \bar{u}_z|_{B^+}$.

$$\begin{aligned} u^-|_{x^n=0} &= -3 + 4u|_{x^n=0} \\ &= u|_{x^n=0} \\ &= u^+|_{x^n=0}; \\ &\forall i \in [n-1], \\ \partial_i u^-|_{x^n=0} &= -3\partial_i u|_{x^n=0} + 4\partial_i u|_{x^n=0} \\ &= \partial_i u|_{x^n=0} \\ &= \partial_i u^+|_{x^n=0} \\ \partial_n u^-|_{x^n=0} &= 3\partial_n u|_{x^n=0} - 4\frac{1}{2}\partial_n u|_{x^n=0} \\ &= u|_{x^n=0} \\ &= \partial_n u^+|_{x^n=0}. \end{aligned}$$

Thus $\bar{u} \in C^1(B)$.
By A2, we have

$$\exists C > 0, \text{ such that } \|\bar{u}_z\|_{W^{1,p}(B)} \leq C\|u\|_{W^{1,p}(B^+)}.$$

Now suppose $\partial U$ is not flat around $z$, we can find $r_1 > 0, \Phi \in C^1(B(z, r_1), \mathbb{R}^n)$, such that $\Phi(\partial U \cap B(z, r_1))$ is in a flat hyperplane, WLOG $\{y_n = 0\}$, and $\det(D\Phi) = \det(D\Psi) = 1$, for $\Psi := \Phi^{-1}$.
Notice that we can find $B(z, r_2) \subset\subset V$ since $V$ is open and $z \in \bar{U} \subseteq V$.
By setting $r = \min(r_1, r_2) > 0$, we can WLOG work with $B(z, r) \subset\subset V$.
Let $z' := \Phi(z), v := u \circ \Psi \in C^1(\Phi(\bar{U})) = C^1(\overline{\Phi(U)})$.
Since $\Phi(B(z, r))$ is open, we can choose some open ball $B := B(z', r') \subseteq \Phi(B(z, r))$. Let $W_z := \Psi(B)$.
Since $\Phi(\partial U \cap B(z, r))$ is in the plane $\{y_n = 0\}$, we have

$$B^+ := B \cap \{y^n \geq 0\} = \Phi(W_z \cap \bar{U}), B^- := B \cap \{y^n \leq 0\} = \Phi(W_z \setminus U).$$

Now we can extend $v$ form $B^+$ to $B$ with

$$\|\bar{v}\|_{W^{1,p}(B)} \leq C\|v\|_{W^{1,p}(B^+)}.$$

Now let $\bar{u}_z := \bar{v} \circ \Psi$, we have that $B = \Phi(W_z), \bar{v} = \bar{u}_z \circ \Phi$, and by 2.32, we have

$$\begin{aligned}
\|\bar{u}_z\|_{W^{1,p}(W_z)} &\leq C_1 \|\bar{v}\|_{W^{1,p}(\Phi(W_z))} \\
&= C_1 \|\bar{v}\|_{W^{1,p}(B)} \\
&\leq C_2 \|v\|_{W^{1,p}(B^+)} \\
&\leq C_3 \|u\|_{W^{1,p}(\Psi(B^+))} \\
&\leq C_3 \|u\|_{W^{1,p}(U)}
\end{aligned}$$

Notice that $\forall z, \Phi(z) \in B \implies z \in W_z$, thus $\{W_z\}_{z \in \partial U}$ forms an open cover for $\partial U$.
Since $\partial U$ is compact, we can find a finite subcover $\{W_i\}_{i=1}^N$.
Notice that $\left(\bar{U} \setminus \bigcup_{i=1}^N W_i\right) \subset U$ is closed, and $U$ is bounded, so we can find $\left(\bar{U} \setminus \bigcup_{i=1}^N W_i\right) \subseteq W_0 \subset\subset U$.
We then have $\bigcup_{i=0}^N W_i = U$.
Now let $(\zeta_i)_{i=0}^N$ be a partition of unity subordinate to $W_i$, such that

$$\forall x \in U \ \sum_{i=0}^N \zeta_i(x) = 1, \ \forall i \geq 0, \begin{cases} 0 \leq \zeta_i \leq 1, \\ \zeta_i \in C^\infty(\mathbb{R}^n), \\ \text{Supp}(\zeta_i) \subseteq W_i. \end{cases}$$

Let $\bar{u} := \sum_{i=0}^N \zeta_i \bar{u}_i$, with $\bar{u}_0 := u$. We have that

$$\begin{aligned}
\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} &\leq \sum_{i=0}^N \|\zeta_i \bar{u}_i\|_{W^{1,p}(\mathbb{R}^n)} \\
&= \sum_{i=0}^N \|\zeta_i \bar{u}_i\|_{W^{1,p}(W_i)} \\
&\leq C_4 \sum_{i=0}^N \|\bar{u}_i\|_{W^{1,p}(W_i)} \\
&= C_5 \|u\|_{W^{1,p}(W_i)},
\end{aligned}$$

since each term is bounded, and we have a finite sum.
We thus define $Eu := \bar{u}$.
We can check that $E$ is linear and bounded. □

**Theorem 2.34** (Extension)**.** *Let $1 \leq p < \infty$. Assume $U$ is bounded, with $\partial U$ be $C^1$. Let $V$ be open and bounded, with $U \subset\subset V$, then there exists a bounded linear operator $E : W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$, such that $\forall u \in W^{1,p}(U)$ :*

1. *$Eu = u$ a.e. in $U$,*
2. *$\text{Supp}(Eu) \subseteq V$,*
3. *$\exists C > 0$, such that $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(U)}$.*

*Proof.* By 2.30, we know $C^\infty(\bar{U}) \subseteq C^1(\bar{U})$ is dense in $W^{1,p}(U)$, and thus $C^1(\bar{U})$ is also dense in $W^{1,p}(U)$.
By 1.22, we can extend the result in the above lemma to get $E : W^{1,p}(U) \to W^{1,p}(\mathbb{R}^n)$.
In addition, since $Eu = \lim_{m \to \infty} Eu_m$ for some $u_m \to u$ in $W^{1,p}(U)$, we also have $Eu = \lim_{m \to \infty} Eu_m = Eu = \lim_{m \to \infty} u_m = u$, a.e..
Also, $\text{Supp}(Eu) \subseteq \bigcup_{m=1}^\infty \text{Supp}(Eu_m) \subseteq V$. □

## 2.6 Traces

**Proposition 2.35** (Young's inequality)**.**

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \forall a, b > 0, \frac{1}{p} + \frac{1}{q} = 1.$$

**Lemma 2.36.** *Let $U$ be bounded, and $\partial U$ is $C^1$, and $1 \leq p < \infty$. Then there exists a bounded linear operator $T : C^1(\bar{U}) \to L^p(\partial U)$; $u \mapsto u|_{\partial U}$ and a constant $C > 0$, such that*

$$\forall u \in C^1(\bar{U}), \|Tu\|_{L^p(\partial U)} \leq C\|u\|_{W^{1,p}(U)}.$$

*Proof.* Consider $z \in \partial U$.
Assume $\partial U$ is flat near $z$ in the hyperplane $\{x^n = 0\}$.
Then there exists an open ball $B_z := B(z, r)$, such that

$$B_z^+ := B \cap \{x^n > 0\} \subseteq \bar{U}, B_z^- := B \cap \{x^n \leq 0\} \subseteq \mathbb{R}^n \setminus U.$$

Since $u$ is $C^1$ and thus continuous, WLOG, we can take $r$ small enough, such that $u$ does not change sign in $B_z$. Namely, $|u| = u \operatorname{sgn}(u(z))$ in $B_z$.
Let $\hat{B}_z := B(z, \frac{r}{2})$, and let $\xi \in C_c^\infty(B_z)$ such that $\xi \geq 0$ in $B_z$, and $\xi = 1$ in $\hat{B}_z$.
Let $\Gamma_z := \hat{B}_z \cap \partial U$, then we have $\operatorname{Supp}(\xi u) \subseteq B_z^+$, and $\xi u = u$ on $T$.
Let $x' := (x^1, \ldots, x^{n-1})$, by Fundamental Theorem of Calculus, we have

$$\int_0^\infty (\xi|u|^p)(x', t)dt = -(\xi|u|^p)(x', 0).$$

In addition, we have

$$\begin{aligned}
\|u\|_{L^p(\Gamma_z)}^p &= \int_{\Gamma_z} |u|^p(x', 0)dx' \\
&\leq \int_{\mathbb{R}^{n-1}} (\xi|u|^p)(x', 0)dx' \\
&= -\int_0^\infty \int_{\mathbb{R}^{n-1}} (\xi|u|^p)(x', t)dx'dt \\
&= -\int_{B_z^+} (\xi|u|^p)(x)dx \\
&= -\int_{B_z^+} \xi_{x_n}|u|^p + \xi p|u|^{p-1}(\operatorname{sgn} u(z))u_{x_n}dx \\
&\leq \int_{B_z^+} |\xi_{x_n}||u|^p + \xi p|u|^{p-1}|u_{x_n}|dx \\
&\leq \int_{B_z^+} |\xi_{x_n}||u|^p + \xi p\left(\frac{(|u|^{p-1})^{\frac{p}{p-1}}}{\frac{p}{p-1}} + \frac{|u_{x_n}|^p}{p}\right)dx \\
&= \int_{B_z^+} |\xi_{x_n}||u|^p + \xi(p-1)|u|^p + \xi|u_{x_n}|^p dx \\
&\leq \int_{B_z^+} (|\xi_{x_n}| + \xi(p-1))|u|^p + \xi|Du|^p dx.
\end{aligned}$$

Since $\xi \in C_c^\infty(B_z)$, by EVT, $|\xi_{x_n}|, \xi$ are all bounded. Thus $\exists C > 0,$ such that $|\xi_{x_n}| + \xi(p-1), \xi \leq C$ in $B_z$. Thus

$$\|u\|_{L^p(\Gamma_z)}^p \leq \int_{B_z^+} C|u|^p + C|Du|^p dx = C\|u\|_{W^{1,p}(B_z^+)}^p \leq C\|u\|_{W^{1,p}(U)}^p.$$

Now if $\partial U$ is not flat near $z$, we can find a $C^1$ diffeomorphism $\Phi$ to make it flat. We still have

$$\|u\|_{L^p(\Gamma_z)} \leq C\|u\|_{W^{1,p}(U)},$$

by the equivalence of Sobolev norms under diffeomorphism 2.32.
Since $\{B_z\}_{z\in\partial U}$ form an open cover for $\partial U$, and $\partial U$ is compact, we can find a finite subcover $\{B_i : x_i \in \partial U\}_{i=1}^N$, and their corresponding $\Gamma_i$.
For each $i \in [N]$, we have that

$$\|u\|_{L^p(\Gamma_i)} \leq C_i \|u\|^p_{W^{1,p}(U)}.$$

We have that

$$\begin{aligned}
\|Tu\|^p_{L^p(\partial U)} &= \int_{\partial U} |u|^p dx \\
&\leq \sum_{i=1}^N \int_{\Gamma_i} |u|^p dx \\
&= \sum_{i=1}^N \|u\|^p_{L^p(\Gamma_i)} \\
&\leq \sum_{i=1}^N C_i \|u\|^p_{W^{1,p}(U)} \\
&= C\|u\|^p_{W^{1,p}(U)},
\end{aligned}$$

by taking $C := \sum_{i=1}^N C_i$. □

**Theorem 2.37.** *Let $U$ be bounded, and $\partial U$ is $C^1$, and $1 \leq p < \infty$. Then there exists a bounded linear operator $T : W^{1,p}(U) \to L^p(\partial U)$ and a constant $C > 0$, such that*

$$\forall u \in W^{1,p}(U) \cap C(\bar{U}), Tu = u|_{\partial U},$$

*and*

$$\forall u \in W^{1,p}(U), \|Tu\|_{L^p(\partial U)} \leq C\|u\|_{W^{1,p}(U)}.$$

*Proof.* By 2.30, we know $C^\infty(\bar{U}) \subseteq C^1(\bar{U})$ is dense in $W^{1,p}(U)$, and thus $C^1(\bar{U})$ is also dense in $W^{1,p}(U)$. By 1.22, we can extend the result in the above lemma to get $T : W^{1,p}(U) \to L^p(\partial U)$. □

**Theorem 2.38.** *Let $U$ be bounded, and $\partial U$ is $C^1$, then for any $u \in W^{1,p}(U)$, we have that*

$$u \in W^{1,p}_0(U) \iff Tu = 0 \text{ on } \partial U.$$

## 2.7 Weak and classical derivatives

*Remark.* For the following part in this subsection, we will use $D^\alpha$ to denote the $\alpha^{th}$ classical derivative of $u$, and $\bar{D}^\alpha$ to be the $\alpha^{th}$ weak derivative of $u$ to avoid confusion.

**Definition 2.19.** A domain $U$ is **path-connected** if $\forall x, y$, there is some continuous path $\gamma : [0, 1] \to U$, such that $\gamma(0) = x, \gamma(1) = y$.

**Proposition 2.39.** *Let $U \subseteq \mathbb{R}^n$ be open and connected, and $1 \leq p \leq \infty$, $u \in W^{1,p}(U)$, then*

$$\bar{D}u = 0 \text{ a.e.} \iff u \text{ is a constant a.e..}$$

*Proof.* Suppose $u$ is a constant a.e., then it means $u$ has a version $\tilde{u}(x) = C$, $\forall\ x \in U$ for some constant $C$. Clearly $\tilde{u}$ is differentiable, and has continuous classical derivative $D\tilde{u}(x) = 0$, $\forall x \in U$. By 2.14, and since $\tilde{u} = u$ a.e., we have that $D\tilde{u}$ is a weak derivative of $v$. Since the weak derivative is unique a.e., we must have $\bar{D}u = 0$ a.e..

On the other hand, suppose $\bar{D}u = 0$ a.e.. We know that by 2.23, for any $\epsilon > 0, x \in U_\epsilon$, and any direction $i \in [n]$,

$$\begin{aligned}(\partial_i u^\epsilon)(x) &= (\eta_\epsilon * \partial_i u)(x) \\ &= \int_U \eta_\epsilon(x)\bar{\partial}_i u(y)dy \\ &= 0,\end{aligned}$$

since $\bar{\partial}_i u(y) = 0$ for a.e. $y \in U$. Since this holds for all $i \in [n]$, we must have $Du^\epsilon(x) = 0$. Thus, we have $Du^\epsilon(x) = 0$ in $U_\epsilon$. Notice that by 2.7, $u^\epsilon \in C^\infty(U_\epsilon)$, which by normal calculus means that $\forall x \in U_\epsilon,\ u_\epsilon(x) = C_\epsilon$ for some constant $C_\epsilon$ that does not depend on $x$.

Again by 2.7, $u^\epsilon \to u$ a.e. as $\epsilon \to 0$. Pick any such $x$ with $u^\epsilon(x) \to u(x)$. Notice that we can find $\delta > 0$, such that $\forall \epsilon \in (0, \delta), x \in U_\epsilon$. Thus we have $u(x) = \lim_{\epsilon\to 0} u_\epsilon(x) = \lim_{\epsilon\to 0} C_\epsilon$. Since $\lim_{\epsilon\to 0} C_\epsilon$ converges, we can call it $C := \lim_{\epsilon\to 0} C_\epsilon$, which is a constant that is independent of $x, \epsilon$.

Now any such $x$ satisfies $u(x) = \lim_{\epsilon\to 0} C_\epsilon = C$, and they are by choice a.e.. □

**Lemma 2.40.** *Consider $1 \le p \le \infty$, and $U = (a_1, b_1) \times \cdots \times (a_n, b_n) \subseteq \mathbb{R}^n$ be an open rectangle. Let $1 \le i \le n$, suppose $u \in W^{1,p}(U)$ has a continuous representative $u^* \in C(U)$, and its $i^{th}$ weak derivative $\bar{\partial}_i u$ has a continuous representative $(\bar{\partial}_i u)^* \in C(U)$, then the regular $i^{th}$ partial derivative*

$$\partial_i(u^*)(x) = (\bar{\partial}_i u)^*(x)\ \forall x \in U$$

*exists and is continuous.*

*Proof.* Pick some $s \in (a_i, b_i)$, let $S := \left\{x \in U : x^i = s\right\}$ be the slice of $U$.
By FTC, there is a unique $v$, defined by

$$v(x^1, \dots, x^n) := u^*(x^1, \dots, x^{i-1}, s, x^{i+1}, \dots x^n) + \int_s^{x^i} (\bar{\partial}_i u)^*(x^1, \dots, x^{i-1}, t, x^{i+1}, \dots x^n)dt,$$

such that $v|_S = u^*|_S$, and the $i^{th}$ normal partial derivative

$$\partial_i v(x) = (\bar{\partial}_i u)^*(x)\ \forall x \in U.$$

We notice that $\bar{\partial}_i v = \partial_i v$a.e. by 2.14.
Thus the weak derivative $\bar{\partial}_i(u^* - v) = \bar{\partial}_i(u^*) - \bar{\partial}_i v = \bar{\partial}_i u - \partial_i v = \bar{\partial}_i u - (\bar{\partial}_i u)^* = 0$a.e..
Fix any $(x^1, \dots, x^{i-1}, x^{i+1}, \dots x^n)$, and denote $w : (a_i, b_i) \to \mathbb{R}$ by

$$w(t) := (u^* - v)(x^1, \dots, x^{i-1}, t, x^{i+1}, \dots x^n).$$

We have that $\bar{D}w = \bar{\partial}_i(u^* - v) = 0$ with respect to $t \in (a_i, b_i)$a.e..
By 2.39, $w(t) = C$a.e. $t \in (a_i, b_i)$ form some constant $C$, since $(a_i, b_i)$ is clearly connected.
Notice that $w$ is continuous, since both $u^*, v$ are continuous on the $x^i$ direction.
Since both $w, C$ are continuous, we have $\forall t \in (a_i, b_i), w(t) = C$.
Since $v|_S = u^*|_S$, we must have $C = w(s) = 0$ and thus

$$\forall t \in (a_i, b_i), u^*(x^1, \dots, x^{i-1}, t, x^{i+1}, \dots x^n) = v(x^1, \dots, x^{i-1}, t, x^{i+1}, \dots x^n).$$

Since this holds for all $(x^1, \dots, x^{i-1}, x^{i+1}, \dots x^n)$, we must have $u^*(x) = v(x)\ \forall x \in U$.
By construction of $v$, we have that

$$\partial_i(u^*)(x) = (\bar{\partial}_i u)^*(x)\ \forall x \in U.$$

□

**Lemma 2.41.** *Consider $1 \le p \le \infty$, and $U \subseteq \mathbb{R}^n$ be open. If $u \in W^{1,p}(U)$ has a continuous representative $u^* \in C(U)$, and its weak derivative $\partial_i u$ has a continuous representative $(\bar{\partial}_i u)^* \in C(U)$, then the regular $i^{th}$ partial derivative*

$$\partial_i(u^*)(x) = (\bar{\partial}_i u)^*(x)\ \forall x \in U$$

*exists and is continuous.*

*Proof.* Notice that any open $U \subseteq \mathbb{R}^n$ can be written as $\bigcup_{j=1}^{\infty} R_j$, where each $R_j$ is an open rectangle.
Fix any $x \in U$, there must be some $R_j \ni x$.
By previous lemma, $\partial_i(u^*)(x) = (\bar{\partial}_i u)^*(x)$.
Since this holds for any $x \in U$, we have our result. □

**Proposition 2.42.** *Consider $1 \leq p \leq \infty, k \geq 0$, and $U \subseteq \mathbb{R}^n$ be open. If $u \in W^{k,p}(U)$ has a continuous representative $u^* \in C(U)$, and all of its weak derivatives $D^\alpha u$ have continuous representatives $(\bar{D}^\alpha u)^* \in C(U)$ for any $|\alpha| \leq k$, then*

$$u^* \in C^k(U),\ D^\alpha(u^*)(x) = (\bar{D}^\alpha u)^*(x)\ \forall x \in U, \forall |\alpha| \leq k.$$

*Proof.* We will use induction on $k$.
The base case is $k = 0$.
Since $|\alpha| = 0$, we trivially have $D^\alpha(u^*)(x) = u^*(x) = (\bar{D}^\alpha u)^*(x)$.
Now, suppose this holds for $k-1$.
Consider any $u \in W^{k,p}(U)$.
If $|\alpha| = 0$, we trivially have $D^\alpha(u^*)(x) = u^*(x) = (\bar{D}^\alpha u)^*(x)$ as before.
Now consider any $|\gamma| = 1$. We know $\gamma = e_i$ for some $1 \leq i \leq n$.
By previous lemma, we have that

$$D^\gamma(u^*)(x) = \partial_i(u^*)(x) = (\bar{\partial}_i u)^*(x) = (\bar{D}^\gamma u)^*(x)\ \forall x \in U.$$

Notice that $\bar{D}^\gamma u \in W^{k-1,p}(U)$, and all of its weak derivatives $\bar{D}^\beta \bar{D}^\gamma u = \bar{D}^{\beta+\gamma} u$ have continuous representatives $(\bar{D}^{\beta+\gamma} u)^* \in C(U)$ for any $|\beta| \leq k-1$. By the induction hypothesis, we have that

$$D^\beta((\bar{D}^\gamma u)^*)(x) = (\bar{D}^{\beta+\gamma} u)^*(x)\ \forall x \in U, \forall |\beta| \leq k-1.$$

For any $1 \leq |\alpha| \leq k$, we can have $\alpha = \beta + \gamma$, where $|\beta| \leq k-1, |\gamma| = 1$.
Now we have $\forall x \in U$,

$$\begin{aligned}(\bar{D}^\alpha u)^*(x) &= (\bar{D}^{\beta+\gamma} u)^*(x) \\ &= D^\beta((\bar{D}^\gamma u)^*)(x) \\ &= D^\beta(D^\gamma(u^*))(x) \\ &= D^{\beta+\gamma}(u^*)(x) \\ &= D^\alpha(u^*)(x).\end{aligned}$$

We have thus proven the result for any $|\alpha| \leq k$.
Since all of its $\alpha^{th}$ derivatives exists and are continuous, we further have that $u^* \in C^k(U)$. □

**Theorem 2.43** (Differentiability almost everywhere)**.** *(Theorem 5.8.5 in Eavan's)*
*Consider $n < p \leq \infty$, and $U \subseteq \mathbb{R}^n$ be open. Assume $u \in W^{1,p}_{loc}(U)$, then $u$ is differentiable a.e. in $U$, and its gradient $Du(x)$ equals its weak gradient $\bar{D}u(x)$ for a.e. $x \in U$.*

## 2.8 Sobolev Inequalities

**Definition 2.20.** For $1 \leq p < n$, the **Sobolev conjugate** of $p$ is $p^* := \frac{np}{n-p}$, with $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$.

**Theorem 2.44** (Gagliardo–Nirenberg–Sobolev)**.** *Let $1 \leq p < n$, then*

$$\exists C > 0,\ \text{such that } \forall u \in C^1_c(\mathbb{R}^n),\ \|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|Du\|_{L^p(\mathbb{R}^n)}.$$

**Corollary 2.45.** *Let $1 \leq p < n$, and $U \subseteq \mathbb{R}^n$, then*

$$\exists C > 0,\ \text{such that } \forall u \in C^1_c(U),\ \|u\|_{L^{p^*}(U)} \leq C\|Du\|_{L^p(U)}.$$

*Proof.* By Gagliardo–Nirenberg–Sobolev's Inequality, there is some $C > 0$, such that

$$\forall v \in C_c^1(\mathbb{R}^n), \|v\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|Dv\|_{L^p(\mathbb{R}^n)}.$$

Notice that for each $u \in C_c^1(U)$, we can extend it by $v(x) := \begin{cases} u(x) & x \in U \\ 0 & x \in \mathbb{R}^n \setminus U \end{cases}$.
Notice that $\text{Supp}(v) \subseteq U$, and $v = u$ on $U$.
Thus $\|v\|_{L^{p^*}(\mathbb{R}^n)} = \|v\|_{L^{p^*}(U)} = \|u\|_{L^{p^*}(U)}$, and $\|Dv\|_{L^p(\mathbb{R}^n)} = \|Du\|_{L^p(U)}$.
In addition, we have that $\lim_{x\to\partial U} D^\alpha u(x) = 0 = \lim_{x\to\partial U} D^\alpha 0, \forall |\alpha| \leq 1$.
Thus this extension is smooth. i.e. $v \in C_c^1(\mathbb{R}^n)$.
We thus have

$$\|u\|_{L^{p^*}(U)} = \|v\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|Dv\|_{L^p(\mathbb{R}^n)} = C\|Du\|_{L^p(U)}.$$

□

**Theorem 2.46** ($W^{1,p}$ embedding into $L^{p^*}$, with $1 \leq p < n$)**.** *Let $1 \leq p < n$, $U \subseteq \mathbb{R}^n$ be open and bounded. If $\partial U$ is $C^1$, then*

$$\exists C > 0, \text{ such that } \forall u \in W^{1,p}(U), \ \|u\|_{L^{p^*}(U)} \leq C\|u\|_{W^{1,p}(U)}.$$

*In addition, since $U$ is bounded, $\forall q \in [1, p^*]$, we have*

$$\exists C > 0, \forall u \in W^{1,p}(U), \ \|u\|_{L^q(U)} \leq C\|u\|_{W^{1,p}(U)}.$$

*Proof.* See Theorem 5.6-2 of Evans and A3Q1. □

**Theorem 2.47** (Poincaré's Inequality)**.** *Let $1 \leq p < n$, $U \subseteq \mathbb{R}^n$ be open and bounded, then*

$$\forall q \in [1, p^*], \exists C \geq 0, \text{ such that } \forall u \in W_0^{1,p}(U), \ \|u\|_{L^q(U)} \leq C\|Du\|_{L^p(U)}.$$

*Proof.* See Theorem 5.6-2 of Evans and A3Q2. □

**Corollary 2.48.** *Let $1 \leq p < n$, $U \subseteq \mathbb{R}^n$ be open and bounded, then $\|Du\|_{L^p(U)}$ and $\|u\|_{W^{1,p}(U)}$ are equivalent norms on $W_0^{1,p}(U)$.*

**Theorem 2.49.** *Let $1 \leq p \leq \infty$, $U \subseteq \mathbb{R}^n$ be open and bounded, then*

$$\exists C \geq 0, \text{ such that } \forall u \in W_0^{1,p}(U), \ \|u\|_{L^p(U)} \leq C\|Du\|_{L^p(U)}.$$

*Proof.* See Theorem 5.6-2 of Evans and A3Q2. □

**Corollary 2.50.** *Let $1 \leq p \leq \infty$, $U \subseteq \mathbb{R}^n$ be open and bounded, then $\|Du\|_{L^p(U)}$ and $\|u\|_{W^{1,p}(U)}$ are equivalent norms on $W_0^{1,p}(U)$.*

*Proof.* See A3Q2. □

**Theorem 2.51.** *($W^{1,p}(U)$ embedding into $C^{0,\gamma}(\bar{U})$, with $n < p \leq \infty$, Morrey's)*
*Let $n < p \leq \infty$, $U \subseteq \mathbb{R}^n$ be open and bounded, such that $\partial U$ is $C^1$ Then there is some constant $C \geq 0$ such that*

$$\forall u \in W^{1,p}(U), \exists \tilde{u} \in C^{0,\gamma}(\bar{U}), \text{ such that } \|\tilde{u}\|_{C^{0,\gamma}(\bar{U})} \leq C\|u\|_{W^{1,p}(U)},$$

*where $\gamma := 1 - \frac{n}{p}$, and $\tilde{u} \in [u]$ is a representative of the equivalence class $[u] \in W^{1,p}(U)$.*

*Remark.* If $p = \infty$, then $\gamma = 1$, and $u^*$ is Lipschitz.

**Theorem 2.52** (Sobolev Inequalities)**.** *Let $U \subseteq \mathbb{R}^n$ be open and bounded, such that $\partial U$ is $C^1$. Let $u \in W^{k,p}(U)$, we have*

1. *If $k < \frac{n}{p}$, we define $q$ by $\frac{1}{q} := \frac{1}{p} - \frac{k}{n}$, then*

$$\|u\|_{L^q(U)} \leq C\|u\|_{W^{k,p}(U)}.$$

*2. If $k > \frac{n}{p}$, we define $t := k - \lfloor \frac{n}{p} \rfloor - 1$, then we have a representative $\tilde{u} \in C^{t,\gamma}(\bar{U})$, such that*

$$\|u^*\|_{C^{t,\gamma}(\bar{U})} \leq C\|u\|_{W^{k,p}(U)},$$

*where $\gamma = \lfloor \frac{n}{p} \rfloor + 1 - \frac{n}{p}$ if $\frac{n}{p} \notin \mathbb{Z}$, and $\gamma$ can be any integer if $\frac{n}{p} \in \mathbb{Z}$.*

*Proof.* See Theorem 5.6-6 of Evans and A3Q3. □

## 2.9 Compactness

**Definition 2.21.** Let $(f_k)_{k=1}^{\infty}$ be a sequence of real-valued functions on $\mathbb{R}^n$. It is **uniformly bounded** if

$$\exists M > 0, \text{ such that } |f_k(x)| \leq M, \ \forall k \in \mathbb{N}_+, x \in \mathbb{R}^n$$

**Definition 2.22.** Let $(f_k)_{k=1}^{\infty}$ be a sequence of real-valued functions on $\mathbb{R}^n$. It is **equicontinuous** if

$$\forall \epsilon > 0, \exists \delta > 0, \text{ such that } \forall x, y \in \mathbb{R}^n, \ \|x - y\| < \delta \implies |f_k(x) - f_k(y)| < \epsilon, \ \forall k \in \mathbb{N}_+$$

**Theorem 2.53.** *(Arzela-Ascoli Compact criterion)*
*Let $(f_k)_{k=1}^{\infty}$ be a sequence of real-valued functions on $\mathbb{R}^n$ such that it is uniformly bounded and equicontinuous, then there exists a subsequence $(f_{k_j})_{j=1}^{\infty}$ and a continuous function $f$ such that $f_{k_j} \to f$ uniformly on compact subsets of $\mathbb{R}^n$.*

**Proposition 2.54.** *(interpolation) Assume $1 \leq s \leq r \leq t \leq \infty$, and $\frac{1}{r} = \frac{\theta}{s} + \frac{1-\theta}{t}$ with $0 \leq \theta \leq 1$. Suppose $u \in L^s(U) \cap L^t(U)$, then $u \in L^r(U)$ and*

$$\|u\|_{L^r(U)} \leq \|u\|_{L^s(U)}^{\theta} \|u\|_{L^t(U)}^{1-\theta}.$$

*Proof.* See AMATH731 A2. □

**Lemma 2.55.** *Let $V \subseteq \mathbb{R}^n$ be open and bounded. Let $1 \leq p < n$, and $(u_m)_{m=1}^{\infty} \subset W^{1,p}(\mathbb{R}^n)$ be any bounded sequence with $\text{Supp}(u_m) \subseteq V$. For $u_m^{\epsilon} := \eta_{\epsilon} * u_m$, we have that for each $\epsilon > 0$, there exists a subsequence $(u_{m_j}^{\epsilon})_{j=1}^{\infty}$ that converges in $L^q(V)$.*

*Proof.*

**Claim 2.55.1.** *The sequence $(u_m^{\epsilon})_{m=1}^{\infty}$ is uniformly bounded.*

*Proof.* Since $(u_m)_{m=1}^{\infty}$ is bounded, there is some $M > 0$, such that $\forall m \in \mathbb{N}_+, \|\hat{u}_m\|_{W^{1,p}(\mathbb{R}^n)} \leq M$.
Consider any $x \in \mathbb{R}^n$, we have

$$\begin{aligned}
|u_m^{\epsilon}(x)| &= \left| \int_{\mathbb{R}^n} \eta_{\epsilon}(x - y) u_m(y) dy \right| \\
&\leq \|\eta_{\epsilon}\|_{L^{\infty}(\mathbb{R}^n)} \left| \int_{\mathbb{R}^n} u_m(y) dy \right| \\
&\leq \|\eta_{\epsilon}\|_{L^{\infty}(\mathbb{R}^n)} \int_{\mathbb{R}^n} |u_m(y)| dy \\
&= \|\eta_{\epsilon}\|_{L^{\infty}(\mathbb{R}^n)} \|u_m\|_{L^1(\mathbb{R}^n)} \\
&= \|\eta_{\epsilon}\|_{L^{\infty}(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} \|u_m\|_{L^p(\mathbb{R}^n)} \\
&= \|\eta_{\epsilon}\|_{L^{\infty}(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} \|u_m\|_{W^{1,p}(\mathbb{R}^n)} \\
&\leq \|\eta_{\epsilon}\|_{L^{\infty}(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} M \\
&= \frac{1}{\epsilon^n} \|\eta\|_{L^{\infty}(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} M \\
&\leq \frac{C}{\epsilon^n} |V|^{1-\frac{1}{p}} M.
\end{aligned}$$

Since $\frac{C}{\epsilon^n} |V|^{1-\frac{1}{p}} M < \infty$ is independent of $m$, we have that the sequence $(u_m^{\epsilon})_{m=1}^{\infty}$ is uniformly bounded. □

**Claim 2.55.2.** *The sequence $(u_m^\epsilon)_{m=1}^\infty$ is equicontinuous.*

*Proof.* Since $(u_m)_{m=1}^\infty$ is bounded, there is some $M > 0$, such that $\forall m \in \mathbb{N}_+, \|\hat{u}_m\|_{W^{1,p}(\mathbb{R}^n)} \leq M$.
By 2.7.2, we have that $\partial_i u_m^\epsilon = (\partial_i \eta_\epsilon) * u_m$.
Thus for any $x \in \mathbb{R}^n, 1 \leq i \leq n$, we have

$$\begin{aligned}
|\partial_i u_m^\epsilon(x)| &= \left| \int_{\mathbb{R}^n} (\partial_i \eta_\epsilon)(x-y) u_m(y) dy \right| \\
&\leq \|\partial_i \eta_\epsilon\|_{L^\infty(\mathbb{R}^n)} \left| \int_{\mathbb{R}^n} u_m(y) dy \right| \\
&\leq \|\partial_i \eta_\epsilon\|_{L^\infty(\mathbb{R}^n)} \|u_m\|_{L^1(\mathbb{R}^n)} \\
&\leq \|\partial_i \eta_\epsilon\|_{L^\infty(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} M \\
\|Du_m^\epsilon(x)\|_1 &\leq \sum_{i=1}^n \|\partial_i \eta_\epsilon\|_{L^\infty(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} M \\
&= \|D\eta_\epsilon\|_{L^\infty(\mathbb{R}^n)} |V|^{1-\frac{1}{p}} M \\
&= \|D\eta_\epsilon\|_{L^\infty(B(0,\epsilon))} |V|^{1-\frac{1}{p}} M \\
&< \infty.
\end{aligned}$$

Since $\|D\eta_\epsilon\|_{L^\infty(B(0,\epsilon))} |V|^{1-\frac{1}{p}} M$ is independent of $x, m$, we have that

$$C := \sup_{m \geq 1} \|Du_m^\epsilon\|_{L^\infty(U)} \leq \|D\eta_\epsilon\|_{L^\infty(B(0,\epsilon))} |V|^{1-\frac{1}{p}} M < \infty.$$

Since $u_m^\epsilon \in C_c^\infty(\mathbb{R}^n)$ by 2.7.1, we have each $u_m^\epsilon$ is Lipschitz with Lipschitz-constant $C$.
Given any $\delta > 0$, we can let $\delta_0 = \frac{\delta}{C}$.
Thus $\forall x, y \in \mathbb{R}^n,$ such that $\|x - y\| < \delta_0$, we have

$$|u_m^\epsilon(x) - u_m^\epsilon(y)| \leq C\|x - y\| < \delta, \ \forall m \in \mathbb{N}_+.$$

Thus the sequence $(u_m^\epsilon)_{m=1}^\infty$ is equicontinuous. □

By the above two lemmas and Arzela-Ascoli Compact criterion 2.53, we know for each $\epsilon > 0$, there exists a subsequence $(u_{m_j}^\epsilon)_{j=1}^\infty$ and a continuous function $u^\epsilon$ such that $u_{m_j}^\epsilon \to f$ uniformly on compact subsets of $\mathbb{R}^n$.
Since $V$ is bounded, $\bar{V}$ is compact, we have that $(u_{m_j}^\epsilon)_{j=1}^\infty$ converges uniformly on $\bar{V}$.
Thus $(u_{m_j}^\epsilon)_{j=1}^\infty$ converges in $L^\infty(V)$.
Thus $(u_{m_j}^\epsilon)_{j=1}^\infty$ converges in $L^q(V)$. □

**Lemma 2.56.** *Let $V \subseteq \mathbb{R}^n$ be open and bounded, such that $\partial V$ is $C^1$. Let $1 \leq p < n$, and $(u_m)_{m=1}^\infty \subset W^{1,p}(\mathbb{R}^n)$ be any bounded sequence with* $\mathrm{Supp}(u_m) \subseteq V$. *For $u_m^\epsilon := \eta_\epsilon * u_m$, we have that $u_m^\epsilon \to u_m$ uniformly in $L^q(V)$ as $\epsilon \to 0$.*

*Proof.* By taking $V'$ to be $V + B(0,1)$ and WLOG consider $\epsilon < 1$, we assume the support of $u_m^\epsilon$ is in $V$.
Since $(u_m)_{m=1}^\infty$ is bounded, there is some $M > 0$, such that $\forall m \in \mathbb{N}_+, \|u_m\|_{W^{1,p}(\mathbb{R}^n)} = \|u_m\|_{W^{1,p}(V)} \leq M$.

**Claim 2.56.1.** *If $u_m$ are smooth, then $\|u_m^\epsilon - u_m\|_{L^1(V)} \leq \epsilon |V|^{1-\frac{1}{p}} M$ for any $\epsilon > 0$.*

*Proof.*

$$\begin{aligned}
u_m^\epsilon(x) - u_m(x) &= (\eta_\epsilon * u_m)(x) - u_m(x) \\
&= \int_{B(0,\epsilon)} \eta_\epsilon(y) u_m(x-y) dy - u_m(x) \int_{B(0,\epsilon)} \eta_\epsilon(y) dy \\
&= \int_{B(0,\epsilon)} \eta_\epsilon(y)(u_m(x-y) - u_m(x)) dy
\end{aligned}$$

Let $z := \frac{y}{\epsilon}$, we have $dy = \epsilon^n dz$. Recall $\eta_\epsilon = \frac{1}{\epsilon^n}\eta\big(\frac{y}{\epsilon}\big)$. We thus have

$$\begin{aligned}
u_m^\epsilon(x) - u_m(x) &= \int_{B(0,\epsilon)} \eta_\epsilon(y)(u_m(x-y) - u_m(x))dy \\
&= \int_{B(0,1)} \frac{\eta(z)}{\epsilon^n} - (u_m(x-\epsilon z)u_m(x))(\epsilon^n dz) \\
&= \int_{B(0,1)} \eta(y)(u_m(x-\epsilon y) - u_m(x))dy \\
&= \int_{B(0,1)} \eta(y)\int_0^1 \frac{d}{dt}u_m(x-\epsilon yt)dtdy \\
&= \int_{B(0,1)} \eta(y)\int_0^1 Du_m(x-\epsilon yt)\cdot(-\epsilon y)dtdy \\
|u_m^\epsilon(x) - u_m(x)| &\le \int_{B(0,1)} \eta(y)\int_0^1 |Du_m(x-\epsilon yt)\cdot(-\epsilon y)|dtdy \\
&= \epsilon\int_{B(0,1)} \eta(y)\int_0^1 |Du_m(x-\epsilon yt)\cdot y|dtdy \\
&\le \epsilon\int_{B(0,1)} \eta(y)\int_0^1 \|Du_m(x-\epsilon yt)\|_1 dtdy,
\end{aligned}$$

since $\|y\|_2 < \epsilon < 1$. Thus

$$\begin{aligned}
\|u_m^\epsilon - u_m\|_{L^1(V)} &= \int_V |u_m^\epsilon(x) - u_m(x)|dx \\
&\le \int_V \epsilon\int_{B(0,1)} \eta(y)\int_0^1 \|Du_m(x-\epsilon yt)\|_1 dtdydx \\
&= \epsilon\int_{B(0,1)} \eta(y)\int_0^1\int_V \|Du_m(x-\epsilon yt)\|_1 dxdtdy \\
&= \epsilon\int_{B(0,1)} \eta(y)\int_0^1\int_{\mathbb{R}^n} \|Du_m(x-\epsilon yt)\|_1 dxdtdy \\
&= \epsilon\int_{B(0,1)} \eta(y)\int_0^1\int_{\mathbb{R}^n} \|Du_m(z)\|_1 dzdtdy \\
&= \epsilon\left(\int_{B(0,\epsilon)} \eta(y)dy\right)\left(\int_0^1 dt\right)\left(\int_{\mathbb{R}^n} \|Du_m(z)\|_1 dz\right) \\
&= \epsilon\int_{\mathbb{R}^n} \|Du_m(z)\|_1 dz \\
&= \epsilon\int_V \|Du_m(z)\|_1 dz \\
&= \epsilon\sum_{i=1}^n \|\partial_i u_m\|_{L^1(V)} \\
&\le \epsilon\sum_{i=1}^n |V|^{1-\frac{1}{p}}\|\partial_i u_m\|_{L^p(V)} \\
&\le \epsilon|V|^{1-\frac{1}{p}}\|u_m\|_{W^{1,p}(V)} \\
&\le \epsilon|V|^{1-\frac{1}{p}}M.
\end{aligned}$$

Notice that this is true for any $\epsilon > 0$. □

Let $\delta > 0$ be given. Since $C^\infty(\bar{V})$ is dense in $W^{1,p}(V)$ by 2.30, we can find some $\bar{u_m} \in W^{1,p}(V)$, such that $\|\bar{u}_m - u_m\|_{W^{1,p}(V)} < \frac{\delta}{3|V|^{1-\frac{1}{p}}}$.
Notice that $\forall m, \|\bar{u}_m\|_{W^{1,p}(V)} \leq M + \frac{\delta}{3|V|^{1-\frac{1}{p}}}$ is bounded. From the claim above, we can find

$$\epsilon_0 := \frac{\delta}{3\left(M + \frac{\delta}{3|V|^{1-\frac{1}{p}}}\right)|V|^{1-\frac{1}{p}}} > 0,$$

such that $\forall 0 < \epsilon < \epsilon_0$, we have

$$\|\bar{u}_m^\epsilon - \bar{u}_m\|_{L^1(V)} < \frac{\delta}{3},\ \forall m \in \mathbb{N}_+.$$

Now $\|u_m - \bar{u}_m\|_{L^1(V)} \leq |V|^{1-\frac{1}{p}}\|u_m - \bar{u}_m\|_{L^p(V)} \leq |V|^{1-\frac{1}{p}}\|u_m - \bar{u}_m\|_{W^1(V)} < \frac{\delta}{3}$.
In addition, by 2.8, we have

$$\begin{aligned}
\|u_m^\epsilon - \bar{u}_m^\epsilon\|_{L^1(V)} &= \|\eta_\epsilon * u_m - \eta_\epsilon * \bar{u}_m\|_{L^1(V)} \\
&= \|\eta_\epsilon * (u_m - \bar{u}_m)\|_{L^1(V)} \\
&\leq \|u_m - \bar{u}_m\|_{L^1(V)} \\
&< \frac{\delta}{3}.
\end{aligned}$$

Now we have

$$\|u_m^\epsilon - u_m\|_{L^1(V)} \leq \|u_m^\epsilon - \bar{u}_m^\epsilon\|_{L^1(V)} + \|\bar{u}_m^\epsilon - \bar{u}_m\|_{L^1(V)} + \|\bar{u}_m - u_m\|_{L^1(V)} < \delta.$$

Notice that this holds for all $\epsilon < \epsilon_0, m \in \mathbb{N}_+$, where the choice of $\epsilon_0$ does not depend on $m$, and thus $\|u_m^\epsilon - u_m\|_{L^1(V)} \to 0$ uniformly when $\epsilon \to 0$.
Now $1 \leq q \leq p^*$, by letting $s = 1, r = q, t = p^*$, we have

$$\begin{aligned}
\|u_m^\epsilon - u_m\|_{L^q(V)} &\leq \|u_m^\epsilon - u_m\|_{L^1(V)}^\theta \|u_m^\epsilon - u_m\|_{L^{p^*}(V)}^{1-\theta} && 2.54 \\
&\leq \|u_m^\epsilon - u_m\|_{L^1(V)}^\theta C^{1-\theta}\|u_m^\epsilon - u_m\|_{W^{1,p}(V)}^{1-\theta} && 2.52 \\
&\leq \|u_m^\epsilon - u_m\|_{L^1(V)}^\theta C^{1-\theta}\left(\|u_m^\epsilon\|_{W^{1,p}(V)} + \|u_m\|_{W^{1,p}(V)}\right)^{1-\theta} \\
&\leq \|u_m^\epsilon - u_m\|_{L^1(V)}^\theta C^{1-\theta}\left(2\|u_m\|_{W^{1,p}(V)}\right)^{1-\theta} && 2.24 \\
&\leq \|u_m^\epsilon - u_m\|_{L^1(V)}^\theta (2CM)^{1-\theta}.
\end{aligned}$$

Given any $\delta > 0$, since $\|u_m^\epsilon - u_m\|_{L^1(V)} \to 0$ uniformly when $\epsilon \to 0$, we can always find some $\epsilon_0 > 0$, such that

$$\forall \epsilon < \epsilon_0, m \in \mathbb{N}_+,\ \|u_m^\epsilon - u_m\|_{L^1(V)} < \left(\frac{\delta}{(2CM)^{1-\theta}}\right)^{1/\theta}.$$

Now for any $m \in \mathbb{N}_+$, we have

$$\|u_m^\epsilon - u_m\|_{L^q(V)} \leq \|u_m^\epsilon - u_m\|_{L^1(V)}^\theta (2CM)^{1-\theta} < \delta.$$

This proves that $u_m^\epsilon \to u_m$ uniformly in $L^q(V)$ as $\epsilon \to 0$. □

**Theorem 2.57** (Rellich-Kondrachov Compactness)**.** *Let $U \subseteq \mathbb{R}^n$ be open and bounded, such that $\partial U$ is $C^1$. Let $1 \leq p < n$, then*

$$W^{1,p}(U) \subset\subset L^q(U)$$

*for any $1 \leq q < p^*$.*

*Proof.* The continuous embedding is done before in 2.52.
Now consider any bounded sequence $(\hat{u}_m)_{m=1}^{\infty} \subset W^{1,p}(U)$.
Thus there is some $M > 0$, such that $\forall m \in \mathbb{N}_+, \|\hat{u}_m\|_{W^{1,p}(U)} \leq M$.
By extension theorem, we may assume $(\hat{u}_m)_{m=1}^{\infty} \subset W^{1,p}(\mathbb{R}^n)$, with $u_m|_U = \hat{u}_m$, and there is some $V$ such that $U \subset\subset V$ and $\forall m \in \mathbb{N}_+, \text{Supp}(u_m) \subseteq V$. In addition,

$$\sup \|u_m\|_{W^{1,p}(\mathbb{R}^n)} = \sup \|u_m\|_{W^{1,p}(V)} \leq \sup C\|\hat{u}_m\|_{W^{1,p}(U)} \leq CM.$$

Thus $(u_m)_{m=1}^{\infty}$ is bounded.
WLOG, we can take $V$ to have $\partial V$ being $C^1$.
Let $u_m^{\epsilon} := \eta_{\epsilon} * u_m$.
By the above lemmas, we know that

1. for each $\epsilon > 0$, there exists a subsequence $(u_{m_j}^{\epsilon})_{j=1}^{\infty}$ that converges in $L^q(V)$, and
2. $u_m^{\epsilon} \to u_m$ uniformly in $L^q(V)$ as $\epsilon \to 0$.

Now given any $\delta > 0$.
By 2, we can find some $\epsilon_0 > 0$, such that $\forall 0 < \epsilon < \epsilon_0$, we have $\forall m \in \mathbb{N}_+, \|u_m^{\epsilon} - u_m\|_{L^q(V)} < \frac{\delta}{3}$.
Now fix some $0 < \epsilon < \epsilon_0$.
By 1, there exists a subsequence $(u_{m_j}^{\epsilon})_{j=1}^{\infty}$ that converges in $L^q(V)$.
In particular, it is Cauchy, and we can find some $N \in \mathbb{N}_+$, such that $\forall i, j \geq N, \left\|u_{m_j}^{\epsilon} - u_{m_i}^{\epsilon}\right\|_{L^q(V)} < \frac{\delta}{3}$.
Now for any $i, j \geq N$, we have that

$$\begin{aligned}
\left\|u_{m_i} - u_{m_j}\right\|_{L^q(V)} &= \left\|u_{m_i} - u_{m_i}^{\epsilon} + u_{m_i}^{\epsilon} - u_{m_j}^{\epsilon} + u_{m_i}^{\epsilon} - u_{m_j}\right\|_{L^q(V)} \\
&\leq \left\|u_{m_i} - u_{m_i}^{\epsilon}\right\|_{L^q(V)} + \left\|u_{m_i}^{\epsilon} - u_{m_j}^{\epsilon}\right\|_{L^q(V)} + \left\|u_{m_i}^{\epsilon} - u_{m_j}\right\|_{L^q(V)} \\
&< \frac{\delta}{3} + \frac{\delta}{3} + \frac{\delta}{3} \\
&= \delta.
\end{aligned}$$

Thus $(u_{m_j})_{j=1}^{\infty}$ is a Cauchy sequence in $L^q(V)$.
Since $L^q(V)$ is complete, there is some $u \in L^q(V)$, such that $\lim_{j\to\infty} \left\|u_{m_j} - u\right\|_{L^q(V)} = 0$.
Since $U \subseteq V$, we have that $\lim_{j\to\infty} \left\|u_{m_j} - u\right\|_{L^q(U)} = 0$.
Since $u + m|_U = \hat{u}_m$, we also have that $\lim_{j\to\infty} \left\|\hat{u}_{m_j} - u\right\|_{L^q(U)} = 0$.
Thus the subsequence $\hat{u}_{m_j}$ converges to some $u \in L^q(V) \subseteq L^q(U)$.
Since $(\hat{u}_m)_{m=1}^{\infty} \subset W^{1,p}(U)$ is any bounded sequence, we have that any bounded subset of $W^{1,p}(U)$ is relative compact in $L^q(U)$. □

**Theorem 2.58.** *Let $U \subseteq \mathbb{R}^n$ be open and bounded, such that $\partial U$ is $C^1$. Let $1 \leq p \leq \infty$, then*

$$W^{1,p}(U) \subset\subset L^p(U).$$

**Theorem 2.59.** *Let $U \subseteq \mathbb{R}^n$ be open and bounded. Let $1 \leq p \leq \infty$, then*

$$W_0^{1,p}(U) \subset\subset L^p(U).$$

## 2.10 Poincare Inequalities

**Definition 2.23.** For a bounded domain $U \subset \mathbb{R}^n$, we denote the average of $u$ over $U$ by

$$(u)_U := \frac{1}{|U|} \int_U u dx.$$

**Theorem 2.60** (Poincaré–Wirtinger's Inequality)**.** *Let $U \subset \mathbb{R}^n$ be open, bounded, and connected, such that $\partial U$ is $C^1$. For any $1 \leq p \leq \infty,\ \exists C > 0$, such that*

$$\forall u \in W^{1,p}(U),\ \|u - (u)_U\|_{L^p(U)} \leq C\|Du\|_{L^p(U)}.$$

*Proof.* Suppose for contradiction it is not true.
Then $\forall k \in \mathbb{N}, \exists u_k \in W^{1,p}(U)$, such that $\|u_k - (u_k)_U\|_{L^p(U)} > k\|Du\|_{L^p(U)}$.
Let $v_k := \frac{u_k - (u_k)_U}{\|u_k - (u_k)_U\|_{L^p(U)}}$.
Notice that

$$\forall k \in \mathbb{N}_+,\ \|v_k\|_{L^p(U)} = 1, (v_k)_U = 0, Dv_k = \frac{Du_k}{\|u_k - (u_k)_U\|_{L^p(U)}}.$$

Thus $\|Dv_k\|_{L^p(U)} = \frac{\|Du_k\|_{L^p(U)}}{\|u_k - (u_k)_U\|_{L^p(U)}} < \frac{1}{k}$.
Which means $\|v_k\|^p_{W^{1,p}(U)} = \|v_k\|^p_{L^p(U)} + \|Dv_k\|^p_{L^p(U)} < 1 + \frac{1}{k^p} \leq 2$.
Since this is true for any $k \in \mathbb{N}_+$, we have that $(v_k)_{k=1}^\infty$ is bounded in $W^{1,p}(U)^p$.
Since $W^{1,p}(U) \subset\subset L^p(U)$, there is a subsequence $(v_{k_j})_{j=1}^\infty$ and some $v \in L^p(U)$, such that

$$\lim_{j\to\infty} \left\|v_{k_j} - v\right\|_{L^p(U)} = 0.$$

Now consider any $1 \leq i \leq k$, and any $\phi \in C_c^\infty(U)$.

$$\begin{aligned}
\left\|v_{k_j}\partial_i\phi - v\partial_i\phi\right\|^p_{L^p(U)} &= \int_U \left|v_{k_j}\partial_i\phi - v\partial_i\phi\right|^p dx \\
&= \int_U |\partial_i\phi|^p \left|v_{k_j} - v\right|^p dx \\
&\leq \|\partial_i\phi\|^p_{L^\infty(U)} \left\|v_{k_j} - v\right\|^p_{L^p(U)}.
\end{aligned}$$

Since $\phi \in C_c^\infty(U)$, we have that $\|\partial_i\phi\|^p_{L^\infty(U)}$ is bounded by some $M > 0$.
Since $\lim_{j\to\infty} \left\|v_{k_j} - v\right\|_{L^p(U)} = 0$, we also have $\lim_{j\to\infty} \left\|v_{k_j}\partial_i\phi - v\partial_i\phi\right\|^p_{L^p(U)} = 0$.
In addition, $\lim_{j\to\infty} \left\|v_{k_j}\partial_i\phi - v\partial_i\phi\right\|_{L^1(U)} \leq \lim_{j\to\infty} |U|^{1-\frac{1}{p}} \left\|v_{k_j}\partial_i\phi - v\partial_i\phi\right\|_{L^p(U)} = 0$.
We have

$$\begin{aligned}
&\lim_{j\to\infty} \int_U \left|v_{k_j}\partial_i\phi - v\partial_i\phi\right| dx = 0 \\
\implies &\lim_{j\to\infty} \int_U (v_{k_j}\partial_i\phi - v\partial_i\phi) dx = 0 \\
\implies &\lim_{j\to\infty} \int_U v_{k_j}\partial_i\phi dx = \lim_{j\to\infty} \int_U v\partial_i\phi dx \\
\implies &-\lim_{j\to\infty} \int_U \partial_i v_{k_j}\phi dx = \int_U v\partial_i\phi dx.
\end{aligned}$$

Yet

$$\begin{aligned}
\left|\lim_{j\to\infty} \int_U \partial_i v_{k_j}\phi dx\right| &\leq \lim_{j\to\infty} \int_U \left|\partial_i v_{k_j}\phi\right| dx \\
&\leq \lim_{j\to\infty} \left\|\partial_i v_{k_j}\right\|_{L^p(U)} \|\phi\|_{L^{\frac{p}{p-1}}(U)} \\
&\leq \lim_{j\to\infty} \left\|Dv_{k_j}\right\|_{L^p(U)} \|\phi\|_{L^{\frac{p}{p-1}}(U)} \\
&\leq \lim_{j\to\infty} \frac{1}{k_j} \|\phi\|_{L^{\frac{p}{p-1}}(U)} \\
&= 0,
\end{aligned}$$

since $\phi \in C_c^\infty(U)$ and $U$ is bounded, which implies $\|\phi\|_{L^{\frac{p}{p-1}}(U)} < \infty$. Thus

$$\int_U v\partial_i \phi dx = -\lim_{j\to\infty} \int_U \partial_i v_{k_j} \phi dx = 0 = -\int_U 0\phi dx.$$

Since this holds for any $\phi \in C_c^\infty(U)$, we must have $\partial_i v = 0$a.e. for any $1 \le i \le n$.
Thus $v \in W^{1,p}(U)$, with $Dv = 0$a.e..
Since $U$ is connected, $v$ is a constant by 2.39.
Since $(v)_U = 0$, we must have $v = 0$a.e..
However, this contradicts with $\|v\|_{L^p(U)} = 1$. □

## 2.11 Sobolev Spaces with Negative Orders

**Definition 2.24.** For any $k \in \mathbb{Z} \setminus \mathbb{N}, p \in [1, \infty]$, we define

$$W^{k,p}(U) := W_0^{-k,p}(U)^*,$$

dual space to $W_0^{-k,p}(U)$. In particular, when $p = 2$, we define

$$H^k(U) := H_0^{-k}(U)^*.$$

**Theorem 2.61.** *Consider any $f \in H^{-1}(U)$.*

1. *There is a tuple $(f^0, \dots, f^n)$ of functions in $L^2(U)$, such that*

$$\forall v \in H_0^1(H),\ \langle f|v\rangle_{H^{-1}(H),H_0^1(H)} = \left\langle f^0, v\right\rangle_{L^2(U)} + \sum_{i=1}^n \left\langle f^i, \partial_i v\right\rangle_{L^2(U)}.$$

   *In this case, we write $f = f^0 - \sum_{i=1}^n \partial_i f^i$.*

2.
$$\|f\|_{H^{-1}(U)} = \inf\left\{ \left( \sum_{i=0}^n \left\|f^i\right\|_{L^2(U)}^2 \right)^{\frac{1}{2}} : (f^i)_{i=0}^n \text{ satisfies 1.} \right\}.$$

*Proof.* 1. Let $f \in H^{-1}(U)$, by the Riesz-Frechet Representation theorem1.24, $\exists! u \in H_0^1(U)$, such that

$$\forall v \in H_0^1(U), \langle f|v\rangle_{H^{-1}(U),H_0^1(U)} = \langle u, v\rangle_{H_0^1(U)},$$

and $\|f\|_{H_0^{-1}(U)} = \|u\|_{H_0^1(U)}$.
Let $f^0 = u, \forall 1 \le n, f^i := \partial_i u$. we have

$$\begin{aligned}\left\langle f^0, v\right\rangle_{L^2(U)} + \sum_{i=1}^n \left\langle f^i, \partial_i v\right\rangle_{L^2(U)} &= \langle u, v\rangle_{L^2(U)} + \sum_{i=1}^n \langle \partial_i u, \partial_i v\rangle_{L^2(U)} \\ &= \langle u, v\rangle_{H_0^1(U)} \\ &= \langle f|v\rangle_{H^{-1}(U),H_0^1(U)}.\end{aligned}$$

2. Consider any $f \in H^{-1}(U)$, from 1, we know that there is such $f^0 = u, \forall 1 \le n, f^i := \partial_i u$, satisfying 1, with

$$\|f\|_{H_0^{-1}(U)} = \|u\|_{H_0^1(U)} = \left( \sum_{i=0}^n \left\|f^i\right\|_{L^2(U)}^2 \right)^{\frac{1}{2}} \ge \inf\left\{ \left( \sum_{i=0}^n \left\|g^i\right\|_{L^2(U)}^2 \right)^{\frac{1}{2}} : (g^i)_{i=0}^n \text{ satisfies 1.} \right\}.$$

Now consider any $g^0, \dots, g^n \in L^2(U)$, such that they satisfies

$$\langle f|v\rangle = \left\langle g^0, v\right\rangle_{L^2(U)} + \sum_{i=1}^{n} \left\langle g^i, v\right\rangle_{L^2(U)}.$$

For any $v \in H_0^1(U)$, we have

$$\begin{aligned}
|\langle f|v\rangle| &= \left| \left\langle g^0, v\right\rangle_{L^2(U)} + \sum_{i=1}^{n} \left\langle g^i, \partial_i v\right\rangle_{L^2(U)} \right| \\
&\le \left| \left\langle g^0, v\right\rangle_{L^2(U)} \right| + \sum_{i=1}^{n} \left| \left\langle g^i, \partial_i v\right\rangle_{L^2(U)} \right| \\
&\le \left\| g^0 \right\|_{L^2(U)} \|v\|_{L^2(U)} + \sum_{i=1}^{n} \left\| g^i \right\|_{L^2(U)} \|\partial_i v\|_{L^2(U)} \\
&\le \left( \sum_{i=0}^{n} \left\| g^i \right\|_{L^2(U)}^2 \right)^{\frac{1}{2}} \left( \|v\|_{L^2(U)}^2 + \sum_{i=1}^{n} \|\partial_i v\|_{L^2(U)}^2 \right)^{\frac{1}{2}} \\
&= \left( \sum_{i=0}^{n} \left\| g^i \right\|_{L^2(U)}^2 \right)^{\frac{1}{2}} \|v\|_{H_0^1(U)}.
\end{aligned}$$

Thus we know

$$\|f\|_{H_0^{-1}(U)} = \sup_{v \in H_0^1(U), v \neq 0} \frac{|\langle f|v\rangle|}{\|v\|_{H_0^1(U)}} \le \inf \left\{ \left( \sum_{i=0}^{n} \left\| g^i \right\|_{L^2(U)}^2 \right)^{\frac{1}{2}} : (g^i)_{i=0}^n \text{ satisfies 1.} \right\}.$$

□

**Corollary 2.62.** *For any $u^* \in L^2(U)^* \subset Lin(L^2(U), \mathbb{R}) \subset Lin(H_0^1(U), \mathbb{R})$, with $u^*$ identified with $u \in L^2(U)$, and any $v \in H_0^1(U) \subseteq L^2(U)$, we have*

$$\langle u^*|v\rangle_{H^{-1}(U), H_0^1(U)} = \langle u^*|v\rangle_{L^2(U)^*, L^2(U)} = \langle u, v\rangle_{L^2(U)}.$$

*In addition, $u^* \in H^{-1}(U)$, and has a representation $(u, 0, \dots, 0)$ as in above theorem, with*

$$\|u^*\|_{H^{-1}(U)} \le \|u\|_{L^2(U)}.$$

*Proof.* The first equality is by definition, and the second equality is by 1.52.

Thus, for any $\|v\|_{H_0^1(U)} = 1$, we have that

$$\begin{aligned}
\left| \langle u^*|v\rangle_{H^{-1}(U), H_0^1(U)} \right| &= \left| \langle u, v\rangle_{L^2(U)} \right| \\
&\le \|u\|_{L^2(U)} \|v\|_{L^2(U)} \\
&\le \|u\|_{L^2(U)} \|v\|_{H_0^1(U)} \\
&= \|u\|_{L^2(U)}.
\end{aligned}$$

Since this holds for any unitary $v \in H_0^1(U)$, we have that

$$\|u^*\|_{H^{-1}(U)} = \sup_{\|v\|_{H_0^1(U)} = 1} \left| \langle u^*|v\rangle_{H^{-1}(U), H_0^1(U)} \right| \le \|u\|_{L^2(U)} < \infty,$$

which proves $u^* \in H^{-1}(U)$.

In addition, $\langle u, v\rangle_{L^2(U)} + \sum_{i=1}^{n} \langle 0, \partial_i v\rangle_{L^2(U)} = \langle u, v\rangle_{L^2(U)} = \langle u^*|v\rangle_{H^{-1}(U), H_0^1(U)}$. □

**Corollary 2.63.** *$\forall u \in H_0^1(U) \subset L^2(U)$, we have $u^* := \langle u, \cdot \rangle_{L^2(U)} \in H^{-1}(U)$, with*

$$\|u^*\|_{H^{-1}(U)} \leq \|u\|_{L^2(U)} \leq \|u\|_{H_0^1(U)}.$$

*In other words, if we identify $u$ with $u^*$, then $H_0^1(U) \subset L^2(U) \subset H^{-1}(U)$ are continuous embeddings.*

*Proof.* Since $u \in H_0^1(U) \subseteq L^2(U)$, by the above corollary, $u^* \in H^{-1}(U)$, and has

$$\|u^*\|_{H^{-1}(U)} \leq \|u\|_{L^2(U)} \leq \|u\|_{H_0^1(U)}.$$

□

**Theorem 2.64.** *For any $k \in \mathbb{Z} \backslash \mathbb{N}$, and $f \in H^{-k}(U)$, there is a tuple $(f^\alpha)_{|\alpha| \leq k}$, such that for all $v \in H^k(U)$,*

$$\langle f | v \rangle_{H^{-k}(U), H^k(U)} = \sum_{|\alpha| \leq k} \langle f^\alpha, D^\alpha v \rangle_{L^2(U)},$$

*with*

$$\|f\|^2_{H^{-k}(U)} = \sum_{|\alpha| \leq k} \|f^\alpha\|^2_{L^2(U)}.$$

## 2.12 Difference Quotients

**Definition 2.25.** Let $U \subset \mathbb{R}^n$ be open, $u \in L^1_{loc}(U), V \subset\subset U$, then for $|h| \in (0, \operatorname{dist}(V, \partial U)), x \in V$, we define:

1. For $i \in [n]$, $u_i^h(x) := u(x + he_i)$
2. For $i \in [n]$, the $i^{th}$ **difference quotient** of size $h$ at $x$ is

$$D_i^h u(x) = \frac{u_i^h(x) - u(x)}{h} = \frac{u(x + he_i) - u(x)}{h}.$$

3.

$$D^h u(x) := (D_1^h u(x), \ldots, D_n^h u(x)).$$

**Proposition 2.65.** *Let $U \subset \mathbb{R}^n$ be open, $u \in L^1_{loc}(U)$, then $\forall i \in [n], |h| > 0$, we have*

$$\operatorname{Supp}(D_i^h u) \subseteq \operatorname{Supp}(u) + \bar{B}(0, |h|).$$

*Thus,*

$$\operatorname{Supp}(D^h u) \subseteq \operatorname{Supp}(u) + \bar{B}(0, |h|).$$

**Proposition 2.66.** *Let $U \subset \mathbb{R}^n$ be open, $u, v \in L^1_{loc}(U), V \subset\subset U$, then $\forall i \in [n], |h| \in (0, \operatorname{dist}(V, \partial U))$, we have*

$$D_i^h(uv) = v_i^h D_i^h u + u D_i^h v.$$

*Proof.* We have

$$\begin{aligned} v_i^h D_i^h u + u D_i^h v &= v_i^h \frac{u_i^h - u}{h} + u \frac{v_i^h - v}{h} \\ &= \frac{v_i^h u_i^h - v_i^h u + u v_i^h - uv}{h} \\ &= \frac{v_i^h u_i^h - uv}{h} \\ &= \frac{(uv)_i^h - uv}{h} \\ &= D_i^h(uv). \end{aligned}$$

□

**Proposition 2.67.** *Let $U \subset \mathbb{R}^n$ be open, $u, v \in L^1_{loc}(U), \mathrm{Supp}(u) \subset V \subset\subset U$, then $\forall i \in [n], |h| \in (0, \frac{1}{3}\operatorname{dist}(V, \partial U))$, we have*

$$\int_U v D_i^{-h} u dx = -\int_U u D_i^h v dx.$$

*Proof.* Notice that $\mathrm{Supp}(D_i^h u) \subseteq \mathrm{Supp}(u) + \bar{B}(0, |h|) \subseteq V + \bar{B}(0, |h|) \subseteq \overline{V + B(0, |h|)}$.
Since $\operatorname{dist}(\overline{V + B(0, |h|)}, \partial U) \geq 2|h|$, we can find $\overline{V + B(0, |h|)} \subset W \subset\subset V$, with $|h| < \operatorname{dist}(W, \partial U)$, where $D_i^{-h} u$ is well-defined in $W$.
In addition, $\mathrm{Supp}(u) \subset V \subset W$, so we can view the integrals as over $W$, by extending $D_i^{-h} u$ to be zero outside of $W$.

$$\begin{aligned}
\int_W v D_i^{-h} u dx &= \int_V v D_i^{-h} u dx \\
&= \int_V v(x) \frac{u(x - he_i) - u(x)}{-h} dx \\
&= -\int_V \frac{v(x)u(x - he_i) - v(x)u(x)}{h} dx \\
&= -\left( \int_V \frac{v(x - he_i + he_i)u(x - he_i)}{h} dx - \int_V \frac{v(x)u(x)}{h} dx \right) \\
&= -\left( \int_{V - he_i} \frac{v(y + he_i)u(y)}{h} dy - \int_V \frac{v(x)u(x)}{h} dx \right) \\
&= -\left( \int_W \frac{v_i^h(y)u(y)}{h} dy - \int_W \frac{v(x)u(x)}{h} dx \right) \\
&= -\int_W \frac{v_i^h(x)u(x) - v(x)u(x)}{h} dx \\
&= -\int_W u \frac{v_i^h - v}{h} dx \\
&= -\int_W u D_i^h v dx.
\end{aligned}$$

□

**Proposition 2.68.** *Let $U \subset \mathbb{R}^n$ be open, $u, D^\alpha u \in L^p_{loc}(U), V \subset\subset U$, then $\forall i \in [n], |h| \in (0, \operatorname{dist}(V, \partial U))$, we have*

$$D^\alpha(u_i^h) = (D^\alpha u)_i^h, \ D^\alpha(D_i^h u) = D_i^h(D^\alpha u) \text{ in } V.$$

*In addition, if $u \in W^{k,p}(U)$, we have $u_i^h, D_i^h u \in W^{k,p}(V)$.*

*Proof.* Given any $i \in [n], |h| \in (0, \operatorname{dist}(V, \partial U))$.

$\forall \phi \in C_c^\infty(V)$, we have $\phi_i^{-h} \in C_c^\infty(V + he_i) \subseteq C_c^\infty(U)$, with $D^\alpha \phi(x) = D^\alpha \phi_i^{-h}(x + he_i)$.

$$\begin{aligned}
\int_V u_i^h(x) D^\alpha \phi(x) dx &= \int_V u(x + he_i) D^\alpha \phi_i^{-h}(x + he_i) dx \\
&= \int_{V+he_i} u(y) D^\alpha \phi_i^{-h}(y) dy \\
&= \int_U u(y) D^\alpha \phi_i^{-h}(y) dy \\
&= (-1)^{|\alpha|} \int_U D^\alpha u(y) \phi_i^{-h}(y) dy \\
&= (-1)^{|\alpha|} \int_{V+he_i} D^\alpha u(y) \phi_i^{-h}(y) dy \\
&= (-1)^{|\alpha|} \int_V D^\alpha u(x + he_i) \phi_i^{-h}(x + he_i) dx \\
&= (-1)^{|\alpha|} \int_V (D^\alpha u)_i^h(x) \phi(x) dx.
\end{aligned}$$

Since this holds for all $\phi \in C_c^\infty(V)$, we must have $D^\alpha(u_i^h) = (D^\alpha u)_i^h$.
In addition,

$$\begin{aligned}
D^\alpha(D_i^h u) &= D^\alpha \left( \frac{u_i^h - u}{h} \right) \\
&= \frac{D^\alpha(u_i^h) - D^\alpha u}{h} \\
&= \frac{(D^\alpha u)_i^h - D^\alpha u}{h} \\
&= D_i^h(D^\alpha u).
\end{aligned}$$

Now suppose $u \in W^{k,p}(U)$.

$$\begin{aligned}
\left\| u_i^h \right\|_{W^{k,p}(V)}^p &= \int_V \sum_{|\alpha| \le k} \left| (D^\alpha(u_i^h))(x) \right|^p dx \\
&= \int_V \sum_{|\alpha| \le k} \left| (D^\alpha u)_i^h(x) \right|^p dx \\
&= \int_V \sum_{|\alpha| \le k} |D^\alpha u(x + he_i)|^p dx \\
&= \int_{V+he_i} \sum_{|\alpha| \le k} |D^\alpha u(y)|^p dy \\
&\le \int_U \sum_{|\alpha| \le k} |D^\alpha u(y)|^p dy \\
&= \|u\|_{W^{k,p}(U)}^p.
\end{aligned}$$

Thus $u_i^h \in W^{k,p}(V)$.
Clearly $u \in W^{k,p}(V)$, so a linear combination of them $D_i^h u \in W^{k,p}(V)$. □

**Theorem 2.69.** *Let $U \subset \mathbb{R}^n$ be open, we have:*

1. *For $p \in [1, \infty)$, and $\forall V \subset\subset U, \exists C > 0$, such that*

$$\left\| D^h u \right\|_{L^p(V)} \le C \|Du\|_{L^p(U)}, \ \forall u \in W^{1,p}(U), \forall |h| \in (0, \operatorname{dist}(V, \partial U)).$$

*2. For $p \in (1,\infty), V \subset\subset U, u \in L^p(V)$, if $\exists C, \delta > 0$, such that $\left\|D^h u\right\|_{L^p(V)} \leq C$, $\forall |h| \in (0,\delta)$, then*

$$u \in W^{1,p}(V),\ \|Du\|_{L^p(V)} \leq C.$$

**Theorem 2.70.** *Let $U \subset \mathbb{R}^n$ be open and bounded, with $\partial U$ being $C^1$, then $u : U \to \mathbb{R}$ is Lipschitz continuous if and only if $u \in W^{1,\infty}(U)$ .*

## 2.13 Weak Derivative Calculus

We here aim to list and prove certain results for weak derivatives. They are not used any where in the rest of this set of notes, but might be useful.

**Proposition 2.71** (Local Product Rule for Weak Derivatives)**.** *Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable, given any $p \in [1,\infty]$, and any $u \in W^{1,p}_{loc}(U)$, $v \in W^{1,\infty}_{loc}(U)$, then $uv \in W^{1,p}_{loc}(U)$ with the $i^{th}$ weak derivative $\partial_i(uv) = u\partial_i v + v\partial_i u$. [4, Proposition 4.1.17]*

**Proposition 2.72.** *Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable, $|\alpha| \leq n$. For $u, v \in L^1_{loc}(U)$, suppose $v|_V = D^\alpha(u|_V)$ on every $V \subset\subset U$, then $v$ is the $\alpha^{th}$ weak derivative of $u$ on $U$ globally. i.e. $v = D^\alpha u$.*

*Proof.* Consider any $\phi \in C_c(U)$, we have that $\text{Supp}(\phi) \subset\subset U$, so we can find a $\text{Supp}(\phi) \subset\subset V \subset\subset U$. Now,

$$\int_U \phi v dx = \int_V \phi v dx = (-1)^{|\alpha|} \int_V D^\alpha \phi v dx = (-1)^{|\alpha|} \int_U D^\alpha \phi v dx,$$

since $v|_V = D^\alpha(u|_V)$ and $\phi$ is constantly 0 outside of $V$. □

**Corollary 2.73** (Product Rule for Weak Derivatives)**.** *Let $U \subseteq \mathbb{R}^n$ be Lebesgue measurable, given any $p \in [1,\infty]$, and any $u \in W^{1,p}(U)$, $v \in W^{1,\infty}(U)$, then $uv \in W^{1,p}(U)$ with the $i^{th}$ weak derivative $\partial_i(uv) = u\partial_i v + v\partial_i u$. Also,*

$$\|uv\|_{W^{1,p}(U)} \leq \left(1 + n^{\frac{p-1}{p}}\right) \|u\|_{W^{1,p}(U)} \|v\|_{W^{1,\infty}(U)}.$$

*[4, Corollary 4.1.18]*

*Remark.* The original statement in [4] claims that $\|uv\|_{W^{1,p}(U)} \leq \|u\|_{W^{1,p}(U)} \|v\|_{W^{1,\infty}(U)}$, which is in general not true.

*Proof.* We firstly note that $\|uv\|_{L^p(U)} \leq \|u\|_{L^p(U)} \|v\|_{L^\infty(U)}$. Also, for all $i \in [n]$, by the above proposition 2.72, we have that $\partial_i(uv) = u\partial_i v + v\partial_i u$ exists globally in $U$. In addition,

$$\begin{aligned}
\|(uv)_i\|_{L^p(U)} &\leq \|u\partial_i v + v\partial_i u\|_{L^p(U)} \\
&\leq \|u\partial_i v\|_{L^p(U)} + \|v\partial_i u\|_{L^p(U)} \\
&\leq \|u\|_{L^p(U)} \|\partial_i v\|_{L^\infty(U)} + \|\partial_i u\|_{L^p(U)} \|v\|_{L^\infty(U)} \\
&< \infty.
\end{aligned}$$

This shows that $(uv)_i \in L^p(U)$. Since this holds for all $i \in [n]$, we have $uv \in W^{1,p}(U)$. In addition,

$$\begin{aligned}
\|uv\|_{W^{1,p}(U)} &= \left( \|uv\|_{L^p(U)}^p + \sum_{i=1}^n \|(uv)_i\|_{L^p(U)}^p \right)^{\frac{1}{p}} \\
&\leq \|uv\|_{L^p(U)} + \sum_{i=1}^n \|(uv)_i\|_{L^p(U)} \\
&\leq \|u\|_{L^p(U)} \|v\|_{L^\infty(U)} + \sum_{i=1}^n \left( \|u\|_{L^p(U)} \|\partial_i v\|_{L^\infty(U)} + \|\partial_i u\|_{L^p(U)} \|v\|_{L^\infty(U)} \right) \\
&\leq \|u\|_{L^p(U)} \left( \|v\|_{L^\infty(U)} + \sum_{i=1}^n \|\partial_i v\|_{L^\infty(U)} \right) + \left( \sum_{i=1}^n \|\partial_i u\|_{L^p(U)} \right) \|v\|_{L^\infty(U)} \\
&\leq \|u\|_{L^p(U)} \|v\|_{W^{1,\infty}(U)} + n^{\frac{p-1}{p}} \left( \sum_{i=1}^n \|\partial_i u\|_{L^p(U)}^p \right)^{\frac{1}{p}} \|v\|_{L^\infty(U)} \\
&\leq \|u\|_{W^{1,p}(U)} \|v\|_{W^{1,\infty}(U)} + n^{\frac{p-1}{p}} \|u\|_{W^{1,p}(U)} \|v\|_{W^{1,\infty}(U)} \\
&= \left( 1 + n^{\frac{p-1}{p}} \right) \|u\|_{W^{1,p}(U)} \|v\|_{W^{1,\infty}(U)}.
\end{aligned}$$

□

**Proposition 2.74** (Product Rule for Weak Derivatives of Bounded Functions)**.** *Given any $p \in [1, \infty]$, and any $u, v \in W^{1,p}(U) \cap L^\infty(U)$, then $uv \in W^{1,p}(U) \cap L^\infty(U)$ with the $i^{th}$ weak derivative $\partial_i(uv) = u\partial_i v + v\partial_i u$. [1, Proposition 9.4]*

**Corollary 2.75** (Quotient Rule for Weak Derivatives of Bounded Functions - Necessary Condition)**.** *Given any $p \in [1, \infty]$, and any $u, v \in W^{1,p}(U) \cap L^\infty(U)$, **such that** $\frac{u}{v} \in W^{1,p}(U) \cap L^\infty(U)$. We have that the $i^{th}$ weak derivative is $\partial_i\left(\frac{u}{v}\right) = \frac{\partial_i uv - u\partial_i v}{v^2}$.*

*Proof.* Since $v\frac{u}{v} = u$, we have by product rule that

$$\begin{aligned}
\partial_i u &= \partial_i\left(\frac{u}{v}\right) v + \frac{u}{v}\partial_i v \\
\partial_i\left(\frac{u}{v}\right) v &= \frac{\partial_i uv - u\partial_i v}{v} \\
\partial_i\left(\frac{u}{v}\right) &= \frac{\partial_i uv - u\partial_i v}{v^2}
\end{aligned}$$

□

*Remark.* Notice that in the above result, it is very important that we know $\frac{u}{v} \in W^{1,p}(U) \cap L^\infty(U)$ to begin with. Namely, we need to know the weak derivative already exists and is in the required space. Thus, this is only a necessary condition. Indeed, consider the counter-example: $u(x) = x^2 \sin(1/x)$, $v(x) = x^2$. Both $u, v \in W^{1,1}(0,1) \cap L^\infty(0,1)$, but even though their quotient $\frac{u}{v} \in L^\infty(0,1)$, it does not lie in $W^{1,1}(0,1)$.

**Proposition 2.76** (Local Chain Rule for Weak Derivatives)**.** *Given any $p \in [1, \infty]$. Suppose $u \in W^{1,p}_{loc}(U)$. Suppose $F \in C^1(\mathbb{R})$ has bounded derivative $F'$, then the post-composition $F \circ u$ lies in $W^{1,p}_{loc}(U)$ and its $i^{th}$ weak derivative is given by $\partial_i(F \circ u) = (F' \circ u) \cdot \partial_i u$. [4, Proposition 4.1.21]*

*Remark.* Since $\frac{1}{x}$ is not $C^1(\mathbb{R})$ (around 0), the quotient rule does not hold in general. Also, although $e^x, x^2 \in C^1(\mathbb{R})$, they don't have a bounded derivative. To this end, we will modify the previous chain rule to get a variation that will weaken the assumption of the behaviour of $F$ over the entire real line.

**Proposition 2.77** (Local Chain Rule Variation for Weak Derivatives)**.** *Given any $p \in [1, \infty]$. Suppose $u \in W^{1,p}_{loc}(U)$. Suppose $F \in C^1(I)$ has bounded derivative $F'$, where $I \subseteq \mathbb{R}$ is an open interval that contains the closure of $u$'s essential image. i.e.*

$$\overline{\bigcap \{J \subseteq \mathbb{R} : (U \setminus u^{-1}(J)) \text{ has measure } 0\}} \subset I.$$

*The post-composition $F \circ u$ lies in $W^{1,p}_{loc}(U)$ and its $i^{th}$ weak derivative is given by $\partial_i(F \circ u) = (F' \circ u) \cdot \partial_i u$.*

*Proof.* Write $I = (a, b)$, where $a < b \in \mathbb{R} \cup \{\pm\infty\}$. WLOG, by redefining $u$ on a measure zero set, we can assume $u(x) \in J$ for all $x \in U$ for some $\overline{J} \subset I$.

Since $\overline{J} \subset I$, we can find some $\delta > 0$, such that $J \subset [a + \delta, b - \delta]$. Indeed, if $a \in \mathbb{R}$, then $J$ is bounded below, and $\inf_{x \in \bar{J}} x$ is achieved by some $x_1$, since $\bar{J}$ is closed. Since $I$ is open, we can find $\delta_1 > 0$, such that $(x_1 - \delta_1, x_1 + \delta_1) \subset I$. Thus $\forall x \in J,\ x - \delta_1 \geq x_1 - \delta_1 \geq a$. If $a = -\infty$, we can pick any $\delta_2 > 0$, and we always have $\forall x \in J, x - \delta_1 \geq a$. Similarly, we can pick $\delta_1 > 0$, such that $\forall x \in J,\ x + \delta_2 < b$. Now pick $\delta := \min(\delta_1, \delta_2) > 0$, we have $J \subset [a + \delta, b - \delta]$.

Consider the function

$$g(x) := \begin{cases} x, & a + \delta \leq x \leq b - \delta \\ b - \delta e^{\frac{1}{\delta}(b - x - \delta)}, & x \geq b - \delta \\ a + \delta e^{\frac{1}{\delta}(x - a - \delta)}, & x \leq a + \delta \end{cases}.$$

We can check that $g$ is well defined and continuous, and

$$g'(x) = \begin{cases} 1, & a + \delta \leq x \leq b - \delta \\ e^{\frac{1}{\delta}(b - x - \delta)}, & x > b - \delta \\ e^{\frac{1}{\delta}(x - a - \delta)}, & x < a + \delta \end{cases}$$

is also continuous and bounded by 1. Thus $g \in C^1(\mathbb{R})$. In addition, $g$ is monotone increasing and $\forall x \in \mathbb{R},\ g(x) \in (a, b) = I$.

Now consider the composition $G := F \circ g$, which is well-defined. Since $F \in C^1(I)$ has bounded derivative $F'$, we have some $M > 0$, such that $\forall s \in I, |F'(s)| \leq M$. We have by calculus chain rule that $G \in C^1(\mathbb{R})$, with $|G'(s)| = |F'(g(s))g'(s)| = |F'(g(s))||g'(s)| \leq M$, which is also bounded. Thus by the above chain rule, we have $G \circ u \in W^{1,p}_{loc}(U)$, and its $i^{th}$ weak derivative id given by

$$\partial_i(G \circ u) = (G' \circ u) \cdot \partial_i u.$$

However, for any $x \in U$, we have $u(x) \in J \subset [a + \delta, b - \delta]$, so $g(u(x)) = u(x)$, and

$$(G \circ u)(x) = F(g(u(x))) = F(u(x)) = (F \circ u)(x).$$

Also,

$$(G' \circ u)(x) = G'(u(x)) = F'(g(u(x)))\cancelto{1}{g'(u(x))} = F'(u(x)) = (F' \circ u)(x).$$

Thus $G \circ u = F \circ u$, and $G' \circ u = F' \circ u$.

This proves $F \circ u \in W^{1,p}_{loc}(U)$, and its $i^{th}$ weak derivative id given by

$$\partial_i(F \circ u) = \partial_i(G \circ u) = (G' \circ u) \cdot \partial_i u = (F' \circ u) \cdot \partial_i u.$$

□

*Remark.* Notice that the above proof requires that $I$ is a connected interval, and $\bar{J} \subseteq I$ to find such a $\delta > 0$.

**Corollary 2.78** (Chain Rule Variation for Weak Derivatives)**.** *Given any $p \in [1, \infty]$. Suppose $u \in W^{1,p}(U)$. Suppose $F \in C^1(I)$ has bounded derivative $F'$, where $I \subseteq \mathbb{R}$ is an open interval that contains the closure of $u$'s essential image, and the post-composition $F \circ u \in L^p(U)$. In this case, $F \circ u$ lies in $W^{1,p}(U)$ and its $i^{th}$ weak derivative is given by $\partial_i(F \circ u) = (F' \circ u) \cdot \partial_i u$.*

*Proof.* Since $u \in W^{1,p}(U) \subseteq W^{1,p}_{loc}(U)$, we have that $F \circ u \in W^{1,p}_{loc}(U)$, with $\partial_i(F \circ u) = (F' \circ u) \cdot \partial_i u$. Thus, $(F' \circ u) \cdot \partial_i u$ is the global $i^{th}$ weak derivative of $F \circ u$ by proposition 2.72.

Now let $M > 0$ be such that $\forall s \in I,\ F'(s) \leq M$. Suppose $p < \infty$,

$$\begin{aligned}
\|F \circ u\|^p_{W^{1,p}(U)} &= \|F \circ u\|^p_{L^p(U)} + \sum_{i\in[n]} \|(F' \circ u) \cdot \partial_i u\|^p_{L^p(U)} \\
&\leq \|F \circ u\|^p_{L^p(U)} + \sum_{i\in[n]} M^p \|\partial_i u\|^p_{L^p(U)} \\
&\leq \|F \circ u\|^p_{L^p(U)} + M^p \|u\|^p_{W^{1,p}(U)} \\
&< \infty.
\end{aligned}$$

Suppose $p = \infty$, we have

$$\begin{aligned}
\|F \circ u\|_{W^{1,\infty}(U)} &= \|F \circ u\|_{L^\infty(U)} + \sum_{i\in[n]} \|(F' \circ u) \cdot \partial_i u\|_{L^\infty(U)} \\
&\leq \|F \circ u\|_{L^\infty(U)} + \sum_{i\in[n]} M \|\partial_i u\|_{L^\infty(U)} \\
&\leq \|F \circ u\|_{L^\infty(U)} + M \|u\|_{W^{1,\infty}(U)} \\
&< \infty.
\end{aligned}$$

Thus, $F \circ u \in W^{1,p}(U)$. □

Notice that we always get $F \circ u \in L^p(U)$ if we assume $U$ is bounded and other assumptions hold. Indeed, we have the following corollary.

**Corollary 2.79** (Chain Rule Variation for Weak Derivatives on Bounded Domain)**.** *Given any $p \in [1, \infty]$. Suppose $u \in W^{1,p}(U)$, where $U \subset \mathbb{R}^n$ is a bounded open set. Suppose $F \in C^1(I)$ has bounded derivative $F'$, where $I \subseteq \mathbb{R}$ is an open interval that contains the closure of $u$'s essential image. In this case, $F \circ u$ lies in $W^{1,p}(U)$ and its $i^{th}$ weak derivative is given by $\partial_i(F \circ u) = (F' \circ u) \cdot \partial_i u$. In addition, when $p = \infty$, the above holds true even when $U$ is not assumed to be bounded.*

*Proof.* Let $M > 0$ be such that $\forall s \in I,\ |F'(s)| < M$. Fix any $s \in I$, and consider any $t \neq s \in I$. By the Mean Value Theorem, there is $r \in (\min(s,t), \max(s,t))$ such that $F'(r) = \frac{F(t)-F(s)}{t-s}$. Thus $F(t) = F(s) + F'(r)(t-s)$, and

$$\begin{aligned}
|F(t)| &\leq |F(s)| + |F'(r)(t-s)| \\
&\leq |F(s)| + M|t-s| \\
&\leq |F(s)| + M|t| + M|s|.
\end{aligned}$$

Let $C := |F(s)| + M|s| < \infty$, we have $\forall t \in I,\ |F(t)| \leq C + M|t|$.

Suppose $p < \infty$, we have

$$\begin{aligned}
\|F \circ u\|^p_{L^p(U)} &= \int_U |F(u(x))|^p dx \\
&\leq \int_U |C + M|u(x)||^p dx \\
&\leq \int_U 2^{p-1}(C^p + M^p |u(x)|^p) dx \\
&= \int_U 2^{p-1} C^p dx + \int_U 2^{p-1} M^p |u(x)|^p dx \\
&= 2^{p-1} C^p |U| + 2^{p-1} M^p \|u\|^p_{L^p(U)} \\
&< \infty,
\end{aligned}$$

since $u \in W^{1,p}(U) \subset L^p(U)$, and $U$ is bounded.

Now suppose $p = \infty$, we have

$$\begin{aligned}\|F \circ u\|_{L^\infty(U)} &= \operatorname*{ess\,sup}_{x\in U} |F(u(x))| \\ &\leq \operatorname*{ess\,sup}_{x\in U} C + M|u(x)| \\ &= C + M \operatorname*{ess\,sup}_{x\in U} |u(x)| \\ &= C + M\|u\|_{L^\infty(U)} \\ &< \infty.\end{aligned}$$

Thus, $F \circ u \in L^p(U)$ and satisfies the above corollary. □

**Corollary 2.80.** *Given any $p \in [1, \infty]$. Suppose $u \in W^{1,\infty}(U)$. In this case, $u^2, e^u$ both lie in $W^{1,\infty}(U)$ and their $i^{th}$ weak derivative is given by $\partial_i(u^2) = 2u\partial_i u,\ \partial_i(e^u) = e^u\partial_i u$.*

*Proof.* Let $M := \|u\|_\infty < \infty$. Notice that the essential image of $u$ lies in $[-M, M]$, which is contained in the open interval $I := (-2M, 2M) \subset \mathbb{R}$.

Both $F(s) := s^2, G(s) := e^s$ are in $C^1(I)$, and $F'(s) = 2s$ and $G'(s) = e^s$ are bounded by $4M, e^{2M}$ on $I$, respectively. □

## 2.14 Fourier Transform

For more on the Fourier Transform, Radon measures, and the Schwartz class $S(\mathbb{R}^n)$, see my notes in Pmath651 Measure Theory.

**Definition 2.26.** Given $f \in L^1(\mathbb{R}^n; \mathbb{C})$, its **Fourier Transform** is $\mathcal{F}(f) := \hat{f} : \mathbb{R}^n \to \mathbb{C}$, where

$$\hat{f}(\omega) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} f(x)e^{-ix\cdot\omega}dx.$$

**Theorem 2.81** (Parseval–Plancherel identity)**.** *For any $f \in L^2(\mathbb{R}^n; \mathbb{C}) \cap L^1(\mathbb{R}^n; \mathbb{C})$, we have that*

$$\|\mathcal{F}f\|_{L^2(\mathbb{R}^n;\mathbb{C})} = \|f\|_{L^2(\mathbb{R}^n;\mathbb{C})}.$$

For more on the unitary and adjoint operators, see my notes in Pmath753 Functional Analysis.

**Corollary 2.82.** *The Fourier transform can be extended to $L^2(\mathbb{R}^n; \mathbb{C})$, and it is unitary. Namely, it preserves the inner product $\langle -, \cdot\rangle_{L^2(\mathbb{R}^n;\mathbb{C})}$, and it is an isomorphism with $\mathcal{F}^{-1} = \mathcal{F}^*$. Also, it is unitary on the Schwartz class $S(\mathbb{R}^n)$ with the $L^2(\mathbb{R}^n)$ inner product.*

**Corollary 2.83.** *The Fourier transform can be extended to $S(\mathbb{R}^n)^*$, where for $\langle u| \in S(\mathbb{R}^n)^*$ we define $\langle \mathcal{F}u| : S(\mathbb{R}^n) \to \mathbb{C}$ by $\psi \mapsto \langle u|\mathcal{F}^{-1}\psi\rangle$. We have that $\langle \mathcal{F}u| \in S(\mathbb{R}^n)^*$. In addition, $\langle \mathcal{F}u|$ is the canonical action defined by $\langle \mathcal{F}u, \cdot\rangle$ when $u \in L^2(\mathbb{R}^n; \mathbb{C})$.*

*Remark.* In some literature, when $u, v \in L^2(\mathbb{R}^n; \mathbb{C})$, the canonical action of $u$ on $v$ is defined to be $\langle u|v\rangle := \int_{\mathbb{R}^n} uvdx$, in order to preserve linearity on both entries. In that case, the Fourier transform of $\Lambda \in S(\mathbb{R}^n)^*$ is defined as $\langle \mathcal{F}\Lambda|\psi\rangle := \langle \Lambda|\mathcal{F}\psi\rangle$. However, we here take the canonical action to be $\langle u|v\rangle := \langle u, v\rangle_{L^2(\mathbb{R}^n;\mathbb{C})} = \int_{\mathbb{R}^n} \bar{u}vdx$, as commonly done with bra-ket notation.

**Theorem 2.84.** *For any $u \in L^2(\mathbb{R}^n), v \in L^2(\mathbb{R}^n; \mathbb{C})$, we have that $v = D^\alpha u \in L^2(\mathbb{R}^n)$ in the weak sense if and only if*

$$(\mathcal{F}v)(\omega) = \prod_{j=1}^n (i\omega_j)^{\alpha_j} (\mathcal{F}u)(\omega),$$

*for a.e. $\omega \in \mathbb{R}^n$ where $\mathcal{F}$ is the Fourier transform.*

*Proof.* Consider any $\phi \in C_c^\infty(\mathbb{R}^n)$. Find $R > 0$ such that $\text{Supp}(\phi) \subset B(0,R)$. We have that

$$\begin{aligned}
\mathcal{F}(D^\alpha \phi)(\omega) &= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} (D^\alpha \phi)(x) e^{-i\omega\cdot x} dx \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{B(0,2R)} (D^\alpha \phi)(x) e^{-i\omega\cdot x} dx \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} (-1)^{|\alpha|} \int_{B(0,2R)} \phi(x) \big(D^\alpha e^{-i\omega\cdot x}\big) dx \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} (-1)^{|\alpha|} \int_{B(0,2R)} \phi(x) \prod_{j=1}^n (-i\omega_j)^{\alpha_j} e^{-i\omega\cdot x} dx \\
&= \frac{1}{(2\pi)^{\frac{n}{2}}} \prod_{j=1}^n (i\omega_j)^{\alpha_j} \int_{B(0,2R)} \phi(x) e^{-i\omega\cdot x} dx \\
&= \prod_{j=1}^n (i\omega_j)^{\alpha_j} \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \phi(x) e^{-i\omega\cdot x} dx \\
&= \prod_{j=1}^n (i\omega_j)^{\alpha_j} (\mathcal{F}\phi)(\omega).
\end{aligned}$$

Now suppose $v = D^\alpha u$ in the weak sense; we have that

$$\begin{aligned}
\langle \mathcal{F}v, \mathcal{F}\phi \rangle_{L^2(\mathbb{R}^n;\mathbb{C})} &= \langle v, \phi \rangle_{L^2(\mathbb{R}^n;\mathbb{C})} \\
&= \langle v, \phi \rangle_{L^2(\mathbb{R}^n)} \\
&= (-1)^{|\alpha|} \langle u, D^\alpha \phi \rangle_{L^2(\mathbb{R}^n)} \\
&= (-1)^{|\alpha|} \langle u, D^\alpha \phi \rangle_{L^2(\mathbb{R}^n;\mathbb{C})} \\
&= (-1)^{|\alpha|} \langle \mathcal{F}u, \mathcal{F}(D^\alpha \phi) \rangle_{L^2(\mathbb{R}^n;\mathbb{C})} \\
&= (-1)^{|\alpha|} \left\langle \mathcal{F}u, \prod_{j=1}^n (i\omega_j)^{\alpha_j} \mathcal{F}\phi \right\rangle_{L^2(\mathbb{R}^n;\mathbb{C})} \\
&= (-1)^{|\alpha|} \int_{\mathbb{R}^n} \overline{(\mathcal{F}u)(\omega)} \prod_{j=1}^n (i\omega_j)^{\alpha_j} (\mathcal{F}\phi)(\omega) d\omega \\
&= (-1)^{|\alpha|} \int_{\mathbb{R}^n} \overline{\prod_{j=1}^n (-i\omega_j)^{\alpha_j} (\mathcal{F}u)(\omega)} (\mathcal{F}\phi)(\omega) d\omega \\
&= \int_{\mathbb{R}^n} \overline{\prod_{j=1}^n (i\omega_j)^{\alpha_j} (\mathcal{F}u)(\omega)} (\mathcal{F}\phi)(\omega) d\omega \\
&= \left\langle \prod_{j=1}^n (i\omega_j)^{\alpha_j} \mathcal{F}u \middle| \mathcal{F}\phi \right\rangle.
\end{aligned}$$

By linearity, the above equation extends to any $\phi + i\psi$, for $\phi, \psi \in C_c^\infty(\mathbb{R}^n)$. Since $C_c^\infty(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n) \supseteq S(\mathbb{R}^n;\mathbb{R})$, and $\mathcal{F}$ is linear and continuous, we have that $\mathcal{F}(C_c^\infty(\mathbb{R}^n) + iC_c^\infty(\mathbb{R}^n))$ is dense in $\mathcal{F}(S(\mathbb{R}^n)) = S(\mathbb{R}^n)$. Thus,

$$\langle \mathcal{F}v | = \left\langle \prod_{j=1}^n (i\omega_j)^{\alpha_j} \mathcal{F}u \right|$$

in $S(\mathbb{R}^n)^*$. Thus, $\mathcal{F}v = \prod_{j=1}^n (i\omega_j)^{\alpha_j} \mathcal{F}u$ a.e.by theorem 2.10.

For the converse, suppose $(\mathcal{F}v)(\omega) = \prod_{j=1}^n (i\omega_j)^{\alpha_j}(\mathcal{F}u)(\omega)$ for a.e. $\omega \in \mathbb{R}^n$. We have that

$$\begin{aligned}(-1)^{|\alpha|}\langle u, D^\alpha\phi\rangle_{L^2(\mathbb{R}^n;\mathbb{C})} &= \int_{\mathbb{R}^n} \overline{\prod_{j=1}^n (i\omega_j)^{\alpha_j}(\mathcal{F}u)(\omega)}(\mathcal{F}\phi)(\omega)d\omega \\ &= \int_{\mathbb{R}^n} \overline{(\mathcal{F}v)(\omega)}(\mathcal{F}\phi)(\omega)d\omega \\ &= \langle \mathcal{F}v, \mathcal{F}\phi\rangle_{L^2(\mathbb{R}^n;\mathbb{C})} \\ &= \langle v, \phi\rangle_{L^2(\mathbb{R}^n;\mathbb{C})}.\end{aligned}$$

Since $u, D^\alpha\phi, \phi$ are all real-valued, so must be $v$. Thus, $v = D^\alpha u$ in the weak sense. □

## 2.15 Sobolev Spaces with Real Orders

**Definition 2.27.** For $p \in [1,\infty)$, and $\sigma \in (0,1)$, we define the **Gagliardo seminorm** for $u \in L^1_{loc}(U)$ to be

$$[u]_{W^{\sigma,p}(U)} = \left(\int_U \int_U \frac{|u(x)-u(y)|^p}{|x-y|^{n+\sigma p}}\,dx\,dy\right)^{1/p}.$$

Also, for any $s = k + \sigma$, where $k \in \mathbb{N}$, we define the **Gagliardo Sobolev norm** to be

$$\|u\|_{W^{s,p}(U)} := \left(\|u\|^p_{W^{k,p}(U)} + \sum_{|\alpha|=k} [D^\alpha u]^p_{W^{\sigma,p}(U)}\right)^{1/p}.$$

We define

$$W^{s,p}(U) := \left\{u \in L^1_{loc}(U) : \|u\|_{W^{s,p}(U)} < \infty\right\}.$$

Similarly,

$$W^{s,p}_0(U) = \overline{C^\infty_c(U)}^{\|\cdot\|_{W^{s,2}(U)}},$$

the closure of $C^\infty_c(U)$. For any $s < 0$, we define

$$W^{s,p}(U) := W_0^{-s,p}(U)^*,$$

the dual space of $W_0^{-s,p}(U)$, equipped with the dual norm.

*Remark.* Notice that the above definition requires $\sigma \in (0,1)$, so it only works for $s \in \mathbb{R} \setminus \mathbb{Z}$.

**Definition 2.28.** For $U = \mathbb{R}^n, p = 2$, and $s \in \mathbb{R}$, we define

$$\|u\|_{H^s(\mathbb{R}^n)} = \left(\int_{\mathbb{R}^n} (1+|\omega|^2)^s |(\mathcal{F}u)(\omega)|^2\,d\omega\right)^{1/2},$$

where $\mathcal{F}u$ is the Fourier transform of $u$. We also define

$$H^s(\mathbb{R}^n) := \left\{u \in S(\mathbb{R}^n)^* : \|u\|_{H^s(\mathbb{R}^n)} < \infty\right\}.$$

*Remark.* For $s \in \mathbb{R}__$, notice that $\mathcal{F}u$ may not be a function, it does not make sense to talk about $\int_{\mathbb{R}^n}(1+|\omega|^2)^s|(\mathcal{F}u)(\omega)|^2\,d\omega$. We are abusing the notation here. Indeed, we are saying $u \in H^s(\mathbb{R}^n)$ if and only if there is some $\tilde{u} \in L^2(\mathbb{R}^n)$, such that for any $\psi \in S(\mathbb{R}^n)$, we have

$$\left\langle \mathcal{F}u \middle| (1+|\omega|^2)^{s/2}\psi \right\rangle = \langle \tilde{u}, \psi\rangle,$$

and $\|u\|_{H^s(\mathbb{R}^n)} := \|\tilde{u}\|_{L^2(\mathbb{R}^n)}$.

**Theorem 2.85.** *Suppose $s = k \in \mathbb{Z}$, then the above definition coincides with the usual definition of Sobolev spaces. In addition, the Sobolev norms $\|\cdot\|_{H^k(\mathbb{R}^n)}$ and $\|\cdot\|_{H^s(\mathbb{R}^n)}$ are equivalent norms.*

*Proof.* We first assume $s = k \in \mathbb{N}$. Pick any $u \in H^k(\mathbb{R}^n)$. For each $|\alpha| \le k$, we have $D^\alpha u \in L^2(\mathbb{R}^n)$. By the Parseval–Plancherel identity, we have

$$\begin{aligned}
\|u\|^2_{H^k(\mathbb{R}^n)} &= \sum_{|\alpha|\le k} \|D^\alpha u\|^2_{L^2(\mathbb{R}^n)} \\
&= \sum_{|\alpha|\le k} \|\mathcal{F}(D^\alpha u)\|^2_{L^2(\mathbb{R}^n)} \\
&= \sum_{|\alpha|\le k} \int_{\mathbb{R}^n} |\mathcal{F}(D^\alpha u)(\omega)|^2 d\omega \\
&= \sum_{|\alpha|\le k} \int_{\mathbb{R}^n} \left| \prod_{j=1}^n (i\omega_j)^{\alpha_j} (\mathcal{F}u)(\omega) \right|^2 d\omega \\
&= \int_{\mathbb{R}^n} \left( \sum_{|\alpha|\le k} \prod_{j=1}^n |\omega_j|^{2\alpha_j} \right) |(\mathcal{F}u)(\omega)|^2 d\omega \\
\|u\|^2_{H^s(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} (1+|\omega|^2)^s |(\mathcal{F}u)(\omega)|^2 \, d\omega \\
&= \int_{\mathbb{R}^n} \left( 1 + \sum_{j=1}^n |\omega_j|^2 \right)^s |(\mathcal{F}u)(\omega)|^2 d\omega \\
&= \int_{\mathbb{R}^n} \left( \sum_{|\alpha|\le s} \binom{s}{\alpha} \prod_{j=1}^n |\omega_j|^{2\alpha_j} \right) |(\mathcal{F}u)(\omega)|^2 d\omega,
\end{aligned}$$

where $\binom{s}{\alpha} := \frac{s!}{(s-|\alpha|)!\alpha_1!\cdots\alpha_n!}$. Thus, $\|u\|^2_{H^s(\mathbb{R}^n)} \le \max_{|\alpha|\le s} \binom{s}{\alpha} \|u\|^2_{H^k(\mathbb{R}^n)} < \infty$.

This shows $H^k(\mathbb{R}^n) \subseteq H^s(\mathbb{R}^n)$.

Now for any $u \in H^s(\mathbb{R}^n)$, we have that for any $|\alpha| \le s$,

$$\begin{aligned}
\int_{\mathbb{R}^n} \left| \prod_{j=1}^n |\omega_j|^{\alpha_j} (\mathcal{F}u)(\omega) \right|^2 d\omega &\le \int_{\mathbb{R}^n} \left( \sum_{|\alpha|\le s} \binom{s}{\alpha} \prod_{j=1}^n |\omega_j|^{2\alpha_j} \right) |(\mathcal{F}u)(\omega)|^2 d\omega \\
&= \|u\|^2_{H^s(\mathbb{R}^n)} \\
&< \infty.
\end{aligned}$$

Thus $\prod_{j=1}^n |\omega_j|^{\alpha_j} \mathcal{F}u \in L^2(\mathbb{R}^n;\mathbb{C})$. Let $u_\alpha := \mathcal{F}^{-1}\left(\prod_{j=1}^n |\omega_j|^{\alpha_j} \mathcal{F}u\right) \in L^2(\mathbb{R}^n;\mathbb{C})$, we have that $u_\alpha = D^\alpha u \in L^2(\mathbb{R}^n)$ in the weak sense. This shows that $u \in H^k(\mathbb{R}^n)$, so $H^s(\mathbb{R}^n) \subseteq H^k(\mathbb{R}^n)$.

The equivalence of norms is given by

$$\|u\|^2_{H^k(\mathbb{R}^n)} \le \|u\|^2_{H^s(\mathbb{R}^n)} \le \max_{|\alpha|\le s} \binom{s}{\alpha} \|u\|^2_{H^k(\mathbb{R}^n)}.$$

Now we will deal with the negative order. Namely, when $s = k \in \mathbb{Z} \setminus \mathbb{N}$. To simplify notation, we will still write $s \in \mathbb{N}_+$, and consider $-s \in \mathbb{Z} \setminus \mathbb{N}$.

For any $u \in H^{-s}(\mathbb{R}^n), v \in S(\mathbb{R}^n) \subseteq H^s(\mathbb{R}^n)$, since $\tilde{u} := (1+|\omega|^2)^{-s/2}(\mathcal{F}u), \tilde{v} := (1+|\omega|^2)^{s/2}(\mathcal{F}v) \in L^2(\mathbb{R}^n;\mathbb{C})$, we have

$$\begin{aligned}
\langle \tilde{u}, \tilde{v} \rangle_{L^2(\mathbb{R}^n;\mathbb{C})} &= \left\langle \mathcal{F}u \middle| (1+|\omega|^2)^{-s/2}(1+|\omega|^2)^{s/2}(\mathcal{F}v) \right\rangle_{H^s(\mathbb{R}^n)^*, H^s(\mathbb{R}^n)} \\
&= \langle \mathcal{F}u | \mathcal{F}v \rangle_{H^s(\mathbb{R}^n)^*, H^s(\mathbb{R}^n)} \\
&= \left\langle u \middle| \mathcal{F}^{-1}\mathcal{F}v \right\rangle_{H^s(\mathbb{R}^n)^*, H^s(\mathbb{R}^n)} \\
&= \langle u | v \rangle_{H^s(\mathbb{R}^n)^*, H^s(\mathbb{R}^n)}.
\end{aligned}$$

Thus,

$$\begin{aligned}\left|\langle u|v\rangle_{H^s(\mathbb{R}^n)^*,H^s(\mathbb{R}^n)}\right| &= |\langle\tilde{u},\tilde{v}\rangle|_{L^2(\mathbb{R}^n;\mathbb{C})}\\ &\le \|\tilde{u}\|_{L^2(\mathbb{R}^n;\mathbb{C})}\|\tilde{v}\|_{L^2(\mathbb{R}^n;\mathbb{C})}\\ &= \|u\|_{H^{-s}(\mathbb{R}^n)}\|v\|_{H^s(\mathbb{R}^n)}.\end{aligned}$$

Since $S(\mathbb{R}^n) \supset C_c^\infty(\mathbb{R}^n)$ is dense in $H^k(\mathbb{R}^n) = H^s(\mathbb{R}^n)$, the above also hold for all $v \in H^s(\mathbb{R}^n)$. Thus $u \in H^s(\mathbb{R}^n)^* = H^k(\mathbb{R}^n)^* = H^{-k}(\mathbb{R}^n)$, with

$$\|u\|_{H^s(\mathbb{R}^n)^*} \le \|u\|_{H^{-s}(\mathbb{R}^n)}.$$

This shows that $H^{-s}(\mathbb{R}^n) \subseteq H^{-k}(\mathbb{R}^n)$.

Now for any $u^* \in H^{-k}(\mathbb{R}^n) = H^k(\mathbb{R}^n)^*$, by Riesz-Frechet Representation theorem 1.24 there is a $u \in H^k(\mathbb{R}^n)$, such that for all $v \in H^k(\mathbb{R}^n)$,

$$\langle u^*|v\rangle_{H^{-k}(\mathbb{R}^n),H^k(\mathbb{R}^n)} = \langle u, v\rangle_{H^k(\mathbb{R}^n)},$$

with $\|u^*\|_{H^{-k}(\mathbb{R}^n)} = \|u\|_{H^k(\mathbb{R}^n)}$. Thus, for any $v \in S(\mathbb{R}^n) \subseteq H^s(\mathbb{R}^n)$,

$$\begin{aligned}\langle\mathcal{F}u^*|\mathcal{F}v\rangle_{S(\mathbb{R}^n)^*,S(\mathbb{R}^n)} &= \langle u^*|v\rangle_{H^{-k}(\mathbb{R}^n),H^k(\mathbb{R}^n)}\\ &= \langle u, v\rangle_{H^k(\mathbb{R}^n)}\\ &= \sum_{|\alpha|\le k}\langle D^\alpha u, D^\alpha v\rangle_{L^2(\mathbb{R}^n)}\\ &= \sum_{|\alpha|\le k}\langle\mathcal{F}(D^\alpha u),\mathcal{F}(D^\alpha v)\rangle_{L^2(\mathbb{R}^n)}\\ &= \sum_{|\alpha|\le k}\left\langle\prod_{j=1}^n(i\omega_j)^{\alpha_j}\mathcal{F}u, \prod_{j=1}^n(i\omega_j)^{\alpha_j}\mathcal{F}v\right\rangle_{L^2(\mathbb{R}^n)}\\ &= \sum_{|\alpha|\le k}\int_{\mathbb{R}^n}\overline{\prod_{j=1}^n(i\omega_j)^{\alpha_j}\mathcal{F}u(\omega)}\prod_{j=1}^n(i\omega_j)^{\alpha_j}\mathcal{F}v(\omega)d\omega\\ &= \sum_{|\alpha|\le k}\int_{\mathbb{R}^n}\overline{\prod_{j=1}^n(-i\omega_j)^{\alpha_j}\prod_{j=1}^n(i\omega_j)^{\alpha_j}\mathcal{F}u(\omega)\mathcal{F}v(\omega)}d\omega\\ &= \int_{\mathbb{R}^n}\overline{\sum_{|\alpha|\le k}\prod_{j=1}^n(\omega_j)^{2\alpha_j}\mathcal{F}u(\omega)\mathcal{F}v(\omega)}d\omega\\ &= \langle p\mathcal{F}u|\mathcal{F}v\rangle.\end{aligned}$$

where $p(\omega) := \sum_{|\alpha|\le k}\prod_{j=1}^n(\omega_j)^{2\alpha_j} \ge 0$. Since $S(\mathbb{R}^n)$ is dense in $H^k(\mathbb{R}^n) = H^s(\mathbb{R}^n)$, for all $v \in H^s(\mathbb{R}^n)$, we have

$$\langle\mathcal{F}u^*|\mathcal{F}v\rangle_{S(\mathbb{R}^n)^*,S(\mathbb{R}^n)} = \langle u, v\rangle_{H^k(\mathbb{R}^n)} = \langle p\mathcal{F}u|\mathcal{F}v\rangle.$$

Now let $\tilde{u} := \frac{p(\omega)}{(1+|\omega|)^{s/2}}\mathcal{F}u$. Since

$$\begin{aligned}0 &\le p(\omega)\\ &\le \sum_{|\alpha|\le k}\prod_{j=1}^n|\omega_j|^{2\alpha_j}\\ &\le \sum_{|\alpha|\le s}\binom{s}{\alpha}\prod_{j=1}^n|\omega_j|^{2\alpha_j}\\ &= (1+|\omega|)^s,\end{aligned}$$

we have that

$$\begin{aligned}
\|\tilde{u}\|^2_{L^2(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} |\tilde{u}|^2 d\omega \\
&= \int_{\mathbb{R}^n} \left| \frac{p(\omega)}{(1+|\omega|)^{s/2}} \mathcal{F}u(\omega) \right|^2 d\omega \\
&= \int_{\mathbb{R}^n} \frac{p(\omega)}{(1+|\omega|)^s} p(\omega) |\mathcal{F}u(\omega)|^2 d\omega \\
&\leq \int_{\mathbb{R}^n} p(\omega) |\mathcal{F}u(\omega)|^2 d\omega \\
&= \langle p\mathcal{F}u | \mathcal{F}u \rangle \\
&= \langle u, u \rangle_{H^k(\mathbb{R}^n)} \\
&= \|u\|^2_{H^k(\mathbb{R}^n)} \\
&= \|u^*\|^2_{H^{-k}(\mathbb{R}^n)} \\
&< \infty.
\end{aligned}$$

This shows $\tilde{u} \in L^2(\mathbb{R}^n)$. For any $v \in S(\mathbb{R}^n)$, we have that $(1+|\omega|)^{-s/2} v \in S(\mathbb{R}^n) = \mathcal{F}(S(\mathbb{R}^n))$, so

$$\begin{aligned}
\left\langle \mathcal{F}u^* \middle| (1+|\omega|)^{-s/2} v \right\rangle &= \left\langle p\mathcal{F}u \middle| (1+|\omega|)^{-s/2} v \right\rangle \\
&= \left\langle \frac{p}{(1+|\omega|)^{s/2}} \mathcal{F}u \middle| v \right\rangle \\
&= \langle \tilde{u} | v \rangle \\
&= \langle \tilde{u}, v \rangle_{L^2(\mathbb{R}^n)}.
\end{aligned}$$

Thus, $u^* \in H^{-s}(\mathbb{R}^n)$, with

$$\|u^*\|_{H^{-s}(\mathbb{R}^n)} = \|\tilde{u}\|_{L^2(\mathbb{R}^n)} \leq \|u^*\|_{H^{-k}(\mathbb{R}^n)}.$$

This proves that $H^{-k}(\mathbb{R}^n) \subseteq H^{-s}(\mathbb{R}^n)$. □

**Theorem 2.86.** *For any $s \in \mathbb{R} \setminus \mathbb{Z}$, when $p = 2$, we have*

$$H^s(U) = W^{s,2}(U).$$

*Also, $\|\cdot\|_{H^s(U)}$ and $\|\cdot\|_{W^{s,2}(U)}$ are equivalent norms.*

# 3 Elliptic PDEs

## 3.1 Weak Solutions

We will consider the model problem: $U \in \mathbb{R}^n$ be open and bounded, with some $f : U \to \mathbb{R}$ be given. We want to find $u : \bar{U} \to \mathbb{R}$, such that $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$.

**Definition 3.1.** A **second order differential operator** is

$$Lu := -\sum_{i,j=1}^{n} \partial_j(a^{ij}(x)\partial_i u) + \sum_{i=1}^{n} b^i(x)\partial_i u + c(x)u.$$

**Definition 3.2.** A **symmetric (uniformly) elliptic second order differential operator** is an $L$ such that $a^{ij} = a^{ji}$, and $\exists \theta > 0,$ such that

$$\sum_{i,j=1}^{n} a^{ij}(x)\xi_i\xi_j \geq \theta\|\xi\|_2^2 \text{ for a.e. } x \in U, \forall \xi \in \mathbb{R}^n.$$

*Remark.* The above definition is equivalent to saying $A(x) = (a^{ij}(x)) \in \mathbb{R}^{n*n}$ is symmetric positive definite for a.e. $x \in U$, with a uniform positive lower bound $\theta > 0$ on their eigenvalues.

**Example 3.1.1.** If we take $a^{ij} = C\delta_{ij}$, we have $Lu = -C\Delta u + b \cdot Du + cu$.

**Definition 3.3.** The **bilinear form associated with** $L$ is given by:

$$B[u, v] := \int_U \left( \sum_{i,j=1}^{n} a^{ij}\partial_i u \partial_j v + \sum_{i=1}^{n} b^i \partial_i uv + cuv \right) dx, \ \forall u, v \in H_0^1(U).$$

**Definition 3.4.** Consider $f = f^0 - \sum_{i=1}^n f^i_{x^i} \in H^{-1}(U)$ as in 2.61.
$u \in H_0^1(U)$ is called a **weak solution** to the BVP $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$, if $u$ satisfies the **weak formulation:**

$$\forall v \in H_0^1(U), B[u, v] = \langle f|v\rangle = \left\langle f^0, v \right\rangle_{L^2(U)} + \sum_{i=1}^{n} \left\langle f^i, \partial_i v \right\rangle_{L^2(U)}.$$

**Definition 3.5.** For $f \in L^2(U)$, we have the special case:
$u \in H_0^1(U)$ is called a **weak solution** to the BVP $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$, if $u$ satisfies the **weak formulation:**

$$\forall v \in H_0^1(U), B[u, v] = \langle f, v\rangle_{L^2(U)}.$$

**Definition 3.6.** If $u \in W^2(U)$ satisfies $Lu(x) = f(x)$ a.e.$x \in U$, we say $u$ is a **strong solution** to the problem $Lu = f$ in $U$.

**Proposition 3.1.** *If a classical solution $u$ exists, i.e $u$ is smooth, and $Lu = f, u|_{\partial U} = 0$, then $u$ is always a strong solution.*

**Proposition 3.2.** *If a strong solution $u$ exists, then $u$ is always a weak solution.*

*Proof.* Firstly consider any $v \in C_c^\infty(U)$, we have

$$\begin{aligned}
\langle f|v\rangle &= \langle Lu, v\rangle \\
&= \int_U Luv dx \\
&= \int_U \left( -\sum_{i,j=1}^n \partial_j(a^{ij}\partial_i u) + \sum_{i=1}^n b^i \partial_i u + cu \right) v dx \\
&= -\sum_{i,j=1}^n \int_U \partial_j(a^{ij}\partial_i u) v dx + \int_U \left( \sum_{i=1}^n b^i \partial_i uv + cuv \right) dx \\
&= \int_U \sum_{i,j=1}^n a^{ij}\partial_i u \partial_j v dx + \int_U \left( \sum_{i=1}^n b^i \partial_i uv + cuv \right) dx \\
&= \int_U \left( \sum_{i,j=1}^n a^{ij}\partial_i u \partial_j v + \sum_{i=1}^n b^i \partial_i uv + cuv \right) dx \\
&= B[u, v].
\end{aligned}$$

Since $H_0^1(U) = \overline{C_c^\infty(v)}$, this holds for any $v \in H_0^1(U)$. □

## 3.2 Existence of weak solution

### 3.2.1 First Existence Theorem

**Theorem 3.3** (Lax-Milgram)**.** *Consider a real Hilbert space $\mathcal{H}$ with $\langle \cdot, \cdot \rangle$ and action $\langle \cdot | \cdot \rangle_{\mathcal{H}^*, \mathcal{H}}$. Assume $B : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ is a bilinear form such that $\exists a, b > 0$ such that $\forall u, v \in \mathcal{H}$,*

$$\begin{aligned}
|B[u, v]| &\leq a\|u\|\|v\| \\
B[u, u] &\geq b\|u\|^2.
\end{aligned}$$

*Then $\forall f \in \mathcal{H}^*, \exists! u \in \mathcal{H}$ such that $\forall v \in \mathcal{H},\ B[u, v] = \langle f|v\rangle$.*

*Proof.* For each $u \in \mathcal{H}$, we define the linear functional $T_u : v \mapsto B[u, v]$.
We have that $|T_u v| = |B[u, v]| \leq a\|u\|\|v\|$, so $\|T_u\|_{\mathcal{H}^*} \leq a\|u\| < \infty$ is bounded. Thus $T_u \in \mathcal{H}^*$.
By the Riesz-Frechet Representation theorem 1.24, $\exists! w \in \mathcal{H}$, such that

$$\forall v \in \mathcal{H}, T_u v = \langle w, v\rangle_{\mathcal{H}},$$

with $\|T_u\|_{\mathcal{H}^*} = \|w\|_{\mathcal{H}}$.
Now define $A : \mathcal{H} \to \mathcal{H}$ by $u \mapsto w$ in the above setting, such that $\forall v \in \mathcal{H}, \langle Au, v\rangle = B[u, v]$.

**Claim 3.3.1.** *For any $u \in \mathcal{H}$, we have that*

$$b\|u\| \leq \|Au\| \leq a\|u\|.$$

*Proof.* We have

$$\|Au\|^2 = \langle Au, Au\rangle = B[u, Au] \leq a\|u\|\|Au\|.$$

If $\|Au\| = 0$, clearly $\|Au\| \leq a\|u\|$.
Otherwise we can divide both side by $\|Au\|$, and get $\|Au\| \leq a\|u\|$.
On the other hand, we have

$$b\|u\|^2 \leq B[u, u] = \langle Au, u\rangle \leq \|Au\|\|u\|.$$

If $\|u\| = 0$, clearly $b\|u\| \leq \|Au\|$.
Otherwise we can divide both side by $\|u\|$, and get $b\|u\| \leq \|Au\|$. □

**Claim 3.3.2.** *We have $A \in B(\mathcal{H})$.*

*Proof.* For any $u_1, u_2, v \in \mathcal{H}, c \in \mathbb{R}$, we have that

$$\begin{aligned}\langle A(u_1 + cu_2), v\rangle &= B[u_1 + cu_2, v] \\ &= B[u_1, v] + cB[u_2, v] \\ &= \langle Au_1, v\rangle + c\langle Au_2, v\rangle \\ &= \langle Au_1 + cAu_2, v\rangle.\end{aligned}$$

Since this holds for all $v \in \mathcal{H}$, we have $A(u_1 + cu_2) = Au_1 + cAu_2$, and thus $A$ is linear.
In addition, we have

$$\|A\|_{B(\mathcal{H})} = \sup_{u\in\mathcal{H}, u\neq 0} \frac{\|Au\|}{\|u\|} \leq \sup_{u\in\mathcal{H}, u\neq 0} \frac{a\|u\|}{\|u\|} = a < \infty.$$

This shows $A$ is bounded, and thus $A \in B(\mathcal{H})$. □

**Claim 3.3.3.** *$A$ is bijective.*

*Proof.* Suppose $Au = 0$, we have that

$$b\|u\| \leq \|Au\| = 0,$$

which means that $u = 0$. Thus $A$ is injective.
Consider any sequence $(y_j)_{j=1}^{\infty} \subset \mathrm{Im}(A)$, such that $\lim_{j\to\infty} y_j = y \in \mathcal{H}$.
We can find $(x_j)_{j=1}^{\infty} \subset \mathcal{H}$, such that $\forall j \geq 1, Ax_j = y_j$.
Since $(y_j)_{j=1}^{\infty}$ is convergent and thus Cauchy, given any $\epsilon > 0$, we can find some $N \geq 1$, such that $\forall i, j \geq N, \|y_j - y_i\| < b\epsilon$.
Now

$$\begin{aligned}\|x_j - x_i\| &\leq \frac{1}{b}\|A(x_j - x_i)\| \\ &= \frac{1}{b}\|Ax_j - Ax_i\| \\ &= \frac{1}{b}\|y_j - y_i\| \\ &< \frac{1}{b}b\epsilon \\ &< \epsilon.\end{aligned}$$

Thus $(x_j)_{j=1}^{\infty}$ is Cauchy.
Since $\mathcal{H}$ is complete, there is some $x \in \mathcal{H}$, such that $\lim_{j\to\infty} x_j = x$.
Since $A$ is bounded and thus continuous, we have that

$$\begin{aligned}Ax &= A\left(\lim_{j\to\infty} x_j\right) \\ &= \lim_{j\to\infty} Ax_j \\ &= \lim_{j\to\infty} y_j \\ &= y.\end{aligned}$$

Thus $y \in \mathrm{Im}(A)$.
This proves that $\mathrm{Im}(A)$ is closed.
Since $A$ is linear, $\mathrm{Im}(A)$ is a closed subspace of $\mathcal{H}$, and thus $\mathcal{H} = \mathrm{Im}(A) \oplus \mathrm{Im}(A)^{\perp}$.
Consider any $w \in \mathrm{Im}(A)^{\perp}$, we must have

$$b\|w\|^2 \leq B[w, w] = \langle Aw, w\rangle = 0.$$

Thus $\mathrm{Im}(A)^{\perp} = \{0\}$, and thus $\mathrm{Im}(A) = \mathcal{H}$.
Thus $A$ is surjective. □

Now by the Bounded Inverse Theorem, $A^{-1}$ exists and is bounded.
By Riesz-Frechet Representation theorem 1.24, given any $f \in \mathcal{H}^*$, we have

$$\exists! w \in \mathcal{H}, \text{ such that } \langle f|v\rangle = \langle w, v\rangle \ \forall v \in \mathcal{H}.$$

Let $u = A^{-1}w$, we have that

$$\forall v \in \mathcal{H},\ B[u,v] = \langle Au, v\rangle = \langle w, v\rangle = \langle f|v\rangle.$$

This proves the existence.
Now suppose there is some $\hat{u}$ such that $\forall v \in \mathcal{H},\ B[\hat{u}, v] = \langle f|v\rangle = B[u,v]$.
We must have $B[u - \hat{u}, v] = 0,\ \forall v \in \mathcal{H}$. Thus

$$b\|u - \hat{u}\| \le B[u - \hat{u}, u - \hat{u}] = 0,$$

and thus $\hat{u} = u$ is unique. □

**Proposition 3.4** (Cauchy's inequality)**.** *For any $a, b, \epsilon > 0$, we have*

$$ab \le \epsilon a^2 + \frac{b^2}{4\epsilon}.$$

**Theorem 3.5** (Energy estimates)**.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $a^{ij}, b^i, c \in L^\infty(U)$, such that $(a^{ij})$ is symmetric positive definite. For the bilinear form defined in 3.3, there exists constants $\alpha, \beta > 0, \gamma \ge 0$, such that $\forall u, v \in H_0^1(U)$,*

$$|B[u,v]| \le \alpha \|u\|_{H^1(U)} \|v\|_{H^1(U)} \tag{1}$$

$$\beta \|u\|^2_{H^1(U)} \le B[u,u] + \gamma \|u\|^2_{L^2(U)}. \tag{2}$$

*Proof.* We have

$$\begin{aligned}
|B[u,v]| &= \left| \int_U \left( \sum_{i,j=1}^n a^{ij} \partial_i u \partial_j v + \sum_{i=1}^n b^i \partial_i u v + cuv \right) dx \right| \\
&\le \sum_{i,j=1}^n \left\|a^{ij}\right\|_{L^\infty(U)} \int_U |\partial_i u||\partial_j v| dx + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} \int_U |\partial_i u||v| dx + \|c\|_{L^\infty(U)} \int_U |u||v| dx \\
&\le \sum_{i,j=1}^n \left\|a^{ij}\right\|_{L^\infty(U)} \|\partial_i u\|_{L^2(U)} \|\partial_j v\|_{L^2(U)} + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} \|\partial_i u\|_{L^2(U)} \|v\|_{L^2(U)} \\
&\quad + \|c\|_{L^\infty(U)} \|u\|_{L^2(U)} \|v\|_{L^2(U)} \\
&\le \sum_{i,j=1}^n \left\|a^{ij}\right\|_{L^\infty(U)} \|u\|_{H^1(U)} \|v\|_{H^1(U)} + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} \|u\|_{H^1(U)} \|v\|_{H^1(U)} \\
&\quad + \|c\|_{L^\infty(U)} \|u\|_{H^1(U)} \|v\|_{H^1(U)} \\
&= \left( \sum_{i,j=1}^n \left\|a^{ij}\right\|_{L^\infty(U)} + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} + \|c\|_{L^\infty(U)} \right) \|u\|_{H^1(U)} \|v\|_{H^1(U)}.
\end{aligned}$$

Taking $\alpha := \sum_{i,j=1}^n \left\|a^{ij}\right\|_{L^\infty(U)} + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} + \|c\|_{L^\infty(U)}$, we notice that $\alpha \ge 0$, and $\alpha = 0 \implies \forall i, j,\ a^{ij} = 0$, which contradicts $(a_{ij})$ is positive definite. Thus $\alpha > 0$, and $|B[u,v]| \le \alpha \|u\|_{H^1(U)} \|v\|_{H^1(U)}$.
On the other hand, consider $\xi = Du \in \mathbb{R}^n$.
We have that

$$\theta \|Du\|_2^2 \le \sum_{i,j=1}^n a^{ij} \partial_i u \partial_j u.$$

Thus

$$\begin{aligned}
\theta\|Du\|^2_{L^2(U)} &= \theta\int_U \|Du\|_2^2 dx \\
&\le \int_U \sum_{i,j=1}^n a^{ij}\partial_i u\partial_j u dx \\
&= B[u,u] - \int_U \left(\sum_{i=1}^n b^i\partial_i uu + cuu\right)dx \\
&\le B[u,u] + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)}\|\partial_i u\|_{L^2(U)}\|u\|_{L^2(U)} + \|c\|_{L^\infty(U)}\|u\|^2_{L^2(U)} \\
&\le B[u,u] + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)}\left(\epsilon\|\partial_i u\|^2_{L^2(U)} + \frac{1}{4\epsilon}\|u\|^2_{L^2(U)}\right) + \|c\|_{L^\infty(U)}\|u\|^2_{L^2(U)} \\
&= B[u,u] + \epsilon\sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)}\|\partial_i u\|^2_{L^2(U)} + \left(\frac{1}{4\epsilon}\sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} + \|c\|_{L^\infty(U)}\right)\|u\|^2_{L^2(U)} \\
&\le B[u,u] + \epsilon\sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)}\|Du\|^2_{L^2(U)} + \left(\frac{1}{4\epsilon}\sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} + \|c\|_{L^\infty(U)}\right)\|u\|^2_{L^2(U)}.
\end{aligned}$$

If $\sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} = 0$, pick any $\epsilon > 0$.
Otherwise choose $\epsilon := \frac{\theta}{2\sum_{i=1}^n \|b^i\|_{L^\infty(U)}} > 0$, and $\gamma := \frac{1}{4\epsilon}\sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} + \|c\|_{L^\infty(U)}$, we have

$$\frac{\theta}{2}\|Du\|^2_{L^2(U)} \le B[u,u] + \gamma\|u\|^2_{L^2(U)}.$$

Since $\|Du\|_{L^p(U)}$ and $\|u\|_{W^{1,p}(U)}$ are equivalent norms on $W_0^{1,p}(U)$ by 2.48, we have that

$$\exists C > 0, \text{ such that } \forall u \in H_0^1(U),\ \|u\|^2_{H^1(U)} \le C\|Du\|^2_{L^p(U)}.$$

Taking $\beta := \frac{\theta}{2C} > 0$, we have

$$\beta\|u\|^2_{H^1(U)} \le \frac{\theta}{2}\|Du\|^2_{L^2(U)} \le B[u,u] + \gamma\|u\|^2_{L^2(U)}.$$

□

**Definition 3.7.** Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\mu \in \mathbb{R}$, we define the operator $L_\mu$ by

$$L_\mu u := Lu + \mu u.$$

We define the bilinear form associated to $L_\mu$ to be $B_\mu$.

**Proposition 3.6.**

$$B_\mu[u,v] = B[u,v] + \int_U \mu uv dx = B[u,v] + \mu\langle u,v\rangle_{L^2(U)}.$$

**Theorem 3.7** (First Existence Theorem)**.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \ge 0$ be the same as in Energy Estimate 3.5. For any $\mu \ge \gamma$ and $\forall f \in H^{-1}(U)$, there is a unique weak solution $u \in H_0^1(U)$ of the BVP:* $\begin{cases} Lu + \mu u = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$

*Proof.* By Energy estimates, we have that $\forall u, v \in H_0^1(U)$,

$$\begin{aligned}
|B_\mu[u,v]| &\le |B[u,v]| + \mu\left|\langle u,v\rangle_{L^2(U)}\right| \\
&\le \alpha\|u\|_{H^1(U)}\|v\|_{H^1(U)} + \mu\|u\|_{L^2(U)}\|v\|_{L^2(U)} \\
&\le (\alpha+\mu)\|u\|_{H^1(U)}\|v\|_{H^1(U)} \\
B_\mu[u,u] &= B[u,u] + \mu\langle u,u\rangle_{L^2(U)} \\
&= B[u,u] + \mu\|u\|_{L^2(U)} \\
&\ge \beta\|u\|^2_{H^1(U)} + (\mu-\gamma)\|u\|^2_{L^2(U)} \\
&\ge \beta\|u\|^2_{H^1(U)}.
\end{aligned}$$

By Lax–Milgram Theorem, for any $f \in H^{-1}(U)$, there is a unique $u \in H_0^1(U)$, such that

$$\forall v \in H_0^1(U), B_\mu[u,v] = \langle f|v\rangle.$$

□

**Corollary 3.8.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \ge 0$ be the same as in Energy Estimate 3.5. For any $\mu \ge \gamma$ and $\forall f \in L^2(U)$, there is a unique weak solution $u \in H_0^1(U)$ of the BVP:* $\begin{cases} Lu + \mu u = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$

### 3.2.2 More Existence Theorems

**Definition 3.8.** Consider $Lu := -\sum_{i,j=1}^n \partial_j(a^{ij}(x)\partial_i u) + \sum_{i=1}^n b^i(x)\partial_i u + c(x)u$, we define its **formal adjoint**

$$L^*v := -\sum_{i,j=1}^n \partial_i(a^{ij}(x)\partial_j v) + \sum_{i=1}^n b^i(x)\partial_i v + c(x)v.$$

For $f \in H^{-1}(U)$, the **adjoint problem** is $\begin{cases} L^*v = f & \text{in } U, \\ v = 0 & \text{on } \partial U \end{cases}$, and the bilinear form associated with it is $B^*[u,v]$.
Notice that $v \in H_0^1(U)$ is a weak solution of the adjoint problem if $v$ satisfies $\forall u \in H_0^1(U), B^*[u,v] = \langle f|u\rangle$.

**Proposition 3.9.**

$$B^*[u,v] := B[v,u].$$

*Remark.* Since $L$ is not bounded, $L^*$ is not its usual adjoint operator. However, when $u, v$ are both smooth, we have that $\langle Lu, v\rangle_{L^2(U)} = B[u,v] = B^*[v,u] = \langle v, L^*u\rangle$.

**Definition 3.9.** For $\mu \in \mathbb{R}$, we can similarly define $L_\mu^* u := L^*u + \mu u$, and the bilinear form associated with it is $B_\mu^*[u,v]$.

**Proposition 3.10.**

$$B_\mu^*[u,v] = B^*[u,v] + \mu\langle u,v\rangle_{L^2(U)} = B[v,u] + \mu\langle v,u\rangle_{L^2(U)} = B_\mu[v,u].$$

**Proposition 3.11.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \ge 0$ be the same as in Energy Estimate 3.5. For any $\mu \ge \gamma$ and $\forall f \in L^2(U)$, there is a unique weak solution $u \in H_0^1(U)$ of the BVP:* $\begin{cases} L^*u + \mu u = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$ *Namely,*

$$\exists! u \in H_0^1(U), \text{ such that } \forall v \in H_0^1(U), B_\mu^*[u,v] = \langle f, v\rangle_{L^2(U)}.$$

*Proof.* For $\alpha, \beta > 0, \gamma \geq 0$ from Energy Estimate 3.5, we have that $\forall u, v \in H_0^1(U)$,

$$\begin{aligned} |B^*[v,u]| &= |B[u,v]| && (3)\\ &\leq \alpha \|u\|_{H^1(U)} \|v\|_{H^1(U)} && (4)\\ \beta \|u\|^2_{H^1(U)} &\leq B[u,u] + \gamma \|u\|^2_{L^2(U)} && (5)\\ &= B^*[u,u] + \gamma \|u\|^2_{L^2(U)}. && (6) \end{aligned}$$

Thus $B$ and $B^*$ have the same energy estimate. By First Existence Theorem 3.7, we have the result. □

**Definition 3.10.** Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \geq 0$ be the same as in Energy Estimate 3.5. For any $\mu \geq \gamma$, we define $L_\mu^{-1} : L^2(U) \to H_0^1(U)$ by $f \mapsto u$, where $u$ is the unique solution to

$$\forall v \in H_0^1(U), B_\mu[u,v] = \langle f, v \rangle_{L^2(U)}$$

given by the First Existence Theorem 3.7.
We can also define $(L_\mu^*)^{-1} : L^2(U) \to H_0^1(U)$ by $f \mapsto u$, where $u$ is the unique solution to

$$\forall v \in H_0^1(U), B_\mu^*[u,v] = \langle f, v \rangle_{L^2(U)}.$$

*Remark.* We notice that by definition $B_\mu[L_\mu^{-1} f, v] = \langle f, v \rangle_{L^2(U)}, \forall v \in H_0^1(U), \forall f \in L^2(U), \forall \mu \geq \gamma$.

**Lemma 3.12.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \geq 0$ be the same as in Energy Estimate 3.5. Then for any $\mu \geq \gamma$, if we let $K = \mu L_\mu^{-1}$, we have that $K : L^2(U) \to H_0^1(U) \subseteq L^2(U)$ is compact.*

*Proof.* Consider any $g \in L^2(U)$, we have that

$$\begin{aligned} \beta \left\| L_\mu^{-1} g \right\|^2_{H^1(U)} &\leq B[L_\mu^{-1} g, L_\mu^{-1} g] + \gamma \left\| L_\mu^{-1} g \right\|^2_{L^2(U)}\\ &\leq B[L_\mu^{-1} g, L_\mu^{-1} g] + \mu \left\| L_\mu^{-1} g \right\|^2_{L^2(U)}\\ &= B_\mu[L_\mu^{-1} g, L_\mu^{-1} g]\\ &= \left\langle g, L_\mu^{-1} g \right\rangle_{L^2(U)}\\ &\leq \|g\|_{L^2(U)} \left\| L_\mu^{-1} g \right\|_{L^2(U)}\\ &\leq \|g\|_{L^2(U)} \left\| L_\mu^{-1} g \right\|_{H^1(U)}\\ &\implies\\ \left\| L_\mu^{-1} g \right\|_{H^1(U)} &\leq \frac{1}{\beta} \|g\|_{L^2(U)}\\ &\implies\\ \|Kg\|_{H^1(U)} &\leq \frac{\mu}{\beta} \|g\|_{L^2(U)}. \end{aligned}$$

Thus, $K : L^2(U) \to H_0^1(U)$ is bounded.
Since $H_0^1(U) \subset\subset L^2(U)$, by 1.35, we have that $K : L^2(U) \to L^2(U)$ is compact. □

**Lemma 3.13.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \geq 0$ be the same as in Energy Estimate 3.5. For any $f \in L^2(U)$, if we let $h := L_\gamma^{-1} f, K = \gamma L_\gamma^{-1}$, we have that $u \in H_0^1(U)$ is a weak solution to $\begin{cases} Lu = f & \text{in } U,\\ u = 0 & \text{on } \partial U \end{cases}$ if and only if $u$ solves $(I - K)u = h$.*

*Proof.* We will firstly show that $u$ solves $\forall v \in H_0^1(U),\ B_\gamma[u,v] = \langle f+\gamma u, v\rangle_{L^2(U)}$, if and only if $u$ solves $u = L_\gamma^{-1}(f+\gamma u)$.
Suppose $\forall v \in H_0^1(U),\ B_\gamma[u,v] = \langle f+\gamma u, v\rangle_{L^2(U)}$.
we have that $u' := L_\gamma^{-1}(f+\gamma u) \in H_0^1(U)$ is the unique solution, such that

$$B_\gamma[u',v] = \langle f+\gamma u, v\rangle,\ \forall v \in H_0^1(U).$$

Thus $u = u' = L_\gamma^{-1}(f+\gamma u)$.
On the other hand, suppose $u = L_\gamma^{-1}(f+\gamma u)$, then we have that

$$\forall v \in H_0^1(U),\ B_\gamma[u,v] = B_\gamma[L_\gamma^{-1}(f+\gamma u), v] = \langle f+\gamma u, v\rangle_{L^2(U)}.$$

Thus, $u \in H_0^1(U)$ is a weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$, if and only if
$u$ solves $\forall v \in H_0^1(U),\ B[u,v] = \langle f, v\rangle_{L^2(U)}$, if and only if
$u$ solves $\forall v \in H_0^1(U),\ B[u,v] + \gamma\langle u, v\rangle_{L^2(U)} = \langle f, v\rangle_{L^2(U)} + \gamma\langle u, v\rangle_{L^2(U)}$, if and only if
$u$ solves $\forall v \in H_0^1(U),\ B_\gamma[u,v] = \langle f+\gamma u, v\rangle_{L^2(U)}$, if and only if
$u$ solves $u = L_\gamma^{-1}(f+\gamma u)$, if and only if
$u$ solves $u = L_\gamma^{-1}f + \gamma L_\gamma^{-1}u$, if and only if
$u$ solves $Iu = h + Ku$, if and only if
$u$ solves $(I-K)u = h$. □

**Theorem 3.14.** *(Second Existence Theorem)*
*Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator.*

1. *Precisely one of the following must be true:*

   *(a) $\forall f \in L^2(U), \exists! u \in H_0^1(U)$, a unique weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$*

   *(b) There is a weak solution $u \neq 0 \in H_0^1(U)$ to the homogeneous problem $\begin{cases} Lu = 0, & \text{in } U, \\ u = 0, & \text{on } \partial U. \end{cases}$*

2. *Let $N \subset H_0^1(U)$ be the solution space of weak solutions to $\begin{cases} Lu = 0, & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$, and let $N^* \subset H_0^1(U)$ be the solution space of weak solutions to $\begin{cases} L^*u = 0, & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$, then $\dim(N) = \dim(N^*) < \infty$.*

3. *The problem $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$ has a weak solution if and only if $f \in (N^*)^\perp \subseteq L^2(U)$.*

*Proof.* Take $\mu = \gamma$.
From the above lemma, we know that for any $f \in L^2(U)$, if we let $K = \gamma L_\gamma^{-1}$, we have that $u \in H_0^1(U)$ is a weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$ if and only if $u$ solves $(I-K)u = L_\gamma^{-1}f$.
We also have shown that $K : L^2(U) \to H_0^1(U) \subseteq L^2(U)$ is compact.

1. By 1.37, we have that exactly one of the following holds:

   (a) $\forall v \in L^2(U),\ \exists! u \in L^2(U),$ such that $(I-K)u = v$.
   In this case, for any $f \in L^2(U),\ \exists! u \in L^2(U),$ such that $(I-K)u = L_\gamma^{-1}f$.
   In addition, since $L_\gamma^{-1}f \in H_0^1(U), Ku = \gamma L_\gamma^{-1}u \in H_0^1(U)$, we must have $u = L_\gamma^{-1}f + Ku \in H_0^1(U)$.
   Thus $u$ is the unique weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$.

(b) $\exists u \neq 0 \in L^2(U)$, such that $(I-K)u = 0 = L_\gamma^{-1}0$.
Similarly, we can see that $u = Ku = \gamma L_\gamma^{-1}u \in H_0^1(U)$.
Thus $u$ is a non-trivial solution to $\begin{cases} Lu = 0 & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$.

2. By the above lemma, $N = \operatorname{Ker}(I-K)$.
By 1.37, we have that $\dim(N) = \dim(\operatorname{Ker}(I-K^*)) < \infty$. Let $L_\gamma^* u := L^* u + \gamma u$.
Consider any $g, h \in L^2(U)$, we have that

$$\begin{aligned} \langle h, K^* g \rangle &= \langle Kh, g \rangle \\ &= \langle g, Kh \rangle_{L^2(U)} \\ &= \gamma \langle g, L_\gamma^{-1} h \rangle_{L^2(U)} \\ &= \gamma B_\gamma^*[(L_\gamma^*)^{-1} g, L_\gamma^{-1} h] \\ &= \gamma B_\gamma[L_\gamma^{-1} h, (L_\gamma^*)^{-1} g] \\ &= \gamma \langle h, (L_\gamma^*)^{-1} g \rangle_{L^2(U)} \\ &= \langle h, \gamma (L_\gamma^*)^{-1} g \rangle_{L^2(U)}. \end{aligned}$$

Since this holds for all $g, h \in L^2(U)$, we have that $K^* = \gamma (L_\gamma^*)^{-1}$.
By the above lemma, we have that $u \in H_0^1(U)$ is a weak solution to $\begin{cases} L^* u = 0 & \text{in } U, \\ u = 0 & \text{on } \partial U \end{cases}$, if and only if $u$ solves $(I-K^*)u = 0$, if and only if $u \in \operatorname{Ker}(I-K^*)$.
Thus $N^* = \operatorname{Ker}(I-K^*)$.

3. (a) $\gamma = 0$.
Notice that $K = 0$, and thus $N^* = \ker(I-K^*) = \ker(I) = \{0\}$.
Thus $(N^*)^\perp = L^2(U)$.
In addition, $N = \ker(I-K) = \ker(I) = \{0\}$, so we must be in case $(a)$.
Thus $\forall f \in (N^*)^\perp$, the problem $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$ has a (unique) weak solution.
The other direction is trivial since $(N^*)^\perp = L^2(U)$ is the whole space.

(b) $\gamma \neq 0$.
By 1.37, we have that $\operatorname{Im}(I-K) = \operatorname{Ker}(I-K^*)^\perp$.
By the above lemma, we have that the problem $\begin{cases} Lu = f & \text{in } U, \\ u = 0 & \text{on } \partial U. \end{cases}$ has a weak solution, if and only if,
there is some $u$ that solves $(I-K)u = L_\gamma^{-1} f$, if and only if,
$L_\gamma^{-1} f \in \operatorname{Im}(I-K) = \operatorname{Ker}(I-K^*)^\perp$, if and only if,
$\forall v \in \operatorname{Ker}(I-K^*) = N^*$,

$$\begin{aligned} \langle L_\gamma^{-1} f, v \rangle &= 0 \\ \frac{1}{\gamma} \langle Kf, v \rangle &= 0 \\ \frac{1}{\gamma} \langle f, K^* v \rangle &= 0 \\ \frac{1}{\gamma} \langle f, K^* v + (I-K^*)v \rangle &= 0 \\ \frac{1}{\gamma} \langle f, v \rangle &= 0 \\ \langle f, v \rangle &= 0, \end{aligned}$$

if and only if $f \in (N^*)^\perp$.

□

**Definition 3.11.** Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. The **spectrum** of $L$ is defined to be

$$\Sigma := \mathbb{R} \setminus \left\{ \lambda \in \mathbb{R} : \forall f \in L^2(U), \exists! u \in H_0^1(U), \text{ such that } \forall v \in H_0^1(U), B_{-\lambda}[u,v] = \langle f, v \rangle_{L^2(U)} \right\}.$$

**Proposition 3.15.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\Sigma$ be the spectrum of $L$.*

1. *$\lambda \notin \Sigma$ if and only if $\begin{cases} Lu = \lambda u + f & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$ has a unique weak solution $u \in H_0^1(U)$ for each $f \in L^2(U)$.*
2. *$\lambda \in \Sigma$ if and only if $\begin{cases} Lu = \lambda u & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$ has a non-trivial weak solution $u \neq 0 \in H_0^1(U)$.*

*Proof.* 1. This is by definition.

2. By Second Existence Theorem 3.14 on $L_{-\lambda}$.

□

**Lemma 3.16.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\gamma \geq 0$ be the same as in Energy Estimate 3.5, and $\Sigma$ be the spectrum of $L$, we always have $\Sigma \subseteq (-\gamma, \infty)$.*

*Proof.* If $\lambda \leq -\gamma$, we have that $-\lambda \geq \gamma$, and by First Existence Theorem 3.7, we have that the problem has a unique weak solution, and thus $\lambda \notin \Sigma$. □

**Theorem 3.17.** *(Third existence theorem)*
*Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator. Let $\Sigma$ be the spectrum of $L$.*

1. *$\Sigma$ is at most countable.*
2. *If $\Sigma$ is infinite, then $\Sigma = \{\lambda_k\}_{k=1}^\infty$ can be arranged in non-decreasing sequence with $\lim_{k\to\infty} \lambda_k = \infty$.*

*Proof.* Let $\gamma' \geq 0$ be the same as in Energy Estimate 3.5, we have $\Sigma \subseteq (-\gamma', \infty) \subseteq (-\gamma, \infty)$ for any $\gamma \geq \gamma'$. We will take some $\gamma > 0$, and consider $\lambda > -\gamma$.

$\lambda \in \Sigma$, if and only if $\begin{cases} Lu = \lambda u & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$ has a non-trivial weak solution $u \neq 0 \in H_0^1(U)$,

if and only if $\begin{cases} Lu + \gamma u = (\lambda + \gamma)u, & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$ has a non-trivial weak solution $u \neq 0 \in H_0^1(U)$.

Suppose $\lambda \in \Sigma$, then let $g = (\lambda + \gamma)u$. By First Existence Theorem 3.7, there is a unique weak solution $(L_\gamma)^{-1}((\lambda + \gamma)u) = \frac{\lambda+\gamma}{\gamma} Ku$ to the problem $\begin{cases} Lu + \gamma u = g, & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$.

Since $u \neq 0 \in H_0^1(U)$ is a weak solution to the problem, we have

$$u = \frac{\lambda + \gamma}{\gamma} Ku.$$

Thus $u \neq 0 \in L^2(U)$ is an eigen-vector for $K$, with corresponding eigenvalue $\frac{\gamma}{\lambda+\gamma}$.
Notice that $\frac{\gamma}{\lambda+\gamma} > 0$, since $\gamma > 0, \lambda > -\gamma$, and thus $\frac{\gamma}{\lambda+\gamma} \in \text{Spec}_p(K) \setminus (\infty, 0]$.
Since this holds for any $\lambda \in \Sigma$, we have $\left\{ \frac{\gamma}{\lambda+\gamma} : \lambda \in \Sigma \right\} \subseteq \text{Spec}_p(K) \setminus (\infty, 0]$.

On the other hand, $\forall \mu \in \mathrm{Spec}_p(K) \setminus \{0\}$, we have that $\lambda' := \frac{\gamma(1-\mu)}{\mu} = -\gamma + \frac{\gamma}{\mu}$ satisfies $\mu = \frac{\gamma}{\lambda'+\gamma}$.
Pick any eigen-vector $u \neq 0 \in L^2(U)$ corresponds to $\mu$, we have that $\frac{\gamma}{\lambda+\gamma} u = Ku$.
Thus $u = (L_\gamma)^{-1}((\lambda'+\gamma)u) \neq 0 \in H_0^1(U)$ is a weak solution to the problem $\begin{cases} Lu + \gamma u = (\lambda'+\gamma)u, & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$.
If $\lambda' > -\gamma \iff \frac{\gamma}{\mu} > 0 \iff \mu > 0$, we have that $\lambda' \in \Sigma$.
Thus, we have $\left\{ \frac{\gamma(1-\mu)}{\mu} : \mu \in \mathrm{Spec}_p(K) \setminus (\infty, 0] \right\} \subseteq \Sigma$.
We have shown that

$$\Sigma = \left\{ \frac{\gamma(1-\mu)}{\mu} : \mu \in \mathrm{Spec}_p(K) \setminus (\infty, 0] \right\}.$$

Since $K$ is compact, by the Spectral theorem 1.39, we have that either

1. $\mathrm{Spec}_p(K) \setminus \{0\} = \{\mu_k\}_{k=1}^N$ is finite, which means $\Sigma \subseteq \left(\lambda_k = \frac{\gamma(1-\mu_k)}{\mu_k}\right)_{k=1}^N$ is finite.
2. $\mathrm{Spec}_p(K) \setminus \{0\} = \{\mu_k\}_{k=1}^\infty$ is countable, and $\lim_{k\to\infty} \mu_k = 0$, which means that $\Sigma \subseteq \left(\lambda_k = \frac{\gamma(1-\mu_k)}{\mu_k}\right)_{k=1}^\infty$ is at most countable.
   In addition, if $\Sigma$ is infinite, it must be $(\lambda_{k_j})_{j=1}^\infty \subseteq (\lambda_k)_{k=1}^\infty$.

   $$\lim_{k\to\infty} |\lambda_k| = \lim_{k\to\infty} \left| \frac{\gamma(1-\mu_k)}{\mu_k} \right| = \lim_{k\to\infty} \left| \frac{\gamma}{\mu_k} \right| = \infty.$$

   Thus $\lim_{k\to\infty} |\lambda_{k_j}| = \lim_{k\to\infty} |\lambda_k| = \infty$.
   Since we have $\forall j, \lambda_{k_j} > -\gamma$, we must have $\lim_{j\to\infty} \lambda_{k_j} = \infty$.

□

**Theorem 3.18.** *(Boundedness of inverse)*
*Let $\Sigma$ be the spectrum of $L$, and $\lambda \notin \Sigma$. Then there is a constant $C > 0$ such that for all $f \in L^2(U)$ and the unique weak solution $u \in H_0^1(U)$ to $\begin{cases} Lu = \lambda u + f & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$, we always have*

$$\|u\|_{L^2(U)} \leq C \|f\|_{L^2(U)}.$$

*Proof.* Consider any $\lambda \notin \Sigma$.
Suppose for contradiction, we can find $(\tilde{u}_k)_{k=1}^\infty \subset H_0^1(U), (\tilde{f}_k)_{k=1}^\infty \subset L^2(U)$, such that $\forall k \geq 1$,

$$\begin{cases} L\tilde{u_k} = \lambda \tilde{u_k} + \tilde{f}_k & \text{in } U, \\ \tilde{u_k} = 0, & \text{on } \partial U \end{cases}$$

and

$$\|\bar{u}\|_{L^2(U)} > k \|\bar{f}\|_{L^2(U)}.$$

Let $u_k := \frac{\tilde{u}_k}{\|\tilde{u}_k\|_{L^2(U)}}, f_k := \frac{\tilde{f}_k}{\|\tilde{u}_k\|_{L^2(U)}}$.
Notice that $\forall k \geq 1, \|u_k\|_{L^2(U)} = 1$, and $\|f_k\|_{L^2(U)} = \frac{\|\tilde{f}_k\|_{L^2(U)}}{\|\tilde{u}_k\|_{L^2(U)}} < \frac{1}{k}$.
In addition, $\forall v \in H_0^1(U)$,

$$\begin{aligned} B[u_k, v] &= \frac{1}{\|\tilde{u}_k\|_{L^2(U)}} B[\tilde{u}_k, v] \\ &= \frac{1}{\|\tilde{u}_k\|_{L^2(U)}} \left( \left\langle \tilde{f}_k, v \right\rangle_{L^2(U)} + \lambda \langle \tilde{u}_k, v \rangle_{L^2(U)} \right) \\ &= \left\langle \frac{\tilde{f}_k}{\|\tilde{u}_k\|_{L^2(U)}}, v \right\rangle_{L^2(U)} + \lambda \left\langle \frac{\tilde{u}_k}{\|\tilde{u}_k\|_{L^2(U)}}, v \right\rangle_{L^2(U)} \\ &= \langle f_k, v \rangle_{L^2(U)} + \lambda \langle u_k, v \rangle_{L^2(U)}. \end{aligned}$$

By Energy Estimate 3.5, we have that

$$\begin{aligned}
\beta\|u_k\|^2_{H^1(U)} &\le B[u_k,u_k]+\gamma\|u_k\|^2_{L^2(U)}\\
&= \langle f_k,u_k\rangle_{L^2(U)}+\lambda\langle u_k,u_k\rangle_{L^2(U)}+\gamma\|u_k\|^2_{L^2(U)}\\
&\le \|f_k\|_{L^2(U)}\|u_k\|_{L^2(U)}+(\lambda+\gamma)\|u_k\|^2_{L^2(U)}\\
&= \|f_k\|_{L^2(U)}+\lambda+\gamma\\
&< \lambda+\gamma+\frac{1}{k}\\
&\le \lambda+\gamma+1.\\
\|u_k\|_{H^1(U)} &\le \sqrt{\frac{\lambda+\gamma+1}{\beta}}
\end{aligned}$$

Thus $(u_k)_{k=1}^\infty$ is a bounded sequence in $H_0^1(U)$.
Since $H_0^1(U)$ is a Hilbert space, and thus reflexive, by 1.28, there $\exists (u_{k_j})_{j=1}^\infty$ a subsequence, and $u\in H_0^1(U)$, such that $u_{k_j}\rightharpoonup u$.
Also, since $H_0^1(U)\subset\subset L^2(U)$, by 1.31, we have that $u_{k_j}\to u$ in $L^2(U)$. Thus,

$$\|u\|_{L^2(U)}=\lim_{j\to\infty}\|u_{k_j}\|_{L^2(U)}=1.$$

Now consider any $v\in H_0^1(U)$, we have that the map $w\mapsto B[w,v]$ is a linear bounded operator, so by weak convergence of $(u_{k_j})_{j=1}^\infty$, we have that

$$\begin{aligned}
B[u,v] &= \lim_{j\to\infty} B[u_{k_j},v]\\
&= \lim_{j\to\infty}\Big(\big\langle f_{k_j},v\big\rangle_{L^2(U)}+\lambda\big\langle u_{k_j},v\big\rangle_{L^2(U)}\Big)\\
&= \lim_{j\to\infty}\big\langle f_{k_j},v\big\rangle_{L^2(U)}+\lambda\Big\langle\lim_{j\to\infty}u_{k_j},v\Big\rangle_{L^2(U)}\\
&\le \lim_{j\to\infty}\|f_{k_j}\|_{L^2(U)}\|v\|_{L^2(U)}+\lambda\langle u,v\rangle_{L^2(U)}\\
&\le \lim_{j\to\infty}\frac{1}{k_j}\|v\|_{L^2(U)}+\lambda\langle u,v\rangle_{L^2(U)}\\
&= \lambda\langle u,v\rangle_{L^2(U)}.
\end{aligned}$$

Namely, $\hat{u}=u$ satisfies $\forall v\in H_0^1(U),\ B_{-\lambda}[\hat{u},v]=0=\langle 0,v\rangle_{L^2(0)}$.
Yet since $\lambda\notin\Sigma$, by definition, we know there is a unique $\hat{u}$ that satisfies the above condition.
Clearly $\hat{u}=0$ satisfies, so by the uniqueness of weak solution, $u=0$.
This contradicts with $\|u\|_{L^2(U)}=1$. □

## 3.3 Regularity

**Theorem 3.19.** *(Interior $H^2$ regularity)*
*Let $U\subseteq\mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}\in C^1(U), b^i,c\in L^\infty(U),\forall i,j\in[n]$. $\forall V\subset\subset U,\exists C>0$, such that for all $f\in L^2(U)$, and $u\in H^1(U)$ being a weak solution to $Lu=f$ in $U$, namely,*

$$\forall v\in H_0^1(U), B[u,v]=\langle f,v\rangle_{L^2(U)},$$

*then*

$$\|u\|_{H^2(V)}\le C\Big(\|f\|_{L^2(U)}+\|u\|_{L^2(U)}\Big).$$

*Thus $u\in H^2_{loc}(U)$.*

*Proof.* Let $V \subset\subset U$ be given.
The idea is to choose a particular $v$, then repeatedly bound all $\left\|D_k^h u\right\|$ from the product rule by $\|Du\|$. The only leftover term will be either $D_k^h(Du)$, or part of $\langle f, v\rangle_{L^2(U)}$ or $\|u\|_H^1(U)$. We thus achieve a bound on $\left\|D_k^h(Du)\right\|$, which allows us to say $u \in H^2_{loc}(U)$.

1. We now fix some $f \in L^2(U)$, and $u \in H^1(U)$ being a weak solution to $Lu = f$ in $U$. We have $\forall v \in H_0^1(U)$,

$$\begin{aligned} B[u,v] &= \langle f, v\rangle_{L^2(U)} \\ \int_U \left(\sum_{i,j=1}^n a^{ij}\partial_i u \partial_j v + \sum_{i=1}^n b^i \partial_i u v + cuv\right) dx &= \int_U fv dx \\ \int_U \left(\sum_{i,j=1}^n a^{ij}\partial_i u \partial_j v\right) dx &= \int_U \tilde{f} v dx \\ \sum_{i,j=1}^n \int_U \left(a^{ij}\partial_i u \partial_j v\right) dx &= \left\langle \tilde{f}, v\right\rangle_{L^2(U)}, \end{aligned}$$

   where $\tilde{f} := f - \sum_{i=1}^n b^i \partial_i u - cu \in L^2(U)$, since $f, \partial_i u, u \in L^2(U), b^i, c \in L^\infty(U)$.

2. Since $V \subset\subset U$, we can choose some $V \subset\subset W \subset\subset U$, and $\zeta \in C_c^\infty(U)$ such that $V < \zeta < W$.
   Choose $|h| > 0$ such that $\operatorname{dist}(V, \partial U) > 8|h|, \operatorname{dist}(W, \partial U) > 6|h|$.
   WLOG, we assume $h > 0$.
   Fix some $k \in [n]$.
   Let $Z := U_{2h} := \{x \in U : \operatorname{dist}(x, \partial U) > 2h\}$ be open.
   Since $U$ is bounded, we have that $Z \subset\subset U, \operatorname{dist}(Z, \partial U) = 2h > |h|$.
   Let $v(x) := \begin{cases} -D_k^{-h}(\zeta^2 D_k^h u)(x) & x \in Z \\ 0 & x \in U \setminus Z \end{cases}$.

   *Remark.* For $x \in V$, we have that

$$\begin{aligned} v(x) &= -D_k^{-h}(D_k^h u)(x) \\ &= -D_k^{-h}\left(\frac{u(x+he_k) - u(x)}{h}\right) \\ &= -\frac{\frac{u(x+he_k-he_k)-u(x-he_k)}{h} - \frac{u(x+he_k)-u(x)}{h}}{h} \\ &= -\frac{2u(x) - u(x+he_k) - u(x-he_k)}{h^2} \\ &= \frac{u(x+he_k) - 2u(x) + u(x-he_k)}{h^2}, \end{aligned}$$

   which is an approximation to $\partial_k^2 u$ if $u$ is smooth.

   Since $u \in H^1(U)$, we have $D_k^h u \in H^1(Z)$.
   Since $\operatorname{Supp}(\zeta) \subset W \subset\subset Z$ is compact, we have $\zeta \in C_c^\infty(Z)$, so $\zeta^2 D_k^h u \in H^1(Z)$.
   Since $U_{4h} \subset\subset Z, \operatorname{dist}(U_{4h}, \partial Z) = 2h > |h|$, we have that $v \in H^1(U_{4h})$.
   In addition, $\operatorname{Supp}(v) \subset \operatorname{Supp}(\zeta^2 D_k^h u) + \bar{B}(0,h) \subseteq W + \bar{B}(0,h) \subseteq U_{6h} + \bar{B}(0,h) \subset U_{4h}$.
   Since $v \in H^1(U_{4h})$ and $\operatorname{Supp}(v) \subset U_{4h}$, we must have $v \in H_0^1(U)$.

3. Now we have

$$\begin{aligned}
\sum_{i,j=1}^{n}\int_U \big(a^{ij}\partial_i u\partial_j v\big)dx &= \sum_{i,j=1}^{n}\int_Z \big(a^{ij}\partial_i u\partial_j v\big)dx \\
&= sum_{i,j=1}^{n}\int_Z \big(a^{ij}\partial_i u\partial_j\big(-D_k^{-h}(\zeta^2 D_k^h u)\big)\big)dx \\
&= -\sum_{i,j=1}^{n}\int_Z \big(a^{ij}\partial_i u D_k^{-h}\big(\partial_j(\zeta^2 D_k^h u)\big)\big)dx \\
&= \sum_{i,j=1}^{n}\int_Z D_k^h(a^{ij}\partial_i u)\big(\partial_j(\zeta^2 D_k^h u)\big)dx \\
&= \sum_{i,j=1}^{n}\int_Z \big(D_k^h(a^{ij})\partial_i u + a^{ij}D_k^h(\partial_i u)\big)\big(\partial_j(\zeta^2)D_k^h u + \zeta^2\partial_j(D_k^h u)\big)\big)dx \\
&= A_1 + A_2 + A_3 + A_4,
\end{aligned}$$

where

$$\begin{aligned}
A_1 &:= \sum_{i,j=1}^{n}\int_Z a^{ij}D_k^h(\partial_i u)\zeta^2\partial_j(D_k^h u)dx, \\
A_2 &:= \sum_{i,j=1}^{n}\int_Z a^{ij}D_k^h(\partial_i u)\partial_j(\zeta^2)D_k^h u dx \\
&= \sum_{i,j=1}^{n}\int_Z a^{ij}D_k^h(\partial_i u)2\zeta(\partial_j\zeta)D_k^h u dx \\
A_3 &:= \sum_{i,j=1}^{n}\int_Z D_k^h(a^{ij})\partial_i u\zeta^2\partial_j(D_k^h u)dx \\
A_4 &:= \sum_{i,j=1}^{n}\int_Z D_k^h(a^{ij})\partial_i u\partial_j(\zeta^2)D_k^h u dx \\
&= \sum_{i,j=1}^{n}\int_Z D_k^h(a^{ij})\partial_i u 2\zeta(\partial_j\zeta)D_k^h u dx
\end{aligned}$$

Now we will examine each term.

$$\begin{aligned}
A_1 &= \sum_{i,j=1}^{n}\int_Z a^{ij}D_k^h(\partial_i u)\zeta^2\partial_j(D_k^h u)dx \\
&= \int_Z \zeta^2\sum_{i,j=1}^{n} a^{ij}\partial_i(D_k^h u)\partial_j(D_k^h u)dx \\
&\geq \int_Z \zeta^2\theta\big\|D(D_k^h u)\big\|_2^2 dx \\
&= \theta\int_Z \zeta^2\big\|D(D_k^h u)\big\|_2^2 dx.
\end{aligned}$$

We also have

$$
\begin{aligned}
|A_2| &\le \sum_{i,j=1}^{n} \int_Z \left|a^{ij} D_k^h(\partial_i u) 2\zeta(\partial_j \zeta) D_k^h u\right| dx \\
&\le \sum_{i,j=1}^{n} \int_Z \left\|a^{ij}\right\|_{L^\infty(U)} \|\partial_j \zeta\|_{L^\infty(U)} \left|D_k^h(\partial_i u) 2\zeta D_k^h u\right| dx \\
&= 2 \sum_{i,j=1}^{n} \left\|a^{ij}\right\|_{L^\infty(U)} \|\partial_j \zeta\|_{L^\infty(U)} \int_Z \left|D_k^h(\partial_i u)\zeta D_k^h u\right| dx \\
&\le 2 \sum_{i,j=1}^{n} \left\|a^{ij}\right\|_{L^\infty(U)} \|\partial_j \zeta\|_{L^\infty(U)} \int_Z \epsilon \left|D_k^h(\partial_i u)\right|^2 \zeta^2 + \frac{1}{4\epsilon} \left|D_k^h u\right|^2 dx \\
&= C_1 \int_Z \epsilon \left|D_k^h(\partial_i u)\right|^2 \zeta^2 + \frac{1}{4\epsilon} \left|D_k^h u\right|^2 dx \\
&= C_1 \int_Z \epsilon \left|\partial_i (D_k^h u)\right|^2 \zeta^2 + \frac{1}{4\epsilon} \left|D_k^h u\right|^2 dx \\
&\le C_1 \int_Z \epsilon \left\|D(D_k^h u)\right\|_2^2 \zeta^2 + \frac{1}{4\epsilon} \left|D_k^h u\right|^2 dx,
\end{aligned}
$$

since $a^{ij} \in L^\infty(U)$, and $\zeta \in C^c(U)$, we have $C_1 := 2\sum_{i,j=1}^{n} \left\|a^{ij}\right\|_{L^\infty(U)} \|\partial_j \zeta\|_{L^\infty(U)} \in (0, \infty)$. Similarly,

$$
\begin{aligned}
|A_3| &\le \sum_{i,j=1}^{n} \int_Z \left|D_k^h(a^{ij}) \partial_i u \zeta^2 \partial_j (D_k^h u)\right| dx \\
&\le \sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \int_Z \left|\partial_i u \partial_j (D_k^h u)\right| \zeta^2 dx \\
&\le \sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \int_Z \|Du\|_2 \left\|D(D_k^h u)\right\|_2 \zeta^2 dx \\
&\le \sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \int_Z \|Du\|_2 \left\|D(D_k^h u)\right\|_2 \zeta dx \\
&\le \sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \int_Z \frac{1}{4\epsilon} \|Du\|_2^2 + \epsilon\zeta^2 \left\|D(D_k^h u)\right\|_2^2 dx \\
&= C_2 \int_Z \frac{1}{4\epsilon} \|Du\|_2^2 + \epsilon\zeta^2 \left\|D(D_k^h u)\right\|_2^2 dx,
\end{aligned}
$$

where

$$
\begin{aligned}
C_2 &:= \sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \\
&\le \frac{1}{h} \sum_{i,j=1}^{n} \left(\left\|a^{ij}\right\|_{L^\infty(Z)} + \left\|a^{ij}\right\|_{L^\infty(Z+he_k)}\right) \\
&\le \frac{1}{h} \sum_{i,j=1}^{n} \left(\left\|a^{ij}\right\|_{L^\infty(U)} + \left\|a^{ij}\right\|_{L^\infty(U)}\right) \\
&\in (0, \infty).
\end{aligned}
$$

Lastly,

$$\begin{aligned}
|A_4| &\le \sum_{i,j=1}^{n} \int_Z \left|D_k^h(a^{ij})\partial_i u 2\zeta(\partial_j\zeta)D_k^h u\right| dx \\
&\le 2\sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \|\partial_j\zeta\|_{L^\infty(U)} \int_Z \left|\partial_i u D_k^h u\right| dx \\
&\le \sum_{i,j=1}^{n} \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \|\partial_j\zeta\|_{L^\infty(U)} \int_Z |\partial_i u|^2 + \left|D_k^h u\right|^2 dx \\
&\le C_3 \int_Z \|Du\|_2^2 + \left|D_k^h u\right|^2 dx,
\end{aligned}$$

where $C_3 := \sum_{i,j=1}^n \left\|D_k^h(a^{ij})\right\|_{L^\infty(Z)} \|\partial_j\zeta\|_{L^\infty(U)} \in (0,\infty)y$ as argued before.
Now

$$\begin{aligned}
&|A_2 + A_3 + A_4| \\
&\le |A_1| + |A_2| + |A_3| \\
&\le \int_Z \epsilon C_1 \left\|D(D_k^h u)\right\|_2^2 \zeta^2 + \frac{C_1}{4\epsilon}\left|D_k^h u\right|^2 + \frac{C_2}{4\epsilon}\|Du\|_2^2 + C_2\epsilon\zeta^2\left\|D(D_k^h u)\right\|_2^2 + C_3\|Du\|_2^2 + C_3\left|D_k^h u\right|^2 dx \\
&= \int_Z (C_1+C_2)\epsilon\left\|D(D_k^h u)\right\|_2^2\zeta^2 + \left(\frac{C_1}{4\epsilon}+C_3\right)\left|D_k^h u\right|^2 + \left(\frac{C_2}{4\epsilon}+C_3\right)\|Du\|_2^2 dx \\
&\le \int_Z (C_1+C_2)\epsilon\left\|D(D_k^h u)\right\|_2^2\zeta^2 + \left(\frac{C_1}{4\epsilon}+C_3\right)\left\|D^h u\right\|_2^2 + \left(\frac{C_2}{4\epsilon}+C_3\right)\|Du\|_2^2 dx \\
&= (C_1+C_2)\epsilon\int_Z \left\|D(D_k^h u)\right\|_2^2\zeta^2 dx + \left(\frac{C_1}{4\epsilon}+C_3\right)\left\|D^h u\right\|_{L^2(U)}^2 + \left(\frac{C_2}{4\epsilon}+C_3\right)\|Du\|_{L^2(U)}^2.
\end{aligned}$$

We know there $\exists C_4 > 0$, such that

$$\left\|D^h u\right\|_{L^2(Z)} \le C_4\|Du\|_{L^2(U)}, \forall |h| \in (0, \operatorname{dist}(Z,\partial U)), \forall u \in H_0^1(U).$$

Thus

$$|A_2 + A_3 + A_4| \le (C_1+C_2)\epsilon\int_Z \left\|D(D_k^h u)\right\|_2^2\zeta^2 dx + \left(\frac{C_2}{4\epsilon} + C_3 + \left(\frac{C_1}{4\epsilon}+C_3\right)C_4^2\right)\|Du\|_{L^2(U)}^2.$$

Taking $\epsilon := \frac{\theta}{2(C_1+C_2)}, C_5(\epsilon) := \frac{C_2}{4\epsilon} + C_3 + \left(\frac{C_1}{4\epsilon}+C_3\right)C_4^2 \in (0,\infty)$, we have

$$\begin{aligned}
\sum_{i,j=1}^{n}\int_U \left(a^{ij}\partial_i u\partial_j v\right)dx &= A_1 + A_2 + A_3 + A_4 \\
&\ge A_1 - |A_2 + A_3 + A_4| \\
&\ge \theta\int_Z \zeta^2\left\|D(D_k^h u)\right\|_2^2 dx - \frac{\theta}{2}\int_Z \left\|D(D_k^h u)\right\|_2^2\zeta^2 dx - C_5\|Du\|_{L^2(U)}^2 \\
&= \frac{\theta}{2}\int_Z \left\|D(D_k^h u)\right\|_2^2\zeta^2 dx - C_5\|Du\|_{L^2(U)}^2.
\end{aligned}$$

4. On the other hand,

$$\begin{aligned}
\left|\left\langle \tilde{f}, v\right\rangle_{L^2(U)}\right| &= \int_U \left|f - \sum_{i=1}^n b^i \partial_i u - cu\right| |v| dx \\
&= \int_U \left(|f| + \sum_{i=1}^n \left|b^i \partial_i u\right| + |cu|\right) |v| dx \\
&\leq \int_U \left(|f| + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} |\partial_i u| + \|c\|_{L^\infty(U)} |u|\right) |v| dx \\
&= \int_U |f||v| dx + \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)} \int_U |\partial_i u||v| dx + \|c\|_{L^\infty(U)} \int_U |u||v| dx \\
&\leq C_6 \left(\int_U |f||v| dx + \int_U |\partial_i u||v| dx + \int_U |u||v| dx\right) \\
&\leq C_6 \left(\int_U \frac{1}{4\epsilon}|f|^2 + \epsilon |v|^2 dx + \int_U \frac{1}{4\epsilon}|\partial_i u|^2 + \epsilon |v|^2 dx + \int_U \frac{1}{4\epsilon}|u|^2 + \epsilon |v|^2 dx\right) \\
&\leq C_6 \int_U \frac{1}{4\epsilon}\left(|f|^2 + |\partial_i u|^2 + |u|^2\right) + 3\epsilon |v|^2 dx \\
&\leq \frac{C_6}{4\epsilon} \int_U |f|^2 + \|Du\|_2^2 + |u|^2 dx + 3C_6 \epsilon \int_U |v|^2 dx,
\end{aligned}$$

where $C_6 := \max\left(1, \sum_{i=1}^n \left\|b^i\right\|_{L^\infty(U)}, \|c\|_{L^\infty(U)}\right) \in (0, \infty)$.
We have shown in step 2 that $\zeta^2 D_k^h u \in H^1(Z), \mathrm{Supp}(\zeta^2 D_k^h u) \subset Z \subset\subset U$, thus $\zeta^2 D_k^h u \in H_0^1(U)$.

$$\begin{aligned}
\int_U |v|^2 dx &= \int_Z |v|^2 dx \\
&= \int_Z \left|-D_k^{-h}(\zeta^2 D_k^h u)\right|^2 dx \\
&\leq \int_Z \left|D^{-h}(\zeta^2 D_k^h u)\right|^2 dx \\
&\leq C_4^2 \int_U \left|D(\zeta^2 D_k^h u)\right|^2 dx \\
&= C_4^2 \int_W \left|D(\zeta^2 D_k^h u)\right|^2 dx \\
&= C_4^2 \int_W \left|D(\zeta^2) D_k^h u + D(D_k^h u)\zeta^2\right|^2 dx \\
&\leq 2C_4^2 \int_W D(\zeta^2)^2 \left|D_k^h u\right|^2 + \left|D(D_k^h u)\right|^2 \zeta^4 dx \\
&\leq 2C_4^2 \int_W \left\|D(\zeta^2)^2\right\|_{L^\infty(U)} \left|D_k^h u\right|^2 + \left|D(D_k^h u)\right|^2 \zeta^2 dx \\
&\leq 2C_4^2 \left\|D(\zeta^2)^2\right\|_{L^\infty(U)} \int_W \left|D_k^h u\right|^2 dx + 2C_4^2 \int_W \left|D(D_k^h u)\right|^2 \zeta^2 dx \\
&\leq 2C_4^4 \left\|D(\zeta^2)^2\right\|_{L^\infty(U)} \int_U \|Du\|_2^2 dx + 2C_4^2 \int_U \left|D(D_k^h u)\right|^2 \zeta^2 dx \\
&\leq C_7 \int_U \|Du\|_2^2 + \left|D(D_k^h u)\right|^2 \zeta^2 dx,
\end{aligned}$$

where $C_7 := 2C_4^2 \max\Big(C_4^2 \big\|D(\zeta^2)^2\big\|_{L^\infty(U)}, 1\Big) \in (0,\infty)$. Thus we have

$$\begin{aligned}
\left|\left\langle \tilde{f}, v\right\rangle_{L^2(U)}\right| &\le \frac{C_6}{4\epsilon}\int_U |f|^2 + \|Du\|_2^2 + |u|^2 dx + 3C_6\epsilon\int_U |v|^2 dx \\
&\le \frac{C_6}{4\epsilon}\int_U |f|^2 + \|Du\|_2^2 + |u|^2 dx + 3C_6C_7\epsilon\int_U \|Du\|_2^2 + \big|D(D_k^h u)\big|^2\zeta^2 dx \\
&\le \left(\frac{C_6}{4\epsilon} + 3C_6C_7\epsilon\right)\Big(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 + \|Du\|_{L^2(U)}^2\Big) + 3C_6C_7\epsilon\int_U \big|D(D_k^h u)\big|^2\zeta^2 dx.
\end{aligned}$$

5. Taking $\epsilon := \frac{\theta}{12C_6C_7} > 0, C_8 := \frac{C_6}{4\epsilon} + 3C_6C_7\epsilon > 0$, we have

$$\begin{aligned}
\sum_{i,j=1}^n \int_U \big(a^{ij}\partial_i u \partial_j v\big) dx &= \left\langle \tilde{f}, v\right\rangle_{L^2(U)} \\
&\le \left|\left\langle \tilde{f}, v\right\rangle_{L^2(U)}\right| \\
&\le C_8\Big(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 + \|Du\|_{L^2(U)}^2\Big) + \frac{\theta}{4}\int_U \big|D(D_k^h u)\big|^2\zeta^2 dx \\
&= C_8\Big(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 + \|Du\|_{L^2(U)}^2\Big) + \frac{\theta}{4}\int_Z \big|D(D_k^h u)\big|^2\zeta^2 dx \\
\sum_{i,j=1}^n \int_U \big(a^{ij}\partial_i u \partial_j v\big) dx &\ge \frac{\theta}{2}\int_Z \big\|D(D_k^h u)\big\|_2^2\zeta^2 dx - C_5\|Du\|_{L^2(U)}^2 \\
\frac{\theta}{4}\int_Z \big\|D(D_k^h u)\big\|_2^2\zeta^2 dx &\le (C_5 + C_8)\Big(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 + \|Du\|_{L^2(U)}^2\Big) \\
\frac{\theta}{4}\int_V \big\|D(D_k^h u)\big\|_2^2 dx &\le (C_5 + C_8)\Big(\|f\|_{L^2(U)}^2 + \|u\|_{L^2(U)}^2 + \|Du\|_{L^2(U)}^2\Big) \\
\int_V \big\|D_k^h(Du)\big\|_2^2 dx &\le C_9\Big(\|f\|_{L^2(U)}^2 + \|u\|_{H^1(U)}^2\Big),
\end{aligned}$$

where $C_9 := \frac{4(C_5+C_8)}{\theta} \in (0,\infty)$.
Notice that for all $j \in [n]$, we have $\partial_j u \in L^2(U)$, and

$$\int_V \big\|D_k^h(\partial_j u)\big\|_2^2 dx \le \int_V \big\|D_k^h(Du)\big\|_2^2 dx \le C_9\Big(\|f\|_{L^2(U)}^2 + \|u\|_{H^1(U)}^2\Big),$$

and this holds for all $k \in [n]$. Thus,

$$\begin{aligned}
\big\|D^h(\partial_j u)\big\|_{L^2(V)}^2 &= \int_V \big\|D^h(\partial_j u)\big\|_2^2 dx \\
&= \int_V \sum_{k=1}^n \big\|D_k^h(\partial_j u)\big\|_2^2 dx \\
&= \sum_{k=1}^n \int_V \big\|D_k^h(\partial_j u)\big\|_2^2 dx \\
&\le \sum_{k=1}^n C_9\Big(\|f\|_{L^2(U)}^2 + \|u\|_{H^1(U)}^2\Big) \\
&= nC_9\Big(\|f\|_{L^2(U)}^2 + \|u\|_{H^1(U)}^2\Big). \\
\big\|D^h(\partial_j u)\big\|_{L^2(V)} &\le \sqrt{nC_9}\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big) \\
&< \infty.
\end{aligned}$$

Since this holds for all $|h| > 0$ such that $\operatorname{dist}(V, \partial U) > 8|h|, \operatorname{dist}(W, \partial U) > 6|h|$, we have $\partial_j u \in H^1(U)$, with

$$\|D(\partial_j u)\|_{L^2(V)} \leq \sqrt{nC_9}\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big).$$

Since this holds for all $j \in [n]$, we have $u \in H^2(V)$, and

$$\begin{aligned}
\left\|D^2 u\right\|_{L^2(V)}^2 &= \int_V \left\|D^2 u\right\|_2^2 dx \\
&= \int_V \sum_{j=1}^n \|\partial_j (Du)\|_2^2 dx \\
&= \sum_{j=1}^n \int_V \|D(\partial_j u)\|_2^2 dx \\
&\leq \sum_{j=1}^n nC_9\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big)^2 \\
&= n^2 C_9\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big)^2 \\
&\implies \\
\|u\|_{H^2(V)}^2 &= \left\|D^2 u\right\|_{L^2(V)}^2 + \|u\|_{H^1(V)}^2 \\
&\leq n^2 C_9\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big)^2 + \|u\|_{H^1(V)}^2 \\
&\leq (n^2 C_9 + 1)\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big)^2.
\end{aligned}$$

Thus we have found $C := \sqrt{n^2 C_9 + 1} \in (0, \infty)$, such that $\|u\|_{H^2(V)} \leq C\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big)$. Since $V$ is arbitrary, we have that $u \in H^2_{loc}(U)$.

6. Notice that the above estimate holds as long as $V \subset\subset U$ and $u \in H^1(U)$. Since $u \in H^1(W)$, we can find some constant $C'$, such that $\|u\|_{H^2(V)}^2 \leq C'\Big(\|f\|_{L^2(W)} + \|u\|_{H^1(W)}\Big)$.
Now consider $v := \xi^2 u \in H_0^1(U)$, we can find $\|Du\|_{L^2(W)} \leq C''\|u\|_{L^2(U)}$ for some $C'' > 0$.
Plugging in will give us
$$\|u\|_{H^2(V)}^2 \leq C\Big(\|f\|_{L^2(U)} + \|u\|_{L^2(U)}\Big).$$

□

**Corollary 3.20.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij} \in C^1(U), b^i, c \in L^\infty(U), \forall i, j \in [n]$. If $f \in L^2(U)$, and $u \in H^1(U)$ is a weak solution to $Lu = f$ in $U$, then $u$ is a strong solution.*

*Proof.* We have that $u \in H^2_{loc}(U)$.
Consider any $V \subset\subset U$, since $a^{ij} \in C^1$, we have $a^{ij} u \in H^2(V)$.

Consider any $v \in C_c^\infty(V)$, we must have

$$\begin{aligned}
\langle f, v\rangle_{L^2(V)} &= B[u,v] \\
&= \int_V \left( \sum_{i,j=1}^n a^{ij}\partial_i u \partial_j v + \sum_{i=1}^n b^i \partial_i uv + cuv \right) dx \\
&= \int_V \left( \sum_{i,j=1}^n a^{ij}\partial_j(\partial_i u) v + \sum_{i=1}^n b^i \partial_i uv + cuv \right) dx \\
&= \int_V \left( \sum_{i,j=1}^n \partial_j(a^{ij}\partial_i u) + \sum_{i=1}^n b^i \partial_i u + cu \right) v dx \\
&= \int_V (Lu) v dx \\
&= \langle Lu, v\rangle_{L^2(V)}.
\end{aligned}$$

Since this holds for all $v \in C_c^\infty(V)$, we must have $Lu(x) = f(x)$ a.e. $x \in V$.
Since this hold for all $V \subset\subset U$, we have that $Lu(x) = f(x)$ a.e. $x \in U$. □

**Theorem 3.21.** *(Higher Interior regularity)*
*Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}, b^i, c \in C^{m+1}(U), \forall i,j \in [n]$ for some $m \in \mathbb{N}$. If $f \in H^m(U), u \in H^1(U)$ is a weak solution to $Lu = f$ in $U$, then $u \in H^{m+2}_{loc}(U)$. In addition, $\forall V \subset\subset U, \exists C > 0$, such that $\forall f \in L^2(U)$, and $u \in H^1(U)$ being a weak solution to $Lu = f$ in $U$, we have*

$$\|u\|_{H^{m+2}(U)} \leq C\Big(\|f\|_{H^m(U)} + \|u\|_{L^2(U)}\Big).$$

**Corollary 3.22.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}, b^i, c \in C^{m+1}(U), \forall i,j \in [n]$ for some $m > \frac{n}{2} - 2 \in \mathbb{N}$. If $f \in H^m(U), u \in H^1(U)$ is a weak solution to $Lu = f$ in $U$, then $u \in C^l(U)$, where $l = m + 2 - \lfloor \frac{n}{2} \rfloor - 1$.*

**Theorem 3.23.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}, b^i, c \in C^\infty(U), \forall i,j \in [n]$. If $f \in C^\infty(U), u \in H^1(U)$ is a weak solution to $Lu = f$ in $U$, then $u \in C^\infty(U)$.*

**Theorem 3.24.** *(Boundary $H^2$ regularity)*
*Let $U \subseteq \mathbb{R}^n$ be bounded and open, with $\partial U$ being $C^2$, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij} \in C^1(\bar{U}), b^i, c \in L^\infty(U), \forall i,j \in [n]$. Then $\exists C > 0$, such that $\forall f \in L^2(U)$ and $u \in H_0^1(U)$ being a weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$, we have*

$$\|u\|_{H^2(U)} \leq C\Big(\|f\|_{L^2(U)} + \|u\|_{L^2(U)}\Big),$$

*and thus $u \in H^2(U)$.*

*Proof.* 1. First prove the case if the boundary is locally flat:

$$U = B(0,1) \cap \{x : x^n > 0\}, V = B(0, \frac{1}{2}) \cap \{x : x^n > 0\}.$$

Similar to the proof of Interior $H^2$ regularity, we first use difference quotients to obtain a bound for derivatives that are not normal to the flat boundary:

$$\sum_{k,l=1, k+l<2n}^n \|\partial_k \partial_l u\|_{L^2(V)} \leq C\Big(\|f\|_{L^2(U)} + \|u\|_{H^1(U)}\Big),$$

where we can transform $\|u\|_{H^1(U)}$ to $\|u\|_{L^2(U)}$.
For the derivative that is normal to the flat boundary $\partial_n\partial_n$, we write the PDE in non divergence form, and use ellipticity to note that $a^{nn} > \theta > 0$ to find:

$$|\partial_n\partial_n| \leq C\left(\sum_{k,l=1,k+l<2n}^{n} |\partial_k\partial_l u| + \|Du\|_2 + |u| + |f|\right) \text{a.e.} x \in U.$$

Thus

$$\|\partial_n\partial_n\|_{L^2(U)} \leq C\left(\sum_{k,l=1,k+l<2n}^{n} \|\partial_k\partial_l u\|_{L^2(U)} + \|Du\|_{L^2(U)} + \|u\|_{L^2(U)} + \|f\|_{L^2(U)}\right).$$

This leads to

$$\|u\|_{H^2(V)} \leq C\Big(\|f\|_{L^2(U)} + \|u\|_{L^2(U)}\Big)$$

2. Take any $x_0 \in \partial U$, let $y = \Phi(x)$ be a $C^2$ straightening map on $B(x_0, r)$ with a $C^2$ inverse $x = \Psi(y)$. Pick some small enough $s$, such that

$$U' = B(0,s) \cap \{y : y^n > 0\} \subseteq \Phi(U \cap B(x_0, r)), V' = B(0, \frac{1}{2}s) \cap \{y : y^n > 0\}.$$

We check the weak formulation is well-defined on $U'$ and that $L'$ satisfies the assumptions of $L$.
Apply step 1 to get

$$\|u'\|_{H^2(V')} \leq C\Big(\|f'\|_{L^2(U')} + \|u'\|_{L^2(U')}\Big).$$

Transform back using $\Psi$.

3. Use compactness to find $V_1, \ldots, V_N$ to cover $\partial U$. Find $V_0 \subset\subset U$ such that $U = \bigcup_{i=0}^{N} V_i$.
Use interior result on $V_0$.
Combine them together.

□

*Remark.* When the solution is unique, we can throw away the $\|u\|_{L^2(U)}$ by boundedness of inverse in the last section.

**Theorem 3.25.** *(Higher boundary regularity)*
*Let $U \subseteq \mathbb{R}^n$ be bounded and open, with $\partial U$ being $C^{m+2}$, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}, b^i, c \in C^{m+1}(\bar{U}), \forall i, j \in [n]$. Then $\exists C > 0$, such that $\forall f \in H^m(U)$ and $u \in H_0^1(U)$ being a weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$, we have*

$$\|u\|_{H^{m+2}(U)} \leq C\Big(\|f\|_{H^m(U)} + \|u\|_{L^2(U)}\Big),$$

*and thus $u \in H^{m+2}(U)$.*

**Corollary 3.26.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, with $\partial U$ being $C^{m+2}$, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}, b^i, c \in C^{m+1}(\bar{U}), \forall i, j \in [n]$ for some $m > \frac{n}{2} - 2 \in \mathbb{N}$. If $f \in H^m(U), u \in H^1(U)$ is a weak solution to $Lu = f$ in $U$, then $u \in C^l(U)$, where $l = m + 2 - \lfloor \frac{n}{2} \rfloor - 1$.*

**Theorem 3.27.** *(Infinite differentiability up to the boundary)*
*Let $U \subseteq \mathbb{R}^n$ be bounded and open, with $\partial U$ being $C^\infty$, and $L$ be a symmetric (uniformly) elliptic second order differential operator, with $a^{ij}, b^i, c \in C^\infty(\bar{U}), \forall i, j \in [n]$. Then $\forall f \in H^\infty(U)$ and $u \in H_0^1(U)$ being a weak solution to $\begin{cases} Lu = f & \text{in } U, \\ u = 0, & \text{on } \partial U \end{cases}$, we have $u \in C^\infty(\bar{U})$.*

# 4 Parabolic PDEs

## 4.1 Spaces Involving Time

### 4.1.1 Bochner Spaces

See more about Bochner Spaces in my Measure Theory Notes.

**Definition 4.1.** Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, a function $u : [0, T] \to X$ is **continuous** at a point $t \in (0, T)$ if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } \forall s, t \in [0, T], |s - t| < \delta \implies \|u(s) - u(t)\| < \epsilon.$$

A function $u$ is continuous if it is continuous at all $t \in (0, 1)$.
$\|u\|_{C([0,T];X)} := \sup_{t\in(0,T)} \|u(t)\|$.

**Theorem 4.1.** *$(C([0, T]; X), \|u\|_{C([0,T];X)})$ is a Banach Space.*

See the definition of Bochner integrable functions in notes of Measure Theory. We will still consider the Lebesgue measure on $[0, T]$.

**Theorem 4.2** (Bochner)**.** *Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, a strongly measurable function $f : [0, T] \to X$ is Bochner integrable if and only if $t \mapsto \|f(t)\|_X$ is integrable. In this case,*

$$\left\| \int_0^T f(t)dt \right\|_X \leq \int_0^T \|f(t)\|_X dt,$$
$$\forall u^* \in X^*, \left\langle u^* \middle| \int_0^T f(t)dt \right\rangle = \int_0^T \langle u^* | f(t) \rangle dt.$$

**Theorem 4.3.** *Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, then Dominated Convergence Theorem, Holder's Inequality, and Minkowski's Inequality still work with the Bochner integral.*

**Theorem 4.4.** *Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, then for any Bochner integrable $f : [0, T] \to X$, we have that $\int_s^t f(\tau)d\tau$ is continuous in both $s, t \in [0, T]$.*

**Theorem 4.5.** *Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space. Suppose $f : [0, T] \to X$ is continuous at $t_0 \in (0, T)$, then*

$$\lim_{h\to 0^+} \frac{1}{h} \int_{t_0}^{t_0+h} f(\tau)d\tau = \lim_{h\to 0^+} \frac{1}{h} \int_{t_0-h}^{t_0} f(\tau)d\tau = f(t_0).$$

*In particular,*

$$\frac{d}{dt} \int_s^t f(\tau)d\tau|_{t=t_0} = f(t_0).$$

*Proof.* Given any $\epsilon > 0$. Since $f$ is continuous at $t_0$, there is some $\delta > 0$, such that $|t - t_0| < \delta \implies |f(t) - f(t_0)| < \epsilon$. Now consider any $h < \delta$.

$$\begin{aligned}
\left| \frac{1}{h} \int_{t_0}^{t_0+h} f(\tau)d\tau - f(t_0) \right| &= \left| \frac{1}{h} \int_{t_0}^{t_0+h} f(\tau)d\tau - \frac{1}{h} \int_{t_0}^{t_0+h} f(t_0)d\tau \right| \\
&= \left| \frac{1}{h} \int_{t_0}^{t_0+h} f(\tau) - f(t_0)d\tau \right| \\
&\leq \frac{1}{h} \int_{t_0}^{t_0+h} |f(\tau) - f(t_0)|d\tau \\
&< \frac{1}{h} \int_{t_0}^{t_0+h} \epsilon d\tau \\
&= \frac{1}{h} h\epsilon \\
&= \epsilon.
\end{aligned}$$

Also, we have that

$$\begin{aligned}\left|\frac{1}{h}\int_{t_0-h}^{t_0} f(\tau)d\tau - f(t_0)\right| &= \left|\frac{1}{h}\int_{t_0-h}^{t_0} f(\tau) - f(t_0)d\tau\right| \\ &\leq \frac{1}{h}\int_{t_0-h}^{t_0} |f(\tau) - f(t_0)|d\tau \\ &< \frac{1}{h}\int_{t_0-h}^{t_0} \epsilon d\tau \\ &= \epsilon.\end{aligned}$$

This proves that

$$\lim_{h\to 0^+}\frac{1}{h}\int_{t_0}^{t_0+h} f(\tau)d\tau = \lim_{h\to 0^+}\frac{1}{h}\int_{t_0-h}^{t_0} f(\tau)d\tau = f(t_0).$$

Now,

$$\begin{aligned}\frac{\int_s^{t_0+h} f(\tau)d\tau - \int_s^{t_0} f(\tau)d\tau}{h} &= \frac{\int_{t_0}^{t_0+h} f(\tau)d\tau}{h}, \\ \frac{\int_s^{t_0} f(\tau)d\tau - \int_s^{t_0-h} f(\tau)d\tau}{h} &= \frac{\int_{t_0-h}^{t_0} f(\tau)d\tau}{h}\end{aligned}$$

Thus

$$\frac{d}{dt}\int_s^t f(\tau)d\tau\bigg|_{t=t_0} = f(t_0).$$

□

Similarly to the Lebesgue Spaces, we can define the Bochner Spaces and Bochner norms:

**Definition 4.2.** Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, and $1 \leq p < \infty$, we define

$$\mathcal{L}^p([0,T];X) := \left\{ f : [0,T] \to X \middle| f \text{ is measurable}, \int_X \|f\|_X^p d\mu < \infty \right\}.$$

In addition, we define

$$\|f\|_{\mathcal{L}^p([0,T];X)} := \left(\int_X \|f\|_X^p d\mu\right)^{\frac{1}{p}}.$$

**Definition 4.3.** Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, we define

$$\mathcal{L}^\infty([0,T];X) := \{f : [0,T] \to X | f \text{ is measurable}, \operatorname{ess\,sup} \|f\|_X < \infty\}.$$

In addition, we define

$$\|f\|_{\mathcal{L}^\infty([0,T];X)} := \operatorname{ess\,sup} \|f\|_X.$$

**Definition 4.4.** Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space. For any $p \in [1, \infty]$, we define

$$L^p([0,T];X) := \mathcal{L}^p([0,T];X)/N,$$

where $N := \{f : [0,T] \to X | f \text{ is measurable}, f = 0 \ \mu - \text{a.e.}\}$. Namely, $[f] \in L^p([0,T];X)$ is the equivalence class of all $g = f$ $\mu$-a.e. for $f \in \mathcal{L}^p([0,T];X)$.
In addition, we define

$$\|[f]\|_{L^p([0,T];X)} := \|f\|_{\mathcal{L}^\infty([0,T];X)}$$

for any representative $f$.

**Theorem 4.6** (Fischer-Riesz-Bochner)**.** *Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space. For all $1 \leq p \leq \infty$, we have that $\left(L^p([0,T];X), \|\cdot\|_{L^p([0,T];X)}\right)$ is a Banach Space.*

Similarly, we can also define $L^p_{loc}(0,T;X), W^{k,p}(0,T;X), H^k(0,T;X)$ and weak derivatives where the test functions are $\phi \in C_c^\infty(0,T;\mathbb{R})$.
We can similarly define the mollification of $f \in L^1_{loc}(0,T;X)$ to be

$$f^\epsilon := \eta_\epsilon * f : (\epsilon, T-\epsilon) \to X;\ t \mapsto \int_{t-\epsilon}^{t+\epsilon} \eta_\epsilon(t-\tau) f(\tau) d\tau.$$

Similarly, we have

**Theorem 4.7.** *Let $f^\epsilon$ be defined as above, we have:*

1. $f^\epsilon \in C^\infty((\epsilon, T-\epsilon); X)$,
2. $\partial_t^k(f^\epsilon) = (\partial_t^k \eta_\epsilon) * f$ *on* $(\epsilon, T-\epsilon)$,
3. $f^\epsilon \to f$ *a.e.* $t \in (0,T)$, *as* $\epsilon \to 0$,
4. *If* $f \in C(0,T;X)$, *we have* $f^\epsilon \to f$ *uniformly on compact subsets of* $U$,
5. *If* $1 \le p < \infty, f \in L^p_{loc}(0,T;X)$, *we have* $f^\epsilon \to f$ *in* $L^p_{loc}(0,T;X)$. *Namely,* $f^\epsilon \to f$ *in* $L^p(V)$, $\forall V \subset\subset (0,T)$.

**Theorem 4.8.** *Let $T > 0$ and $(X, \|\cdot\|)$ be a Banach Space, $p \in [1,\infty]$, and $u \in W^{1,p}(0,T;X)$, then*

1. $u(t) = u(s) + \int_s^t u'(\tau) d\tau$ *for a.e.* $0 \le s \le t \le T$;
2. *There is a representative* $\tilde{u} \in C([0,T], X)$ *of* $u$. *In particular,* $\tilde{u}(t) = \tilde{u}(s) + \int_s^t u'(\tau) d\tau$ *for any* $0 \le s \le t \le T$;
3. $\exists C > 0$ *such that* $\forall u \in W^{1,p}(0,T;X), \sup_{t\in[0,T]} \|u(t)\|_X \le C \|u\|_{W^{1,p}(0,T;X)}$.

*Proof.* We will prove for $p \in [1,\infty)$.

1. Let $u^\epsilon := \eta_\epsilon * u$, we have that $u^\epsilon \in C^\infty((\epsilon, T-\epsilon); X)$, and $\partial_t(u^\epsilon) = (\partial_t \eta_\epsilon) * u$ on $(\epsilon, T-\epsilon)$.
   We also have $f^\epsilon(t) \to f(t)$ a.e. $t \in (0,T)$.
   Similar to 2.23, we can show that $\partial_t(u^\epsilon) = \eta_\epsilon * \partial_t u = (\partial_t u)^\epsilon$ on $(\epsilon, T-\epsilon)$.
   Since $u \in W^{1,p}(0,T;X)$, we know that $\partial_t u \in L^p_{loc}(0,T;X)$, so $(\partial_t u)^\epsilon \to \partial_t u$ in $L^p_{loc}(0,T;X)$.
   Since $|(0,T)| = T < \infty$, we have that $\partial_t(u^\epsilon) \to \partial_t u$ in $L^1_{loc}(0,T;X)$, which means

$$\forall [s,t] \subset (0,T),\ \lim_{\epsilon\to 0} \int_s^t \|(\partial_t(u^\epsilon))(\tau) - (\partial_t u)(\tau)\|_X d\tau = 0.$$

   We have that $\left\| \int_s^t (\partial_t(u^\epsilon))(\tau) - (\partial_t u)(\tau) d\tau \right\|_X \le \int_s^t \|(\partial_t(u^\epsilon))(\tau) - (\partial_t u)(\tau)\|_X d\tau$ for any fixed $[s,t] \subset (0,T)$ and $\epsilon < \min(s, T-t)$. Thus

$$\lim_{\epsilon\to 0} \left\| \int_s^t (\partial_t(u^\epsilon))(\tau) - (\partial_t u)(\tau) d\tau \right\|_X = 0$$

   for any $[s,t] \subset (0,T)$.
   Now $u^\epsilon(t) = u^\epsilon(s) + \int_s^t (\partial_t(u^\epsilon))(\tau) d\tau$ for any $[s,t] \subset (\epsilon, T-\epsilon)$ by FTC, since $u^\epsilon \in C^\infty((\epsilon, T-\epsilon); X)$.
   We have

$$\begin{aligned}
&\left\| -u(t) + u(s) + \int_s^t (\partial_t u)(\tau) d\tau \right\|_X \\
=&\left\| u^\epsilon(t) - u(t) - u^\epsilon(s) + u(s) - \int_s^t (\partial_t(u^\epsilon))(\tau) d\tau + \int_s^t (\partial_t u)(\tau) d\tau \right\|_X \\
\le&\|u^\epsilon(t) - u(t)\|_X + \|u^\epsilon(s) - u(s)\|_X + \left\| \int_s^t (\partial_t(u^\epsilon))(\tau) d\tau - \int_s^t (\partial_t u)(\tau) d\tau \right\|_X \\
\le&\|u^\epsilon(t) - u(t)\|_X + \|u^\epsilon(s) - u(s)\|_X + \left\| \int_s^t (\partial_t(u^\epsilon))(\tau) - (\partial_t u)(\tau) d\tau \right\|_X
\end{aligned}$$

for any $s, t, \epsilon$ such that $[s,t] \subset (\epsilon, T-\epsilon)$.
Since each term goes to 0 as $\epsilon \to 0$ for a.e. $0 \le s \le t \le T$, we must have

$$\left\| -u(t) + u(s) + \int_s^t (\partial_t u)(\tau) d\tau \right\|_X = 0$$

for a.e. $0 \le s \le t \le T$.
We thus have

$$u(t) = u(s) + \int_s^t (\partial_t u)(\tau) d\tau$$

for a.e. $0 \le s \le t \le T$.

2. Fix any representative for $u$.
Notice that the set $N$ where the above property does not hold has measure 0.
Now fix some point $s \in [0,T] \setminus N$, we define

$$\tilde{u}(t) := \begin{cases} u(s) - \int_t^s u'(\tau) d\tau & t < s \\ u(s) + \int_s^t u'(\tau) d\tau & t \ge s \end{cases}.$$

For any $t \in [0,T] \setminus N$, we have that

$$u(t) := u(s) + \int_s^t u'(\tau) d\tau = \tilde{u}(t)$$

if $t \ge s$, and

$$u(s) = u(t) + \int_t^s u'(\tau) d\tau \implies u(t) = u(s) - \int_t^s u'(\tau) d\tau = u(s)$$

if $t < s$.
Thus $\tilde{u} = u$ a.e. $t \in [0,T]$, which means $\tilde{u}$ is a representative of $u$.
In addition, $\tilde{u}(t)$ is continuous since $\int_t^s u'(\tau) d\tau$ and $\int_s^t u'(\tau) d\tau$ are both continuous in $t$, and

$$\lim_{t \to s^-} \tilde{u}(t) = \lim_{t \to s^-} \left( \tilde{u}(t) u(s) - \int_t^s u'(\tau) d\tau \right) = u(s) = u(s) + \int_s^s u'(\tau) d\tau = \tilde{u}(s).$$

3. See A5Q2.

□

**Proposition 4.9.** *Suppose $\mathcal{H}$ is a Hilbert Space, and $u, v \in C^1(0,T;\mathcal{H})$, then we have*

$$\forall t \in [0,T], \ \frac{d}{dt} \langle u(t), v(t) \rangle_{\mathcal{H}} = \langle u'(t), v(t) \rangle_{\mathcal{H}} + \langle v'(t), u(t) \rangle_{\mathcal{H}},$$

*where $u'(t) := \lim_{h \to 0} \frac{u(t+h) - u(t)}{t}$ is the classical derivative in $X$.*

*Proof.* We have

$$\begin{aligned}
\frac{d}{dt} \langle u(t), v(t) \rangle_{\mathcal{H}} &= \lim_{h \to 0} \frac{\langle u(t+h), v(t+h) \rangle_{\mathcal{H}} - \langle u(t), v(t) \rangle_{\mathcal{H}}}{h} \\
&= \lim_{h \to 0} \frac{\langle u(t+h), v(t+h) \rangle_{\mathcal{H}} - \langle u(t+h), v(t) \rangle_{\mathcal{H}} + \langle u(t+h), v(t) \rangle_{\mathcal{H}} - \langle u(t), v(t) \rangle_{\mathcal{H}}}{h} \\
&= \lim_{h \to 0} \frac{\langle u(t+h), v(t+h) \rangle_{\mathcal{H}} - \langle u(t+h), v(t) \rangle_{\mathcal{H}}}{h} + \lim_{h \to 0} \frac{\langle u(t+h), v(t) \rangle_{\mathcal{H}} - \langle u(t), v(t) \rangle_{\mathcal{H}}}{h} \\
&= \lim_{h \to 0} \frac{\langle u(t+h), v(t+h) - v(t) \rangle_{\mathcal{H}}}{h} + \lim_{h \to 0} \frac{\langle u(t+h) - u(t), v(t) \rangle_{\mathcal{H}}}{h} \\
&= \left\langle \lim_{h \to 0} u(t+h), \lim_{h \to 0} \frac{v(t+h) - v(t)}{h} \right\rangle_{\mathcal{H}} + \left\langle \lim_{h \to 0} \frac{u(t+h) - u(t)}{h}, v(t) \right\rangle_{\mathcal{H}} \\
&= \langle u(t), v'(t) \rangle_{\mathcal{H}} + \langle u'(t), v(t) \rangle_{\mathcal{H}}.
\end{aligned}$$

□

### 4.1.2 Sobolev Spaces In Time

Now we consider the cases where $X$ might be any of the tuple $H_0^1(U) \subset L^2(U) \cong L^2(U)^* \subset H^{-1}(U)$, and see what is the relationship between the time weak derivatives in each space.

**Lemma 4.10.** *Suppose $\boldsymbol{u}, \boldsymbol{u}' \in L^1\big(0,T;H_0^1(U)\big)$, then we must have $\boldsymbol{u}'$ is also the time weak derivative of $\boldsymbol{u}$ in $L^1\big(0,T;L^2(U)\big)$.*

*Proof.* Firstly, for each $t \in [0,T]$, we have $\mathbf{u}(t), \mathbf{u}'(t) \in H_0^1(U) \subset L^2(U)$, so $\mathbf{u}, \mathbf{u}'$ are indeed functions $[0,T] \to L^2(U)$.
In addition,

$$\begin{aligned}\int_0^T \|\mathbf{u}(t)\|_{L^2(U)} dt &\le \int_0^T \|\mathbf{u}(t)\|_{H^2(U)} dt \\ &= \|\mathbf{u}\|_{L^1\left(0,T;H_0^1(U)\right)} \\ &< \infty.\end{aligned}$$

Similarly for $\int_0^T \|\mathbf{u}'(t)\|_{L^2(U)} dt < \infty$.
Thus $\mathbf{u}, \mathbf{u}' \in L^1\big(0,T;L^2(U)\big)$.
Now $\forall \phi \in C_c^\infty(0,T)$, we have

$$\begin{aligned}0 &\le \left\| \int_0^T \phi'(t)\mathbf{u}(t)dt + \int_0^T \phi(t)\mathbf{u}'(t)dt \right\|_{L^2(U)} \\ &\le \left\| \int_0^T \phi'(t)\mathbf{u}(t)dt + \int_0^T \phi(t)\mathbf{u}'(t)dt \right\|_{H_0^1(U)} \\ &= 0.\end{aligned}$$

Thus $\int_0^T \phi'(t)\mathbf{u}(t)dt = -\int_0^T \phi(t)\mathbf{u}'(t)dt$ in $L^2(U)$ for any $\phi \in C_c^\infty(0,T)$, which shows that $\mathbf{u}'$ is also the time weak derivative of $\mathbf{u}$ in $L^1\big(0,T;L^2(U)\big)$. □

**Lemma 4.11.** *Let $\boldsymbol{u} \in L^1\big(0,T;H_0^1(U)\big), \boldsymbol{v} \in L^1\big(0,T;H^{-1}(U)\big)$, we have $\boldsymbol{v} = (\boldsymbol{u}^*)' \iff$*

$$\forall \phi \in C_c^\infty(0,T), w \in H_0^1(U),\ \int_0^T \phi'(t)\langle \boldsymbol{u}(t), w\rangle_{L^2(U)} dt = -\int_0^T \phi(t)\langle \boldsymbol{v}(t)|w\rangle_{H^{-1}(U),H_0^1(U)} dt,$$

*where $\boldsymbol{u}^*(t) := \langle \boldsymbol{u}(t), \cdot\rangle_{L^2(U)} \in H^{-1}(U)$ as usual.*

*Proof.* $(\mathbf{u}^*)' = \mathbf{v}$ by definition means

$$\forall \phi \in C_c^\infty(0,T),\ \int_0^T \phi'(t)\mathbf{u}^*(t)dt = -\int_0^T \phi(t)\mathbf{v}(t)dt$$

in $H^{-1}(U)$.
Consider any $w \in H_0^1(U)$, we have $\langle \cdot|w\rangle_{H^{-1}(U),H_0^1(U)} \in \big(H^{-1}(U)\big)^*$.
By Bochner's Theorem 4.2 and linearity of the duality pairing, we have

$$\begin{aligned}\left\langle \int_0^T \phi'(t)\mathbf{u}^*(t)dt \middle| w \right\rangle_{H^{-1}(U),H_0^1(U)} &= \left\langle -\int_0^T \phi(t)\mathbf{v}(t)dt \middle| w \right\rangle_{H^{-1}(U),H_0^1(U)} \\ \int_0^T \langle \phi'(t)\mathbf{u}^*(t)|w\rangle_{H^{-1}(U),H_0^1(U)} dt &= -\int_0^T \langle \phi(t)\mathbf{v}(t)|w\rangle_{H^{-1}(U),H_0^1(U)} dt \\ \int_0^T \phi'(t)\langle \mathbf{u}^*(t)|w\rangle_{H^{-1}(U),H_0^1(U)} dt &= -\int_0^T \phi(t)\langle \mathbf{v}(t)|w\rangle_{H^{-1}(U),H_0^1(U)} dt \\ \int_0^T \phi'(t)\langle \mathbf{u}(t), w\rangle_{L^2(U)} dt &= -\int_0^T \phi(t)\langle \mathbf{v}(t)|w\rangle_{H^{-1}(U),H_0^1(U)} dt.\end{aligned}$$

□

**Lemma 4.12.** *Suppose $\boldsymbol{u} \in L^1\big(0,T;H_0^1(U)\big)$, and $\boldsymbol{u}'$ is its time weak derivative in $L^1\big(0,T;L^2(U)\big)$, then we must have the action function*

$$(\boldsymbol{u}')^* := t \mapsto \langle \boldsymbol{u}(t), \cdot \rangle_{L^2(U)}$$

*is the time weak derivative of $\boldsymbol{u}^* := t \mapsto \langle \boldsymbol{u}(t), \cdot \rangle_{L^2(U)}$ in $L^1\big(0,T;H^{-1}(U)\big)$. Namely,*

$$(\boldsymbol{u}')^* = (\boldsymbol{u}^*)'.$$

*Proof.* Consider any $\phi \in C_c^\infty(0,T)$, by definition of weak derivative, we have

$$\int_0^T \phi'(t)\mathbf{u}(t)dt = -\int_0^T \phi(t)\mathbf{u}'(t)dt.$$

in $L^2(U)$.
Now for any $w \in H_0^1(U) \subset L^2(U)$, we have $\langle \cdot, w \rangle_{L^2(U)} \in \big(L^2(U)\big)^*$.
By Bochner's theorem 4.2 and linearity of the inner product, we have

$$\begin{aligned}
\left\langle \int_0^T \phi'(t)\mathbf{u}(t)dt, w \right\rangle_{L^2(U)} &= \left\langle -\int_0^T \phi(t)\mathbf{u}'(t)dt, w \right\rangle_{L^2(U)} \\
\int_0^T \langle \phi'(t)\mathbf{u}(t), w \rangle_{L^2(U)} dt &= -\int_0^T \langle \phi(t)\mathbf{u}'(t), w \rangle_{L^2(U)} \\
\int_0^T \phi'(t)\langle \mathbf{u}(t), w \rangle_{L^2(U)} dt &= -\int_0^T \phi(t)\langle \mathbf{u}'(t), w \rangle_{L^2(U)} dt \\
\int_0^T \phi'(t)\langle \mathbf{u}(t), w \rangle_{L^2(U)} dt &= -\int_0^T \phi(t)\langle (\mathbf{u}')^*(t) | w \rangle_{H^{-1}(U), H_0^1(U)} dt.
\end{aligned}$$

Thus $(\mathbf{u}')^* = (\mathbf{u}^*)'$ from the above lemma. □

**Corollary 4.13.** *Suppose $\boldsymbol{u}, \boldsymbol{u}' \in L^1\big(0,T;H_0^1(U)\big)$, then we must have the action function*

$$(\boldsymbol{u}')^* := t \mapsto \langle \boldsymbol{u}(t), \cdot \rangle_{L^2(U)}$$

*is the time weak derivative of $\boldsymbol{u}^* := t \mapsto \langle \boldsymbol{u}(t), \cdot \rangle_{L^2(U)}$ in $L^1\big(0,T;H^{-1}(U)\big)$.*

*Remark.* Recall $H_0^1(U) \subset L^2(U) \cong L^2(U)^* \subset H^{-1}(U)$, and we can identify $\mathbf{u}(t) \in H_0^1(U)$ with $\mathbf{u}^*(t) := \langle \mathbf{u}(t), \cdot \rangle_{L^2(U)} \in H^{-1}(U)$. The above lemmas allow us to further abuse this notation and identify $\mathbf{u}'$ with $(\mathbf{u}')^* = (\mathbf{u}^*)'$.

**Definition 4.5.** Suppose $\mathbf{u} \in L^1\big(0,T;H_0^1(U)\big)$, we abuse the notation and denote

$$\mathbf{u}' := \mathbf{v} \in L^1\big(0,T;H^{-1}(U)\big)$$

to be the time weak derivative of $\mathbf{u}$, if $\mathbf{v} = (\mathbf{u}^*)'$ is the time weak derivative of the action function

$$\mathbf{u}^* := t \mapsto \langle \mathbf{u}(t), \cdot \rangle_{L^2(U)}$$

in $L^1\big(0,T;H^{-1}(U)\big)$.

*Remark.* This is a further extension of the original definition of the weak derivative, since $\mathbf{u}'$ may not exist in $L^1\big(0,T;H_0^1(U)\big)$ even if such a $\mathbf{v} = (\mathbf{u}^*)'$ exists in $L^1\big(0,T;H^{-1}(U)\big)$ or even $L^1\big(0,T;L^2(U)^*\big)$.

We also have the following results:

**Theorem 4.14.** *The dual space of $L^2\big(0,T;H_0^1(U)\big)$ is $L^2\big(0,T;H^{-1}(U)\big)$, and the dual space of $L^2\big(0,T;H^{-1}(U)\big)$ is $L^2\big(0,T;H_0^1(U)\big)$. The contraction map is defined to be $\forall \boldsymbol{u}\in L^2\big(0,T;H_0^1(U)\big), \boldsymbol{v}\in L^2\big(0,T;H^{-1}(U)\big)$,*

$$\langle \boldsymbol{u}|\boldsymbol{v}\rangle_{L^2\left(0,T;H_0^1(U)\right),L^2(0,T;H^{-1}(U))} := \langle \boldsymbol{v}|\boldsymbol{u}\rangle_{L^2(0,T;H^{-1}(U)),L^2\left(0,T;H_0^1(U)\right)} := \int_0^T \langle \boldsymbol{v}(t)|\boldsymbol{u}(t)\rangle_{H^{-1}(U),H_0^1(U)}dt.$$

*Proof.* We can quickly show one side of inclusion:

$$\begin{aligned}\int_0^T \langle \mathbf{v}(t)|\mathbf{u}(t)\rangle_{H^{-1}(U),H_0^1(U)}dt &\le \int_0^T \|\mathbf{v}(t)\|_{H^{-1}(U)}\|\mathbf{u}(t)\|_{H_0^1(U)}dt\\ &\le \left(\int_0^T \|\mathbf{v}(t)\|_{H^{-1}(U)}^2 dt\right)^{\frac{1}{2}}\left(\int_0^T \|\mathbf{u}(t)\|_{H_0^1(U)}^2 dt\right)^{\frac{1}{2}}\\ &= \|u\|_{L^2\left(0,T;H_0^1(U)\right)}\|v\|_{L^2(0,T;H^{-1}(U))},\end{aligned}$$

which shows $L^2\big(0,T;H_0^1(U)\big)\subseteq L^2\big(0,T;H^{-1}(U)\big)^*$, and $L^2\big(0,T;H^{-1}(U)\big)\subseteq L^2\big(0,T;H_0^1(U)\big)^*$. □

**Lemma 4.15.** *Suppose $\boldsymbol{u}\in L^1\big(0,T;H_0^1(U)\big), \boldsymbol{u}'\in L^1\big(0,T;H^{-1}(U)\big)$, then $\forall\epsilon>0$,*

$$\boldsymbol{u}'^{\epsilon} = \Big(t\mapsto \langle (\boldsymbol{u}^\epsilon)'(t),\cdot\rangle_{L^2(U)}\Big),$$

*where $(\boldsymbol{u}^\epsilon)'$ is the classical derivative of $\boldsymbol{u}^\epsilon$ in $C^\infty\big(\epsilon,T-\epsilon;H_0^1(U)\big)$.*

*Proof.* Since $\mathbf{u}'$ is the time weak derivative of $\mathbf{u}^*$ in $L^2\big((0,T);H^{-1}(U)\big)$, we can show that

$$\mathbf{u}'^{\epsilon} = \eta_\epsilon * \mathbf{u}' = (\eta_\epsilon * \mathbf{u}^*)'$$

in $L^2\big((0,T);H^{-1}(U)\big)$ similarly as in 2.23.
Also, one can see that $\forall t\in[\epsilon,T-\epsilon], v\in H_0^1(U)$,

$$\begin{aligned}\langle(\eta_\epsilon*\mathbf{u}^*)(t)|v\rangle_{H^{-1}(U),H_0^1(U)} &= \left\langle \int_0^T \eta_\epsilon(t-\tau)\mathbf{u}^*(\tau)d\tau \middle| v\right\rangle_{H^{-1}(U),H_0^1(U)}\\ &= \int_0^T \eta_\epsilon(t-\tau)\langle \mathbf{u}^*(\tau)|v\rangle_{H^{-1}(U),H_0^1(U)}d\tau\\ &= \int_0^T \eta_\epsilon(t-\tau)\langle \mathbf{u}(\tau),v\rangle_{L^2(U)}d\tau\\ &= \left\langle \int_0^T \eta_\epsilon(t-\tau)\mathbf{u}(\tau)d\tau, v\right\rangle_{L^2(U)}\\ &= \langle \mathbf{u}^\epsilon(t),v\rangle_{L^2(U)}.\end{aligned}$$

Since $(\mathbf{u}^\epsilon)'$ exists as the weak derivative of $\mathbf{u}^\epsilon$, we have

$$(\eta_\epsilon*\mathbf{u}^*)' = \Big(t\mapsto\langle(\mathbf{u}^\epsilon)'(t),\cdot\rangle_{L^2(U)}\Big)$$

in $L^2\big((0,T);H^{-1}(U)\big)$. □

**Theorem 4.16.** *Suppose $\boldsymbol{u}\in L^2\big(0,T;H_0^1(U)\big), \boldsymbol{u}'\in L^2\big(0,T;H^{-1}(U)\big)$.*

1. *There is a representative $\tilde{\boldsymbol{u}}\in C\big([0,T];L^2(U)\big)$ of $\boldsymbol{u}$;*
2. *For any $\boldsymbol{v}\in L^2\big(0,T;H_0^1(U)\big), \boldsymbol{v}'\in L^2\big(0,T;H^{-1}(U)\big)$, the mapping $t\mapsto\langle\tilde{\boldsymbol{u}}(t),\tilde{\boldsymbol{v}}(t)\rangle_{L^2(U)}$ is absolutely continuous, and for a.e. $t\in[0,T]$, we have*

$$\frac{d}{dt}\langle\tilde{\boldsymbol{u}}(t),\tilde{\boldsymbol{v}}(t)\rangle_{L^2(U)} = \langle\boldsymbol{u}'(t)|\boldsymbol{v}(t)\rangle_{H^{-1}(U),H_0^1(U)} + \langle\boldsymbol{v}'(t)|\boldsymbol{u}(t)\rangle_{H^{-1}(U),H_0^1(U)}.$$

3. *The mapping* $t \mapsto \|\tilde{\boldsymbol{u}}(t)\|^2_{L^2(U)}$ *is absolutely continuous, and for a.e.* $t \in [0,T]$, *we have*

$$\frac{d}{dt}\|\boldsymbol{u}(t)\|^2_{L^2(U)} = 2\langle \boldsymbol{u}'(t) | \boldsymbol{u}(t)\rangle_{H^{-1}(U),H^1_0(U)}.$$

4. $\exists C > 0$, *such that* $\forall \boldsymbol{u} \in L^2\big(0,T;H^1_0(U)\big)$, $\boldsymbol{u}' \in L^2\big(0,T;H^{-1}(U)\big)$,

$$\sup_{t\in[0,T]} \|\tilde{\boldsymbol{u}}(t)\|_{L^2(U)} \le C\Big(\|\boldsymbol{u}\|_{L^2\left(0,T;H^1_0(U)\right)} + \|\boldsymbol{u}'\|_{L^2(0,T;H^{-1}(U))}\Big),$$

*where the constant* $C$ *only depends on* $T$.

*Proof.* 1. We can extend $\mathbf{u}$ to $[-\sigma, T+\sigma]$ for an $\delta > 0$ by reflection and cut off as done in 2.34.
Now for any $\epsilon, \delta \in (0,\sigma)$, we can define $\mathbf{u}^\epsilon := \eta_\epsilon * \mathbf{u}, \mathbf{u}^\delta := \eta_\delta * \mathbf{u}$, both well-defined on $[0,T]$.
By 4.7, $\mathbf{u}^\epsilon, \mathbf{u}^\delta \in C^\infty\big([0,T];H^1_0(U)\big)$, so we have

$$\mathbf{u}^\epsilon, \mathbf{u}^\delta \in C^\infty\big([0,T];H^1_0(U)\big) \subset C^\infty\big([0,T];L^2(U)\big).$$

Now for any $t \in [0,T]$, we have that

$$\begin{aligned}
&\frac{d}{dt}\big\|\mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t)\big\|_{L^2(U)} \\
&= \frac{d}{dt}\big\langle \mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t), u^\epsilon(t) - \mathbf{u}^\delta(t)\big\rangle_{L^2(U)} \\
&= \big\langle (\mathbf{u}^\epsilon - \mathbf{u}^\delta)'(t), \mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t)\big\rangle_{L^2(U)} + \big\langle \mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t), (\mathbf{u}^\epsilon - \mathbf{u}^\delta)'(t)\big\rangle_{L^2(U)} \\
&= 2\big\langle (\mathbf{u}^\epsilon)'(t) - (\mathbf{u}^\delta)'(t), \mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t)\big\rangle_{L^2(U)},
\end{aligned}$$

where $(\mathbf{u}^\epsilon)', (\mathbf{u}^\delta)'$ are the classical derivatives as functions $[0,T] \to L^2(U)$.
Also, since $\mathbf{u}^\epsilon, \mathbf{u}^\delta \in C^\infty\big([0,T];H^1_0(U)\big)$, we have that their weak derivatives exist in $L^2\big((0,T);H^1_0(U)\big) \subset L^1\big((0,T);H^1_0(U)\big)$, and by above lemma, are also weak derivatives in $L^1\big((0,T);L^2(U)\big)$.
Since any weak derivative is a.e. equal to the classical derivative if the latter exists, we will just use $(\mathbf{u}^\epsilon)', (\mathbf{u}^\delta)'$ to represent the weak derivatives in $L^2\big((0,T);H^1_0(U)\big)$, and the above equality still holds for a.e. $t \in [0,T]$.
Integrating both sides on any $[s,t] \subseteq [0,T]$, we get

$$\begin{aligned}
&\big\|\mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t)\big\|_{L^2(U)} - \big\|\mathbf{u}^\epsilon(s) - \mathbf{u}^\delta(s)\big\|_{L^2(U)} \\
&= \int_s^t \frac{d}{d\tau}\big\|\mathbf{u}^\epsilon(\tau) - \mathbf{u}^\delta(\tau)\big\|_{L^2(U)} d\tau \\
&= \int_s^t 2\big\langle (\mathbf{u}^\epsilon)'(\tau) - (\mathbf{u}^\delta)'(\tau), \mathbf{u}^\epsilon(\tau) - \mathbf{u}^\delta(\tau)\big\rangle_{L^2(U)} d\tau \\
&= \int_s^t 2\big\langle (\mathbf{u}^\epsilon)'(\tau) - (\mathbf{u}^\delta)'(\tau) \big| \mathbf{u}^\epsilon(\tau) - \mathbf{u}^\delta(\tau)\big\rangle_{H^{-1}(U),H^1_0(U)} d\tau \\
&\le \int_s^t 2\big\|(\mathbf{u}^\epsilon)'(\tau) - (\mathbf{u}^\delta)'(\tau)\big\|_{H^{-1}(U)}\big\|\mathbf{u}^\epsilon(\tau) - \mathbf{u}^\delta(\tau)\big\|_{H^1_0(U)} d\tau \\
&\le \int_s^t \big\|(\mathbf{u}^\epsilon)'(\tau) - (\mathbf{u}^\delta)'(\tau)\big\|^2_{H^{-1}(U)} + \big\|\mathbf{u}^\epsilon(\tau) - \mathbf{u}^\delta(\tau)\big\|^2_{H^1_0(U)} d\tau \\
&= \big\|(\mathbf{u}^\epsilon)' - (\mathbf{u}^\delta)'\big\|^2_{L^2(0,T;H^{-1}(U))} + \big\|\mathbf{u}^\epsilon - \mathbf{u}^\delta\big\|^2_{L^2\left(0,T;H^1_0(U)\right)},
\end{aligned}$$

where we again identify $(\mathbf{u}^\epsilon)'(\tau), (\mathbf{u}^\delta)'(\tau) \in H^1_0(U)$ with $\langle (\mathbf{u}^\epsilon)'(\tau), \cdot\rangle_{L^2(U)}, \big\langle (\mathbf{u}^\delta)'(\tau), \cdot\big\rangle_{L^2(U)} \in H^{-1}(U)$ as usual.
By 4.7, we have $\mathbf{u}^\epsilon, \mathbf{u}^\delta \to \mathbf{u}$ in $L^2\big((0,T);H^1_0(U)\big)$, so

$$\begin{aligned}
\big\|\mathbf{u}^\epsilon - \mathbf{u}^\delta\big\|_{L^2\left(0,T;H^1_0(U)\right)} &= \big\|\mathbf{u}^\epsilon - \mathbf{u} + \mathbf{u} - \mathbf{u}^\delta\big\|_{L^2\left(0,T;H^1_0(U)\right)} \\
&\le \|\mathbf{u}^\epsilon - \mathbf{u}\|_{L^2\left(0,T;H^1_0(U)\right)} + \big\|\mathbf{u} - \mathbf{u}^\delta\big\|_{L^2\left(0,T;H^1_0(U)\right)} \\
&\to 0
\end{aligned}$$

as $\epsilon, \delta \to 0$.
Since $\mathbf{u}'$ is the time weak derivative of $\mathbf{u}^*$ in $L^2\big((0,T);H^{-1}(U)\big)$, we can again show that $(\eta_\epsilon * \mathbf{u}^*)' = \eta_\epsilon * \mathbf{u}'$ in $L^2\big((0,T);H^{-1}(U)\big)$.
By 4.7 and lemma, we have $(\mathbf{u}^\epsilon)', (\mathbf{u}^\delta)' \to \mathbf{u}'$ in $L^2\big(0,T;H^{-1}(U)\big)$.
Similar as above, we have $\big\|(\mathbf{u}^\epsilon)' - (\mathbf{u}^\delta)'\big\|_{L^2(0,T;H^{-1}(U))} \to 0$ as $\epsilon, \delta \to 0$.
In addition, for a.e. $s \in [0,T]$, we must have $\big\|\mathbf{u}^\epsilon(s) - \mathbf{u}^\delta(s)\big\|_{L^2(U)} \to 0$.
Pick any of these $s$, we have

$$\big\|\mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t)\big\|_{L^2(U)} \le \big\|\mathbf{u}^\epsilon(s) - \mathbf{u}^\delta(s)\big\|_{L^2(U)} + \big\|(\mathbf{u}^\epsilon)' - (\mathbf{u}^\delta)'\big\|^2_{L^2(0,T;H^{-1}(U))} + \big\|\mathbf{u}^\epsilon - \mathbf{u}^\delta\big\|^2_{L^2\left(0,T;H_0^1(U)\right)}.$$

Since this holds for any $t \in [0,T]$, we have

$$\begin{aligned}
&\big\|\mathbf{u}^\epsilon - \mathbf{u}^\delta\big\|_{C([0,T];L^2(U))} \\
=& \sup_{t\in[0,T];L^2(U)} \big\|\mathbf{u}^\epsilon(t) - \mathbf{u}^\delta(t)\big\|_{L^2(U)} \\
\le& \big\|\mathbf{u}^\epsilon(s) - \mathbf{u}^\delta(s)\big\|_{L^2(U)} + \big\|(\mathbf{u}^\epsilon)' - (\mathbf{u}^\delta)'\big\|^2_{L^2(0,T;H^{-1}(U))} + \big\|\mathbf{u}^\epsilon - \mathbf{u}^\delta\big\|^2_{L^2\left(0,T;H_0^1(U)\right)} \\
\to& 0.
\end{aligned}$$

as $\epsilon, \delta \to 0$, since each term goes to 0.
This shows that $\mathbf{u}^\epsilon$ is a Cauchy sequence in $C\big([0,T];L^2(U)\big)$, and since it is a Banach Space, there must be some

$$\tilde{\mathbf{u}} := \lim_{\epsilon\to 0} \mathbf{u}^\epsilon \in C\big([0,T];L^2(U)\big).$$

Now since for a.e. $t \in [0,T]$, $\mathbf{u}(t) = \lim_{\epsilon\to 0} \mathbf{u}^\epsilon(t)$, and $\forall t \in [0,T]$, $\tilde{\mathbf{u}}(t) = \lim_{\epsilon\to 0} \mathbf{u}^\epsilon(t)$, we have that $\tilde{\mathbf{u}} = \mathbf{u}$ for a.e. $t \in [0,T]$ is a representative of $\mathbf{u}$.

2. Similar as above, we can show that for a.e. $t \in [0,T]$, we have

$$\frac{d}{dt}\langle \mathbf{u}^\epsilon(t), \mathbf{v}^\epsilon(t)\rangle_{L^2(U)} = \langle (\mathbf{u}^\epsilon)'(t), \mathbf{v}^\epsilon(t)\rangle_{L^2(U)} + \langle (\mathbf{v}^\epsilon)'(t), \mathbf{u}^\epsilon(t)\rangle_{L^2(U)}.$$

Integrating over any $(s,t) \subset [0,T]$ gives

$$\langle \mathbf{u}^\epsilon(t), \mathbf{v}^\epsilon(t)\rangle_{L^2(U)} = \langle \mathbf{u}^\epsilon(s), \mathbf{v}^\epsilon(s)\rangle_{L^2(U)} + \int_s^t \langle (\mathbf{u}^\epsilon)'(\tau), \mathbf{v}^\epsilon(\tau)\rangle_{L^2(U)} + \langle (\mathbf{v}^\epsilon)'(\tau), \mathbf{u}^\epsilon(\tau)\rangle_{L^2(U)} d\tau.$$

Now

$$
\begin{aligned}
&\left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau), \mathbf{v}^\epsilon(\tau)\rangle_{L^2(U)} d\tau - \int_s^t \langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| \\
&= \left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau)|\mathbf{v}^\epsilon(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau - \int_s^t \langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| \\
&\le \left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau)|\mathbf{v}^\epsilon(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau - \int_s^t \langle (\mathbf{u}^\epsilon)'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| \\
&\quad + \left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau - \int_s^t \langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| \\
&= \left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau)|\mathbf{v}^\epsilon(\tau)-\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| + \left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau)-\mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| \\
&\le \int_s^t \|(\mathbf{u}^\epsilon)'(\tau)\|_{H^{-1}(U)} \|\mathbf{v}^\epsilon(\tau)-\mathbf{v}(\tau)\|_{H_0^1(U)} d\tau + \int_s^t \|(\mathbf{u}^\epsilon)'(\tau)-\mathbf{u}'(\tau)\|_{H^{-1}(U)} \|\mathbf{v}(\tau)\|_{H_0^1(U)} d\tau \\
&\le \sqrt{\int_s^t \|(\mathbf{u}^\epsilon)'(\tau)\|^2_{H^{-1}(U)} d\tau}\sqrt{\int_s^t \|\mathbf{v}^\epsilon(\tau)-\mathbf{v}(\tau)\|^2_{H_0^1(U)} d\tau} \\
&\quad + \sqrt{\int_s^t \|(\mathbf{u}^\epsilon)'(\tau)-\mathbf{u}'(\tau)\|^2_{H^{-1}(U)} d\tau}\sqrt{\int_s^t \|\mathbf{v}(\tau)\|^2_{H_0^1(U)} d\tau} \\
&\le \sqrt{\int_0^T \|(\mathbf{u}^\epsilon)'(\tau)\|^2_{H^{-1}(U)} d\tau}\sqrt{\int_0^T \|\mathbf{v}^\epsilon(\tau)-\mathbf{v}(\tau)\|^2_{H_0^1(U)} d\tau} \\
&\quad + \sqrt{\int_0^T \|(\mathbf{u}^\epsilon)'(\tau)-\mathbf{u}'(\tau)\|^2_{H^{-1}(U)} d\tau}\sqrt{\int_0^T \|\mathbf{v}(\tau)\|^2_{H_0^1(U)} d\tau} \\
&= \|(\mathbf{u}^\epsilon)'\|_{L^2(0,T;H^{-1}(U))} \|\mathbf{v}^\epsilon - \mathbf{v}\|_{L^2\left(0,T;H_0^1(U)\right)} + \|(\mathbf{u}^\epsilon)' - \mathbf{u}'\|_{L^2(0,T;H^{-1}(U))} \|\mathbf{v}\|_{L^2\left(0,T;H_0^1(U)\right)},
\end{aligned}
$$

by Holder's Inequality.
Notice that

$$\|(\mathbf{u}^\epsilon)'\|_{L^2(0,T;H^{-1}(U))} \to \|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))} < \infty, \|(\mathbf{u}^\epsilon)' - \mathbf{u}'\|_{L^2(0,T;H^{-1}(U))} \to 0,$$

since we have shown $(\mathbf{u}^\epsilon)' \to \mathbf{u}'$ in $L^2\big(0,T;H^{-1}(U)\big)$.
Also,

$$\|\mathbf{v}^\epsilon - \mathbf{v}\|_{L^2\left(0,T;H_0^1(U)\right)} \to 0,$$

since $\mathbf{v}^\epsilon \to \mathbf{v}$ in $L^2\big(0,T;H_0^1(U)\big)$.
Thus, $\lim_{\epsilon\to 0}\left|\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau), \mathbf{v}^\epsilon(\tau)\rangle_{L^2(U)} d\tau - \int_s^t \langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau\right| = 0$, which means

$$\lim_{\epsilon\to 0}\int_s^t \langle (\mathbf{u}^\epsilon)'(\tau), \mathbf{v}^\epsilon(\tau)\rangle_{L^2(U)} d\tau = \int_s^t \langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau.$$

Similarly, we also have

$$\lim_{\epsilon\to 0}\int_s^t \langle (\mathbf{v}^\epsilon)'(\tau), \mathbf{u}^\epsilon(\tau)\rangle_{L^2(U)} d\tau = \int_s^t \langle \mathbf{v}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau.$$

Taking the limit of $\epsilon \to 0$, we have

$$\begin{aligned}
&\langle \tilde{\mathbf{u}}(t), \tilde{\mathbf{v}}(t) \rangle_{L^2(U)} \\
&= \lim_{\epsilon \to 0} \langle \mathbf{u}^\epsilon(t), \mathbf{v}^\epsilon(t) \rangle_{L^2(U)} \\
&= \lim_{\epsilon \to 0} \left( \langle \mathbf{u}^\epsilon(s), \mathbf{v}^\epsilon(s) \rangle_{L^2(U)} + \int_s^t \langle (\mathbf{u}^\epsilon)'(\tau), \mathbf{v}^\epsilon(\tau) \rangle_{L^2(U)} + \langle (\mathbf{v}^\epsilon)'(\tau), \mathbf{u}^\epsilon(\tau) \rangle_{L^2(U)} d\tau \right) \\
&= \lim_{\epsilon \to 0} \langle \mathbf{u}^\epsilon(s), \mathbf{v}^\epsilon(s) \rangle_{L^2(U)} + \lim_{\epsilon \to 0} \int_s^t \langle (\mathbf{u}^\epsilon)'(\tau), \mathbf{v}^\epsilon(\tau) \rangle_{L^2(U)} d\tau + \lim_{\epsilon \to 0} \int_s^t \langle (\mathbf{v}^\epsilon)'(\tau), \mathbf{u}^\epsilon(\tau) \rangle_{L^2(U)} d\tau \\
&= \langle \tilde{\mathbf{u}}(s), \tilde{\mathbf{v}}(s) \rangle_{L^2(U)} + \int_s^t \langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} d\tau + \int_s^t \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} d\tau \\
&= \langle \tilde{\mathbf{u}}(s), \tilde{\mathbf{v}}(s) \rangle_{L^2(U)} + \int_s^t \Big( \langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} + \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} \Big) d\tau.
\end{aligned}$$

In particular, if we take $s = 0$, we have

$$\langle \tilde{\mathbf{u}}(t), \tilde{\mathbf{v}}(t) \rangle_{L^2(U)} = \langle \tilde{\mathbf{u}}(0), \tilde{\mathbf{v}}(0) \rangle_{L^2(U)} + \int_0^t \Big( \langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} + \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} \Big) d\tau.$$

Also, by Holder's Inequality,

$$\begin{aligned}
&\int_0^T \Big| \langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} + \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} \Big| d\tau \\
&\leq \int_0^T \Big| \langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} \Big| d\tau + \int_0^t \Big| \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} \Big| d\tau \\
&\leq \int_0^T \|\mathbf{u}'(\tau)\|_{H^{-1}(U)} \|\mathbf{v}(\tau)\|_{H_0^1(U)} d\tau + \int_0^T \|\mathbf{v}'(\tau)\|_{H^{-1}(U)} \|\mathbf{u}(\tau)\|_{H_0^1(U)} d\tau \\
&\leq \sqrt{\int_0^T \|\mathbf{u}'(\tau)\|_{H^{-1}(U)}^2 d\tau} \sqrt{\int_0^T \|\mathbf{v}(\tau)\|_{H_0^1(U)}^2 d\tau} + \sqrt{\int_0^T \|\mathbf{v}'(\tau)\|_{H^{-1}(U)}^2 d\tau} \sqrt{\int_0^T \|\mathbf{u}(\tau)\|_{H_0^1(U)}^2 d\tau} \\
&= \|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))} \|\mathbf{v}\|_{L^2\left(0,T;H_0^1(U)\right)} + \|\mathbf{v}'\|_{L^2(0,T;H^{-1}(U))} \|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)} \\
&< \infty.
\end{aligned}$$

We have shown that $\langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} + \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)}$ is Lebesgue integrable. By Royden-Fitzpatrick's Theorem, we have that $t \mapsto \langle \tilde{\mathbf{u}}(t), \tilde{\mathbf{v}}(t) \rangle_{L^2(U)}$ is absolutely continuous, and for a.e. $t \in [0, T]$,

$$\frac{d}{dt} \langle \tilde{\mathbf{u}}(t), \tilde{\mathbf{v}}(t) \rangle_{L^2(U)} = \langle \mathbf{u}'(\tau) | \mathbf{v}(\tau) \rangle_{H^{-1}(U), H_0^1(U)} + \langle \mathbf{v}'(\tau) | \mathbf{u}(\tau) \rangle_{H^{-1}(U), H_0^1(U)}.$$

Since $\tilde{\mathbf{u}}(t) = \mathbf{u}(t)$ for a.e. $t \in [0, T]$, we have the result.

3. Take $\mathbf{v} = \mathbf{u}$ in 2.

4. Integrate $\|\tilde{\mathbf{u}}(t)\|_{L^2(U)} = \|\tilde{\mathbf{u}}(s)\|_{L^2(U)} + \int_s^t 2\langle \mathbf{u}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau$ for $0 \le s \le T$, we have

$$\begin{aligned}
T\|\tilde{\mathbf{u}}(t)\|_{L^2(U)}^2 &= \int_0^T \|\mathbf{u}(t)\|_{L^2(U)}^2 ds \\
&= \int_0^T \|\tilde{\mathbf{u}}(s)\|_{L^2(U)}^2 ds + 2\int_0^T \int_s^t \langle \mathbf{u}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)} d\tau ds \\
&\le \int_0^T \|\tilde{\mathbf{u}}(s)\|_{H_0^1(U)}^2 ds + 2\int_0^T \int_s^t \|\mathbf{u}'(\tau)\|_{H^{-1}(U)} \|\mathbf{u}(\tau)\|_{H_0^1(U)} d\tau ds \\
&\le \|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + 2\int_0^T \int_0^T \|\mathbf{u}'(\tau)\|_{H^{-1}(U)} \|\mathbf{u}(\tau)\|_{H_0^1(U)} d\tau ds \\
&= \|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + 2T\int_0^T \|\mathbf{u}'(\tau)\|_{H^{-1}(U)} \|\mathbf{u}(\tau)\|_{H_0^1(U)} d\tau \\
&\le \|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + T\int_0^T \|\mathbf{u}'(\tau)\|_{H^{-1}(U)}^2 + \|\mathbf{u}(\tau)\|_{H_0^1(U)}^2 d\tau \\
&= (T+1)\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + T\|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}^2 \\
\|\tilde{\mathbf{u}}(t)\|_{L^2(U)}^2 &\le \frac{T+1}{T}\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + \|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}^2 \\
&\le C^2\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + C^2\|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}^2 \\
&\le C^2\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}^2 + C^2\|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}^2 + 2C^2\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)}\|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))} \\
&= \Big(C\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)} + C\|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}\Big)^2 \\
\|\tilde{\mathbf{u}}(t)\|_{L^2(U)}^2 &\le C\Big(\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)} + \|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}\Big),
\end{aligned}$$

where we take $C^2 := \max\left(1, \frac{T+1}{T}\right) > 1$, which is independent of $\mathbf{u}$ and $\mathbf{u}'$.
Since this holds for any $t \in [0,T]$, we have

$$\sup_{t\in[0,T]} \|\tilde{\mathbf{u}}(t)\|_{L^2(U)} \le C\Big(\|\mathbf{u}\|_{L^2\left(0,T;H_0^1(U)\right)} + \|\mathbf{u}'\|_{L^2(0,T;H^{-1}(U))}\Big).$$

□

## 4.2 Second Order Parabolic Equations

**Definition 4.6.** Let $U \subseteq \mathbb{R}^n$ be open and bounded, we define $U_T := U \times (0,T]$ for $T > 0$.

**Definition 4.7.** An **initial boundary value problem** is: given $f : U_T \to \mathbb{R}, g : U \to \mathbb{R}$, we want to find $u(x,t) : \bar{U_T} \to \mathbb{R}$, such that

$$\begin{cases} u_t + Lu = f & \text{in } U_T \\ u = 0 & \text{on } \partial U \times [0,T] \\ u = g & \text{on } U \times \{t = 0\} \end{cases},$$

where

$$Lu := -\sum_{i,j=1}^n \partial_j(a^{ij}(\cdot,t)\partial_i u) + \sum_{i=1}^n b^i(\cdot,t)\partial_i u + c(\cdot,t)u$$

for some $a^{ij}, b^i, c : U_T \to \mathbb{R}$.
We say that the partial differential operator $\partial_t + L$ is an **symmetric (uniformly) parabolic second order differential operator** if $a^{ij} = a^{ji}$, and $\exists \theta > 0,$ such that

$$\sum_{i,j=1}^n a^{ij}(x,t)\xi_i\xi_j \ge \theta\|\xi\|_2^2, \ \forall (x,t) \in U_T, \xi \in \mathbb{R}^n.$$

**Definition 4.8.** The **parabolic assumptions** are:

1. $U \subseteq \mathbb{R}^n$ is bounded and open
2. $T > 0$
3. $a^{ij}, b^i, c \in L^\infty(U_T)$
4. $f \in L^2(U_T), g \in L^2(U)$
5. $\partial_t + L$ is a symmetric (uniformly) parabolic second order differential operator.

**Definition 4.9.** Given a function $u : U_T \to \mathbb{R}$, we want to consider $\mathbf{u} : t \mapsto u(\cdot, t)$, for any $t \in [0, T]$.

**Proposition 4.17.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, $T > 0$, then $f \in L^2(U_T) \iff \boldsymbol{f} \in L^2\big(0, T; L^2(U)\big)$.*

*Proof.* We have

$$\begin{aligned}\|\mathbf{f}\|^2_{L^2(0,T;L^2(U))} &= \int_0^T \|\mathbf{f}(t)\|^2_{L^2(U)} dt \\ &= \int_0^T \int_U |f(x,t)|^2 dx dt \\ &= \|f\|_{L^2(U_T)}.\end{aligned}$$

□

**Definition 4.10.** $\mathbf{u} \in L^2\big(0, T; H^1_0(U)\big)$, identified with its continuous representative $\tilde{\mathbf{u}} \in C\big([0, T]; L^2(U)\big)$ as in 4.16.1, with the time weak derivative $\mathbf{u}' \in L^2\big(0, T; H^{-1}(U)\big)$, is a **weak solution** of the IBVP if

$$\begin{gathered}\forall v \in H^1_0(U),\ \langle \mathbf{u}'(t) | v \rangle_{H^{-1}(U), H^1_0(U)} + B[\mathbf{u}(t), v; t] = \langle \mathbf{f}(t), v \rangle_{L^2(U)},\ \text{a.e. } t \in [0, T], \\ \mathbf{u}(0) = g,\end{gathered}$$

where bilinear form associated to the above problem is

$$\forall w, v \in H^1_0(U),\ B[w, v; t] := \int_U \left( \sum_{i,j=1}^n a^{ij}(\cdot, t) \partial_i w \partial_j v + \sum_{i=1}^n b^i(\cdot, t) \partial_i w v + c(\cdot, t) w v \right) dx.$$

## 4.3 Galerkin Method

**Definition 4.11.** Let $(w_k)_{k=1}^\infty$ be an orthogonal basis of $H^1_0(U)$, and also an orthonormal basis of $L^2(U)$. For $m \in \mathbb{N}_+$, we define $V_m := \operatorname{Span}\left(\{w_j\}_{j=1}^m\right) \subset H^1_0(U)$ be a subspace. A function $\mathbf{u}_m := t \mapsto \sum_{k=1}^m d_m^k(t) w_k$ is a **weak solution of the problem in** $V_m$ if $\forall v \in V_m$,

$$\begin{gathered}\left\langle \sum_{k=1}^m {d_m^k}'(t) w_k, v \right\rangle_{L^2(U)} + B[\mathbf{u}_m(t), v; t] = \langle \mathbf{f}(t), v \rangle_{L^2(U)},\ \text{for a.e. } t \in [0, T], \\ \langle \mathbf{u}_m(0), v \rangle_{L^2(U)} = \langle g, v \rangle_{L^2(U)}.\end{gathered}$$

**Definition 4.12.** We define the **ODE system associated to the problem** to be: $\forall j \in [m]$,

1. $d_m^j : [0, T] \to \mathbb{R}$ is absolutely continuous.
2. For a.e. $t \in [0, T]$, ${d_m^j}'(t) = -\sum_{k=1}^m e_k^j(t) d_m^k(t) + f^j(t)$
3. $d_m^j(0) = \langle g, w_j \rangle_{L^2(U)}$,

where $e_k^j(t) := B[w_k, w_j; t]$, $f^j(t) := \langle \mathbf{f}(t), w_j \rangle_{L^2(U)}$.

**Proposition 4.18.** *$\boldsymbol{u}_m(t) = \sum_{k=1}^m d_m^k(t) w_k$ is a weak solution in $V_m$ if and only if $\vec{d}_m$ is a solution to the ODE system.*

*Proof.* Since $(w_k)_{k=1}^\infty$ is an orthonormal basis of $V_m$ in $\langle \cdot, \cdot \rangle_{L^2(U)}$, we have

$$\begin{aligned}
\left\langle \sum_{k=1}^m {d_m^k}'(t) w_k, v \right\rangle_{L^2(U)} + B[\mathbf{u}_m(t), v; t] &= \langle \mathbf{f}(t), v \rangle, && \forall v \in V_m \\
&\iff \\
\left\langle \sum_{k=1}^m {d_m^k}'(t) w_k, v \right\rangle_{L^2(U)} + B\left[\sum_{k=1}^m d_m^k(t) w_k, v; t\right] &= \langle \mathbf{f}(t), v \rangle, && \forall v \in V_m \\
&\iff \\
\left\langle \sum_{k=1}^m {d_m^k}'(t) w_k, w_j \right\rangle_{L^2(U)} + B\left[\sum_{k=1}^m d_m^k(t) w_k, w_j; t\right] &= \langle \mathbf{f}(t), w_j \rangle, && \forall j \in [m] \\
&\iff \\
\sum_{k=1}^m {d_m^k}'(t) \langle w_k, w_j \rangle_{L^2(U)} + \sum_{k=1}^m d_m^k(t) B[w_k, w_j; t] &= \langle \mathbf{f}(t), w_j \rangle, && \forall j \in [m] \\
&\iff \\
\sum_{k=1}^m {d_m^k}'(t) \delta_k^j + \sum_{k=1}^m d_m^k(t) e_k^j(t) &= f^j(t), && \forall j \in [m] \\
&\iff \\
{d_m^j}'(t) + \sum_{k=1}^m d_m^k(t) e_k^j(t) &= f^j(t), && \forall j \in [m].
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\langle \mathbf{u}_m(0), v \rangle_{L^2(U)} &= \langle g, v \rangle_{L^2(U)}, && v \in V_m \\
&\iff \\
\langle \mathbf{u}_m(0), w_j \rangle_{L^2(U)} &= \langle g, w_j \rangle_{L^2(U)}, && \forall j \in [m] \\
&\iff \\
\left\langle \sum_{k=1}^m d_m^k(0) w_k, w_j \right\rangle_{L^2(U)} &= \langle g, w_j \rangle_{L^2(U)}, && \forall j \in [m] \\
&\iff \\
\sum_{k=1}^m d_m^k(0) \langle w_k, w_j \rangle_{L^2(U)} &= \langle g, w_j \rangle_{L^2(U)}, && \forall j \in [m] \\
&\iff \\
\sum_{k=1}^m d_m^k(0) \delta_k^j &= \langle g, w_j \rangle_{L^2(U)}, && \forall j \in [m] \\
&\iff \\
d_m^j(0) &= \langle g, w_j \rangle_{L^2(U)}, && \forall j \in [m].
\end{aligned}$$

□

**Theorem 4.19.** *Since $f^i, e_j^k$ are locally integrable, there is a unique absolutely continuous solution $\vec{d}_m$ to the ODE system.*

**Corollary 4.20.** *For each $m \in \mathbb{N}_+$, there is a unique weak solution $\boldsymbol{u}_m$ of the form $t \mapsto \sum_{k=1}^m d_m^k(t) w_k$ of the problem in $V_m$.*

**Proposition 4.21.** *The weak solution $\boldsymbol{u}_m$ also satisfies $\forall v \in V_m$,*

$$\langle \boldsymbol{u}_m'(t), v \rangle_{L^2(U)} + B[\boldsymbol{u}_m(t), v; t] = \langle \boldsymbol{f}(t), v \rangle_{L^2(U)}, \text{ for a.e. } t \in [0,T],$$
$$\langle \boldsymbol{u}_m(0), v \rangle_{L^2(U)} = \langle g, v \rangle_{L^2(U)}.$$

*In addition,*

$$\|\boldsymbol{u}_m(0)\|_{L^2(U)} \leq \|g\|_{L^2(U)},$$

*and*

$$\lim_{m \to \infty} \boldsymbol{u}_m(0) = g$$

*in $L^2(U)$.*

*Proof.* Consider any $\phi \in C_c^\infty(0,T)$. By linearity of the Bochner integral, we have

$$\begin{aligned}
\int_0^T \phi'(t) \mathbf{u}_m(t) dt &= \int_0^T \left( \phi'(t) \sum_{k=1}^m d_m^k(t) w_k \right) dt \\
&= \sum_{k=1}^m \left( \int_0^T \phi'(t) d_m^k(t) w_k dt \right) \\
&= \sum_{k=1}^m \left( \int_0^T \phi'(t) d_m^k(t) dt \right) w_k \\
&= \sum_{k=1}^m \left( \int_0^T \phi(t) d_m^{k\,\prime}(t) dt \right) w_k \\
&= \int_0^T \phi(t) \left( \sum_{k=1}^m d_m^{k\,\prime}(t) w_k \right) dt.
\end{aligned}$$

Thus it has a weak derivative $\mathbf{u}_m'(t) = \sum_{k=1}^m d_m^{k\,\prime}(t) w_k$.
We then plug that in the definition of weak solution of the problem in $V_m$.
On the other hand, since $\mathbf{u}_m(0) \in V_m$, we have that

$$\begin{aligned}
\|\mathbf{u}_m(0)\|_{L^2(U)}^2 &= \langle \mathbf{u}_m(0), \mathbf{u}_m(0) \rangle_{L^2(U)} \\
&= \langle g, \mathbf{u}_m(0) \rangle_{L^2(U)} \\
&\leq \|g\|_{L^2(U)} \|\mathbf{u}_m(0)\|_{L^2(U)}.
\end{aligned}$$

Since $\|\mathbf{u}_m(0)\|_{L^2(U)} \geq 0$, we have $\|\mathbf{u}_m(0)\|_{L^2(U)} \leq \|g\|_{L^2(U)}$.
Also, since $(w_k)_{k=1}^\infty$ is an orthonormal basis of $L^2(U)$, we have

$$\begin{aligned}
\lim_{m \to \infty} \mathbf{u}_m(0) &= \lim_{m \to \infty} \sum_{k=1}^m d_m^k(t) w_k \\
&= \lim_{m \to \infty} \sum_{k=1}^m \langle g, w_j \rangle_{L^2(U)} w_k \\
&= g.
\end{aligned}$$

□

**Proposition 4.22.** *The weak solution $\boldsymbol{u}_m \in L^2\big(0,T; H_0^1(U)\big)$.*

*Proof.*

$$\begin{aligned}
\int_0^T \|\mathbf{u}_m\|^2_{H_0^1(U)} dt &= \int_0^T \left\| \sum_{k=1}^m d_m^k(t) w_k \right\|^2_{H_0^1(U)} dt \\
&= \int_0^T \left\langle \sum_{k=1}^m d_m^k(t) w_k, \sum_{j=1}^m d_m^j(t) w_j \right\rangle_{H_0^1(U)} dt \\
&= \int_0^T \sum_{k=1}^m \sum_{j=1}^m d_m^k(t) d_m^j(t) \langle w_k, w_j \rangle_{H_0^1(U)} dt \\
&= \int_0^T \sum_{k=1}^m {d_m^k}^2(t) \|w_k\|^2_{H_0^1(U)} dt \\
&= \sum_{k=1}^m \|w_k\|^2_{H_0^1(U)} \int_0^T {d_m^k}^2(t) dt \\
&= \sum_{k=1}^m \|w_k\|^2_{H_0^1(U)} \left\|d_m^k\right\|^2_{L^2(0,T)} \\
&\leq \infty,
\end{aligned}$$

since each $w_k \in H_0^1(U)$, and each $d_m^k$ is absolutely continuous (thus continuous, thus in $L^2(0,T)$). □

**Proposition 4.23.** *The weak solution* $\boldsymbol{u}_m \in C\big([0,T]; L^2(U)\big)$.

*Proof.* Given any $\epsilon > 0$.
For any $1 \leq k \leq m$, since $d_m^k$ is is absolutely continuous (thus continuous), we can find $\delta_k > 0$, such that

$$\forall s,t \in [0,T],\ |t-s| < \delta_k \implies \left|d_m^k(t) - d_m^k(s)\right| < \frac{\epsilon}{\sqrt{m}}.$$

Now take $\delta := \min_{k\in[m]} \delta_k > 0$, we have that $\forall s,t \in [0,T],$ such that $|t-s| < \delta_k$,

$$\begin{aligned}
\|\mathbf{u}_m(t) - \mathbf{u}_m(s)\|^2_{L^2(U)} &= \left\| \sum_{k=1}^m {d_m^k}'(t) w_k - \sum_{k=1}^m {d_m^k}'(s) w_k \right\|^2_{L^2(U)} \\
&= \left\| \sum_{k=1}^m \Big({d_m^k}'(t) - {d_m^k}'(s)\Big) w_k \right\|^2_{L^2(U)} \\
&= \sum_{k=1}^m \Big({d_m^k}'(t) - {d_m^k}'(s)\Big)^2 \\
&< \sum_{k=1}^m \frac{\epsilon^2}{m} \\
&= \epsilon^2,
\end{aligned}$$

since $(w_k)_{k=1}^\infty$ is an orthornormal basis for $L^2(U)^2$.
Thus $\|\mathbf{u}_m(t) - \mathbf{u}_m(s)\|_{L^2(U)} < \epsilon$ and since this works for any $\epsilon > 0$, we have that $\mathbf{u}_m \in C\big([0,T]; L^2(U)\big)$. □

Notice that the above two propositions says that $\mathbf{u}_m$ is the representative $\tilde{u_m}$ as in 4.16, and we thus have the following result:

**Corollary 4.24.** *The weak solution* $\boldsymbol{u}_m$ *satisfies that the mapping* $t \mapsto \|\boldsymbol{u}_m(t)\|^2_{L^2(U)}$ *is absolutely continuous, and for a.e.* $t \in [0,T]$*, we have*

$$\frac{d}{dt}\|\boldsymbol{u}_m(t)\|^2_{L^2(U)} = 2\langle \boldsymbol{u}_m'(t) | \boldsymbol{u}_m(t)\rangle_{H^{-1}(U), H_0^1(U)} = 2\langle \boldsymbol{u}_m'(t), \boldsymbol{u}_m(t)\rangle_{L^2(U)}.$$

**Theorem 4.25** (Gronwall's inequality)**.** *Let $t_0 \in \mathbb{R}$, and let $J = [a, b] \ni t_0$ be a bounded close interval. Suppose $\eta : J \to \mathbb{R}_+$ is nonnegative and absolutely continuous, and $\phi, \psi : J \to \mathbb{R}_+$ are both nonnegative summable functions. Suppose*

$$\eta'(t) \leq \phi(t)\eta(t) + \psi(t) \text{ a.e. } t \in J,$$

*then*

$$\eta(t) \leq \left(\eta(t_0) + \left|\int_{t_0}^{t} \psi(s)ds\right|\right) \exp\left|\int_{t_0}^{t} \phi(s)ds\right|, \ \forall t \in J.$$

*Proof.* For $t \in [t_0, b]$, let $F(t) := \eta(t_0) + \int_{t_0}^{t} \psi(s) + \phi(s)\eta(s)ds$. We have that $F$ is absolutely continuous, and that

$$\begin{aligned}
\eta(t) &= \eta(t_0) + \int_{t_0}^{t} \eta'(s)ds \\
&\leq \int_{t_0}^{t} \phi(s)\eta(s) + \psi(s)ds \\
&= F(t). \\
F'(t) &= \psi(t) + \phi(t)\eta(t) \\
&\leq \psi(t) + \phi(t)F(t).
\end{aligned}$$

Let $G(t) := \exp\left(-\int_{t_0}^{t} \phi(s)ds\right) \leq 1$, we have

$$\begin{aligned}
G(t)F'(t) - G(t)\phi(t)F(t) &\leq G(t)\psi(t) \\
\frac{d}{dt}(G(t)F(t)) &\leq \psi(t).
\end{aligned}$$

Since $G(t_0) = 1$, and $F(t_0) = \eta(t_0)$, integrating both sides gives

$$\begin{aligned}
G(t)F(t) &= G(t_0)F(t_0) + \int_{t_0}^{t} \frac{d}{ds}(G(s)F(s))ds \\
&\leq \eta(t_0) + \int_{t_0}^{t} \psi(s)ds \\
\eta(t) &\leq F(t) \\
&\leq \left(\eta(t_0) + \int_{t_0}^{t} \psi(s)ds\right) \exp\left(\int_{t_0}^{t} \phi(s)ds\right).
\end{aligned}$$

For $t \in [a, t_0]$, let $F(t) := \eta(t_0) - \int_{t_0}^{t} \psi(s) + \phi(s)\eta(s)ds$, and apply the analogous argument. □

**Theorem 4.26** (Energy Estimate)**.** *Let $U \subseteq \mathbb{R}^n$ be bounded and open, $T > 0$, $a^{ij}, b^i, c \in L^\infty(U_T)$, and $\partial_t + L$ be a symmetric (uniformly) parabolic second order differential operator. There exists $C > 0$ that only depends on $U, T, L$, such that $\forall f \in L^2(U_T),\ g \in L^2(U),\ m \in \mathbb{N}_+$,*

$$\sup_{0 \leq t \leq T} \|\boldsymbol{u}_m(t)\|_{L^2(U)} + \|\boldsymbol{u}_m\|_{L^2\left(0,T;H_0^1(U)\right)} + \|\boldsymbol{u}_m'\|_{L^2(0,T;H^{-1}(U))} \leq C\Big(\|\boldsymbol{f}\|_{L^2(0,T;L^2(U))} + \|g\|_{L^2(U)}\Big),$$

*where $\boldsymbol{u}_m$ are the weak solutions in $V_m$ as in above.*

*Proof.* We will bound each term on the left hand side.

1. Consider any $m \in \mathbb{N}_+$, we have that the $\mathbf{u}_m$ satisfies $\forall v \in V_m$,

$$\begin{aligned}
\langle \mathbf{u}_m'(t), v\rangle_{L^2(U)} + B[\mathbf{u}_m(t), v; t] &= \langle \mathbf{f}(t), v\rangle_{L^2(U)}, \text{ for a.e. } t \in [0, T], \\
\langle \mathbf{u}_m(0), v\rangle_{L^2(U)} &= \langle g, v\rangle_{L^2(U)},
\end{aligned}$$

In particular, $\mathbf{u}_m(t) = \sum_{k=1}^m d_m^k(t) w_k \in V_m$.
Thus for a.e. $t \in [0,T]$, we have

$$\langle \mathbf{u}_m'(t), \mathbf{u}_m(t) \rangle_{L^2(U)} + B[\mathbf{u}_m(t), \mathbf{u}_m(t); t] = \langle \mathbf{f}(t), \mathbf{u}_m(t) \rangle_{L^2(U)}.$$

By a similar proof as in 3.5, there exists constants $\beta > 0, \gamma \geq 0$ that only depends on $U$ and the coefficients of $L$, such that $\forall u \in H_0^1(U)$, and for a.e. $t \in [0,T]$,

$$\beta \|u\|_{H^1(U)}^2 \leq B[u,u;t] + \gamma \|u\|_{L^2(U)}^2.$$

We thus have

$$\begin{aligned}
\langle \mathbf{u}_m'(t), \mathbf{u}_m(t) \rangle_{L^2(U)} + B[\mathbf{u}_m(t), \mathbf{u}_m(t); t] &\geq \langle \mathbf{u}_m'(t), \mathbf{u}_m(t) \rangle_{L^2(U)} + \beta \|\mathbf{u}_m(t)\|_{H^1(U)}^2 - \gamma \|\mathbf{u}_m(t)\|_{L^2(U)}^2 \\
&= \frac{1}{2} \frac{d}{dt} \|\mathbf{u}_m(t)\|_{L^2(U)}^2 + \beta \|\mathbf{u}_m(t)\|_{H^1(U)}^2 - \gamma \|\mathbf{u}_m(t)\|_{L^2(U)}^2 \\
\langle \mathbf{f}(t), \mathbf{u}_m(t) \rangle_{L^2(U)} &\leq \|\mathbf{f}(t)\|_{L^2(U)} \|\mathbf{u}_m(t)\|_{L^2(U)} \\
&\leq \frac{1}{2} \|\mathbf{f}(t)\|_{L^2(U)}^2 + \frac{1}{2} \|\mathbf{u}_m(t)\|_{L^2(U)}^2 \\
\frac{d}{dt} \|\mathbf{u}_m(t)\|_{L^2(U)}^2 + 2\beta \|\mathbf{u}_m(t)\|_{H^1(U)}^2 &\leq \|\mathbf{f}(t)\|_{L^2(U)}^2 + \|\mathbf{u}_m(t)\|_{L^2(U)}^2 + 2\gamma \|\mathbf{u}_m(t)\|_{L^2(U)}^2 \\
&= \|\mathbf{f}(t)\|_{L^2(U)}^2 + (1+2\gamma) \|\mathbf{u}_m(t)\|_{L^2(U)}^2.
\end{aligned}$$

Notice that this inequality hold for any $f \in L^2(U_T),\ g \in L^2(U),\ m \in \mathbb{N}_+$, and the corresponding weak solutions $\mathbf{u}_m$ in $V_m$.

2. Since $2\beta \|\mathbf{u}_m(t)\|_{H^1(U)}^2 \geq 0$, we have

$$\frac{d}{dt} \|\mathbf{u}_m(t)\|_{L^2(U)}^2 \leq \|\mathbf{f}(t)\|_{L^2(U)}^2 + (1+2\gamma) \|\mathbf{u}_m(t)\|_{L^2(U)}^2.$$

Take $\eta(t) := \|\mathbf{u}_m(t)\|_{L^2(U)}^2$, which is nonnegative and absolutely continuous.
Also, take $\psi(t) := \|\mathbf{f}(t)\|_{L^2(U)}^2,\ \phi(t) := 1 + 2\gamma$, which are both nonnegative and summable.
By Gronwall's inequality, we have that $\forall t \in [0,T]$,

$$\begin{aligned}
\|\mathbf{u}_m(t)\|_{L^2(U)}^2 &\leq \exp\left( \int_0^t (1+2\gamma) ds \right) \left( \|\mathbf{u}_m(0)\|_{L^2(U)}^2 + \int_0^t \|\mathbf{f}(s)\|_{L^2(U)}^2 ds \right) \\
&= \exp(t(1+2\gamma)) \left( \|\mathbf{u}_m(0)\|_{L^2(U)}^2 + \int_0^t \|\mathbf{f}(s)\|_{L^2(U)}^2 ds \right) \\
&\leq \exp(T(1+2\gamma)) \left( \|g\|_{L^2(U)}^2 + \int_0^T \|\mathbf{f}(s)\|_{L^2(U)}^2 ds \right) \\
&= C_1 \Big( \|g\|_{L^2(U)}^2 + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}^2 \Big) \\
&\leq C_1 \Big( \|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))} \Big)^2,
\end{aligned}$$

where we take $C_1 := \exp(T(1+2\gamma)) > 0$ that only depends on $T, \gamma$.
We thus have shown that $\forall t \in [0,T], \|\mathbf{u}_m(t)\|_{L^2(U)} \leq \sqrt{C_1} \Big( \|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))} \Big)$.
Thus,

$$\sup_{0 \leq t \leq T} \|\mathbf{u}_m(t)\|_{L^2(U)} \leq \sqrt{C_1} \Big( \|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))} \Big),$$

which bounds the first term in what we want.

3. From step 1, we have that for a.e. $t \in [0,T]$,

$$\frac{d}{dt}\|\mathbf{u}_m(t)\|^2_{L^2(U)} + 2\beta\|\mathbf{u}_m(t)\|^2_{H^1(U)} \le \|\mathbf{f}(t)\|^2_{L^2(U)} + (1+2\gamma)\|\mathbf{u}_m(t)\|^2_{L^2(U)}.$$

Integrating over $[0,T]$ gives

$$\int_0^T \frac{d}{dt}\|\mathbf{u}_m(t)\|^2_{L^2(U)}dt + 2\beta\int_0^T \|\mathbf{u}_m(t)\|^2_{H^1(U)}dt \le \int_0^T \|\mathbf{f}(t)\|^2_{L^2(U)}dt + (2\gamma+1)\int_0^T \|\mathbf{u}_m(t)\|^2_{L^2(U)}dt$$

$$\|\mathbf{u}_m(T)\|^2_{L^2(U)} - \|\mathbf{u}_m(0)\|^2_{L^2(U)} + 2\beta\|\mathbf{u}_m\|^2_{L^2\left(0,T;H^1_0(U)\right)} \le \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))} + (2\gamma+1)\int_0^T \|\mathbf{u}_m(t)\|^2_{L^2(U)}dt.$$

Notice that $\|\mathbf{u}_m(T)\|^2_{L^2(U)} \ge 0$, and in step 2, we have shown that for a.e. $t \in [0,T]$,

$$\|\mathbf{u}_m(t)\|^2_{L^2(U)} \le C_1\Big(\|g\|^2_{L^2(U)} + \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))}\Big).$$

$$\begin{aligned}
2\beta\|\mathbf{u}_m\|^2_{L^2\left(0,T;H^1_0(U)\right)} &\le \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))} + (2\gamma+1)\int_0^T \|\mathbf{u}_m(t)\|^2_{L^2(U)}dt + \|\mathbf{u}_m(0)\|^2_{L^2(U)} \\
2\beta\|\mathbf{u}_m\|^2_{L^2(0,T;H^1(U))} &\le \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))} + (2\gamma+1)\int_0^T C_1\Big(\|g\|^2_{L^2(U)} + \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))}\Big)dt + \|g\|^2_{L^2(U)} \\
&= \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))} + (2\gamma+1)TC_1\Big(\|g\|^2_{L^2(U)} + \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))}\Big) + \|g\|^2_{L^2(U)} \\
&= ((2\gamma+1)TC_1+1)\Big(\|g\|^2_{L^2(U)} + \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))}\Big) \\
\|\mathbf{u}_m\|^2_{L^2\left(0,T;H^1_0(U)\right)} &\le C_2\Big(\|g\|^2_{L^2(U)} + \|\mathbf{f}\|^2_{L^2(0,T;L^2(U))}\Big) \\
&\le C_2\Big(\|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}\Big)^2,
\end{aligned}$$

where $C_2 := \frac{(2\gamma+1)TC_1+1}{\beta} > 0$ only depends on $\beta, \gamma, T$.
We thus have bounded the second term

$$\|\mathbf{u}_m\|_{L^2\left(0,T;H^1_0(U)\right)} \le \sqrt{C_2}\Big(\|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}\Big).$$

4. Now fix any function $v \in H^1_0(U)$ with $\|v\|_{H^1_0(U)} = 1$, write $v = \sum_{j=1}^\infty \hat{v}_j w_j = v_1 + v_2$, where $v_1 := \sum_{j=1}^m \hat{v}_j w_j \in V_m, v_2 := \sum_{j=m+1}^\infty \hat{v}_j w_j \in V_m^\perp$ is the unique decomposition of $v$ in $H^1_0(U) = V_m \oplus V_m^\perp$.
Notice that $\|v\|^2_{H^1(U)} = \|v_1\|^2_{H^1(U)} + \|v_2\|^2_{H^1(U)}$.
Thus $\|v_1\|_{H^1(U)} \le 1$.
Since $v_1 \in V_m$, we have for a.e. $t \in [0,T]$, $\langle \mathbf{u}_m'(t), v_1\rangle_{L^2(U)} + B[\mathbf{u}_m(t), v_1; t] = \langle \mathbf{f}(t), v_1\rangle_{L^2(U)}$.
Now since $\mathbf{u}_m'(t) \in V_m$, we have $\langle \mathbf{u}_m'(t), v_2\rangle_{L^2(U)} = 0$, so

$$\begin{aligned}
\langle \mathbf{u}_m'(t)|v\rangle_{H^{-1}(U),H^1_0(U)} &= \langle \mathbf{u}_m'(t), v\rangle_{L^2(U)} \\
&= \langle \mathbf{u}_m'(t), v_1\rangle_{L^2(U)} + \langle \mathbf{u}_m'(t), v_2\rangle_{L^2(U)} \\
&= \langle \mathbf{u}_m'(t), v_1\rangle_{L^2(U)} \\
&= \langle \mathbf{f}(t), v_1\rangle_{L^2(U)} - B[\mathbf{u}_m(t), v_1; t].
\end{aligned}$$

Again by a similar proof as in 3.5, there exists constants $\alpha > 0$ that only depends on $U$ and the coefficients of $L$, such that $\forall u, v \in H^1_0(U)$, and for a.e. $t \in [0,T]$,

$$|B[u,v;t]| \le \alpha\|u\|_{H^1(U)}\|v\|_{H^1(U)}.$$

We thus have

$$\begin{aligned}\langle \mathbf{u}_m'(t)|v\rangle_{H^{-1}(U),H_0^1(U)} &= \langle \mathbf{f}(t), v_1\rangle_{L^2(U)} - B[\mathbf{u}_m(t), v_1; t]\\ &\le \|\mathbf{f}(t)\|_{L^2(U)}\|v_1\|_{L^2(U)} + \alpha\|\mathbf{u}_m(t)\|_{H^1(U)}\|v_1\|_{H^1(U)}\\ &\le \|\mathbf{f}(t)\|_{L^2(U)}\|v_1\|_{H^1(U)} + \alpha\|\mathbf{u}_m(t)\|_{H^1(U)}\|v_1\|_{H^1(U)}\\ &\le \|\mathbf{f}(t)\|_{L^2(U)} + \alpha\|\mathbf{u}_m(t)\|_{H^1(U)}.\end{aligned}$$

Since this holds for any $v \in H_0^1(U)$ with $\|v\|_{H_0^1(U)} = 1$, we have

$$\|\mathbf{u}_m'(t)\|_{H^{-1}(U)} = \sup_{v\in H_0^1(U) \text{ such that } \|v\|_{H_0^1(U)}=1} \langle \mathbf{u}_m'(t)|v\rangle_{H^{-1}(U),H_0^1(U)} \le \|\mathbf{f}(t)\|_{L^2(U)} + \alpha\|\mathbf{u}_m(t)\|_{H^1(U)}.$$

Squaring and integrating this over $[0,T]$, we have

$$\begin{aligned}\int_0^T \|\mathbf{u}_m'(t)\|_{H^{-1}(U)}^2 dt &\le \int_0^T \Big(\|\mathbf{f}(t)\|_{L^2(U)} + \alpha\|\mathbf{u}_m(t)\|_{H^1(U)}\Big)^2 dt\\ \|\mathbf{u}_m'\|_{L^2(0,T;H^{-1}(U))}^2 &\le \int_0^T 2\Big(\|\mathbf{f}(t)\|_{L^2(U)}^2 + \alpha^2\|\mathbf{u}_m(t)\|_{H^1(U)}^2\Big)dt\\ &\le 2\int_0^T \|\mathbf{f}(t)\|_{L^2(U)}^2 dt + 2\alpha^2\int_0^T \|\mathbf{u}_m(t)\|_{H^1(U)}^2 dt\\ &= 2\|\mathbf{f}\|_{L^2(0,T;L^2(U))}^2 + 2\alpha^2\|\mathbf{u}_m\|_{L^2\left(0,T;H_0^1(U)\right)}^2\\ &\le 2\|\mathbf{f}\|_{L^2(0,T;L^2(U))}^2 + 2\alpha^2 C_2\Big(\|g\|_{L^2(U)}^2 + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}^2\Big)\\ &\le C_3\Big(\|g\|_{L^2(U)}^2 + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}^2\Big)\\ &\le C_3\Big(\|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}\Big)^2,\end{aligned}$$

where we take $C_3 := 2\alpha^2 C_2 + 1 > 0$ that only depends on $\beta, \gamma, \alpha, T$.
We thus have bounded the third term $\|\mathbf{u}_m'\|_{L^2(0,T;H^{-1}(U))} \le \sqrt{C_3}\Big(\|g\|_{L^2(U)} + \|\mathbf{f}\|_{L^2(0,T;L^2(U))}\Big)$.

Now let us take $C := \sqrt{\max\{C_1, C_2, C_3\}} > 0$, which only depends on $U, T, L$, we have that

$$\sup_{0\le t\le T}\|\mathbf{u}_m(t)\|_{L^2(U)} + \|\mathbf{u}_m\|_{L^2\left(0,T;H_0^1(U)\right)} + \|\mathbf{u}_m'\|_{L^2(0,T;H^{-1}(U))} \le C\Big(\|\mathbf{f}\|_{L^2(0,T;L^2(U))} + \|g\|_{L^2(U)}\Big).$$

□

**Theorem 4.27.** *There is a weak solution to the IBVP, namely, $\exists \boldsymbol{u} \in L^2\big(0,T;H_0^1(H)\big)$, identified with its continuous representative $\tilde{\boldsymbol{u}} \in C\big([0,T];L^2(U)\big)$, such that*

$$\begin{aligned}\langle \boldsymbol{u}'(t)|v\rangle + B[\boldsymbol{u}(t), v; t] &= \langle \boldsymbol{f}(t), v\rangle_{L^2(U)},\ \forall v \in H_0^1(U),\ \text{a.e. } t \in [0,T]\\ \boldsymbol{u}(0) &= g\end{aligned}$$

*Proof.* By energy estimate, we have that $(\mathbf{u}_m)_{m=1}^\infty$ is bounded in $L^2\big(0,T;H_0^1(U)\big)$.
By 1.28, there is a subsequence $(\mathbf{u}_{m_j})_{j=1}^\infty$ and $\mathbf{u} \in L^2\big(0,T;H_0^1(U)\big)$ such that $\mathbf{u}_{m_j} \rightharpoonup \mathbf{u}$. WLOG, we will consider its continuous representative $\mathbf{u} \in C\big(0,T;L^2(U)\big)$ by 4.16.1.
Similarly, $(\mathbf{u}_m')_{m=1}^\infty$ is bounded in $L^2\big(0,T;H^{-1}(U)\big)$, so is $(\mathbf{u}_{m_j}')_{j=1}^\infty$. Thus there is a subsequence $(\mathbf{u}_{m_{j_l}}')_{l=1}^\infty$ and $\mathbf{w} \in L^2\big(0,T;H^{-1}(U)\big)$ such that $\mathbf{u}_{m_{j_l}}' \rightharpoonup \mathbf{w}$.
Since $\mathbf{u}_{m_j} \rightharpoonup \mathbf{u}$, we must have $\mathbf{u}_{m_{j_l}} \rightharpoonup \mathbf{u}$ as well. By A5Q3, $\mathbf{w} = \mathbf{u}'$.
Now we would like to show that $\mathbf{u}$ is indeed a a weak solution to the IBVP.
Consider $\mathbf{v} \in C^1\big([0,T];H_0^1(U)\big)$ of the form $\mathbf{v}(t) = \sum_{k=1}^N d^k(t)w_k$, where $N > 0$ is an integer, $(d^k(t))_{k=1}^N$ are

smooth functions, and $(w_k)_{k=1}^{\infty}$ be a basis as before.
We can show that these $\mathbf{v}$ are dense in $L^2\big(0,T;H_0^1(U)\big)$.
For any such $\mathbf{v}$, if we choose any $m \geq N$, we have the weak solution $\vec{u}_m$ in $V_m$ satisfies

$$\langle \mathbf{u}'_m(t)|w_k\rangle_{H^{-1}(U),H_0^1(U)} + B[\mathbf{u}_m(t),w_k;t] = \langle \mathbf{f}(t),w_k\rangle_{L^2(U)},\ \forall k\in[m],\ \text{for a.e. } t\in[0,T].$$

Multiplying by $d^k(t)$ and summing over $k\in[N]$, we have that

$$\langle \mathbf{u}'_m(t)|\mathbf{v}(t)\rangle + B[\mathbf{u}_m(t),\mathbf{v}(t);t] = \langle \mathbf{f}(t),\mathbf{v}(t)\rangle_{L^2(U)},\ \text{for a.e. } t\in[0,T].$$

Integrating over $t\in[0,T]$, we have

$$\int_0^T \langle \mathbf{u}'_m(t)|\mathbf{v}(t)\rangle_{H^{-1}(U),H_0^1(U)}dt + \int_0^T B[\mathbf{u}_m(t),\mathbf{v}(t);t]dt = \int_0^T \langle \mathbf{f}(t),\mathbf{v}(t)\rangle_{L^2(U)}dt.$$

Since $\mathbf{v}\in C^1\big([0,T];H_0^1(U)\big)\subset L^2\big(0,T;H_0^1(U)\big)\cong\big(L^2\big(0,T;H^{-1}(U)\big)\big)^*$, and $\mathbf{u}'_{m_{j_l}}\rightharpoonup\mathbf{v}$, we have

$$\int_0^T \Big\langle \mathbf{u}'_{m_{j_l}}(t)\Big|\mathbf{v}(t)\Big\rangle_{H^{-1}(U),H_0^1(U)}dt \to \int_0^T \langle \mathbf{u}'(t)|\mathbf{v}(t)\rangle_{H^{-1}(U),H_0^1(U)}dt.$$

Also, if we consider the operator $T_v:\mathbf{w}\mapsto\int_0^T B[\mathbf{w}(t),\mathbf{v}(t);t]dt$ for any $\mathbf{w}\in L^2\big(0,T;H_0^1(U)\big)$, we can see that

$$\begin{aligned}
|T_v\mathbf{w}| &= \left|\int_0^T B[\mathbf{w}_m(t),\mathbf{v}(t);t]dt\right| \\
&\leq \int_0^T |B[\mathbf{w}_m(t),\mathbf{v}(t);t]|dt \\
&\leq \int_0^T \alpha\|\mathbf{w}_m(t)\|_{H^1(U)}\|\mathbf{v}(t)\|_{H^1(U)}dt \\
&\leq \alpha\left(\int_0^T \|\mathbf{w}_m(t)\|^2_{H^1(U)}\right)^{\frac{1}{2}}\left(\int_0^T \|\mathbf{v}(t)\|^2_{H^1(U)}\right)^{\frac{1}{2}} \\
&= \alpha\|\mathbf{w}_m\|_{L^2(H^1(U))}\|\mathbf{v}\|_{L^2(H^1(U))}.
\end{aligned}$$

For some $\alpha>0$ that only depends on $U,L$.
Thus $\|T_v\|_{\left(L^2\left(0,T;H_0^1(U)\right)\right)^*}\leq\alpha\|\mathbf{v}\|_{L^2(H^1(U))}$, so $T_v\in\big(L^2\big(0,T;H_0^1(U)\big)\big)^*$.
Since $\mathbf{u}_{m_{j_l}}\rightharpoonup\mathbf{u}$, we have that

$$\int_0^T B[\mathbf{u}_{m_{j_l}}(t),\mathbf{v}(t);t]dt\to\int_0^T B[\mathbf{u}(t),\mathbf{v}(t);t]dt.$$

We now have

$$\begin{aligned}
\int_0^T \langle \mathbf{f}(t),\mathbf{v}(t)\rangle_{L^2(U)}dt &= \lim_{l\to\infty}\int_0^T \langle \mathbf{f}(t),\mathbf{v}(t)\rangle_{L^2(U)}dt \\
&= \lim_{l\to\infty}\left(\int_0^T \Big\langle \mathbf{u}'_{m_{j_l}}(t)\Big|\mathbf{v}(t)\Big\rangle_{H^{-1}(U),H_0^1(U)}dt + \int_0^T B[\mathbf{u}_{m_{j_l}}(t),\mathbf{v}(t);t]dt\right) \\
&= \lim_{l\to\infty}\int_0^T \Big\langle \mathbf{u}'_{m_{j_l}}(t)\Big|\mathbf{v}(t)\Big\rangle_{H^{-1}(U),H_0^1(U)}dt + \lim_{l\to\infty}\int_0^T B[\mathbf{u}_{m_{j_l}}(t),\mathbf{v}(t);t]dt \\
&= \int_0^T \langle \mathbf{u}'(t)|\mathbf{v}(t)\rangle_{H^{-1}(U),H_0^1(U)}dt + \int_0^T B[\mathbf{u}(t),\mathbf{v}(t);t]dt
\end{aligned}$$

Since such $\mathbf{v}$ are dense in $L^2\big(0,T;H_0^1(U)\big)$ and both sides of the above equality are continuous, we can extend it so that $\forall \mathbf{v}\in L^2\big(0,T;H_0^1(U)\big)$,

$$\int_0^T \langle \mathbf{f}(t),\mathbf{v}(t)\rangle_{L^2(U)}dt=\int_0^T \langle \mathbf{u}'(t)|\mathbf{v}(t)\rangle_{H^{-1}(U),H_0^1(U)}dt+\int_0^T B[\mathbf{u}(t),\mathbf{v}(t);t]dt.$$

Now consider any $v\in H_0^1(U)$ and any $\phi\in C_c^\infty(0,T)$, we always have $v\phi\in C\big(0,T;H_0^1(U)\big)\subset L^2\big(0,T;H_0^1(U)\big)$. Thus we have

$$\int_0^T \langle \mathbf{f}(t),v\phi(t)\rangle_{L^2(U)}dt=\int_0^T \langle \mathbf{u}'(t)|v\phi(t)\rangle_{H^{-1}(U),H_0^1(U)}dt+\int_0^T B[\mathbf{u}(t),v\phi(t);t]dt$$
$$\int_0^T \phi(t)\langle \mathbf{f}(t),v\rangle_{L^2(U)}dt=\int_0^T \phi(t)\Big(\langle \mathbf{u}'(t)|v\rangle_{H^{-1}(U),H_0^1(U)}+B[\mathbf{u}(t),v;t]\Big)dt$$

Since this works for all $\phi\in C_c^\infty(0,T)$, we must have for a.e. $t\in[0,T]$,

$$\langle \mathbf{f}(t),v\rangle_{L^2(U)}=\langle \mathbf{u}'(t)|v\rangle_{H^{-1}(U),H_0^1(U)}+B[\mathbf{u}(t),v;t].$$

Now consider any $\mathbf{v}\in C^1\big(0,T;H_0^1(U)\big)$ such that $\mathbf{v}(T)=0$.
By IBP 4.16.2, we have

$$\begin{aligned}0&=\langle \mathbf{u}(T),\mathbf{v}(T)\rangle_{L^2(U)}\\&=\langle \mathbf{u}(0),\mathbf{v}(0)\rangle_{L^2(U)}+\int_0^T\Big(\langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}+\langle \mathbf{v}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}\Big)d\tau\\0&=\Big\langle \mathbf{u}_{m_{j_l}}(T),\mathbf{v}(T)\Big\rangle_{L^2(U)}\\&=\Big\langle \mathbf{u}_{m_{j_l}}(0),\mathbf{v}(0)\Big\rangle_{L^2(U)}+\int_0^T\left(\Big\langle \mathbf{u}'_{m_{j_l}}(\tau)\Big|\mathbf{v}(\tau)\Big\rangle_{H^{-1}(U),H_0^1(U)}+\Big\langle \mathbf{v}'(\tau)\Big|\mathbf{u}_{m_{j_l}}(\tau)\Big\rangle_{H^{-1}(U),H_0^1(U)}\right)d\tau.\end{aligned}$$

Since $\mathbf{v}\in L^2\big(0,T;H_0^1(U)\big)\cong L^2\big(0,T;H^{-1}(U)\big)^*$ and $\mathbf{u}'_{m_{j_l}}\rightharpoonup \mathbf{u}'$ in $L^2\big(0,T;H^{-1}(U)\big)$, we have

$$\lim_{l\to\infty}\int_0^T\Big\langle \mathbf{u}'_{m_{j_l}}(\tau)\Big|\mathbf{v}(\tau)\Big\rangle_{H^{-1}(U),H_0^1(U)}d\tau=\int_0^T\langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}d\tau.$$

Similarly,

$$\lim_{l\to\infty}\int_0^T\Big\langle \mathbf{v}'(\tau)\Big|\mathbf{u}_{m_{j_l}}(\tau)\Big\rangle_{H^{-1}(U),H_0^1(U)}d\tau=\int_0^T\langle \mathbf{v}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}d\tau.$$

Also, since $\lim_{m\to\infty}\mathbf{u}_m(0)=g$ in $L^2(U)$, we have that the subsequence $\lim_{l\to\infty}\mathbf{u}_{m_{j_l}}(0)=g$ as well.
Thus

$$\lim_{l\to\infty}\Big\langle \mathbf{u}_{m_{j_l}}(0),\mathbf{v}(0)\Big\rangle_{L^2(U)}=\langle \mathbf{u}(T),\mathbf{v}(T)\rangle_{L^2(U)}.$$

Thus we have

$$\begin{aligned}0&=\langle \mathbf{u}(0),\mathbf{v}(0)\rangle_{L^2(U)}+\int_0^T\Big(\langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}+\langle \mathbf{v}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}\Big)d\tau\\&=\lim_{l\to\infty}\left(\Big\langle \mathbf{u}_{m_{j_l}}(0),\mathbf{v}(0)\Big\rangle_{L^2(U)}+\int_0^T\left(\Big\langle \mathbf{u}'_{m_{j_l}}(\tau)\Big|\mathbf{v}(\tau)\Big\rangle_{H^{-1}(U),H_0^1(U)}+\Big\langle \mathbf{v}'(\tau)\Big|\mathbf{u}_{m_{j_l}}(\tau)\Big\rangle_{H^{-1}(U),H_0^1(U)}\right)d\tau\right)\\&=\langle g,\mathbf{v}(0)\rangle_{L^2(U)}+\int_0^T\Big(\langle \mathbf{u}'(\tau)|\mathbf{v}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}+\langle \mathbf{v}'(\tau)|\mathbf{u}(\tau)\rangle_{H^{-1}(U),H_0^1(U)}\Big)d\tau.\end{aligned}$$

We thus have $\langle \mathbf{u}(0),\mathbf{v}(0)\rangle_{L^2(U)}=\langle g,\mathbf{v}(0)\rangle_{L^2(U)}$ for any such $\mathbf{v}$.
Notice that for any $v\in H_0^1(U)$, we can simply let $\mathbf{v}(t):=\frac{T-t}{T}v$, which satisfies the requirement, and $\mathbf{v}(0)=v$.
Thus $\langle \mathbf{u}(0),v\rangle_{L^2(U)}=\langle g,v\rangle_{L^2(U)}$ for any $v\in H_0^1(U)$.

Since $H_0^1(U)$ is dense in $L^2(U)$, we have that $\langle \mathbf{u}(0), v\rangle_{L^2(U)} = \langle g, v\rangle_{L^2(U)}$ for any $v \in L^2(U)$.
This proves

$$\mathbf{u}(0) = g.$$

□

**Theorem 4.28.** *A weak solution to our IBVP is unique.*

*Proof.* Assume $\mathbf{u}_1, \mathbf{u}_2$ are both weak solutions to our IBVP.
Then $\forall v \in H_0^1(U)$, for a.e. $t \in [0,T]$,

$$\langle \mathbf{u}_1'(t)|v\rangle + B[\mathbf{u}_1, v; t] = \langle \mathbf{u}_2'(t)|v\rangle + B[\mathbf{u}_2, v; t] = \langle \mathbf{f}(t), v\rangle_{L^2(U)},$$

and

$$\mathbf{u}_1(0) = g = \mathbf{u}_2(0).$$

Let $\mathbf{u} := \mathbf{u}_1 - \mathbf{u}_2$, we have that

$$\langle \mathbf{u}'(t)|v\rangle + B[\mathbf{u}, v; t] = 0, \ \forall v \in H_0^1(U), \text{for a.e. } t \in [0,T],$$

and

$$\mathbf{u}(0) = 0.$$

Choosing $v = \mathbf{u}(t) \in H_0^1(U)$, we have that

$$\langle \mathbf{u}'(t)|\mathbf{u}(t)\rangle + B[\mathbf{u}(t), \mathbf{u}(t); t] = 0.$$

By 4.16.3, we have that for a.e. $t \in [0,T]$,

$$\begin{aligned}
\frac{d}{dt}\|\mathbf{u}(t)\|_{L^2(U)}^2 &= 2\langle \mathbf{u}'(t)|\mathbf{u}(t)\rangle \\
&= -2B[\mathbf{u}(t), \mathbf{u}(t); t] \\
&\leq 2\gamma\|\mathbf{u}(t)\|_{L^2(U)}^2 - 2\beta\|\mathbf{u}(t)\|_{H^1(U)}^2 \\
&\leq 2\gamma\|\mathbf{u}(t)\|_{L^2(U)}^2,
\end{aligned}$$

where $\gamma \geq 0, \beta > 0$ are constants similar in 3.5.
Take $\eta(t) := \|\mathbf{u}(t)\|_{L^2(U)}^2$, by Gronwall's inequality, we have that $\forall t \in [0,T]$,

$$\|\mathbf{u}(t)\|_{L^2(U)}^2 \leq \exp(2t)\|\mathbf{u}(0)\|_{L^2(U)}^2 = 0.$$

Thus $\mathbf{u}(t) = 0$ and so $\mathbf{u}_1 = \mathbf{u}_2$ is unique. □

# 5 Semi-group Theory

This follows from Section 7.4 in [2], with more details.

**Definition 5.1** (Strongly Continuous Semigroup)**.** Let $X$ be a Banach space and $B(X)$ be the space of bounded linear operators on $X$. A family of operators $(S(t))_{t\geq 0} \subset B(X)$ is called a *strongly continuous semigroup* (or a $C_0$-*semigroup*) if it satisfies the following conditions:

1. **Identity at zero:**
$$S(0) = I$$
where $I$ is the identity operator on $X$.

2. **Semigroup property:**
$$S(t+s) = S(t)S(s) \quad \text{for all } t, s \geq 0.$$

3. **Strong continuity:** For every $x \in X$, the map
$$[0,\infty) \ni t \longmapsto S(t)x \in X$$
is continuous with respect to the norm topology of $X$. That is, for all $t_0 \geq 0$,
$$\lim_{t\to t_0} \|S(t)x - S(t_0)x\| = 0.$$

The **infinitesimal generator** $A$ of $(S(t))_{t\geq 0}$ is the linear operator defined by:
$$Ax = \lim_{t\to 0^+} \frac{S(t)x - x}{t}$$
The **domain** of $A$, denoted by $D(A)$, is the set of all elements $x \in X$ for which this limit exists in the norm topology of $X$:
$$D(A) = \left\{ x \in X : \lim_{t\to 0^+} \frac{S(t)x - x}{t} \text{ exists} \right\}$$

**Example 5.0.1.** Let $X = \mathbb{R}$. Consider $S(t)x := e^{\lambda t}x$ for some $\lambda \in \mathbb{R}$. We have that $S(0)x = x = Ix$. Also, $S(t+s) = e^{\lambda(t+s)} = e^{\lambda t}e^{\lambda s} = S(t)S(s)$. Also, $\|T(t)x - T(t_0)x\| = \left|e^{\lambda t} - e^{\lambda t_0}\right||x| \to 0$. Thus, $(S(t))_{t\geq 0}$ is a strongly continuous semigroup. Now consider $Ax := \lambda x$, we have that
$$\lim_{t\to 0^+} \frac{S(t)x - x}{t} = \lim_{t\to 0^+} \frac{e^{\lambda t} - e^{\lambda 0}}{t} x = \frac{d}{dt}(e^{\lambda t})|_{t=0}x = \lambda e^{\lambda t}|_{t=0}x = \lambda x = Ax.$$
Thus, $A$ is the infinitesimal generator of $S(t)$. Also, $D(A) = \mathbb{R} = X$.

Notice that $\mathbf{x}(t) = S(t)x = e^{\lambda t}x$ is the solution of the ODE
$$\frac{d}{dt}\mathbf{x} = \lambda \mathbf{x} = Ax.$$

Indeed, the strongly continuous semigroup theory is developed as a generalization of this example. Thus, we sometimes abuse the notation, and write $S(t)$ as $e^{tA}$.

**Theorem 5.1** (Differential properties of semigroups)**.** *[2, Theorem 7.4.1] Let $(S(t))_{t\geq 0}$ is a strongly continuous semigroup, and $A$ be its infinitesimal generator. Suppose $u \in D(A)$, then the following holds:*

1. *For any $t \geq 0$, $S(t)u \in D(A)$, with $AS(t)u = S(t)Au$.*
2. *For any $t > 0$, the mapping $t \mapsto S(t)u$ is in $C^1(0,\infty)$ with $\frac{d}{dt}S(t)u = AS(t)u$.*

*Proof.* 1. We have that

$$\begin{aligned}\lim_{s\to 0^+}\frac{S(s)S(t)u-S(t)u}{t} &= \lim_{s\to 0^+}\frac{S(t)S(s)u-S(t)u}{t}\\ &= \lim_{s\to 0^+} S(t)\frac{S(s)u-u}{t}\\ &= S(t)\lim_{s\to 0^+}\frac{S(s)u-u}{t}\\ &= S(t)Au,\end{aligned}$$

where the first step is due to the semigroup property, and in the last two steps we used the fact that $S(t)\in B(X)$, so it is linear and continuous. Thus $S(t)u\in D(A)$, and $AS(t)u = S(t)Au$.

2. We have that

$$\begin{aligned}&\lim_{h\to 0^+}\left\|\frac{S(t)u-S(t-h)u}{t}-AS(t)u\right\|\\ =&\lim_{h\to 0^+}\left\|\frac{S(t-h)S(h)u-S(t-h)u}{t}-AS(t)u\right\|\\ =&\lim_{h\to 0^+}\left\|S(t-h)\frac{S(h)u-u}{t}+S(t-h)Au-S(t-h)Au+AS(t)u\right\|\\ \le&\lim_{h\to 0^+}\left\|S(t-h)\left(\frac{S(h)u-u}{t}-Au\right)\right\|+\|S(t)Au-S(t-h)Au\|\\ \le&\lim_{h\to 0^+}\left\|\frac{S(h)u-u}{t}-Au\right\|+\lim_{h\to 0^+}\|S(t)Au-S(t-h)Au\|\\ =&\,0.\end{aligned}$$

Similarly, $\lim_{h\to 0^+}\left\|\frac{S(t+h)u-S(t)u}{t}-AS(t)u\right\| = 0$. Thus, $\frac{d}{dt}S(t)u = AS(t)u = S(t)Au$, which is continuous.

□

**Theorem 5.2** (Properties of generators)**.** *[2, Theorem 7.4.2] Let $(S(t))_{t\ge 0}$ is a strongly continuous semigroup, and $A$ be its infinitesimal generator. We have that $A$ is a closed operator, and the domain $D(A)$ is dense in $X$.*

*Proof.* Fix any $u\in X$, and define $u^t := \int_0^t S(s)uds$ for $t>0$. Since $S(s)u$ is continuous at 0, we have that

$$\lim_{t\to 0^+}\frac{u^t}{t} = \lim_{t\to 0^+}\frac{1}{t}\int_0^{0+t} S(s)uds = S(0)u = u.$$

Now we claim that each $u^t$ is in $D(A)$. □

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